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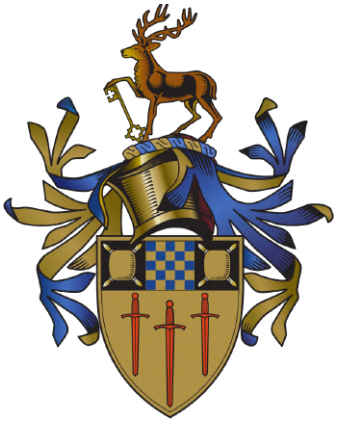
Muir, S. H. (2023). New Works for Classical Ukulele [University of Surrey].
https://doi.org/10.15126/thesis.900753
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New Works for Classical Ukulele

by

Samantha Muir



Submitted for the Degree of Doctor of Philosophy

School of Arts
Faculty of Arts and Social Sciences

Supervisors:

Stephen Goss
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New Works for Classical Ukulele

A portfolio and thesis on the theory and practice of the classical ukulele

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Abstract

To date, despite its rising popularity, no academic research has been undertaken on the classical ukulele style. That is to say there is no academic context or foundation for the techniques and repertoires that define the classical ukulele. Thus, while the overall purpose of this research inquiry was forward looking in its aim of creating new works, it was also backward looking in its desire to better understand and draw on the instrument's European heritage. The research outcomes presented in this portfolio are, therefore, both practical (scores and recordings) and theoretical (papers and articles).

The practical outcomes are contained in portfolio folders B (Scores) and C (Recordings) and meet the primary aim of the research inquiry to create new, idiomatic works for classical ukulele in order to help establish a contemporary classical repertoire for the instrument. The new works include original pieces, arrangements and commissioned pieces. A number of didactic works are also included as evidence of my continued engagement with the ukulele community and ongoing commitment to promote classical ukulele. The publication of my *The Classical Ukulele Method* in 2019 was in fact the first ukulele method to offer a systematic approach to learning the techniques associated with the classical ukulele style.

The COVID-19 pandemic and resulting lockdowns affected my research in unexpected, and surprisingly positive, ways. With all live events and face-to-face teaching cancelled, new opportunities arose through social media and online platforms such as Zoom. During this period of isolation and restriction my engagement with the broader ukulele community increased. Despite the many frustrations of unstable internet connections and latency issues, which made playing together impossible, many workshops and performances were held online. In 2020 I created a Patreon channel offering pre-recorded tutorials and live workshops on classical ukulele techniques and repertoire. Connecting with many aspiring classical ukulele players across the globe (USA, Canada, Australia, Japan, EU and UK) highlighted the importance of developing a robust curriculum for the classical ukulele. Details of all events are contained in Folder D (Extra Material).

The thesis is in two sections. The first section explores 'The Classical Ukulele in Theory' and the second 'The Classical Ukulele in Practice'. The theoretical material includes research into

the Madeiran machete, one of the direct forerunners of the ukulele. The culmination of this research was an article, commissioned and published by *Revista Portuguesa de Educação Artística* (2022), which contributed new insights into the musical career of Clara Phelps, a student of the nineteenth century Madeiran composer and machete player Candido Drummond. Prior to writing the article I was commissioned by the Associação Regional de Educação Artística, Madeira to record the first full CD of pieces for machete and guitar by Candido Drummond (*The Music of Candido Drummond de Vasconcelos, 2019).*

The second part of 'The Classical Ukulele in Theory' contains a detailed discussion of the classical ukulele style and looks at the historical use of guitar techniques on the ukulele. This discussion provides a background and a context for Section 2 'The Classical Ukulele in Practice'. This section then explores the different techniques used in the new works for classical ukulele and illustrates how many of these techniques were adapted from the classical guitar, the four course Renaissance guitar and the five course Baroque guitar. The full texts of the theoretical papers and articles which are not included in the body of the thesis are contained in Folder A (Published Papers and Presentations).

Although classical ukulele is often regarded as a new and more formal approach to the ukulele and one that is largely driven by players with a background in classical guitar, my research findings demonstrate how many of the earliest players also approached the ukulele from a guitar perspective. In this context the current classical ukulele movement is perhaps best viewed as a reawakening, revitalising and reimagining of old techniques, methods and repertoires, driven by a vision to create contemporary classical works for the instrument.

Acknowledgements

I would like to begin by thanking my principal supervisor Stephen Goss for his constant support and guidance throughout this project. Thanks also to my secondary supervisor Milton Mermikides for his advice and participation. I would also like to thank the many people who assisted me with various aspects of my research, in particular Giovanni Albini, Jim Tranquada and Paulo Esteireiro. Special thanks to the International Guitar Research Centre at the University of Surrey, and the Cambridge Consortium for Guitar Research for giving me a platform to present my work and for offering critical advice and moral support. My heartfelt gratitude goes to all the composers I have had the pleasure and the privilege of working with over the last five years and who contributed invaluable new and innovative compositions to the portfolio. Special mention to all the classical ukulele fans, from near and far, who continue to support my work. Their dedication and enthusiasm are helping to shine a light on the charm and unique potential of classical ukulele. I am also grateful to *Ukulele* magazine for including my arrangements and helping to promote classical ukulele to the broader international ukulele community. Last, but by no means least, I would like to thank my family and friends for their steadfast belief in me.

Declaration of Originality

This thesis and the work to which it refers are the results of my own efforts. Any ideas, data, images or text resulting from the work of others (whether published or unpublished) are fully identified as such within the work and attributed to their originator in the text, bibliography or in footnotes. This thesis has not been submitted in whole or in part for any other academic degree or professional qualification. I agree that the University has the right to submit my work to the plagiarism detection service TurnitinUK for originality checks. Whether or not drafts have been so assessed, the University reserves the right to require an electronic version of the final document (as submitted) for assessment as above.

New Works for Classical Ukulele

by

Samantha Muir

Submitted in partial fulfilment for the degree of
Doctor of Philosophy (PhD) in Music

Department of Music & Media, School of Arts

University of Surrey
March 2023

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'If the Hardwick Hall guitar is shown actual size, it is indeed a small instrument: a double-course Tudor ukulele.'

Christopher Page, 2015
The Guitar in Tudor England
CUP

'It is not probable that the machete will ever emigrate from Madeira.'

John Dix, 1851
A Winter in Madeira, A Summer in Florence & Spain
Holdredge, NY

INTRODUCTION

This thesis and the accompanying portfolio, though born of heuristic exploration and research inquiries, rather than specific questions, (Nelson, 2013) is unified by a motivation to illuminate the historical roots, pedagogical development and enriching of repertoire (through transcription and original composition) of the classical ukulele. The guide, which is included as a separate document, provides an overview of the submission.

The introduction states the research inquiry and its overall aim and related objectives. As the reader may be unacquainted with the term 'classical ukulele' a brief definition is given - although it should be stressed that the thesis and the accompanying portfolio will demonstrate, explore and make explicit the exact, or inexact, nature of classical ukulele. The summary will re-evaluate the term 'classical ukulele' in the context of the research findings. It should be established, however, that the term 'classical ukulele' comes from John King's pioneering book/CD *The Classical Ukulele* (Flea Market Music) published in 2004. As King's work is widely considered the catalyst for the current classical ukulele movement, and the author has not located any previous publications using this specific term, the author's literature review focused on publications for classical ukulele from 2004. An overview of the literature review (see p.11) is presented in order to demonstrate the lack of resources for classical ukulele in 2017 (at the commencement of this PhD) and to establish the author's reason for embarking on this project. It also helps to illustrate the author's contribution, not only to our knowledge of classical ukulele from musical, technical and historical perspectives, but to its repertoire and the techniques that it encompasses.

To the best of the author's knowledge this is the first PhD on classical ukulele. As such, an essential function of this portfolio is to create an academic standing for the instrument. The new works embrace the term original in the broadest context: as something that springs, or is derived, from the 'origin' or source (OED). Thus, the aim of creating an original, contemporary classical repertoire (supported by a selection of didactic material) for the ukulele proposes a future for the instrument as a solo concert instrument. A theoretical discussion of the techniques used in the new works is included in Section 2 (see 'The Classical Ukulele in Practice'). Meanwhile, the four papers included in the portfolio of new works offer a fuller understanding of the instrument's origins; in particular the ukulele's European heritage and its connection to the machete de Braga (or machete), as well as insights into the techniques used in some of the earliest ukulele methods.

The Research Inquiry

In his book *Practice as Research in the Arts,* Robin Nelson suggests the term *research inquiry* may be preferable to *research question*. His reasoning for this is that ‘questions may imply answers and the kinds of work typically undertaken in the PaR [Practice as Research] PhD context, while they yield findings, do not typically produce solutions to problems in the mode of answers.’ (Nelson, 2013, 96) Conducting a research inquiry allowed a degree of artistic and creative freedom which yielded unexpected outcomes such as my exploration into tintinnabuli technique and my discovery that the machete (a forerunner of the ukulele) had been played in England in the late nineteenth century. The portfolio should be viewed as a collection of theoretical and practical research findings based on the theme of classical ukulele. The contribution to knowledge can be summarised as follows:

1. Contributing to our understanding of the history of the ukulele and its various stylistic strands.
2. Contributing through transcription and pedagogical material to the repertoire and technique of classical ukulele.
3. Contributing through the commissioning and composition of contemporary repertoire for the instrument.

Nelson suggests that in a complex or multilayered inquiry, such as this, the examiner be given a ‘clue’ to help them navigate the different components. He uses the archaic form *clew* (clue) to denote a thread that guides the examiner through the various findings of the inquiry (Nelson 2013, 10). To this end the thread which binds all the elements of this portfolio is classical ukulele. How the thread binds the intricacies of the portfolio is fully explained in the section on methodology. An explanation of the term classical ukulele is given after the aim and objectives of the research inquiry are stated.

Following Nelson’s model, the core of the inquiry is articulated by an overall aim. A series of objectives then cover the different activities undertaken in order to achieve that aim.

Aim:

- To create original, contemporary and idiomatic works for classical ukulele through the application and adaptation of techniques, both historical and modern, commonly associated with the guitar.

Objectives:

- Define the term classical ukulele and understand its context within the ukulele scene.
- Explore the history of the ukulele in order to better understand its rich cultural and musical heritage and use this historical knowledge to inform my practice.
- Document any new knowledge that may further our understanding of the ukulele.
- Explore, through compositions and arrangements, how new and perhaps old guitar techniques might be applied to the ukulele.
- Provide documented evidence of the findings.
- Engage with other ukulele players to promote the development of classical ukulele and determine ways in which this can be achieved.
- Create new didactic works to aid and encourage the development of classical ukulele.

A Brief Definition of Classical Ukulele

As this submission is an exploration into the classical ukulele style it is perhaps disingenuous to be too specific in defining the term classical ukulele from the outset as this inquiry is intended to deepen and broaden our understanding rather than to limit our expectations. Nevertheless, some guidelines are offered here to provide a context for classical ukulele and enable the reader to follow the 'clew' or thread that binds this portfolio together.

The term classical ukulele first came into use following the publication of John King's (1953-2009) groundbreaking book/CD *The Classical Ukulele* published in 2004 by Flea Market Music. The style and content of King's book offered a surprising deviation for a publisher best known for producing collections of ukulele songbooks with chords and lyrics. In 2004 there were no comparable publications for the ukulele. King's arrangements were diverse; ranging from *Jesu, Joy of Man's Desiring* by JS Bach to a hula (Alekoki) by David Kalakaua, the last King of Hawaii. At the time of publication *The Classical Ukulele* (2004) may have seemed like an oddity, but its enduring legacy is made explicit both within, and by, this thesis. Although not specifically about John King's work, this research inquiry explores and extends the approach and techniques first laid down by King. It is, therefore, inevitable that many references are made to his work.

Unfortunately, King did not provide a comprehensive written framework for what he thought constituted the classical ukulele style and, until this PhD, no academic study has been undertaken to better understand, or develop, his approach. Perhaps King thought his arrangements spoke for themselves, or perhaps he simply enjoyed the enigma of playing works by JS Bach,

Mozart, Beethoven and other celebrated composers on a small, four string guitar-shaped instrument that is most commonly regarded as either a child's toy, a comedy prop, or an instrument that is only ever strummed. At this stage, the most logical way of understanding King's technical and stylistic approach is to draw a parallel between classical guitar and classical ukulele. King, like many classical ukulele players who have followed him (including the author of this thesis), came from a classical guitar background and its influence is a constant, inescapable presence. In his introduction to *The Classical Ukulele*, King went as far as stressing that tone production was 'central to the classic aesthetic.' (King, 2004, 6) The pure, rounded tone associated with classical guitar is certainly one of the outstanding features aspired to be classical ukulele players. Other characteristics, such as hand positions and stylistic approaches are equally important. One notable difference is that while the term classical guitar is also used to identify a particular type of instrument suited to playing in the classical guitar style, classical ukulele can be played on most standard ukuleles. Furthermore, suggesting that classical ukulele simply parallels classical guitar, greatly undermines how, in recent years, a new vanguard of ukulele players (including the author) have used the idiomatic features of the instrument to create highly innovative and original music.

As no academic study has yet been undertaken on classical ukulele this research inquiry has, at times, felt like taking the lid off the proverbial can of worms. Nevertheless, it is the aim of this portfolio to explore and document how guitar techniques can be used as a basis for creating original contemporary repertoire for the ukulele. Having made the comparison with classical guitar, the more general term guitar techniques will be now be used in order to extend the discussion to include techniques used by historically related instruments such as the Baroque guitar.[1] Although the word 'classical' may appear contradictory in the context of creating new, contemporary repertoire for the instrument the term classical ukulele is used in the interests of clarity and continuity, as well as an acknowledgement to the pioneering work of John King. At this stage there does not appear to be a better way of indicating the classical ukulele approach. If the research findings suggest otherwise this will addressed in the conclusion. The main purpose of the inquiry, however, is not to debate nomenclature but to explore through practice if and how guitar techniques may provide a framework that realises and elucidates the unique qualities of the ukulele while recognising its rich cultural and musical heritage. Thus, the new works include theoretical and practical elements and are intended to provide substantial insights rather than definitive conclusions. (Nelson, 2013, 29)

[1] John King made extensive use of the Baroque guitar technique of campanella in his arrangements. See Chapter 2.1 for a full discussion on campanella.

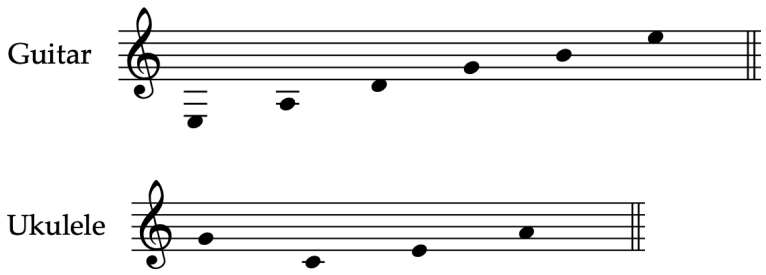
Methodologies

Tool & Technique

'…instruments may know things for their users.' (De Souza, 2017, 24)

I came to the ukulele with the technique of a trained and experienced classical guitarist.[2] From the moment I picked up the ukulele I knew how to pluck the strings to make a pleasing sound and I knew how and where to place my fingers on the fretboard in order to produce different notes and chords. This is because the fretboards of the guitar and ukulele have the same layout. The frets are a semitone apart on both instruments and as I play up the fingerboard the higher frets produce higher sounds. But, on the ukulele, the positions of the notes do not correspond to the sounds I am used to hearing on the the guitar. For example, when I press the first fret of the second string (from here on Jonathan de Souza's method of indicating fret then string as 1,2) the note I expect, in my guitar brain, is C. But the note I hear on the ukulele is F. The explanation is simple: the ukulele is tuned a fourth higher than the guitar (see figure 1). By this reasoning I might expect that if I placed a capo on the fifth fret of the guitar and ignored the lower two strings, it would now become a ukulele. It does not. For one thing, the smaller body of the ukulele produces a different timbre. There is also the matter of the woods used to make the instrument and the different techniques used by the luthiers. The real issue, however, is that with the capo positioned on the fifth fret of the guitar, the fourth string sounds an octave lower than the open fourth string of a ukulele.

Figure 1. The tunings of the guitar and the ukulele



As every ukulele player knows, the soprano, concert and tenor ukuleles all use re-entrant tuning and, consequently, the fourth string is tuned an octave higher than the corresponding string on the

[2] In sections discussing my own first-hand experience I use first-person rather than 'the author'.

linear tuned guitar.[3] That might seem to be stating the obvious. But the creative possibilities of re-entrant tuning and, to some extent the mental disruption caused by it, are essential to this research inquiry. I want to play the ukulele using the technique that feels most natural to me. In other words using the fingerpicking technique of a classical guitarist. Like John King, I enjoy the beautiful harp-like sound I can produce on the ukulele. I do not want to change the nature of the instrument to suit my linear expectations, I want to know what the upside down world of re-entrant tuning can teach me.

In *Music At Hand* Jonathan de Souza discusses how knowledge may not only belong to the user of a tool, but to the tool itself. In this sense 'instruments may know things for their users.' (De Souza, 2017, 24) In de Souza's model the instrument is viewed as a type of prosthesis: an addition or extension of the musician. But an instrument is not there to simply assist the player to *make* music, it also there to help the player *find* music. By understanding the unique qualities of an instrument, its hidden knowledge is revealed. To this end, when the technique of the player, and the unique qualities of the instrument come together, they enter into a symbiotic relationship. At this point 'The instrument, together with my action, reveals a world.' (De Souza, 2017, 23) Put in the context of this research inquiry there are two expected outcomes:

- The re-entrant tuning, smaller scale, and different timbral nuances of the ukulele will alter and disrupt my guitar knowledge.
- Meanwhile, my technical facility will undergo a process of change, adaptation, and innovation which will create new opportunities and enable the ukulele to *speak* in new ways.

Although I cannot escape from my background as a classical guitarist, and I have no intention of pretending to do so, my purpose is not simply to impose a classical guitar approach on the ukulele. My purpose is to utilise and develop the skills I already possess in order to extend the possibilities of the classical ukulele approach. This will be achieved through research into historically related instruments, including the machete de Braga and the Renaissance guitar. The techniques and repertoires of these instruments will, I anticipate, inform and deepen my understanding of the ukulele and further illuminate its potential. My approach is, therefore, a two-way process which will enable us (myself and the instrument) to work together to *find* new music.

[3] The differences between linear and re-entrant are discussed throughout this thesis.

The Quartet of Selves

At an early stage I decided that my research inquiry would make a more substantial contribution to knowledge if I incorporated all areas of my practice to include performance, composition/ arrangement, teaching, and research. In order to clarify these four roles I devised a Practice as Research (PaR) persona which I called the Quartet of Selves. My PaR persona operates like a string quartet. The performer-self is like the first violinist who leads the quartet but, like a string quartet, all 'selves' are vital, interconnected and in constant dialogue with each other. The PaR persona allowed a holistic approach to my research inquiry.

In the Quartet of Selves each self interacts with the other 'selves' in order to perceive, realise, clarify and articulate the research insights. Each member of the quartet may at times take the role of the soloist but the other selves remain as passive participants ready to enter into the dialogue as needed. How the 'selves' complement each other is illustrated by the following example: The teacher-self structures and delivers a workshop on *pima* finger-picking techniques. The composer-self contributes a suitable study piece using *pima* style. The performer-self takes the lead by demonstrating the piece to the class. Finally, in what might be perceived as the coda, the researcher-self then reflects on, and articulates, the insights in the commentary. Collectively, the 'selves' form a community which is unified and motivated by the common aim of creating new works to help further our understanding of classical ukulele and aid its development.

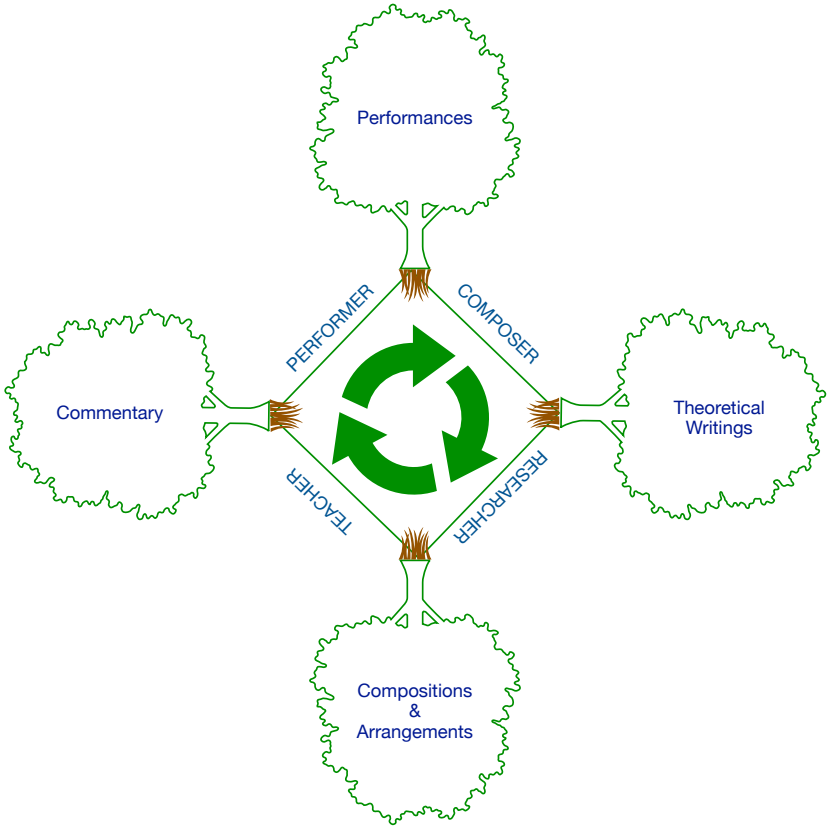
The Mycorrhizal Research Model

Having established the Quartet of Selves as way of conceptualising my different roles as performer, composer/arranger, teacher and researcher it was then necessary to find a way of representing the symbiotic nature of the research outcomes. Nelson's idea of a conceptual metaphor, such as Deleuze and Guattari's rhizome[4] 'with a tangle of interconnected thought' which allows for multiple strands of inquiry, (Nelson, 2013, 54) provided the framework. In botany the rhizome is 'an elongated, usually horizontal, subterranean stem which sends out roots and leafy shoots at intervals along its length' (OED online). Applied to research it allows an ever expanding narrative as new ideas constantly spring to mind and forge new lines of inquiry. For a creative practitioner this idea is compelling as it allows artistic free thinking. But, while Nelson argues that this model leads to an 'opening out' of disciplines, (Nelson, 2013, 54) I would argue that it may also result in a tangled web of research outcomes where the clew, or thread, is difficult to unravel. For this reason I decided

[4] In the writings of Deleuze and Guattari (see particularly *A Thousand Plateaus*, 1980) a rhizomatic theory, with its possibilities of making multiple connections or networks, is proposed as an alternative to hierarchical ways of thinking.

to develop my own model that borrowed from the idea of the rhizome but offered a more controlled approach.

Figure 2. An illustration of the author's 'Mycorrhizal Research Model'



My model (see figure 2) was based on the research of biologist and tree ecologist Suzanne Simard.[5] In the 1990s Simard's research found that forests should be regarded as super-organisms and not, as previously thought, a disparate collection of individual trees all competing for resources. Trees, in fact, share resources and communicate with each other through complex underground networks of roots and fungi called mycorrhiza. The word mycorrhiza refers to this symbiotic relationship between tree roots and fungi.

In the mycorrhizal research model (see figure 2) the portfolio forms a small, interconnected forest with the outcomes represented by four trees. Operating as a quartet of practitioners 'playing'

[5] Simard is a professor & faculty member of the Department of Forest and Conservation Sciences at the University of British Columbia. Her groundbreaking research on mycorrhizal networks has opened a new field of study and changed our perception of how trees interact.

to the theme of classical ukulele the mycorrhizal network conceptualises how the constant sharing and reprocessing of resources enables all the ‘trees’ to grow and flourish.

Summary of the Literature Review

Owing to newness of this field of research, and the fact that the primary aim of this inquiry was to create new works for classical ukulele, the literature review focused on collating the existing publications for classical and fingerstyle ukulele. The findings helped to determine the need for a holistic approach led by the Quartet of Selves.

When I first started to play the ukulele in 2012 there were very few publications for classical ukulele. Furthermore, the main publications for classical ukulele appeared to be by classical guitarists. Another factor in helping to shape my research proposal was the obvious lack of original material for classical ukulele. The purpose of my literature review was, therefore, to either substantiate, or refute, these points and better understand the existing material for classical ukulele.

The review took the form of a list of publications for classical or fingerstyle ukulele.[6] Due to the rather nebulous nature of the term classical ukulele it seemed logical to include all fingerstyle publications. Hence, *The Baroque Ukulele* (2012) by Tony Mizen falls easily into the category of classical ukulele while *Bluegrass Fingerstyle Solos* (2011) by Ondrej Sarek would seem to be stretching the boundaries. The list is not definitive but it is comprehensive enough to highlight the lack of material (print or digital) and establish the reason for not only creating new concert works for classical ukulele, but incorporating a range of levels.

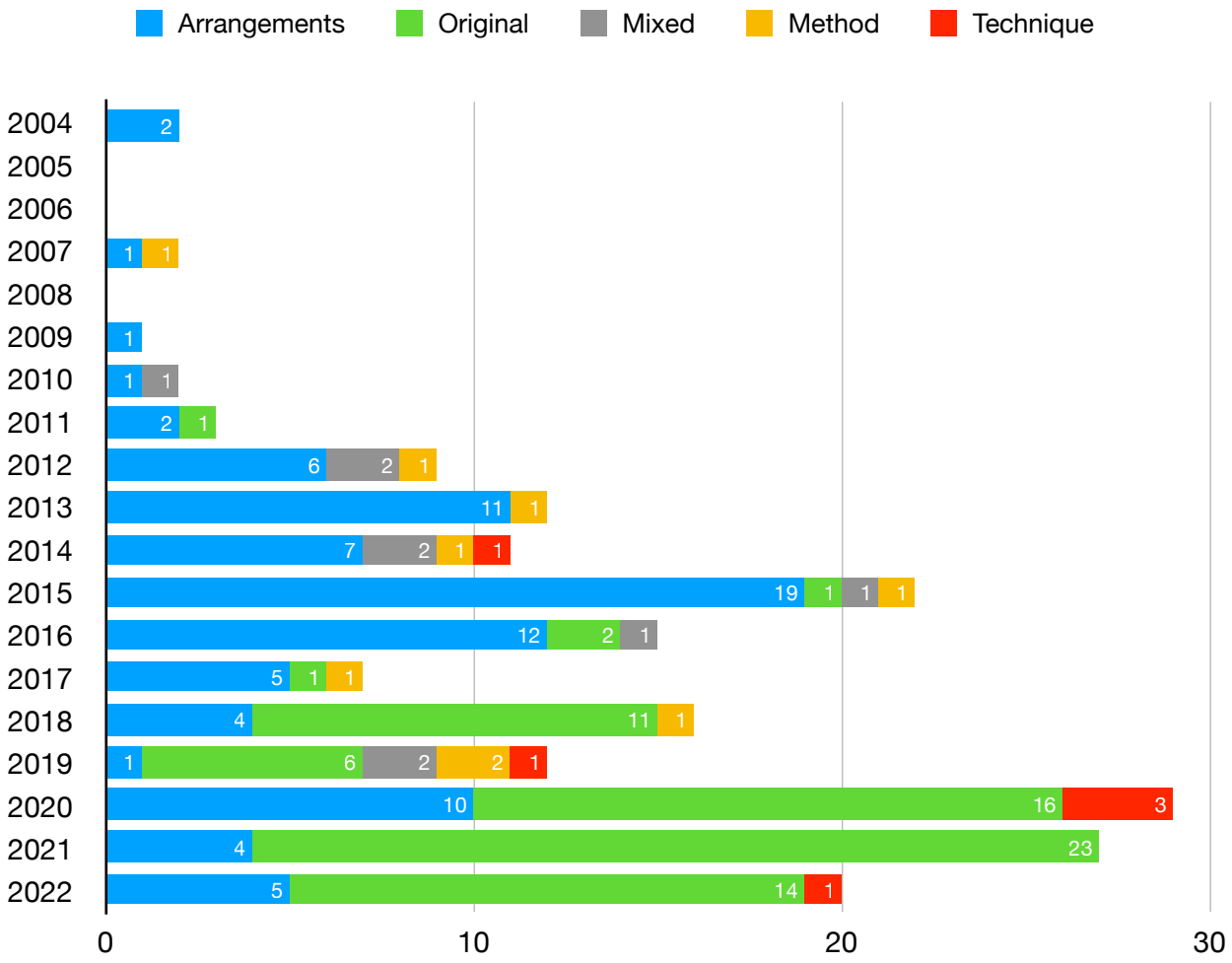
As shown in figure 3, the results were striking. In 2004 there were just two publications for classical or fingerstyle ukulele, both of which were by John King. Up until 2013 the number of publications per year did not exceed single figures. Since 2013 there has been an overall growth in numbers with twenty-nine publications in 2020 and twenty-seven in 2021. It is also interesting to note the shift in 2018, from arrangements to original works. In 2015 there were nineteen books of arrangements and just one original publication. By 2018 the tide had turned and there were just four books of arrangements compared to eleven original works.

In regard to didactic material the overall figures show that between 2004 and 2022 there were only nine methods and four publications on technical development. The only method dedicated specifically to classical ukulele technique is the author’s *The Classical Ukulele Method* (2019). The literature review illustrates the growth, and type, of classical and fingerstyle

[6] The full list, in alphabetical order, is published on my website & regular updated: https://iloveclassicalukulele.com/list-of-classical-finger-style-publications/

publications from 2004 to 2022. It also clarifies the author's contribution to the classical ukulele repertoire (a total of eleven publications)[7] and the fact that classical ukulele is still at an early stage of development.

Figure 3. The growth of ukulele publications between 2004 & 2022 shown in graph form



[7] Folder B.1 or https://productionsdoz.com/en/nos-artistes/muir-samantha

1. The Classical Ukulele in Theory

The theoretical material includes three published papers and one published article (see Folders A.1 & A.2). Numerous presentations of the theoretical material have also been given during the course of my studies (see Folder D). Owing to my interest in the historical connection between the machete and the ukulele, and how my research revealed new findings about the machete, a brief abstract of the article 'The Machete of Madeira: A British Perspective' is given below in order to outline the significance of the research.[8] The research is supported by the CD *The Music of Candido Drumond de Vasconcelos* for machete and guitar and a short slideshow of images called *The Clara Polka*.[9] The slideshow contains images of the machete from the nineteenth century and helps to establish its diminutive size and physical similarities to the ukulele.

The full text (with some modifications) of the paper 'Classical Approaches to the Ukulele: from John King to Ernest Ka'ai', which was originally presented at the first Ukulele International Conference (Milan, 2021), is included here as it provides a context for classical ukulele and lays a foundation for the discussion in Section 2 'The Classical Ukulele in Practice'.

1.1. The Machete of Madeira: A British Perspective

This article built on the work Portuguese academics Paulo Esteireiro and Manuel Morais, and ukulele historians Jim Tranquada and John King, by offering new insights into the machete de Braga, one of forerunners of the ukulele from the Portuguese island of Madeira. The author's research discovered how Clara Phelps, a British student of the celebrated Madeiran composer Candido Drummond (c.1841 - c.1880-3), went on to give numerous performances in England in the late nineteenth century.

The author is one of the few people outside Madeira to play the machete and its unique nineteenth century repertoire. The CD *The Music of Candido Drummond* for machete (Samantha Muir) and guitar (Lara Taylor) was commissioned and published by Associação Regional de Educação Artística, Madeira (2019). The CD contains a selection of pieces from Drummond's hand written manuscript of 1846 and is the first CD dedicated to the works of Candido Drummond.

Article abstract

As one of the direct forerunners of the ukulele, part of my research has been devoted to better understanding the machete (or machete de Braga) from the Portuguese island of Madeira. This

[8] Full text available here: https://rpea.madeira.gov.pt/index.php/rpea/article/view/185/169?fbclid=IwAR2NbxgbwGFxnk0S-lO8luOjPquhRVhB3YMGxCDAu9CUJPrmUYknjaIcSyY

[9] See Folder A.1

research has been both practical and theoretical. The practical element involved playing, performing and recording some of the unique classical repertoire of the nineteenth century Madeiran composer Candido Drummond de Vasconcelos. The very existence of this repertoire is evidence of the ukulele's classical potential. As one of the few people outside Madeira to play the machete, my performances are helping to promote this unique instrument and its repertoire while illuminating an important part of the ukulele's heritage. The theoretical research, presented in this article, was driven by a desire to know more about the British interest in the machete in the nineteenth century. The work of Portuguese scholars Paulo Esteireiro[10] and Manuel Morais revealed how the majority of information on the machete from this time is found in the travelogues and diaries of English speaking residents and visitors to the island. The British were particularly curious about this tiny guitar with four strings and some even learnt how to play it. While Esteireiro and Morais focused their research on Madeira, the homeland of the machete, my own research explored whether, and to what extent, the machete had been used in Victorian England. The research revealed new and previously unknown information about the machete; the findings of which are presented in the article.[11]

1.2. Classical Approaches to the Ukulele: from John King to Ernest Ka'ai

This paper fulfils the objective of defining the term classical ukulele while providing an historical context for classical ukulele. As the first academic research undertaken on classical ukulele it contributes to our knowledge and understanding by providing a theoretical foundation for a style of playing the ukulele which continues to gain popularity.

Abstract

This paper explores the origins of the term classical ukulele and discusses the techniques and repertoires associated with the classical ukulele style. The term classical ukulele has only become widely used since the publication of John King's *The Classical Ukulele* in 2004 and is an approach largely influenced by classical guitarists. As such it is easy to assume that classical ukulele is a relatively new phenomenon. This paper, however, will review this idea by examining the classical approaches used in of some of the earliest ukulele methods. By exploring these early methods, and analysing excerpts from the influential *Crescendo* magazine, this paper will offer a new reading of the ukulele's formative years. What emerged during the course of this research inquiry was that despite not using the term 'classical', a number of early ukulele players who also played the guitar approached the ukulele as a classical instrument. It is also noteworthy that many of the controversies surrounding the first

[10] The article was written at the invitation of Paulo Esteireiro: Diretor de Serviços de Investigação, Comunicação, Edições e Formação & Secretaria Regional de Educação, Ciência e Tecnologia Madeira

[11] Published in *Revista Portuguesa de Educação Artística Vol,11 No.2* (2021). The date of publication was actually December 2022.

ukulele methods, such as the use of tablature and chord diagrams, exist to this day. The ukulele, it would seem, has always challenged conventional approaches to how music is taught and notated. On the one hand the ukulele's 'free spirit' has enabled it to thrive as a popular instrument, while on the other it has impeded its acceptance into mainstream classical music. Looking at the ukulele in this context led to a reassessment of the recent classical ukulele movement and the realisation that classical ukulele should not, after all, be viewed as a new approach but as a reawakening, revitalising and reimagining of old techniques, methods and repertoires driven by a vision of creating new, challenging and contemporary works.

Introduction

Today, the term classical ukulele is widely used to denote a technique and a repertoire which finds its roots in western classical music. The first known use of the term 'classical ukulele' was by John King with the publication of his book/CD *The Classical Ukulele* in 2004. Undoubtedly King drew on his classical guitar background, not only in his choice of title but in his technical, musical and historical approaches to the ukulele. King's arrangements included works by Bach, Chopin, Mozart and Beethoven. A parallel can thus be drawn with classical guitar masters including Francisco Tárrega (1874-1909) and Andrés Segovia (1983-1987) who, in their efforts to develop a concert repertoire for the guitar, arranged works by composers including Chopin, Bach, Granados and Albéniz. Was King simply applying his classical guitar knowledge to the ukulele, or was he seeking a new 'classical' vision for the ukulele? And if so, what exactly was this vision? Sadly, King passed away in 2009 and *The Classical Ukulele* and *Famous Solos and Duets for the 'Ukulele* (Mel Bay, 2004) remain his only two publications. Nevertheless, both of these books, particularly *The Classical Ukulele,* have had a major impact on the ukulele scene. Today classical ukulele is a thriving movement of players, teachers, arrangers and composers. Many of these players, like King, come from classical guitar backgrounds and have applied classical guitar technique and repertoire to the ukulele. Although this has proved hugely popular it has also led to the suggestion that such players are simply using the ukulele as a small guitar. This paper will, therefore, explore the classical approach to the ukulele in order to determine what it is, when it was first initiated, and whether it should be viewed as a new approach or an integral part of the ukulele's rich and varied history.

John King and the 'Classic Aesthetic'

King was quick to point out in his introduction to *The Classical Ukulele* that playing classical music on a little four string instrument was nothing new.

> With the recent discovery and publication of a manuscript of duets for machete and guitar the argument against the validity of performing classical music on the little four string instrument has been rendered moot. (King, 2004, 4)

The machete (or machete de Braga) is one of the instruments from the Portuguese island of Madeira that was taken to Hawaii in 1879 and subsequently transformed into the ukulele. The discovery in the 1990s of a manuscript of duets for machete and guitar by Candido Drummond de Vasconcelos dated 1846 proved that the machete, once regarded as a simple folk instrument, had also achieved recognition as a sophisticated instrument. In the second half of the nineteenth century the machete featured in concerts and other musical events both in Madeira and in England (Muir, 2022). Only in recent years has the machete's unique repertoire and its historical importance started to become known.

King was one of the first people outside Madeira to perform, in recent times, Drummond's pieces on the machete. Despite the machete being a traditional instrument of Madeira[12] the majority of pieces in the collection were composed in the form of European ballroom dances including waltzes, polkas, marches, bolleras and quadrilles. Robert White, a British visitor to the islands, described the machete's repertoire in his guidebook of 1859 as 'the fashionable music of our ballrooms.' (White, 1859, 38) The word 'fashionable' is an important reminder to view Drummond's music in an historical context. As White observed, Drummond was composing popular music in order to appeal to current tastes and fashions. Furthermore, Drummond's music was original and unique to the machete. King, on the other hand, borrowed historical repertoire from instruments such as the piano, the harpsichord and the cello in order to create a *new* repertoire for the ukulele. In so doing he demonstrated the potential of the ukulele as a classical instrument, but was creating a 'classical' repertoire retrospectively. As innovative as his arrangements were, the material in *The Classical Ukulele* was neither original nor contemporary.

Publisher Jim Beloff heralded *The Classical Ukulele* as a 'first-of-its-kind book'. (King, 2004, 2) He went on to explain that while ukulele players often included one classical piece in their repertoire as a novelty item, King's innovation was to create an entire collection (nineteen pieces) of classical pieces for the ukulele. Despite being called 'classical' the collection was not entirely classical in its repertoire selection. Traditional tunes from Japan (*Sakura*), England (*Greensleeves*), Ireland (*Danny Boy*) and Hawaii (*Alekoki*, *Pūpū A'o 'Ewa* and *Ahe Lau Makani*) featured alongside

[12] The machete is related to the Portuguese cavaquinho and is thought to have originated in the Braga region of Portugal, hence its full name machete de Braga. Since the twentieth century the name braguinha has been used in Madeira with the name machete used to refer to instruments from the 19th century.

Bach (*Prelude BWC 846*, *Prelude BWV 1007* and *Jesu, Joy of Man's Desiring*), Mozart (A*h! Vous Dirai-je, Maman*), Beethoven (*Für Elise*), and Couperin (*Les Barricades Mystérieuses*). The final piece strayed even further from the classical idiom by offering an arrangement of Scott Joplin's jazz standard *The Entertainer* for ukulele and guitar.

While some of the pieces in *The Classical Ukulele* fit neatly into the genre of classical music, others seem to deliberately contradict it. The question of what is legitimate classical ukulele repertoire is often fiercely debated on social media. Should a video of an instrumental arrangement of *Amazing Grace* be allowed in a classical ukulele group? Some say yes, and some no. I defer to John King whose inclusion of traditional pieces is a further reflection of his guitar background. Many classical guitarists incorporate arrangements of folk, pop and traditional tunes in their repertoires. Even classical guitar purist Andrés Segovia performed arrangements of traditional tunes with one of his most popular pieces being Miguel Llobet's arrangement of the traditional Catalan tune *El Noi de la Mare*. Numerous other classical guitarists from John Williams (b.1941) to Miloš Karadaglić (b.1983) have included instrumental arrangements of songs by the Beatles, folk songs and other popular hits in their concerts and recordings. In fact, many classical musicians today, including cellists, flautists, violinists and pianists, incorporate arrangements of popular music in their repertoire. While the classical music purists may despair, it highlights how the term 'classical' is not necessarily genre specific.

King's repertoire selection indicates he agreed with this relaxed stylistic approach. How then should we define classical ukulele? While King does not specify this he does provide clues. In his introductory notes on *The Classical Ukulele* he stresses the importance of sound. 'The most important aspect of the right-hand [picking hand] technique is tone production. A round, clear tone free from extraneous noise is central to the classic aesthetic.' (King, 2004, 6) Thus, the guiding principal of classical ukulele, as King saw it, is tone production. He never, however, suggested how this could be achieved on the ukulele. That work has been left to an increasing number of classical ukulele players, the majority of whom have a background in classical guitar. In recent years an increasing number of ukulele books have been dedicated to improving finger picking techniques and tone production.[13]

The modern approach to playing classical guitar is to use the nails of the picking hand to pluck the strings. King was more open minded about the ukulele. He suggested nails could be used, or the fleshy tips of the fingers, or even a plectrum. If nails are used 'they should be filed and

[13] For example: R.MacKillop, 2011, *Progressive Fingerstyle Studies for Ukulele*, Mel Bay; D.Ward, 2017, *Arpeggio Meditations*, self-published; E.Pfeiffer, 2022, *Rhythmic Fingerstyle Etudes*, self-published; S.Muir, 2019, *100 Arpeggio Exercises for Ukulele*, Les Productions d'Oz

polished so that they do not produce rough and scratchy sounds'. (King, 2004, 6) Once again King was emphasising the importance of a round, clear tone. It would, however, be disingenuous to suggest that sound quality is only a feature of classical ukulele players as it also an attribute of other ukulele instrumentalists such as Herb Ohta-San (b.1934) and Jake Shimabukuro (b.1976) whose roots are in popular music.

What emerges in any discussion of classical ukulele is the realisation that the boundaries defining it are fluid rather than fixed. At the time of publication *The Classical Ukulele* (2004) was a bold and audacious move towards a new way of playing and perceiving the ukulele. As publisher Jim Beloff noted in his foreword, the ukulele is 'normally associated with happy island melodies' rather than sophisticated solo repertoire. Both the title and the content offered something totally different from a publisher (Flea Market Music) well known for producing ukulele song books with chords and lyrics. An important feature of *The Classical Ukulele* was that the arrangements were presented in both notation and tablature. The inclusion of tablature meant the material was accessible to players who could not read music and established a precedent for future classical and fingerstyle publications. Nevertheless, the use of tablature is not without controversy and warrants further discussion. Firstly, however, we will explore how King's arrangements, while new in concept, were actually making full and clever use of the idiomatic features of the ukulele.

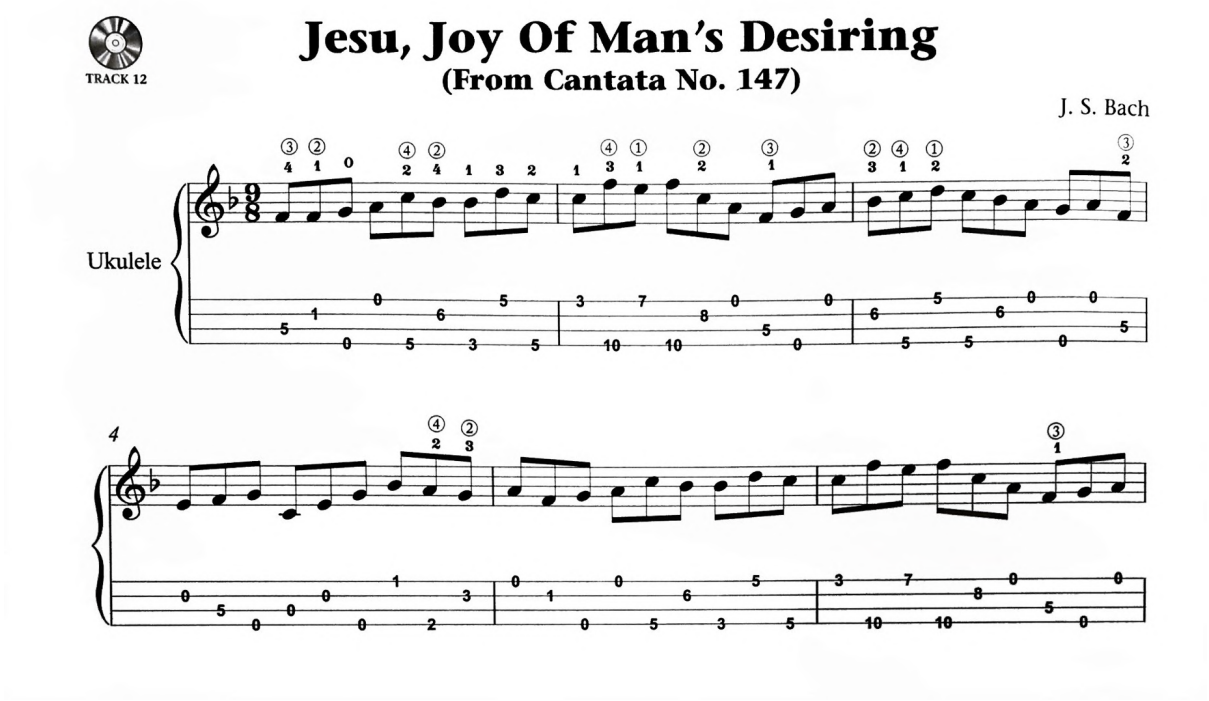
Campanella

During his classical guitar studies King had learnt about the five course Baroque guitar and how composers such as Gaspar Sanz (1640-1710) and Francisco Corbetta (1615-1681) had utilised that instrument's re-entrant tuning to create novel over-ringing effects called campanellas. The term campanella, also *campanelle* (French) and *campanela* (Spanish), comes from the Italian word meaning 'a small bell' (OED). By placing the notes of short melodic, or scale-like, passages on alternating strings the over-lapping notes produce a sound reminiscent of small ringing bells, hence the term campanellas (see figure 1.2.1).

Like the Baroque guitar, the original tuning of the ukulele is re-entrant. Realising this, King decided to try campanella on the ukulele. His first experiment with campanella is described on his Nalu Music blog.

> I taught myself an appropriate Bach tune using campanella technique and applied it to my mother's pineapple[14] [ukulele]. It was a revelation. The instrument had a voice and when I played, it sang to me.[15]

Figure 1.2.1. An example of John King's campanella approach with the notes placed on alternating strings



King's use of campanella technique highlighted the ukulele's European roots and helped to forge a new vision for the ukulele based on historic performance practice. He went on to record an entire CD of pieces by Bach arranged in campanella style - *JS Bach Partita No.3 BWV 1006 for Unaccompanied Ukulele* (2006). The CD also includes a selection of pieces from Bach's cello suites as well as *Jesu, Joy of Man's Desiring* (see figure 1.2.1). King received wide acclaim for his harp-like sound and virtuosic playing which elevated the ukulele to the realm of classical music. A number of pieces from the Bach CD were included in *The Classical Ukulele.* One of the most innovative and distinctive features of King's use of campanella was that he applied it to entire pieces. This was a markedly different approach from the Baroque guitarists who used campanella sparingly and only to create a short, novel effect within a piece.[16]

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[14] Pineapple refers to a type of ukulele that is shaped like a pineapple rather than waisted like a guitar.

[15] The Nalu Music website is, unfortunately, no longer available.

[16] See Chapter 2.1 for further discussion of campanella technique.

What is evident in King's work is that he regarded campanella as an integral part of the classical ukulele approach. The constant movement of notes across the strings and along the fingerboard posed new and unfamiliar technical challenges for ukulele players. The picking hand utilises the thumb, index, middle and occasionally ring fingers while the fretting hand requires carefully worked out fingerings. For an instrument commonly associated with chords and strumming patterns these technical demands are more typical of the guitar and may help to explain why so many classical ukulele players come from a guitar background. As the first person to consistently and consciously use campanella style on the ukulele King was a modern pioneer. His arrangements and performances, which can be viewed on YouTube, have influenced a new generation of classical ukulele players, many of whom are committed to advancing the repertoire and status of the ukulele. While King certainly sparked a new movement of classical ukulele players, he was not, as will be shown, the first guitarist to promote classical approaches to the ukulele.

The Ukulele in the BMG Movement in America in the Early 20th Century

In order to appreciate King's use of the term classical ukulele it is first necessary to understand the term classical guitar. The term classical guitar was first used in the early twentieth century by guitarists, most notably Andrés Segovia, who sort to distance the guitar from its folkloric traditions and elevate it to a serious concert instrument. Increasingly the term classic, or classical, guitar was used to replace the term Spanish guitar. One of the earliest uses was by Vahdah Olcott-Bickford (1885-1980) in her *Method For Classic Guitar* first published in 1921. Olcott-Bickford was one of the most prominent figures of the Banjo, Mandolin and Guitar (BMG) movement in North America.[17] In fact, Jeffrey Noonan in his book *The Guitar in America* (2008) considers her and William Foden (1860-1947) 'the most important American players and apologists for the guitar in the first half of the twentieth century.' (Noonan, 2008, 148) Olcott-Bickford was well known as a performer, composer, arranger and teacher. At different times she wrote columns for the two most prominent magazines of the BMG movement - *Cadenza* and *Crescendo*. In the early twentieth century it was normal for fretted instrumentalists to play the guitar, the banjo and the mandolin. With the rising popularity of the ukulele on mainland America, particularly following the Panama-

[17] The BMG movement gained prominence in North America in the late 19th to early 20th centuries. It incorporated fretted string instruments, specifically the banjo, the mandolin and the guitar, and later included the ukulele. Its members were dedicated to introducing these instruments to America's elite musical establishments. The Guild of American Banjoists, Mandolinists and Guitarists held regular meetings and concerts for its members in order to promote new techniques and repertoires. The two main publications of the BMG movement were the *Crescendo* and the *Cadenza* magazines. These magazines included articles, reviews and repertoire pieces.

Pacific International Exposition of 1915, the BMG movement soon adopted the ukulele as a kindred instrument. Vahdah Olcott-Bickford was one of the most prominent classical guitarists to take up the ukulele.

In 1920, the year before the publication of her guitar method, Olcott-Bickford and her husband Zahr Myron-Bickford (1876-1961) published *The Bickford Method for the Ukulele*. Their aim was to 'present all of the various styles of playing' (Bickford, 1920, 3) which included 'the ordinary Stroke Method' used by the Hawaiians, 'the guitar, or picking style, and the use of the plectrum or felt pick'. (Bickford, 1920, 3) Their comments are a salient reminder that as a new instrument this was a time of development and experimentation. The three techniques in use at this time were strumming, finger picking and the use of the felt pick. Even the tuning of the ukulele was not standardised with both the original C6 tuning and the newer D6 tuning in use.[18] *The Bickford Method* uses the D6 tuning. (See figures 1.2.2. & 1.2.3.)

Figure 1.2.2. The original C6 tuning

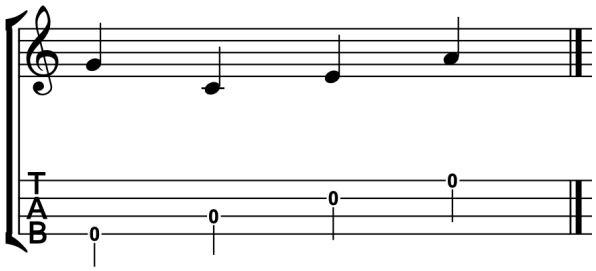
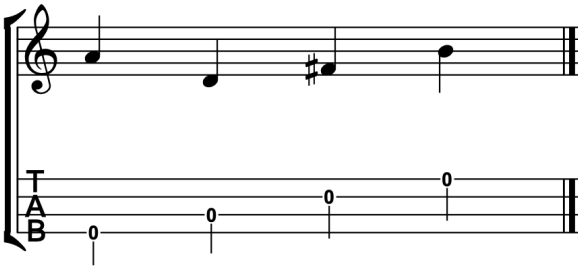


Figure 1.2.3. The D6 tuning used in The Bickford Method



[18] The C6 tuning is given as the tuning of the ukulele in the earliest known (Hawaiian) method for either the ukulele or the taro-patch E.C. Holstein's *Chords of the Taro-Patch Guitar* of 1894, published by Hawaiian News Co.

Although much of the method is dedicated to strumming techniques, referred to at the time as the strokes, it also reflects the influence of the guitar and the desire for a more formal, classical approach to the ukulele. This is apparent in the determination of the Bickfords to present a ‘real system of study’. (Bickford, 1920, 3) The method begins with a detailed section on the rudiments of music as well as sections on how to hold, and how to tune, the ukulele. The method only uses staff notation. Diagrams, such as tablature and chord boxes, were viewed as ‘a great detriment to the musical advancement of the pupil’. (Bickford, 1920, 3) An advertisement in the *Crescendo* in September 1920 cited no less than eighteen reasons why the authors regarded their method as ‘superior to all the others’.[19] One of their first points was that the ‘guitar, or finger style of playing is thoroughly explained and illustrated, a feature almost entirely lacking in existing Methods’.[20] The importance of the guitar, or finger style of playing, was stressed in the introduction to the method, with the authors noting that ‘This is a very effective manner of playing the ukulele and should be thoroughly cultivated.’ (Bickford, 1920, 9)

Figure 1.2.4. Little Guitar from The Bickford Method



As shown in the arpeggio technique employed in *Little Guitar*, the Bickfords were keen to draw a parallel with the guitar. The introductory text to this piece gives a brief instruction on using the third, or ring finger of the picking hand. ‘The third finger is often very effective in the guitar style of playing and its use should be cultivated.’ (Bickford, 1920, 57) When using the third finger (i.e. the ring or *a* finger) the player is instructed to avoid placing the little finger on the top of the instrument ‘since it would interfere with the freedom of the [third] finger.’ (Bickford, 1920, 57) In

[19] *Crescendo* Volume 13 Number 3, September 1920

[20] *Crescendo* Volume 13 Number 3, September 1920

Little Guitar the picking hand plays a repeated arpeggio pattern with the thumb, index and middle fingers. The melodic note, situated on the first string, is played by the ring *(a)* finger, thus mirroring a technique common with guitarists. Today the picking hand is normally indicated using the *pima* system but at that time the thumb was indicated with a cross, the index finger with one dot, the middle finger with two dots and the ring finger with three dots. As shown in figure 1.2.4 each finger of the picking hand is assigned to a string in order to create a fluid arpeggio pattern using *p-i-m-a*.

Little Guitar is followed by a short piece that utilises another guitar technique and is called *Special Study for Right Hand* (see figure 1.2.5) The study is a three note tremolo exercise using thumb, middle and index fingers *(p-m-i)*. The piece, and the technique, are very typical of the guitar and might easily be played on that instrument.

Figure 1.2.5. Special Study for the the Right Hand from The Bickford Method



Other pieces, such as *Cupid's Pranks* (see figure 1.2.6) and *Dance of the Dolls*, incorporate both melodic and chordal passages with many of the chords being plucked with the fingers in the style of classical guitar rather than strummed. The final chord in figure 1.2.6 contains a symbol familiar to guitarists. The squiggled line indicates the chord should be rolled, or arpeggiated. In every way *Cupid's Pranks* looks and sounds like it belongs on the guitar.

Given the influence of Vahdah and Zahr Bickford in the BMG movement it is perhaps surprising that the notations and symbols used in *The Bickford Method* did not set a precedent for future ukulele methods. Many ukulele players today, however, are not familiar with these symbols or the layout of a score. Only recently is a more standardised approach to ukulele scores, using both notation and tablature, being pursued by publishers such as Flea Market Music, Les Productions d'Oz and many independent publishers.

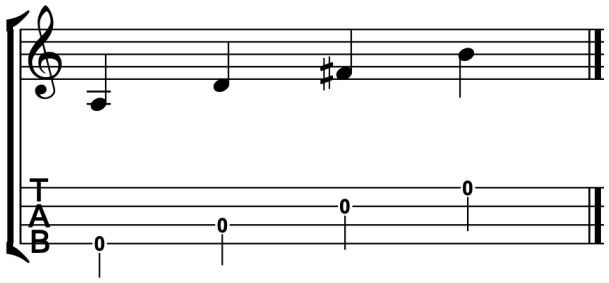
Figure 1.2.6. Cupid's Pranks from The Bickford Method



The Bickford Method is an important early work for the ukulele both in its vision of promoting the ukulele as a serious instrument and the fact that it was co-written by one of the pre-eminent guitarists of the time. At over seventy pages long the method is comprehensive. The authors tackled many of the challenges and controversies which the ukulele faced - particularly surrounding the use of tablature. The Bickfords were unequivocal in their support of staff notation.

> Any instrument which is worthy of having an instruction book written for it, is of course worthy of being written for in the legitimate musical notation, hence the diagrams so frequently given in connection with ukulele music and methods, have been eliminated from the technical part of the Method. (Bickford, 1920, 3)

Figure 1.2.7. The Bickford Method notated the A an octave lower to avoid confusion in the voicing



Due to the re-entrant tuning of the ukulele the voicing can often be confusing. *The Bickford Method*, as with several other methods of the time, did use re-entrant (D6) tuning but chose to notate the fourth string A an octave lower than it sounds. Their reason for this was 'to avoid confusion in reading, particularly in connection with chords employing all four strings.' (Bickford, 1920, 8)

As shown in figure 1.2.7 the A written below the staff clarifies the position of the fourth string notes. This approach, however, was not without controversy. Charles S. DeLano of Los Angeles, a well known ukulele and steel guitar teacher, as well as a columnist for the *Crescendo* magazine, considered this method too confusing.

> This style of writing the fourth-string notes one octave lower was first introduced by a Native Hawaiian in his published book.[21] Many purchased it and, not being able to understand it, came to me for instruction. Not in a single case were they satisfied with this way of writing the fourth string, as they all preferred the notes at actual pitch.[22]

DeLano makes a good point. The notes might look right on paper, but they did not sound right. A more recent solution has been to use low G tuning so that the music both looks and sounds correct. The problem with using a linear tuning is that the ukulele loses its natural voice and the potential for employing campanella technique is reduced.

As one of the first BMG members to capitalise on the growing popularity of the ukulele and the steel guitar on the mainland DeLano provides some interesting insights. His correspondence course *Instructions for Playing the Hawaiian Ukulele* (1914) was frequently advertised in the *Crescendo*. Realising his target clientele were largely BMG teachers, DeLano promised that 'You can increase your business 100% by teaching them [ukulele and steel guitar]'.[23] In 1916 when the *Crescendo* fully committed to the ukulele and the steel guitar by launching its 'Hawaiian Round Table', DeLano became the main protagonist for the ukulele.

In his article of November 1917 DeLano mentioned Ernest Ka'ai[24] as being 'one of the foremost of all ukulele players and teachers of the Islands [Hawaii]'. On one of Ka'ai's trips to the Pacific coast DeLano had 'had the pleasure of being with him [Ka'ai] and secured from him many

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[21] He is probably referring to George Kia's *Self Instructor for the Ukulele and Taro-Patch Fiddle* of 1914

[22] *Crescendo* Volume 10 Number 4, October 1917

[23] *Crescendo* Volume 7 Number 8, February 1915

[24] The Ukulele Hall of Fame website considers Ernest Ka'ai the first Hawaiian ukulele virtuoso

of his fancy strokes all of which are published in my ukulele instructor'.[25] Despite DeLano's willingness to plagiarise Ka'ai's work, he disapproved of Ka'ai's method of teaching the ukulele.

> He has manuscript paper with a staff of four lines representing the four strings of the instrument. Crossing these diagrams are lines indicating the frets. Dark dots on the vertical lines give the location of the fingers on the strings. An open note represents an open string.[26]

DeLano went on to say that Ka'ai had admitted 'his book was not satisfactory and had no published solos in note form…He informed me he was going to publish a new one, but to the present time I have no knowledge of such a publication.'[27] This article was written in 1917 and appears to be referring to Ka'ai's original method of 1906 (revised 1910). In *The Ukulele: A Hawaiian Guitar and How To Play It* Ka'ai used a rudimentary form of tablature in which the four strings were indicated by lines, and circles were placed on the appropriate strings. A hollow circle indicated an open string and a black circle indicated a fret. The fret number was then written above the circle. What DeLano omitted to say in his article of 1917 is that Ka'ai had by this time released a new and updated method called *The Ukulele and How It[']s Played* (1916). This method used both tablature and staff notation and included a number of Ka'ai's highly innovative solos. These will be discussed shortly. DeLano was, therefore, either oblivious to Ka'ai's new method or deliberately omitting to mention it in order to protect his own business interests. By 1916 DeLano was one of the most prominent teachers and players on the Pacific coast. The 'DeLano Hawaiian Steel Guitar and Ukulele Sextet' had played in nearly all the principal cities on the Pacific coast from San Francisco south, including a recital at the Panama-Pacific Exposition of 1915.[28] Despite regarding Ka'ai as 'a fine singer and expert ukulele player', DeLano concluded his article with a further swipe at the Hawaiian.

> The point I wish to make is that the solos by the diagram system in this case were total failures and led the pupil to believe the ukulele could be mastered in three or four lessons. This resulted in a complete misunderstanding of the possibilities of this popular instrument.[29]

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[25] *Crescendo* Volume 9 Number 10, April 1917

[26] *Crescendo* Volume 9 Number 10, April 1917

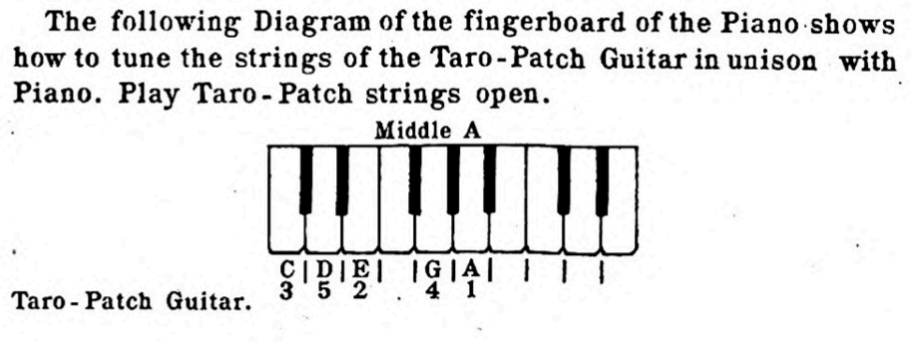
[27] *Crescendo* Volume 9 Number 10, April 1917

[28] *Crescendo* Volume 9 Number 3, September 1916

[29] *Crescendo* Volume 9 Number 10, April 1917

Numerous early ukulele methods including *The Bickford Method* (1920), the *Kamiki Ukulele Method* (originally published in 1915), the *Eureka Method* (1924), Geo Kia's *Self Instructor for The Ukulele and Taro-Patch Fiddle* (1914), as well as DeLano's correspondence course (1914) were published in staff notation only. They did, however, often include diagrams for chords. As E.C. Holstein had noted in his *Chords of the Taro-Patch Guitar* (1894) the use of chord diagrams represented 'A new system for learning to play the chords of the taro-patch guitar without a teacher.' (Holstein, 1894, 2) Even those who rigorously supported the use of staff notation, such as the Bickfords, obviously agreed that chord diagrams were a useful shortcut; despite their insistence on using 'legitimate musical notation' (Bickford, 1920, 3) they also included chord diagrams in the appendix of their method.

Figure 1.2.8. The taro-patch tuning as shown in Holstein's method



As the earliest known method for either the taro-patch or the ukulele Holstein's method warrants a brief mention if for no other reason than to clarify the use of the term taro-patch and its tuning. In 1894, at the time of publication, the instrument known as the taro-patch was, in fact, the Hawaiian name for the five string rajão from Madeira; not the later version of the taro-patch, referred to in Kia's method of 1914, which had four sets of double strings. As Holstein's method confirms, the taro-patch to which he refers had 5 single strings. As shown in figure 1.2.8 both the fourth and the fifth strings were re-entrant.

The relevance of the original taro-patch (rajão) in the early history of the ukulele is often over-shadowed by discussions of the machete. DeLano, however, makes an interesting reference to it in one of his first articles. In a discussion on the origins of the ukulele he noted that, 'The taro patch, a single five-stringed instrument of larger size [than the ukulele] was first in use.'[30] He goes

[30] *Crescendo* Volume 9 Number 8, February 1917

on to say, ‘A little later the fifth string was omitted and then the present sized ukulele was chosen as much more convenient in size and sweeter in tone.’[31] DeLano makes no mention of the machete and seems to be implying that the ukulele was an adaption of the taro-patch, or the Madeiran rajão.[32]

Today it is generally accepted that the tuning of the ukulele comes from the rajão while its diminutive size comes from the machete (Tranquada & King, 2012). Holstein’s method, while rudimentary in content, is useful as it confirms that both instruments were in use in Hawaii in 1894. ‘The Ukulele Guitar is smaller than the Taro-Patch Guitar and has but four strings, the tuning of which is the same as [the] Taro-Patch deprived of the fifth string.’ (Holstein, 1894, 3) As Jim Tranquada notes in an article published by *Ukulele Magazine*[33] (2019), images from the 1890s show taro-patches missing the fifth string in order, presumably, to be played as slightly larger bodied ukuleles.[34] Around 1914 the original taro-patch was phased out and its name was applied to an eight string instrument comprising four sets of double strings (C6 tuning).

Notation Systems: The Debate Continues

As the first method for either the ukulele or the taro-patch Holstein’s method set the precedent for using chord diagrams which are still popular today. Meanwhile, Ka’ai established the use of tablature but, as seen with DeLano and the Bickfords, this system remains controversial. The main argument against the use of tablature is its limitation in only showing *where* the notes are situated on the fingerboard. Staff notation, on the other hand, is a universal system of writing music which tells musicians *what* the notes are. If the ukulele is to be accepted as a serious instrument worthy of academic study then the ability to read music is undoubtedly essential. There are, however, other considerations which support the use of pictorial forms of notation such as tablature and chord diagrams. The accessibility of these systems facilitates ease of learning and enables many ukulele hobbyists of all ages to participate in music making. Currently, many composers, arrangers and publishers are including both staff notation and tablature in their scores in order to cater for all players.

Another argument for the use of tablature is that notating for the re-entrant tuning of the ukulele has, right from the earliest methods, proved problematic. As previously discussed, many

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[31] *Crescendo* Volume 9 Number 8, February 1917

[32] Both Manuel Nunes and Augusto Dias advertised themselves as makers of guitars, machetes and stringed instruments in the Portuguese Hawaiian newspaper *O Luso Hawaiiano* in 1885. Much later they specified themselves as makers of ukuleles and taro-patch guitars. They are not known to have used the term rajão.

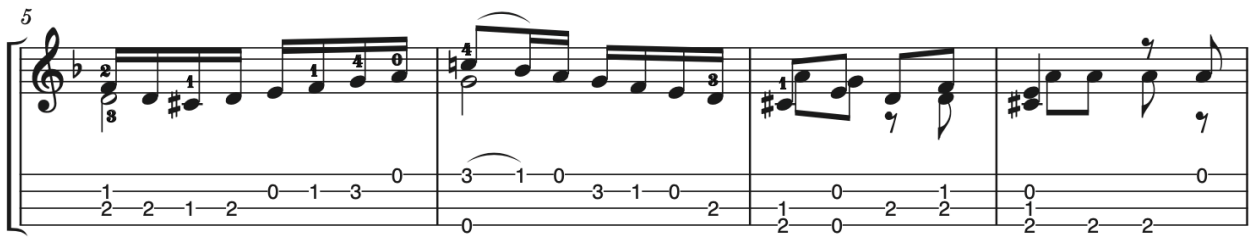
[33] https://ukulelemagazine.com/stories/ukulele-history-140-years-ago-the-ukulele-as-we-know-it-arrives-in-hawaii

[34] Later, larger bodied ukuleles became known as concert and tenor ukuleles.

early methods opted to notate the fourth string an octave lower than it actually sounds; a solution which enabled the score to look correct while allowing the ukulele to maintain its characteristic re-entrant voice. The discrepancy between the notated pitch and the actual pitch can, however, be misleading (as DeLano noted) not only in the sound but by implying that the open fourth string G, for example, cannot be moved to another position on the fingerboard. When the fourth string notes are notated at the correct (re-entrant) pitch, however, other problems arise.

The following example illustrates one of the challenges in determining the role of the high G fourth string when notated at the correct pitch. The first example shows a passage from *Lesson 2* of Sor's *Progressive Lessons Opus 31* (Muir, 2018) arranged for re-entrant tuned ukulele. In bars 7 and 8 (see figure 1.2.9) the fourth string notes are shown with tails down although they often sound higher than the melodic notes (tails up).

Figure 1.2.9. An example of voicing problems created by the re-entrant tuning (see bars 7 & 8)



The example in figure 1.2.9 both looks and *is* technically incorrect in the staff notation. In order to create the correct voicing the As and the G should be written, and played, an octave lower. If the stems of these notes were on the right they would seem to be the melodic notes - which they are not. Meanwhile, the tablature clearly shows that these notes are played on the fourth string and their role is to provide the accompaniment. The re-entrant tuning, thus, creates a conundrum. The re-entrant sound is both characteristic of, and idiomatic to, the ukulele but is difficult to notate according to the conventions of music theory. The problem can be solved with the use of low G tuning, as the resulting linear order of the strings allows the notation to both look and sound correct. The question is: *should the ukulele abandon its natural voice in order to comply to musical convention? Is there another solution?*

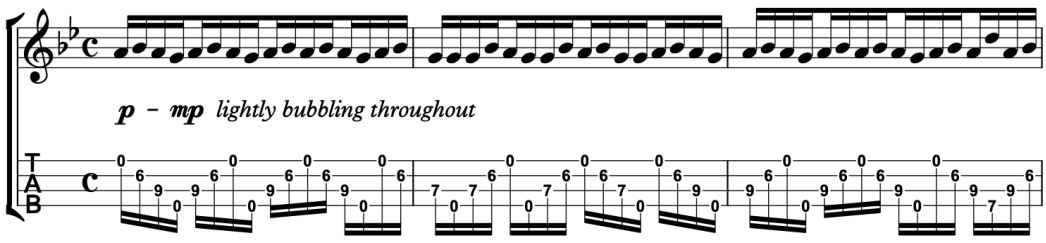
The simplest solution for the the re-entrant tuning is to use tablature but then we arrive back at the problem of ukulele players not being able to read music. The debate is ongoing. My own

preference is to use both tablature and staff notation and suggest that both systems should be incorporated in ukulele instruction. It is also worth remembering that despite its bad press, tablature has been used as a notation system for fretted instruments for many hundreds of years. Instruments including the four course Renaissance guitar, the five course Baroque guitar, the vihuela, the lute and the theorbo all use some form of tablature. In fact, anyone studying these instruments is expected to read both staff notation and tablature. It would seem to make sense for the ukulele to also embrace both systems.

Two further advantages of tablature are that it allow instruments to be tuned at different pitches and that it clearly illustrates campanella passages. On the ukulele campanella has developed into a unique style with many composers and arrangers using it for entire pieces. Using tablature avoids the need for detailed and cumbersome fingerings and clarifies the intentions of the arranger or composer.

In figure 1.2.10, showing the opening of James Else's *Parting Voices*,[35] the tablature clearly indicates how the notes, particularly the repeated notes, are intended to be played on different strings to create timbral nuances. The subtle shadings created by playing an open A followed by an A at 9,3 are essential to the mood and character of the piece. In the third bar (figure 1.2.10) the high D is played at 7,4. As the highest note in the bar it might be expected to be positioned on the first string at 5,1 but by positioning it on the fourth string, Else maintains the movement of the notes across all the strings to maximise the over-ringing. This means the D is played with the thumb rather than a finger resulting in a different attack and timbral effect. This piece is an excellent example of how the composer is immediately able to communicate his intentions to the performer. If notation with fingering indicators were used the score would be cluttered and less immediate. Thus, the benefit of presenting a score in both tablature and notation is that it tells the performer both *what* the notes are and *where* they should be played.

Figure 1.2.10. From Parting Voices by James Else. The tab clearly indicates where the notes are positioned.



[35] For score see Folder B.4. For audio see Folder C.2.

Ernest Ka’ai’s Use of Campanella Effects

Although campanella style is a relatively modern approach on the ukulele I was surprised and delighted to find campanella effects in many pieces by Ernest Ka’ai. As previously mentioned Ka’ai was one of the first ukulele players to use staff notation and tablature in his 1916 method, *The Ukulele and How It[’]s Played.* As a pioneering player, arranger and composer Ka’ai is often underrated. The solos in his method represent some of the most important early works for ukulele. Ka’ai, like many of the early ukulele players, also played the guitar and his pieces incorporated both strumming and finger picking techniques. While there are no early recordings of Ka’ai his use of tablature enables us to see how he utilised the re-entrant tuning to create over-ringing (campanella) effects.

In the opening triplet of *Haere*[36] (see figure 1.2.11) the notes move from 0,4 to 0,2 to 0,4 to create a campanella effect between the fourth and second strings. The Gs might easily have been played at 3,2 but, as the tablature shows, Ka’ai clearly wanted the notes to ring on. The triplet figure leads to the C at 3,1 thus completing the tonic triad of C. By using the notes of the C major triad Ka’ai created a simple effect that is both harmonic and melodic. The triplet idea is repeated three times with the G and E repeated and the first string note falling from C to B to A. The falling top notes subtly alter the harmonic effect (C, C major 7, C6). The final time the triplet is stated it is followed by a dotted rhythm leading back to the tonic, this time played as a full C major chord. The idea is simple but idiomatic and highly effective.

Figure 1.2.11. The opening of Haere by Ernest Ka’ai with the tablature showing the over-ringing effects



It is also interesting to note that although many methods at this time were using the D6 tuning, Ka’ai used the original C6 tuning inherited from the Madeiran rajão, or taro-patch. Furthermore, many of his pieces, despite having Hawaiian titles, used European forms such as the waltz (*Hene Waltz*, *Lauia Waltz*), the schottische (*Banjo Schottische)* and the mazurka (*Kawehi*). Thus many of

[36] For audio see Folder C.3.

Ka'ai's pieces fall into the category of European ballroom dance music and are curiously reminiscent of the works of the Madeiran composer Candido Drummond who also made extensive use of European dance forms in his repertoire for the machete. (Muir, 2022)

Although much has been written about how the machete was taken to Hawaii in 1879 and quickly integrated into Hawaiian culture, very little is known about the music the first Madeiran immigrants played. An article in the *Hawaiian Gazette* in September 1879 announces the arrival of the Madeiran instruments and gives some insight into the repertoire. 'During the past week a band of Portuguese, composed of Madeira Islanders recently arrived here, have been delighting the people with nightly street concerts.'[37] The article goes on to describe 'their strange instruments' as being a cross between a guitar and a banjo. The music is described as 'very sweet' and 'soothing'. Unfortunately, no machete manuscripts have ever been found in Hawaii and there is no evidence to suggest that the music of Candido Drummond migrated with the machete. What cannot be denied, however, is that the influence of European ballroom dance forms is evident in the music of both Drummond and Ka'ai and both men recognised the potential to play classical repertoire on their instruments.

Echoes of the Machete

While researching the *Crescendo* magazine the only mention of the machete I was able to find was in an article by F. Landry Berthoud in his 'Mandolinist's Round Table', of 1918. In the article Berthoud wrote of an encounter with a Portuguese tailor by the name of Mr Millar. Mr Millar was born on the island of Madeira and played a small instrument which he (Millar) called a 'treble guitar' or 'nachette'. This instrument was 'about the size of an ukulele but simply a miniature guitar, finer and more delicate lines than a ukulele.' During the nineteenth century the machete was often spelt machette (Muir, 2022) and Berthoud is most likely mistaking the 'm' for an 'n'. Furthermore, as he tells us the instrument was tuned D4-G4-B4-D5 it is almost certainly a machete.

Mr Millar's repertoire was played entirely from memory and included 'fantasies on various operas, dainty little gavottes and mazurkas, some dashing little polkas, a march or two and a great number of Spanish and other characteristic songs'.[38] The article appeared in 1918 and Berthoud concluded by saying that he had met Mr Millar 'twenty years ago' and the tailor had since died. This would date his encounter around 1898. Whether Mr Millar played any of the Drummond

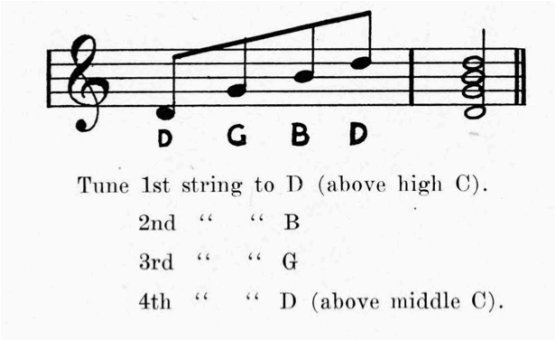
[37] *Hawaiian Gazette* 3 September 1879

[38] *Crescendo* Volume 10 Number 7, January 1918

pieces is impossible to know, but the article did reveal that repertoire of similar form was being played on the machete in North America at the end of the century.

In 1915 the ukulele's connections with Madeira were re-established with the publication of a method by A.A. Santos and Angeline Nunes. Their *Original Method and Self-Instructor on the Ukulele* claimed that 'the original way of tuning the Ukulele has not been put into print, [and] consequently it is unknown to most people'. As figure 1.2.12 shows, Santos and Nunes used the traditional tuning of the machete.

Figure 1.2.12. The 'original' way of tuning the ukulele according to Santos & Nunes



According to Santos and Nunes this was 'the true and original method on the ukulele'. (Santos & Nunes, 1915, 3). It was an extraordinary claim. While they made no mention of the machete it is most likely that Angeline Nunes had learnt about the machete from her father-in-law Manuel Nunes - one of the original ukulele makers from Madeira.[39] Nunes first made machetes in Hawaii, but later focused on making the more popular ukulele. What suddenly inspired Santos and Nunes to declare that the original tuning of the ukulele was the same as the machete is uncertain, but they regarded 'the true and original method' to be superior. Their method was, they claimed, for anyone who wanted 'to learn to play the Ukulele beyond just the mere strumming.' (Santos & Nunes, 1915, 3) It is not clear, however, what repertoire they had in mind as much of their method is dedicated to chords and strumming techniques. The few solo pieces consist of just a single melodic line. Although the method promised much the content is somewhat disappointing and, unsurprisingly, their 'original' tuning never caught on. Even so, it is an interesting and salient reminder of the ukulele's connection to the Madeiran machete.

[39] Manuel Nunes, a Portuguese immigrant from Madeira, was one of the first men to make ukuleles in Hawaii. Initially, however, Nunes made machetes and guitars. In 1885 his advertisement in the Portuguese Hawaiian newspaper *O Luso Hawaiiano* offered "instrumentos de corda, violas e machets" (string instruments, guitars and machetes). By 1909 he had declared himself the 'Inventor of the Ukulele' but at the time he was just about to open a new ukulele shop in Honolulu with his sons so this was more of a marketing ploy than a statement of fact.

Perhaps Santos and Nunes saw their method as a way of establishing the ukulele as a solo instrument based on stories they had heard from Manuel Nunes about the sophisticated repertoire of the machete. While none of the pieces in their method show any influence of the works of Candido Drummond, they did include a 'note' to say that the authors were 'arranging for piano and ukulele famous Hawaiian songs, classical and popular music which they intend compiling into an inexpensive book'. (Santos & Nunes, 1915, 16) Unfortunately, no record of any such book exists.

Jennie Durkee and her 'American Way' of Playing the Ukulele

One of the most interesting and innovative ukulele players who came from a guitar background was Chicago based Jennie Durkee. In 1917 Durkee was described by the *Crescendo* 'as one of America's greatest guitar soloists.'[40] Her repertoire on the guitar included substantial concert works including *Il Trovatore* by Mertz and *Grand Valse Caprice* by her teacher William Foden (1860-1947). As typical of the BMG movement Durkee also played mandolin and banjo. By 1917 she had taken up the ukulele as well. An editorial in the *Crescendo* of 1918 described her transition from not being able to listen to a ukulele, to becoming a 'ukulele fiend'.[41] The transition was thanks to Durkee's innovative approach which she called the 'American way' of playing the ukulele. This involved the use of a thick, felt pick which was moved rapidly across the strings to create a tremolo effect.

Durkee arranged numerous classical pieces for her tremolo technique, including *Barcarole* from *Tales of Hoffman* by Offenbach, *Serenade* by Moszkowski, *Funiculi Funicula* by Denza and *Sextet* from *Lucia di Lammermoor* by Donnizetti. Her performances on the ukulele received wide acclaim. In one concert Durkee reportedly 'played *Sextet* from *Lucia* on the ukulele before an audience of over five hundred masons and their wives and friends and they were most enthusiastic over her work.'[42] Her most popular pieces, including *Sextet* from *Lucia*, were published in 1917 in *The Artist's Collection of Ukulele Songs*. The book was originally self-published but promoted by Lyon & Healy, a company for whom her father had worked as an inventor and designer. Lyon & Healy also sold Durkee's special felt picks and, as an advertisement from 1918 shows, were obviously keen to promote her American approach.

> No longer is it [the ukulele] the insignificant 'strumming' accompaniment, but rather an interest commanding solo instrument destined to take its place in popular favor with

[40] *Crescendo* Volume 10 Number 5, November 1917

[41] *Crescendo* Volume 10, Number 10, April 1917

[42] *Crescendo* Volume 10 Number 10, April 1918

> Mandolin, Guitar, etc. Every teacher will unhesitatingly add the Ukulele to his curriculum if he hears the instrument played the 'American Way'.[43]

In 1919, due to the popularity of her collection of solos, Durkee produced a method called *The American Way: Instructions for Playing Solos*. In 1920 she combined both books to form one publication called *The American Way of Playing Ukulele Solos*. This edition was published by Lyon & Healy. The success of her book was lauded by the *Crescendo* magazine in October 1920. 'In less than two years her book has reached its sixth edition, and Lyon & Healy have sold 45,000 models of the little plectrum she devised.'[44] In 1921 the *Crescendo* called her 'the world's greatest ukulele soloist'[45] following a successful tour of Denver, Colorado, where she reportedly played to packed houses. One concert at the Knight-Campbell Music Store proved so popular that a 'second concert was given the same evening, and still hundreds of people were turned away.'[46] The programme included her popular ukulele solos *Dear Old Pal of Mine* by Rice, *Ave Maria* by Bach-Gounod, *Sextet* from *Lucia* by Donizetti and *Quartet* from *Rigoletto* by Verdi. The accolades continued and in December 1921 the *Crescendo* gave another glowing account of Durkee's ability.

> Miss Durkee also consented to appear at the noon concerts which are given at the Denver Auditorium, which has a seating capacity of 12,000. Although it is never completely filled at these noon concerts, she did appear before 1,500 people, playing with the big organ, and her ukulele was heard in every part of the hall. She was enthusiastically received here also, and had to respond to two encores. An excellent example of Miss Durkee's artistry; also proof that the ukulele, when properly played, can be heard in the largest halls.[47]

If her ukulele really could be heard in every part of such a large auditorium she must have produced an impressive sound. As well as her public concerts and demonstrations Durkee also appeared on the new medium of radio and on 27th April 1922 she made three trials recordings for the Victor Talking Machine Company. The Discography of American Historical Recordings website lists their status as unknown. The recordings were not publicly released and presumably have been lost. The

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[43] *Crescendo* Volume 11 Number 1, July 1918

[44] *Crescendo* Volume 13 Number 4, October 1920

[45] *Crescendo* Volume 14 Number 5, November 1921

[46] *Crescendo* Volume 14 Number 6, December 1921

[47] *Crescendo* Volume 14 Number 6, December 1921

descriptions are all we have to tell us that *Sextet* from *Lucia* was performed as a ukulele solo with piano accompaniment, while *Dear Old Pal of Mine* and *Humoresque* were both ukulele solos.

Throughout the 1920s Durkee was frequently featured on the new medium of radio. In May 1922 she also featured on Chicago K-Y-W in a musical programme courtesy of Lyon & Healy. On this occasion she was accompanied by Wayne Wadhams on the guitar. On 23 September 1927 the *Santa Ana, California*, reported that 'Louise Howatt, the "happiness girl of radio" from station KMTR, and Jennie Durkee, ukulele wizard, proved to be the most popular artists on the program presented by Walter Biddick.'[48]

A final mention of Durkee was made in *Van Nuys News* in December 1931 when she appeared as the guest artist at the Christmas concert of the Hollywood Conservatory of Music and Arts. The article described her use of the felt pick as 'a means whereby the ukulele can be used as a high-class solo instrument.' The article also mentioned her recent tour of Buffalo, Syracuse and Detroit culminating at the American Guild Convention in New York. During the course of the tour Durkee had reportedly played to over seventy-five thousand people.

> Rarely had the public given thought to the interpretation of the work of masters such as Chopin, Beethoven, Mozart and Rossini on such a small instrument as a ukulele, and such a feat accomplished in the delightful manner for which Miss Durkee is noted, is an inspiration to any audience.[49]

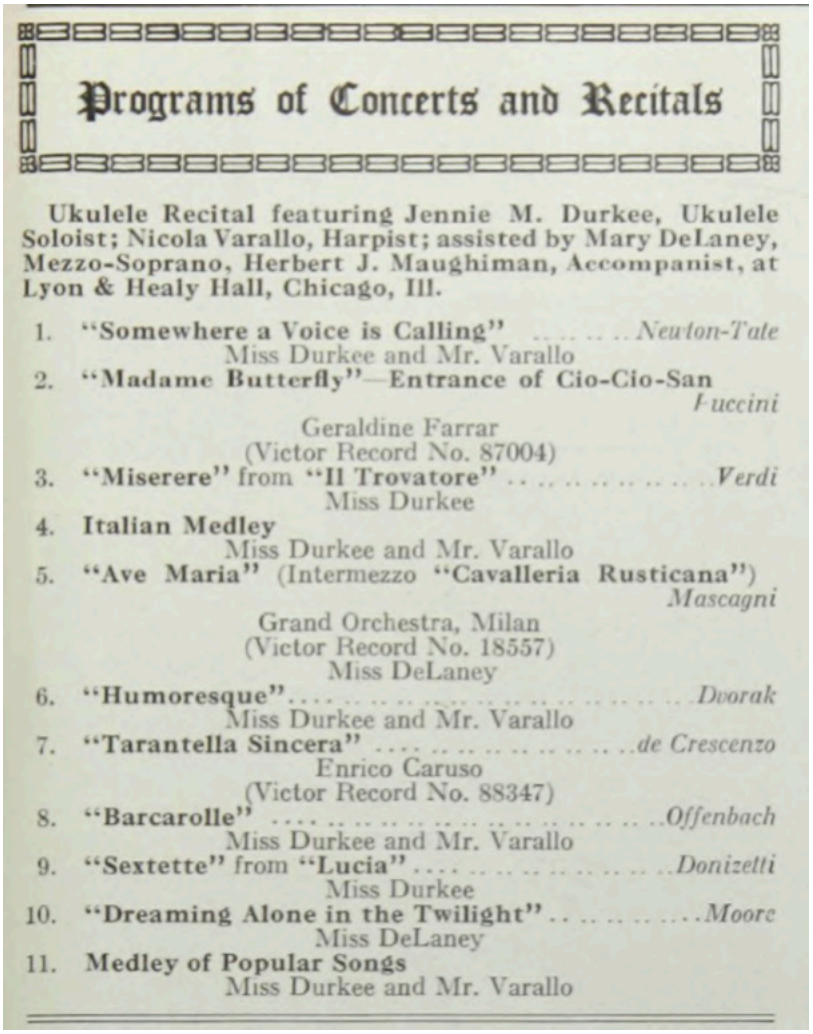
Despite her popularity and the many accolades she received, Durkee's method of playing the ukulele did not endure. Today she is virtually unknown, having been over-shadowed by the likes of May Singhi Breen, Roy Smeck, Cliff Edwards and George Formby. Nevertheless, Durkee remains an important early figure in the history of the ukulele. She was a true pioneer, not only in her use of the felt pick but in her arrangements of classical pieces by composers such as Donizetti, Offenbach, Verdi, Dvorak and Rossini. (see figure 1.2.13)

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[48] Accessed from *Newspaper Archive* on 16 June 2022

[49] *Van Nuys News*, December 17, 1931

Figure 1.2.13. The programme from one of Jennie Durkee's concerts



Programs of Concerts and Recitals

Ukulele Recital featuring Jennie M. Durkee, Ukulele Soloist; Nicola Varallo, Harpist; assisted by Mary DeLaney, Mezzo-Soprano, Herbert J. Maughiman, Accompanist, at Lyon & Healy Hall, Chicago, Ill.

1. "Somewhere a Voice is Calling" ..... *Newton-Tate*
   Miss Durkee and Mr. Varallo
2. "Madame Butterfly"—Entrance of Cio-Cio-San *Puccini*
   Geraldine Farrar
   (Victor Record No. 87004)
3. "Miserere" from "Il Trovatore" ..... *Verdi*
   Miss Durkee
4. Italian Medley
   Miss Durkee and Mr. Varallo
5. "Ave Maria" (Intermezzo "Cavalleria Rusticana") *Mascagni*
   Grand Orchestra, Milan
   (Victor Record No. 18557)
   Miss DeLaney
6. "Humoresque" ..... *Dvorak*
   Miss Durkee and Mr. Varallo
7. "Tarantella Sincera" ..... *de Crescenzo*
   Enrico Caruso
   (Victor Record No. 88347)
8. "Barcarolle" ..... *Offenbach*
   Miss Durkee and Mr. Varallo
9. "Sextette" from "Lucia" ..... *Donizetti*
   Miss Durkee
10. "Dreaming Alone in the Twilight" ..... *Moore*
    Miss DeLaney
11. Medley of Popular Songs
    Miss Durkee and Mr. Varallo

Changing Attitudes

The early influence of guitarists, including Ernest Ka'ai, Vahdah Olcott-Bickford, and Jennie Durkee is evident in their ukulele methods both in the application of fingerstyle techniques and in their choice of repertoire. The Bickfords were pragmatic in their approach to the ukulele. They recognised its rising popularity and the growing market for methods whilst remaining lukewarm in their enthusiasm. Their foreword was condescending in stating that 'No attempt has been made to make more out of the ukulele than its capacity warrants.' (Bickford, 1920, 3) The ukulele could not, in their opinion, take the place of 'an instrument with greater power, tonal and harmonic possibilities'. (Bickford, 1920, 3) No doubt they were referring to the greater range and volume of the guitar.

Durkee on the other hand embraced the ukulele wholeheartedly; her unique approach and distinctive repertoire suggest an inquisitive and daring nature. By drawing on her knowledge and experience of the guitar, the mandolin, and the banjo she offered a unique vision for the new and emerging ukulele. Although many members of the BMG movement initially embraced the ukulele as a kindred instrument, attitudes were soon to change. With the growing reputation of guitar virtuoso Andrés Segovia, who made his American debut in 1928, more players began to focus specifically on the classical guitar. When Sophocles Papas took over *Cadenza's* 'The Guitar and Steel Guitar Round Table' from Vahdah Olcott-Bickford in March 1929 he promised that 'the Hawaiian guitar will receive equal attention and consideration as the Spanish.' (*Crescendo* February 1929/Noonan, 2008, 166) But, as Noonan pointed out, 'Papas encouraged American guitarists to follow Segovia's model, a model that had no use for mandolins, banjos, and ensembles of these instruments.' (Noonan, 2008, 169) This undoubtedly included the ukulele.[50] As the push towards classical guitar gathered momentum the ukulele found its niche in entertainment and comedy with performers such as Roy Smeck (USA) and George Formby (UK).

Virtuosic strumming patterns, novelty techniques such as twirling the ukulele and in the case of Formby humorous and risqué lyrics, became the new norm for the ukulele on both mainland America and the UK. Only in its homeland of Hawaii was the ukulele taken more seriously with notable players including Herb Ohta, Eddie Kamae, Jesse Kalima, Israel Kamakawiwa'ole and more recently Daniel Ho and Jake Shimabukuro. In 2016 Ho teamed up with classical guitar legend Pepe Romero to record their groundbreaking album *Aloha España.* The album featured duet arrangements for guitar and ukulele of guitar classics such as *Recuerdos de la Alhambra* by Tárrega, *Canarios* by Sanz and *Asturias* by Albéniz. In the liner notes Romero made special reference to John King for 'championing classical ukulele [and being] a major inspiration for me to do this project.'[51]

Looking Back to Move Forward

John King's *The Classical Ukulele* (2004) has undoubtedly had a huge impact in steering the ukulele towards a fingerstyle approach based on classical guitar techniques and a repertoire which finds its roots in Western classical music. In so doing the ukulele has reconnected with the aspirations of some of the earliest players including Ernest Ka'ai, Vahdah Olcott-Bickford and

[50] Noonan refers to the ukulele on several occasions but always as 'this faddish instrument' (Noonan, 2008, 67) or as an 'insubstantial fad'. (Noonan, 2008, 129) His dismissal of the ukulele seems to reflect the growing elitism of classical guitarists in the 20th century and the move away from the early BMG ethos of regarding the fretboard instruments as a family.

[51] *Aloha España* CD, 2016, Pepe Romero & Daniel Ho

Jennie Durkee. King took the ukulele to new levels with his virtuosic playing, immaculate tone, and historical knowledge. His campanella arrangements of Bach demonstrated how a Baroque guitar technique could be applied to many masterpieces to create a new vision for the ukulele. Although King arranged works for the ukulele, composers today including Choan Gálvez, James Else, Loretta Notareschi, Tony Mizen, David John Roche and the author have gone one step further by using campanella techniques to create new and original repertoire for the ukulele.

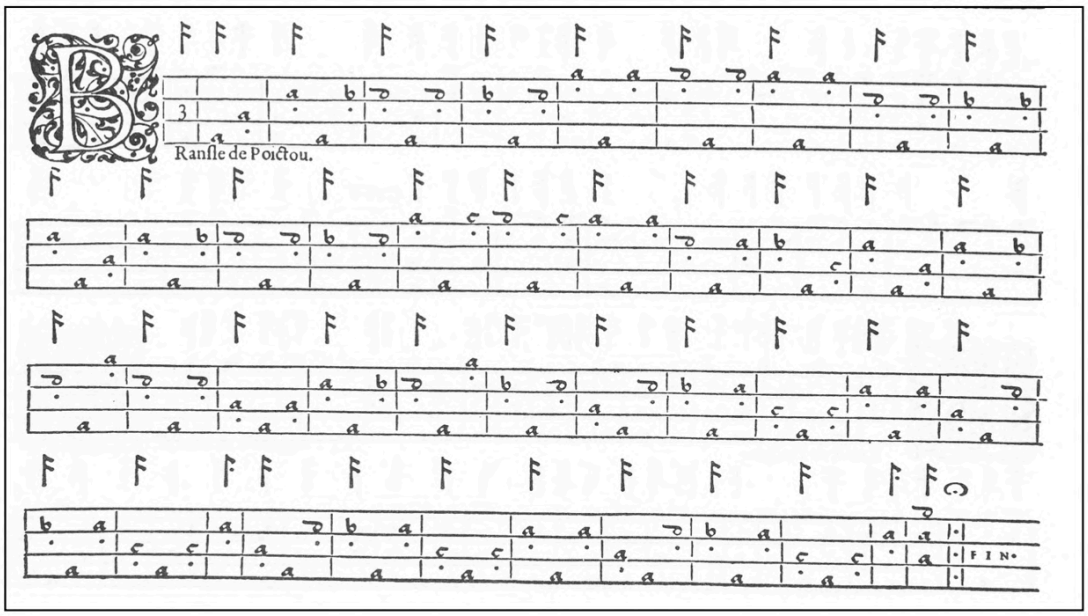
By continuing to explore the ukulele's European roots a fuller understanding of the musical and technical possibilities of the instrument are being made known. Many players including Davide Donelli (Italy), Elisabeth Pfeiffer (Germany), Donald Bousted (1957 - 2021, UK) and the author (UK), have been performing four course Renaissance guitar repertoire on the ukulele, thus illuminating the historical connections between these two instruments. Figure 1.2.14 shows the original tablature of *Bransle de Poitou* from Adrian Le Roy's *Tiers Livre De Tablature de Guiterre* published in Paris in 1552.[52] The French tablature uses letters rather than numbers to represent the frets but the premise is the same as our modern tablature. The piece is easily played on the ukulele using the original tablature. Thus, the ukulele can be seen as both a time traveller revisiting and revitalising the roots of the guitar, and a trailblazer searching for a new future.

Conclusion

Although the ukulele was only created after 1879 it is the embodiment of an instrument which is both new and ancient, Hawaiian and European, strummed and plucked. An ever growing number of classical ukulele players, particularly those with an interest in researching the instrument, are bringing more awareness to the potential of the ukulele as a serious instrument. This is reflected in the development of its repertoire and the inclusion of the ukulele in university courses such as that conducted by Giovanni Albini at the Conservatorio Antonio Vivaldi in Alessandria, Italy. Many early ukulele methods are testimony to how the instrument has always been influenced by guitarists. In this context the classical ukulele movement is perhaps best viewed as a reawakening, revitalising and reimagining of old techniques, methods and repertoires, driven by a vision to create new and contemporary works. In its brief history the ukulele has enjoyed waves of popularity followed by periods of ignominy. It is hoped that by developing an academic foundation and robust curriculum for the ukulele the instrument will continue to enjoy success.

[52] For audio see Folder C.4. Playing four course guitar repertoire on the ukulele is discussed further in Section 2 'The Classical Ukulele in Practice'.

Figure 1.2.14. The original tablature of Bransle de Poitou by Adrian le Roy, 1552





An advert for Durkee's 'American Way' of playing the ukulele with her innovative felt pick. From the *Crescendo* Vol.11 No.1

2. The Classical Ukulele in Practice

This section provides a commentary on the techniques used in the new works which include original arrangements and compositions, as well as commissioned pieces. For those who have access the full scores are in Folder B and the recordings in Folder C. For those who do not have access to these folders please see Spotify, Bandcamp and YouTube for the audio and video recordings, and the individual composer's websites for scores. My published scores are available from Les Productions d'Oz.[53] The commentary, however, includes musical excerpts from the scores. Each chapter discusses a different technique and includes examples of how these techniques have been applied to the new works.

The chapters are:

2.1. Campanella
2.2.a. Tintinnabuli
2.2.b. Combining Tintinnabuli & Campanella
2.3. Arpeggios
2.4. Chord-melody
2.5. Approaches to Playing Chords
2.6. Tremolo
2.7. Harmonics
2.8. Microtonal Ukulele
2.9. Ukulele & Electronics
2.10. Damping/Muting Effects
2.11. Alternate Tunings
2.12. A Ukulele Concerto
2.13. Overview of the Didactic Material

As one of the first composers to use campanella and tintinnabuli in my original works for the ukulele the chapters on these techniques are the most substantial; providing not only an overview of these techniques but how and why they were employed. Many of the author's compositions and arrangements combine a variety of techniques, thus offering new and idiomatic ways of composing

[53] https://productionsdoz.com/nos-artistes/muir-samantha?lg=en_US

and arranging for the ukulele. During the course of this research many of the new works were published by Les Productions d'Oz, thus forming a substantial body of new repertoire for classical ukulele. The publications include three books of arrangements, one book of technical exercises, one method, and seven original works. The commentary also refers to a number of unpublished and commissioned scores, all of which are catalogued at the end. The new works fulfil the over-riding aim of this research inquiry:

- To create original, contemporary and idiomatic works for classical ukulele through the application and adaptation of techniques, both historical and modern, commonly associated with the guitar.

2.1. Campanella

Introduction

Seventeenth century guitar composers, including Gaspar Sanz (1640 - 1710) and Francisco Corbetta (1615 - 1681), sometimes used the re-entrant tuning (see figure 2.1.1) of the Baroque guitar to place melodic notes on the fourth and fifth courses. In linear tunings (such as the modern guitar) these strings are tuned an octave lower and would most commonly be used for bass notes. But, Baroque guitar composers found that the higher tunings allowed melodic notes to be placed on alternating strings to create novel over-ringing effects. James Tyler gives three possible tunings for the five-course Baroque guitar (Tyler, 2011, 5). The tuning most relevant to this discussion is shown in figure 2.1.1. Here the fourth and fifth courses are tuned as re-entrant unisons to create a tuning which is ideal for campanella playing.

Figure 2.1.1. The re-entrant, or campanella, tuning of the Baroque guitar (from Tyler, 2011, 5)

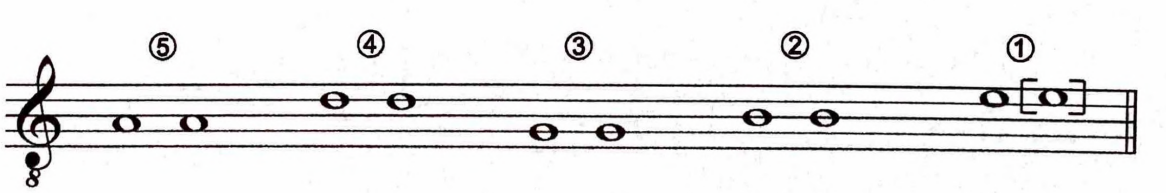
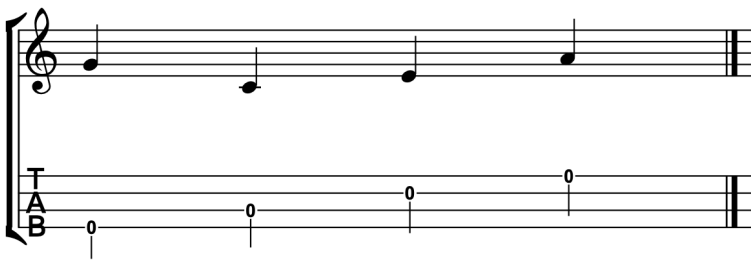


Figure 2.1.2 shows the standard re-entrant tuning of the ukulele. The four single strings are tuned to the intervals of a falling 5th, a rising 3rd and a rising 4th thus creating the same pitch relationships as the Baroque guitar excluding the fifth course.

Figure 2.1.2. The re-entrant tuning of the ukulele

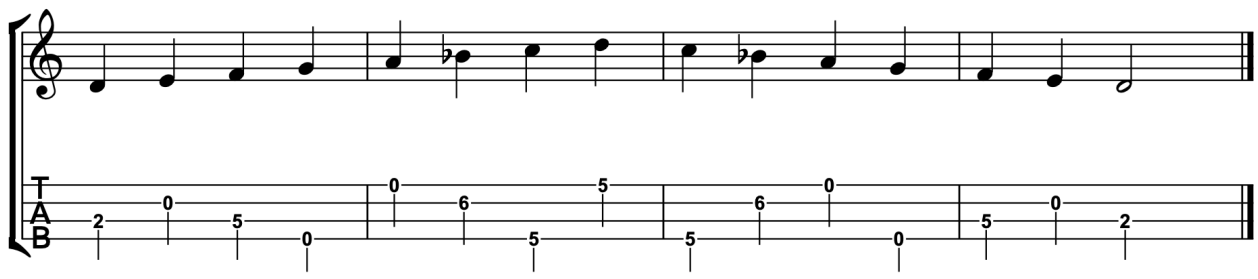


The term campanella, also *campanelle* (French) and *campanela* (Spanish), comes from the Italian word meaning ‘a small bell’ (OED). James Tyler and Paul Sparks in *The Guitar and Its Music* describe ‘campanelas’ as ‘cascading scales’. (Sparks & Tyler, 2002,153) According to Sparks and Tyler the technique was first used by Angelo Michele Bartolotti (c.1615-81) in his *Libro primo di chitarra spagnola* of 1640. ‘Another innovation is Bartolotti’s use of scale passages wherein the individual notes are divided up amongst all the courses, giving the harp - or bell-like effect of notes cascading of ringing one into the other.’ (Sparks & Tyler, 2002, 69) The term ‘campanelas’ was, however, first used by Gaspar Sanz some thirty-four years later. In *A Guide to Playing the Baroque Guitar* (OUP, p.111, 2002) Tyler elaborates further:

> It [campanella] is performed by plucking each note of a scale (or other melodic passage) on a different course and employing as many open strings as possible, thus allowing each scale note to ring on longer than would otherwise be possible. The effect is similar to playing a rapid scale on the open strings of a harp without dampening any of them; successive notes ring on and blend into the previous ones. (Tyler, 2011, 24)

When applied to the ukulele a campanella scale (see figure 2.1.3) can be played with the notes moving across all four strings, thus maximising the over-ringing effect. In campanella playing open strings feature prominently for both musical and technical reasons. An open note will ring-on naturally until either the string is fretted or the note naturally decays. Meanwhile, the fretting hand uses the open note as an opportunity to shift position. Playing these shifts often require a leap up the fingerboard. For example in figure 2.1.3 the first finger plays the opening D. As the next open string E is played the first finger moves to fifth position ready for the F.[54]

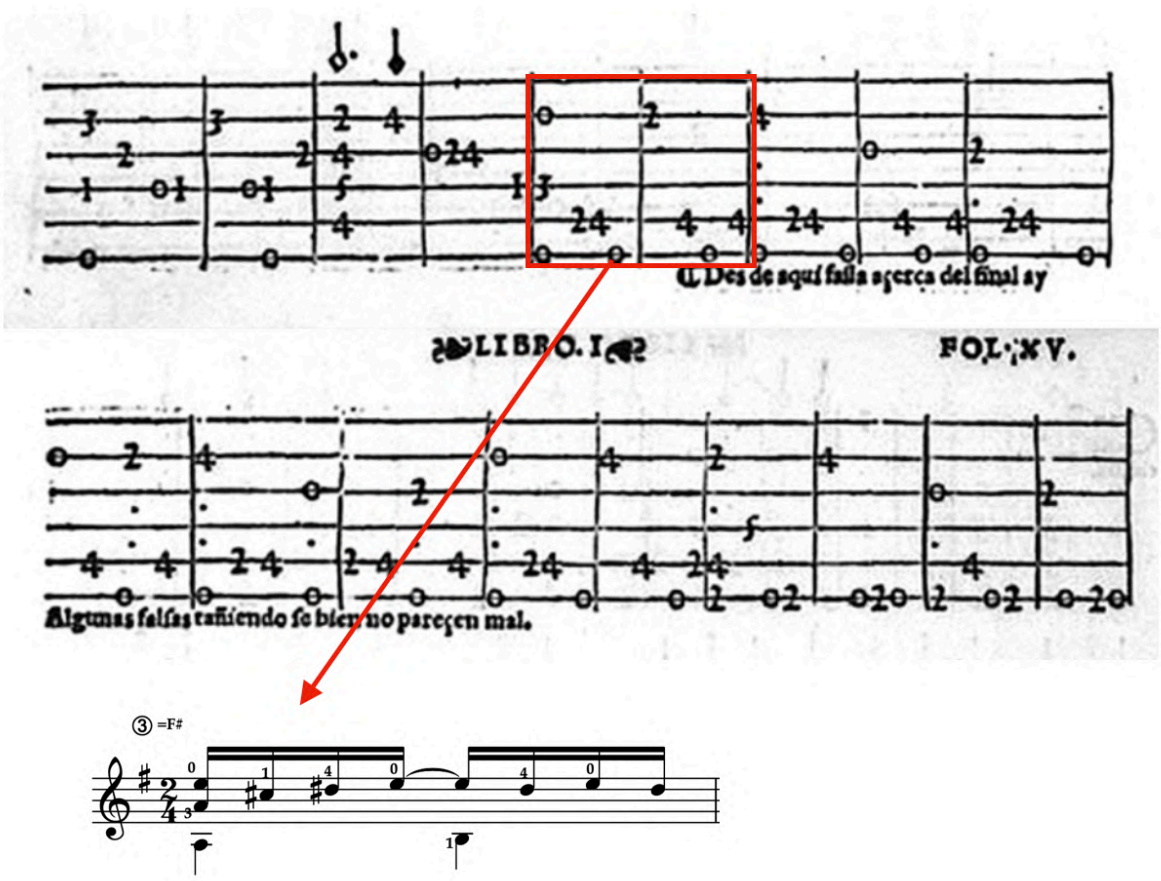
Figure 2.1.3. The D natural minor scale arranged in campanella style as the tab clearly shows.



[54] For audio see Folder C.5.

Tyler's comparison to the harp is worth exploring as it brings to mind a much earlier piece by Spanish composer and vihuelist Alonso Mudarra (c.1510-80). Mudarra's well known *Fantasia* (often called *Fantasia no.X*), which 'imitates the harp in the style of Luduvico',[55] was first published in *Tres Libros de Musica* in 1546.[56] His reference to the harp is often misconstrued as an early example of campanella playing. Mudarra's intention, however, was not to create campanella effects but to imitate the chromatic dissonances associated with the well known harpist Ludovico.

Figure 2.1.4. Showing part of the Fantasia by Mudarra from the original manuscript



Towards the end of the *Fantasia* the score contains the direction, 'From here to near the end there are some dissonant notes; played well, they are not displeasing.' (see figure 2.1.4) The piece would have sounded very modern for its time and perhaps explains why it is still popular with classical guitarists today. The main point here is that the reference to the harp was not, as is sometimes interpreted, in relation to over-lapping or harp-like effects (campanellas) but in the deliberate use of dissonance on the guitar, or vihuela. Figure 2.1.4 shows a section of the *Fantasia* from the original

[55] The harpist was most likely Ludovico, so this appears to be a spelling mistake by Mudarra.

[56] The vihuela is a 15th century guitar shaped instrument from Spain with either 5 or 6 double strings.

manuscript with an example of the tablature in notation to highlight the dissonance created between the D# played on the second string and the open E string. While the objective is to create a short and specific dissonance rather than the cascading effect of a campanella scale, the over-ringing effect is, nevertheless, campanella-like. This suggests that the deliberate use of over-ringing notes is an idiomatic feature of plucked stringed instruments and is perhaps as old as the guitar itself. It was the Baroque guitar composers, however, who recognised campanella as a deliberate compositional device.

In his book *Italian Guitar Music of the 17th Century: Battuto and Pizzicato*, Lex Eisenhardt regards campanella as 'part of the palette of standard formulas' for guitarists during the Baroque period.[57] He also points out that by 1648 Corbetta was using very few campanellas and had perhaps come to view the effect as 'somewhat hackneyed'. (Eisenhardt, 2015, 139) Nevertheless, campanella has remained an idiomatic and favourite guitar technique that is still used by guitarists, and subsequently ukulele players, today.

Campanella: An Affordance Issue

In his article 'Guitar Thinking'[58] Jonathan De Souza uses the term *affordances*[59] to describe 'an object's…possibilities for action'. (De Souza, 2021, 5) He uses the example of a guitar which *affords* playing and a chair which *affords* sitting. As De Souza points out, however, a chair may also be used in other ways, such as to prop up a music book, and a guitar cannot be played by a goldfish. The possibilities of an object, therefore, depend on certain factors such as a person's abilities or an object's conventions such as the tuning of a guitar. To this end the tuning of a guitar *affords* the use of campanella.[60] But, as will be further demonstrated, the use of re-entrant tunings further extends the possibilities for campanella, often allowing it to occur naturally as an idiomatic feature of the instrument.

[57] Generally considered to be from approximately 1600 to 1750.

[58] De Souza, Jonathan. 2021. 'Guitar Thinking' Soundboard Scholar 7, (1). https://digitalcommons.du.edu/sbs/vol7/iss1/1

[59] De Souza's use of *affordances* is based on the work of JJ Gibson, *The Theory of Affordances, in Perceiving, Acting, and Knowing: Toward an Ecological Psychology*, ed. Robert Shaw & John Bransford (Hillsdale, NJ: Erlbaum, 1979), 67-82

[60] A brief mention should be made of the 'harp effect' used in the original compositions of French-Algerian guitarist Pierre Bensusan in which melodic notes are placed on different strings to prolong the resonance of each string. He notes the importance of using a combination of open strings and fretted notes to achieve the best effect. See P. Bensusan, 1985. *The Guitar Book*, Hal Leonard, USA

Campanella on the Ukulele

As previously shown the standard tuning of the ukulele is re-entrant, thus making the instrument more characteristic of a Baroque guitar than a standard classical guitar. Although campanella was not intentionally used on the ukulele until the work of John King in the 1990s, the use of over-ringing effects have occurred naturally since the instrument first gained popularity in the early twentieth century. Campanella can occur idiomatically or be employed intentionally on the ukulele. John King's pioneering work in campanella style will be discussed in due course. Firstly, however, the use of over-ringing effects by the Hawaiian virtuoso Ernest Ka'ai (1881-1962) will be used to demonstrate how campanella is a natural and idiomatic feature of the ukulele (for the purposes of this chapter re-entrant tuning will be assumed).

The Natural Use of Campanella on the Ukulele

The Ukulele Hall of Fame website regards Ernest Ka'ai as the first Hawaiian ukulele virtuoso. He was also the author of the first known ukulele method. *The Ukulele A Hawaiian Guitar and How To Play It* was first published in 1906[61] and revised in 1910. In 1916 Ka'ai released a new and expanded method called *The Ukulele and How It[']s Played.* Several pieces from this method employ campanella effects.

For example, *Haere* (see figure 2.1.5) begins with the melodic notes placed across the strings to maximise the over-ringing. The open G *could* be played at 3,2 but, as the tablature clearly indicates, Ka'ai specifies an open G (0,4). The resulting over-lapping notes form little triplet figures of G-E-G followed by a crotchet. The crotchets form a descending melodic line of C-B-A-G, subtly altering the harmony (C major , C major 7, A minor 7) until the final beat of bar 2 when the phrase concludes with a full C major chord. It is a simple, but effective, way of creating both melodic and

Figure 2.1.5. The opening bars of Haere by Ernest Ka'ai



[61] No known copies of the 1906 edition exist, but John King did locate an edition from 1910.

harmonic interest without the voice leading capacity of a lower register. It also feels quite natural to play the G as an open note (0,4) with the thumb while the open E (0,2) is struck with the index finger. Meanwhile, first string notes naturally fall under the middle finger to create a picking pattern of *i-p-i-m*.

Similarly in *Banjo Schottische* the Gs in bars 7 and 15 are indicated as open strings (0,4). The over-ringing effects shown in figures 2.1.6 and 2.1.7 create a fuller, more resonant sound. Employing the open G (0,4) to create these cross-string resonances appears to be intentional. In *El Recuerdo* Ka'ai used a G but this time specified that it should be played at 3,2 (see figure 2.1.8) to form a linear scale. It makes sense both musically and technically to play this scale as an ascending line with the closed G, rather than as a campanella with an open G. The scale can, therefore, either be played with *p-i* or *i-m*.

Figure 2.1.6. Bars 5 to 7 of Banjo Schottische by Ernest Ka'ai



Figure 2.1.7. Bar 15 of Banjo Schottische by Ernest Ka'ai

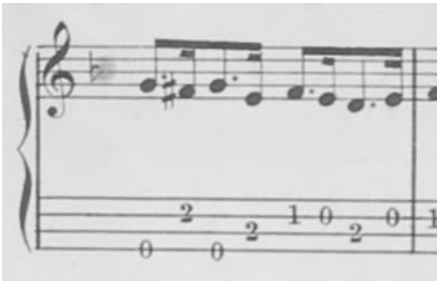


Figure 2.1.8. Bar 1 of El Recuerdo by Ernest Ka'ai



Haere and *Banjo Schottische* show Ka'ai's clever and deliberate use of the re-entrant tuning to create a fuller, more pleasing sound. There are, however, a number of anomalies in his method which raise some questions. For example, the opening of *Kawehi* features a triplet pattern not unlike the one used in *Haere*. But, as can be seen in figure 2.1.9, this time the tablature indicates that the notes are to be played along the first string. This seems odd given that the triplet notes are all derived from the opening chord and the logical thing would be to maintain the chord shape and play them as an arpeggio. From a practical point of view it does not makes sense to move B ♭ from 6,2 to 1,1 - as is shown in the tablature. The shift would be a musical and technical disruption and it seems highly unlikely that Ka'ai would actually play it this way. Similarly in bars 2 and 3 it makes more sense to maintain the chord shapes and play the triplets across the strings.

Figure 2.1.9. The opening of Kawehi from Ka'ai's method (1910)



John King's arrangement in *Famous Solos and Duets* (2004) places the notes across the strings in campanella style (see figure 2.1.10).

Figure 2.1.10. John King's arrangement of Ka Wehi using the campanella effect



This would seem to be the most natural solution, both technically and musically, and it is hard to imagine that Ka'ai would not play these notes across the strings. Given that the method contains a number of mistakes, such as the D♯ in bar 2 of the notation, which should be an E (see figure 2.1.9), it is not unreasonable to suggest that the tablature was printed incorrectly.

Although Ka'ai is not known to have used the term campanella, or to have specifically applied it as an intentional device, the above examples demonstrate how he made use of the *affordances* of the ukulele to create music which, as Jonathan De Souza describes in *Music at Hand*, both sounds and feels good. (De Souza, 2017) Ka'ai's use of the open G (0,4) in *Haere* and the *Banjo Schottische* are examples of the performer/composer 'thinking with his hands and thinking with his instrument' (De Souza, 2017, 27). I would argue that it was Ka'ai's innate knowledge of the ukulele that naturally led him to create campanella effects. As will be demonstrated, however, this was something quite different from King's intentional use of campanella as a stylistic technique.

John King's Intentional Use of Campanella on the Ukulele

The deliberate use of campanella as a technical device on the ukulele was first advocated by John King in the 1990s.[62] King had learnt about campanella while studying classical guitar. When he realised that the ukulele, like the Baroque guitar, also uses re-entrant tuning he decided to try campanella on the ukulele. Sadly, King passed away in 2009 but his experiment was recorded on his Nalu Music website.

> I taught myself an appropriate Bach tune using campanella technique and applied it to my mother's pineapple [ukulele]. It was a revelation. The instrument had a voice and when I played, it sang to me.[63]

King's campanella arrangements - which ranged from works by JS Bach to traditional Irish, Scottish and Hawaiian tunes - helped to bring new technical and musical challenges to the ukulele. Nowhere is this more clearly illustrated than on his groundbreaking CD *JS Bach: Partita No.3 BWV 1006 for Unaccompanied Ukulele* (2006). The CD also includes a selection of pieces from Bach's cello suites as well as *Jesu, Joy of Man's Desiring*. The distinguishing feature of the CD is that all of the pieces are arranged in campanella style. King received wide acclaim for his harp-like sound, virtuosic technique and ability to bring the ukulele into the realm of classical music.

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[62] Although John King's approach has already been discussed in Chapter 1.2 'Classical Approaches to the Ukulele' a brief overview is provided here to give context to this discussion.

[63] Unfortunately the Nalu Music website is no longer available.

A number of pieces on the CD were included in King's *The Classical Ukulele* book/CD (2004). This book, which also includes arrangements of pieces by Mozart, Beethoven and Couperin, helped to spark an approach to the ukulele that finds its roots in classical music and has become known as classical ukulele. King's campanella approach was not, however, limited to classical repertoire but embraced traditional music from Hawaii, Scotland and Ireland. His innovation was to apply campanella technique to an entire piece. This was a very different approach to the Baroque composers who used campanella as a device to create a limited, novel effect within a specific piece. By applying campanella to an entire piece, King treated campanella as a genre or style in itself.

Using Campanella in My Arrangements

Campanella is one of the most effective, interesting and challenging techniques to use on the ukulele. Like King, it was the musical potential and technical challenge of this technique that initially attracted me[64] to the ukulele. I have made extensive use of campanella in many of my arrangements, including works by Bach, Bartok and Sor as well as in many traditional and Renaissance pieces.

Muir v King: Two Different Campanella Approaches to Prelude BWV 1007 by JS Bach

My arrangement of the *Prelude BWV 1007*[65] from the 1st cello suite by Bach, based on the manuscript of Anna Magdalena, offered a different approach to that used by John King. While King applied campanella to every note I found it more effective to use campanella to highlight specific passages. Figure 2.1.11 shows the opening of King's arrangement[66] with all the notes placed on alternating strings. Meanwhile figure 2.1.12 shows my approach with the higher notes placed on one string.

In bar 1 King placed the Gs at 3,2 while I decided to play them as open notes (0,4). In his version the notes E-D-E are played across the strings in the following pattern: 7,1-7,4-7,1. Meanwhile, I opted to play all three notes on the first string with a slur between the E and the D (see figure 2.1.12). Playing the notes on alternate strings creates an over-ringing effect between E and D while playing them on the same string creates a more linear effect. Both approaches are effective in their own ways; King's over-lapping notes result in subtle dissonances while my more linear version makes use of slurs to create a more fluid approach.

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[64] As this discussion is about my own work I will use first person rather than 'the author'.

[65] For Audio see Folder C.5. For Full Score see Folder B.3.

[66] King 2007

Figure 2.1.11. The opening bars of John King's arrangement of the Prelude BWV 1007 by JS Bach

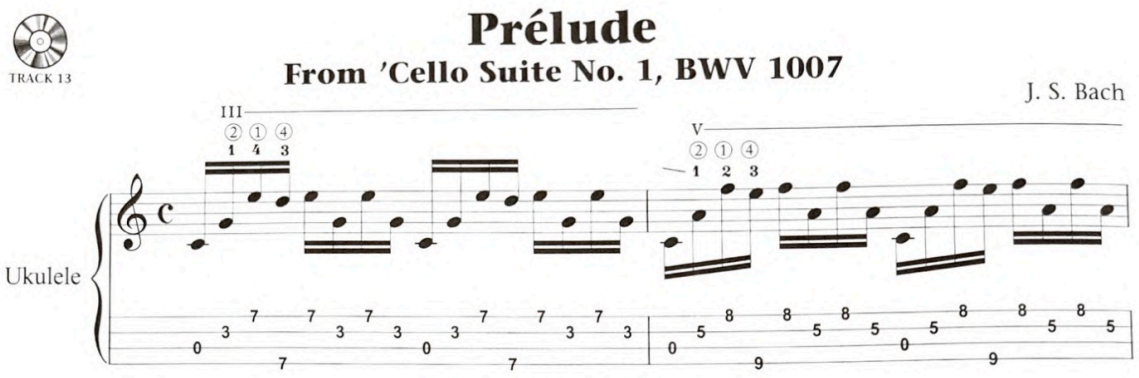
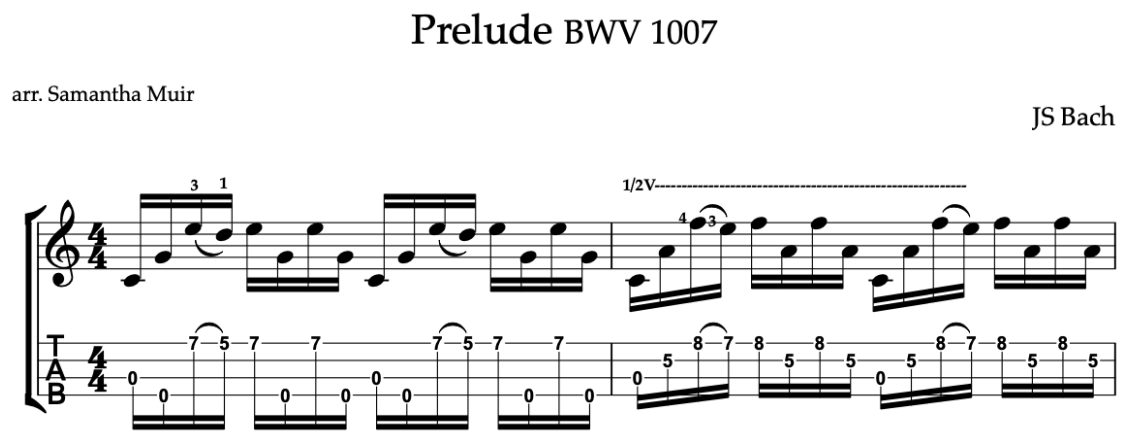


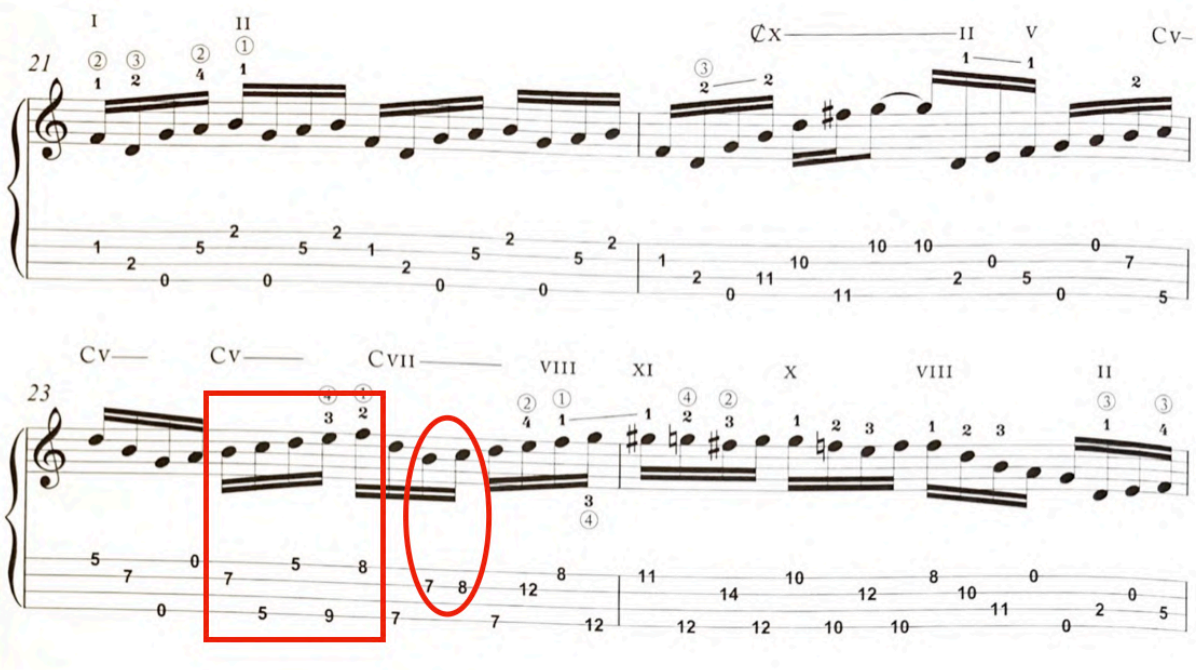
Figure 2.1.12. The opening bars of my arrangement of the Prelude BWV 1007 by JS Bach



In the second section, beginning midway through bar 22, (see figures 2.1.13 & 2.1.14) both arrangements utilise campanella approaches but in different ways. In bar 23 I chose to play the B on beat three at 11,3 while King placed it at 7,2. In King's version the following C (5,4) and D (5,1) are then played with a barre at the fifth fret (see the notes in the red square in figures 19 and 20). I placed the C at 8,2 and the D at 7,4. The next two notes (E and F) are then played on the first string at 7,1 and 8,1. King, however, maintained the campanella effect by playing the E at 9,4 and the F at 8,1. We both placed the following D and B at 7,1 and 7,2 respectively but while I play them with fretting fingers 1 and 2, King suggests using a barre at the seventh fret. Given that finger 1 is

barring the seventh fret the most obvious solution for King, therefore, is to let finger 2 drop onto the C at the 8,2, thus breaking the campanella (see the notes circled in red in figures 2.1.13 and 2.1.14). Although my arrangement was more flexible in its approach to campanella I did opt to play the B and C on different strings - 7,2 - 5,4. I would argue, however, that these fingering decisions were technical as much as musical as both King and myself were seeking a way to transition into the chromatic scale passage in bar 24.

Figure 2.1.13. Bars 21-24 of JK's arrangement of the Prelude BWV 1007 by JS Bach with the relevant passages highlighted in red.



To return to King's version, on beat four of bar 23 finger 1 is still positioned as a barre on the seventh fret. After the D at 7,4 is played, finger 4 stretches up to play the E at 12,2. At this point the barre is released and finger 1 moves to play the F at 8,1. Finger 3 then stretches across to play the G at 12,4 and finger 1 slides up to play the G♯ (11,1) at the beginning of bar 24. In my version I avoid the barre by using finger 1 on the D at 7,4 and then sliding it back to fifth position ready to play the C at 5,4. Meanwhile finger 4 drops onto 7,2 to play the B. Following the C at 5,4, the D at 5,1 is played with finger 2. Finger 2 then slides up ready to play the F at 8,1 and the E is played at 9,4 with finger 3. This prepares the hand to play the chromatic scale passage in bar 24 on the first string. By comparison King's fingering moves the hand into eleventh position (meaning finger 1 plays the G♯ at 11,1) ready to play the chromatic scale passage as a campanella. Played this way means that at the beginning of bar 24, although the chromatic notes are descending in pitch from G♯

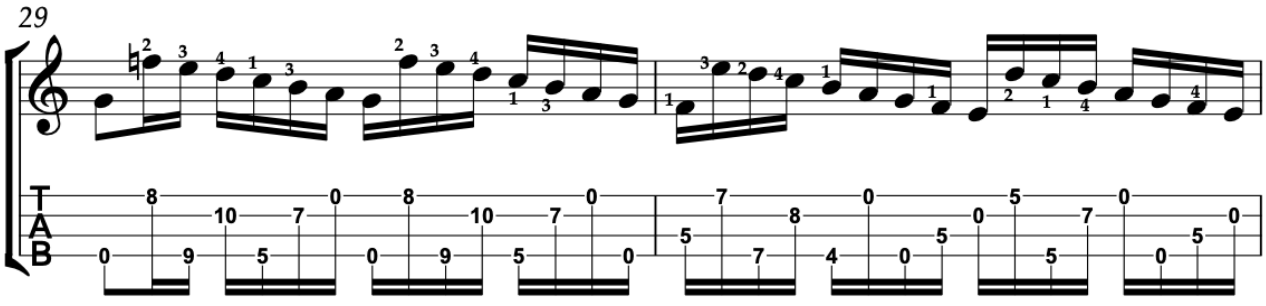
(11,1) to G (12,4) to F♯ (14,2), they are now ascending in position - moving from frets eleven to twelve to fourteen (refer to figure 2.1.14). The campanella also sounds very dissonant and modern which is why I opted to use step-wise motion for this passage.

Figure 2.1.14. Bars 21-24 of my arrangement of the Prelude BWV 1007 by JS Bach with the relevant passages highlighted in red.



Like King I did opt to play the cascading scale passages in bars 29 and 30 (see figure 2.1.15) to evoke the sound of 'little bells'. In the final analysis both arrangements offer unique and valid solutions to playing this 'Prelude' on the ukulele.

Figure 2.1.15. Bars 29 - 30 of my arrangement of the Prelude BWV 1007 by JS Bach



Brian Boru's March

Brian Boru (941-1014) was High King of Ireland from 1002 to 1014[67] and is celebrated for helping to unite Ireland. Brian is thought to have played the harp[68] and is considered a patron of the arts. My arrangement of the tune *Brian Boru's March*[69] is a useful example of how campanella can be used to alter the mood and dynamic of a piece. The march-like A section (see figure 2.1.16) uses a chord melody approach to evoke a strident mood. The contrasting B section is gentler and more lyrical and ideally suited to campanella treatment (see figure 2.1.17).

Figure 2.1.16. The A section of Brian Boru's March



In the A section the first six bars begin with three note chords (Dm, Dm, C, C, Dm, Dm). Plucked with the *p-i-m* fingers these punchy chords help to establish the march feel. The climactic bar 7 contains two major chords - B ♭ followed by a full A major chord. The darker D minor tonality then returns in bar 8 to end the A section.

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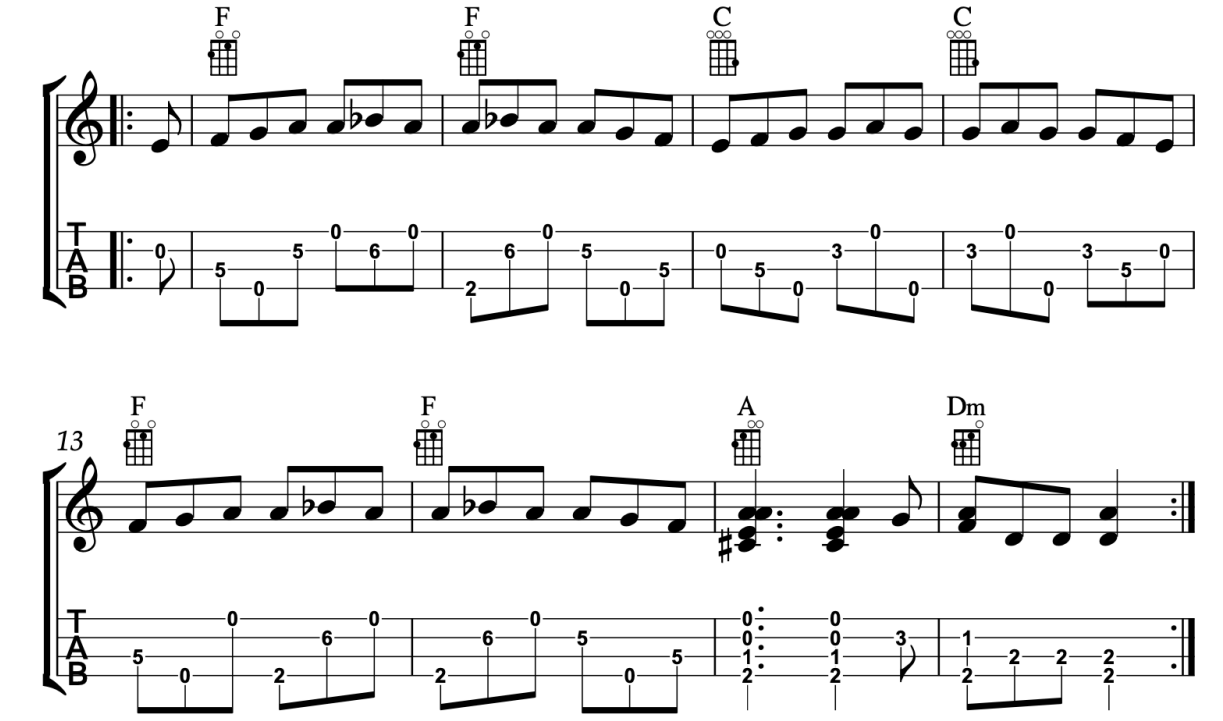
[67] Sourced from britannica.com online

[68] The Brian Boru harp, also known as the Trinity College Harp, is the oldest harp in Ireland and is the official symbol of the Irish government. It was donated to Trinity College in 1782 and, according to legend, it belonged to Brian Boru.

[69] For audio see Folder C.5. For full score see either Folder B.3. or *12 Traditional Tunes for Fingerstyle Style,* pub. Les Productions d'Oz, 2019

The contrasting mood of the B section (see figure 2.1.17) is indicated by the change in the harmonic structure; the sultry D minor tonality is now replaced by its brighter relative major of F (see bars 8 & 9 and bars 13 & 14). The melody is a succession of flowing eighth notes which, when placed across the strings, create a harp-like sound. The over-lapping minor seconds (E to F and A to Bb) add interest by sparking subtle but distinctive dissonances. Using campanella in this section made use of the natural features of the ukulele to add nuance and contrast, and suggested the mythical harp of Brian Boru.

Figure 2.1.17. The B section of Brian Boru's March arranged in campanella style



This arrangement of *Brian Boru's March* will be included in the forthcoming London College of Music at the University of West London's grade 6 syllabus. As it is such a short piece I suggested coupling it with another Irish tune, *The Blarney Pilgrim*, which was also arranged in campanella style.[70]

[70] Sourced from *O'Neill's Music of Ireland*, 1903

The Blarney Pilgrim[71]

This well known Irish jig in the D mixolydian mode is in three sections. The fast moving melody in 6/8 time is ideally suited to being played in campanella style (see figure 2.1.18). With the notes constantly moving across the strings the resulting over-ringing effects require no further accompaniment. There are, however, a couple of places which do not use campanella. In bar 1 for example, (see figure 2.1.18) the D and E could have been played on alternating strings (2,3 and 0,2) but slurs were an effective way of kick starting the rhythm and establishing a jaunty feel.

Figure 2.1.18. The A section of The Blarney Pilgrim



Some of the technical challenges of campanella are highlighted in this piece. In particular the scenario where strong beats are played by a weaker finger and vice versa. For example, the open Gs (0,4) at the end of bar 1 and again in bar 2 (see figure 2.1.18) fall on weaker beats but both are

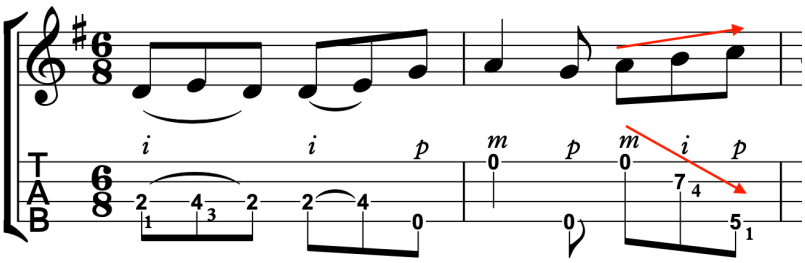
[71] For audio see Folder C.5. For full score see either Folder B.3. or *The Little Book of Irish Tunes,* pub. Muir, 2016

played by the thumb - the naturally stronger digit. Meanwhile, in bar 2 the weaker *m* finger plays on the stronger first and fourth beats. The resulting Gs (0,4), which are all the more dominant by being played as open strings, might easily be overplayed and create a lopsided feel to the rhythm. Moving them to a closed string would, however, mean the campanella is less effective.

Similarly in bar 3 the strong beats fall to the *i* (beat one) and *m* (beat four) fingers while *p* plays the weaker third and fifth beats. Consequently the *i* and *m* fingers need to play with more emphasis in order to maintain the pulse while the thumb is required to use a lighter touch. When playing a melody in campanella style, as this piece demonstrates, the use of the picking hand thumb is essential but its role is distinctively different from guitar technique where it is most frequently assigned the role of playing the bass. Furthermore, campanella lacks the security of repeated patterns which are a feature of arpeggios, and thus, the use of the thumb in ukulele picking technique requires careful consideration of balance, tone control and rhythmical emphasis.

With regard to the fretting hand campanella often creates counter-intuitive patterns, as previously seen in the Bach *Prelude BWV 1007*, where the notes are descending in pitch but ascending in position. In *The Blarney Pilgrim* the notes in the second part of bar 2 (see figure 2.1.19) can be seen in the notation to be rising in pitch - A, B, C. In the tablature, however, the notes appear to be moving in a downward motion from strings one to two to four.

Figure 2.1.19. Detail of bar 2 from The Blarney Pilgrim showing the counter-intuitive nature of campanella



Generally speaking, ascending figures on fretted instruments either move from the lower strings to the higher strings or along the fingerboard from left to right. It is for this reason that any movement either across the fingerboard from the string closest to the player's chin to their feet, or from left to right along the fingerboard, is referred to as *up*. The re-entrant tuning of the ukulele challenges our perception of the fingerboard, particularly when campanella is employed. To return to the three notes in figure 2.1.19. The open A (0,1) is followed by a seven fret leap to B at 7,2 despite these

notes only being a tone apart. The following note, C, despite only being a semitone (one fret) higher than the B at 7,2 now moves back down the fingerboard to 5,4. If the C was played at 8,2, one fret up from the B at 7,2, the movement would be more intuitive but the over-ringing effect would be lost.

My musical brain, both aurally and physically, would happily play all three notes in figure 2.1.19 in first position on the first string from 0,1 to 2,1 to 3,1. In fact, I would hardly need to think about it as my brain and my hands are, after so many years of playing, easily synced to both hear and *feel* these notes as an ascending line. Alternatively, I could play them either in second position at 2,4, 4,4, 5,4 or in fifth position at 5,2, 7,2, 8,2 or in ninth position at 9,3, 11,3, 12,3 (see figure 2.1.20). In all of these scenarios the notes are ascending sequentially along one string from left to right. Once the first note (A) is located my hand *knows* that the B is two frets higher and the C one fret higher than the B.

Figure 2.1.20. The different linear positions for the notes A, B, C



Playing these note as a campanella (0,1, 7,2, 5,4) can be likened to what De Souza describes an act of ‘voluntary self-sabotage’. (De Souza, 2017, 83) Rather than playing the notes in a pattern that feels automatic after so many years of playing, they are now literally peppered across three strings

and seven frets. Suddenly there is no obvious pattern or close relationship between the notes. The highest note C (5,4), is situated on the lowest physical string despite that string sounding as the second highest open string. Furthermore, the C (5,4) is two frets *down* the fingerboard even though the B at 7,2 is a step lower. Although I still recognise the sound of each note I now have to actively think about their positions on the fingerboard. In my musical brain the notes are still sonic neighbours but when played as a campanella they are in different neighbourhoods of the fingerboard.

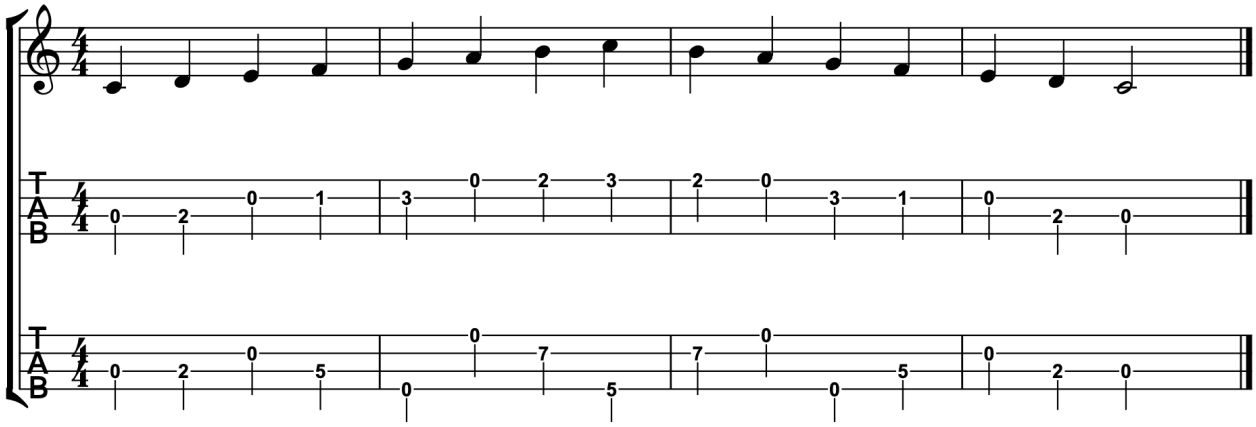
In De Souza's model of 'self-sabotage' he uses the example of retuning the strings of a guitar, or physically altering the instrument, in order to change the interface. By doing so, as in the case of guitarist Kurt Rosenwinkel, this is seen as a deliberate disruption of 'predictable patterns' (De Souza, 2017, 88) in order to '*hear* the music properly.' (De Souza, 2017, 88) In other words Rosenwinkel wanted to break away from well known patterns in order to find new creative outcomes. When playing campanella on the ukulele, the fingerboard (which in De Souza's terms is an interface) remains unchanged but the placement of the notes now feels chaotic. My practiced, and therefore automatic, approach to a scale has to be un-practised and relearnt. When I play in campanella style my brain literally *hurts* as I have to actively think about the position of notes rather than feel my way to them or rely on learnt patterns. The tablature line becomes an essential visual aid in locating the notes as it presents a clear picture of where the notes are placed, thus reducing the need for a lot of detailed fingerings. When playing in campanella style I can no longer *feel* my way to the notes but have to constantly adjust to positional shifts and move my hands accordingly.

This sense of disruption is felt in both the fretting hand and the picking hand. As previously seen in *The Blarney Pilgrim* the picking hand fingering has to constantly adjust to the movement across the strings. As can be seen in figure 2.1.18 the fingering changes in every bar. There are no repeating patterns, such as are typical of arpeggios, and, thus, both hands are in a constant state of flux. To this end playing the ukulele has never simply been a matter of applying my guitar skills to the smaller fingerboard but is a constant process of relearning and adapting.

The topsy-turvy world of re-entrant tuning offers unique and idiomatic possibilities for playing the ukulele. Overcoming the *brain ache* of playing in campanella style has forced me to break away from the habits and practised patterns of a guitarist. This has resulted in the creation of new arrangements and, as will be discussed in the next section, new compositions which explore and exploit the natural features of the instrument. Although much of this discussion has centred on the technical demands of playing in campanella style, the most important aspect of this approach is

the simple beauty created by over-ringing notes. It is difficult to describe the sparkling beauty of campanella and the *frisson* of all the subtle dissonances that are created when even a simple C major scale is played as a campanella (see figure 2.1.21). It is for this reason that players, such as myself, are dedicated to promoting campanella style and re-entrant tuning as idiomatic features of the ukulele.

Figure 2.1.21. The C major scale from linear and campanella perspectives



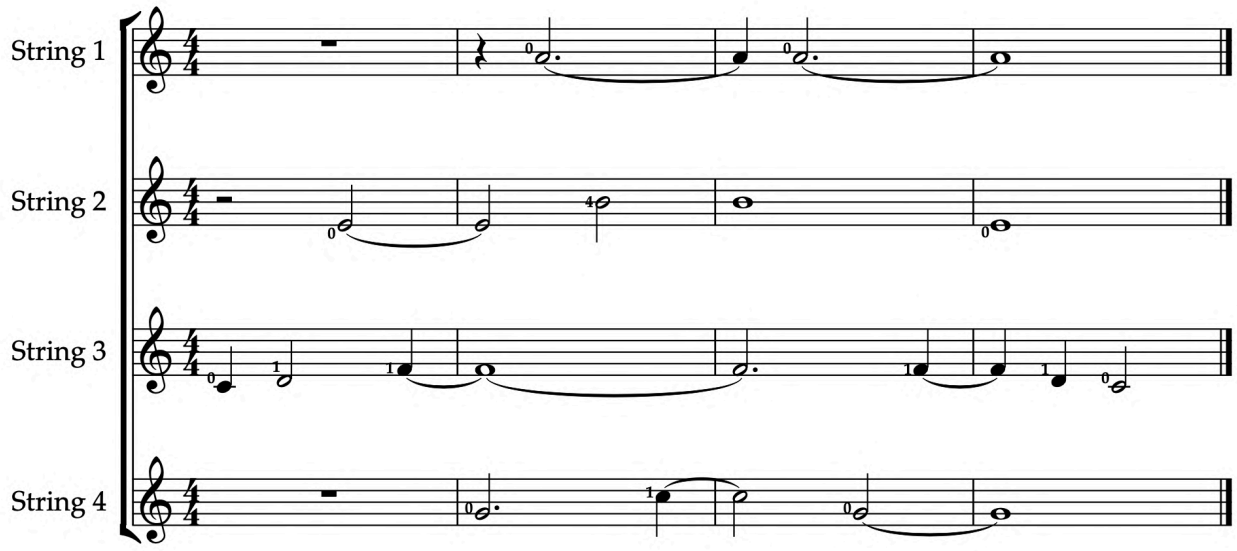
When demonstrating campanella in workshops I often use the C major scale[72] played first as a linear scale across the strings in first position and then as a campanella (see figure 2.1.21). Participants are asked if they can hear a difference. The initial reaction is often, *no*. But after repeating the two scales, and asking participants to listen very carefully, they begin to hear the subtle nuances of the campanella version. I notice how heads gradually lean in as participants begin to focus on the sound. Each note is vitally important as, in order to maximise the over-ringing effects, the player (and for that matter the listener) must pay close attention to the subtle interplay of consonance and dissonance.

In figure 2.1.22 a stave is assigned to each string on order to demonstrate the over-ringing potential of the C major scale. Of course, factors such as tempo, acoustic and instrument will effect the actual duration of each note so the example is offered purely as an illustration. In bar 2, if the F (5,3) is released too soon the over-ringing effect, or dissonance between the overlapping minor 2nd (0,2 to 5,3), will be lost - even though in reality the effect is fleeting. The main point here is to illustrate the importance of the holding and releasing of the fretting fingers in order to achieve the

[72] For audio samples of the linear and campanella scales see Folder C.5

desired over-ringing effect. To this end, campanella playing is not just about repositioning the notes on the fingerboard but in aligning technique and listening skills to maximise the effect.

Figure 2.1.22. The C major scale written on 4 staves to demonstrate the over-ringing potential of campanella



Using Campanella in Original Compositions

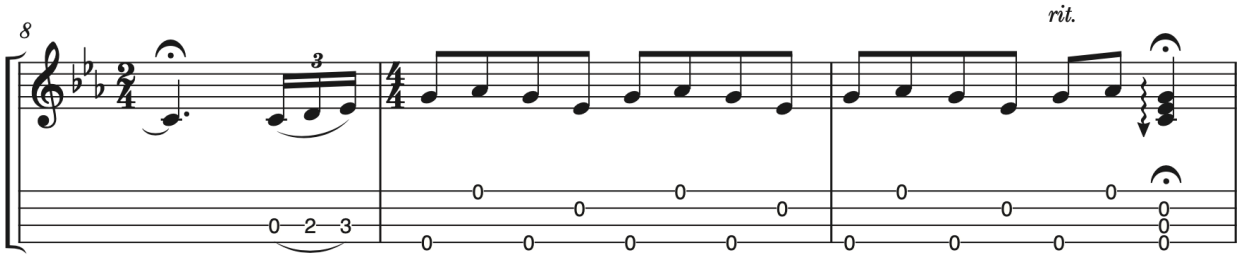
The primary aim of this portfolio is to explore how campanella can be used to create new, original and idiomatic repertoire for the ukulele, thus building upon the ideas of John King and extending the boundaries of classical ukulele. As previously discussed King was an innovative arranger. My own work has focused on how to create new contemporary classical repertoire for the ukulele. This section looks at how I have applied campanella in my own compositions and how it has been used by other composers in commissioned pieces.

The Falling Rain

The Falling Rain,[73] my first major work for the ukulele, experimented with various techniques including arpeggios, harmonics and campanella. One of the biggest innovations, however, was the use of an alternate tuning in which the first and second strings are lowered a semitone to A ♭ and E ♭ in order to sound an open minor tonality. At nearly ten minutes long this is a substantial solo work by ukulele standards and is characterised by a number of contrasting sections. The piece was an opportunity to combine campanella effects with other techniques such as arpeggios and slurs.

Campanella first occurs in bars 9 and 10 (see figure 2.1.23) with a simple repeated idea comprising just four notes G-A ♭-G-E ♭. With the altered tuning all four notes are played as open strings. The over-ringing intervals of minor 2nd (G and A ♭) and minor 3rd (G to E ♭) create an ominous mood and firmly establish the change from the usual happy major tonality of the open strings to a darker sounding minor tonality. This is further emphasised by the final C minor chord which is played by dragging the finger across the strings in the order 2-3-4 to create a reverse roll effect. In the notation the notes appear to be played in the order G-E ♭-C, but as the tablature shows they actually sound E ♭-C -G.

Figure 2.1.23. Bars 9 & 10 of The Falling Rain

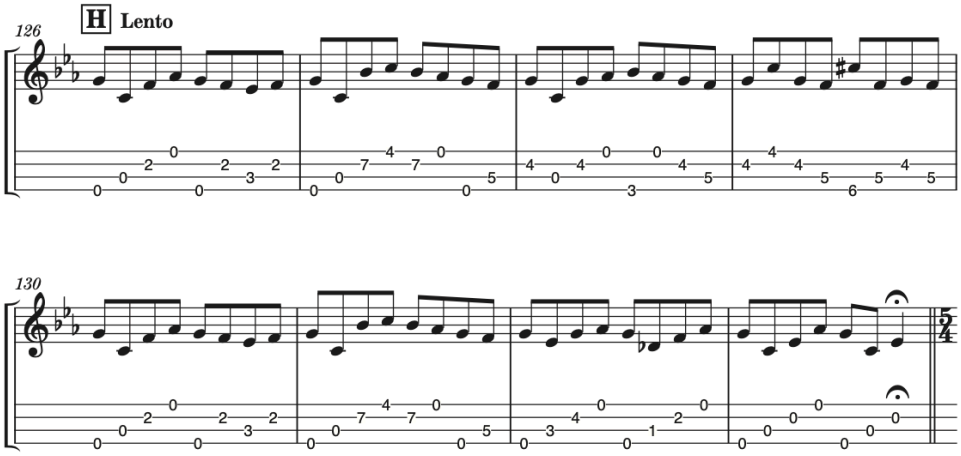


The piece comprises nine contrasting sections with each section representing the different intensity of a Spanish *gota fria*,[74] or rainstorm. The Lento section (see figure 2.1.24), which is repeated three times, represents a lull in the deluge. The single melodic line is played as a campanella in order to create a more delicate and spacious atmosphere. Each note of the melody represents a single raindrop hitting the earth and shattering into a myriad of tiny splashes, or over-ringing effects.

[73] For audio see Folder C.1. For full Score see Folder B.1.

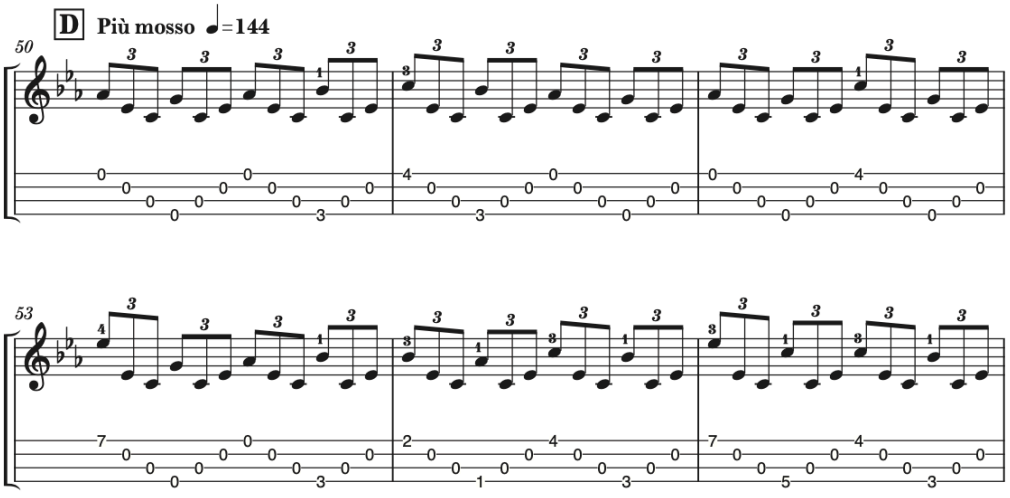
[74] Literally ‘cold drop’. Gota fria is the Spanish name for an intense rain event on the Mediterranean coast.

Figure 2.1.24. The Lento section of The Falling Rain with the melody played as a campanella



The faster Section D (see figure 2.1.25) is an example of campanella being woven into an arpeggio pattern in order to blend melodic and harmonic effects. The melody moves between strings 1 and 4 to create the campanella effect while the inner strings, played as open notes (C and E ♭), form a repeated, accompanying arpeggio. The picking pattern for each bar is *a-m-i-p-i-m-a-m-i-p-i-m* with the melody falling under *a* (string 1) and *p* (string 4). In *The Classical Ukulele* John King suggests that on the ukulele the *a,* or ring finger, is rarely used. (King, 2004, 6) But, as this passage demonstrates, assigning a picking finger to each string is an effective was of creating a fast and fluid pattern which both feels and sounds good. While the idea for this picking pattern stemmed from my knowledge of guitar technique, it demonstrates how a guitar technique can lead to the

Figure 2.1.25. The opening of the D section which blends campanella & arpeggio techniques



development of new, yet idiomatic, approaches to the ukulele - in this case the merging of campanella and arpeggio techniques.

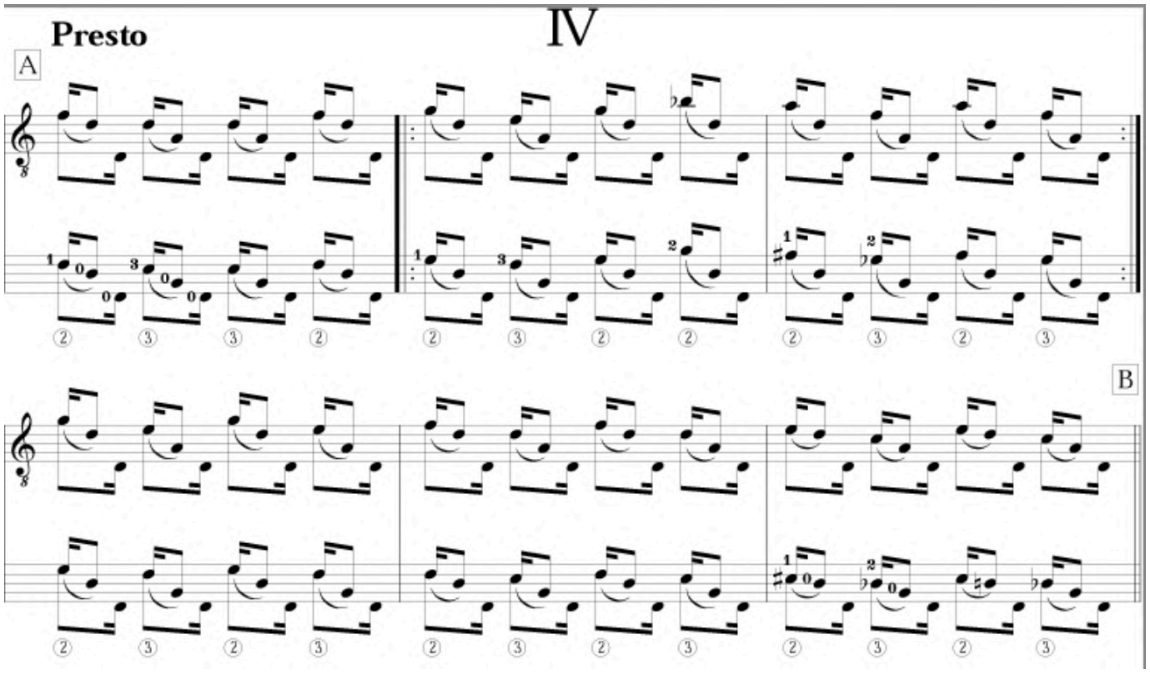
The climactic Presto section (see figure 2.1.26) of *The Falling Rain* was a further exploration into how campanella can be adapted and developed on the ukulele. My aim was to create a shimmering soundscape similar to the final section of Carlo Domeniconi's *Koyunbaba Suite* for guitar (1985). In Domeniconi's piece the final section (Presto) represents the wind rushing through the grass (see figure 2.1.27). By adapting Domeniconi's idea of rapid pull-offs, both across the strings and along the fingerboard, my aim was to suggest the relentless deluge of the *gota fria*.

Figure 2.1.26. The opening of the Presto section of The Falling Rain



In *Koyunbaba* each triplet pattern begins with a plucked note, followed by a pull-off to an open string and then a plucked open string. As shown in figure 2.1.27 Domeniconi alternates this pattern between strings 2 and 3. I maintained his idea of the triplet pattern but moved the triplets across three strings in the order 3-2-1-2-3 (see figure 2.1.26) to create additional over-ringing effects. The triplets are intended to sound exhilarating but controlled. *Koyunababa* was one of the pieces I performed in my final recital at the Royal College of Music and has always been a piece which tested my technical facility. In adapting this technique to the ukulele I wanted to create a concert solo that was both virtuosic and contemporary. Having performed *The Falling Rain* numerous times I regard it as one of the most challenging and successful pieces in the portfolio. The other techniques used in this piece will be discussed in the relevant sections of this commentary.

Figure 2.1.27. The opening of the Presto section of Koyunbaba by Carlo Domeniconi



Variations on The Dowie Dens of Yarrow

This haunting Scottish ballad became the basis for a set of nine short variations.[75] Once again campanella features prominently throughout and, as in *The Falling Rain*, is often combined with other techniques. As shown in figure 2.1.28 the theme begins in chord melody style but also makes use of campanella ideas. At the end bar 1 the melodic note C is moved from the first string (3,1) to the fourth string (5,4) to create a campanella effect. In bar 2 the melodic notes are still treated as campanellas despite being part of a chord (see the notes circled in red in figure 2.1.28).

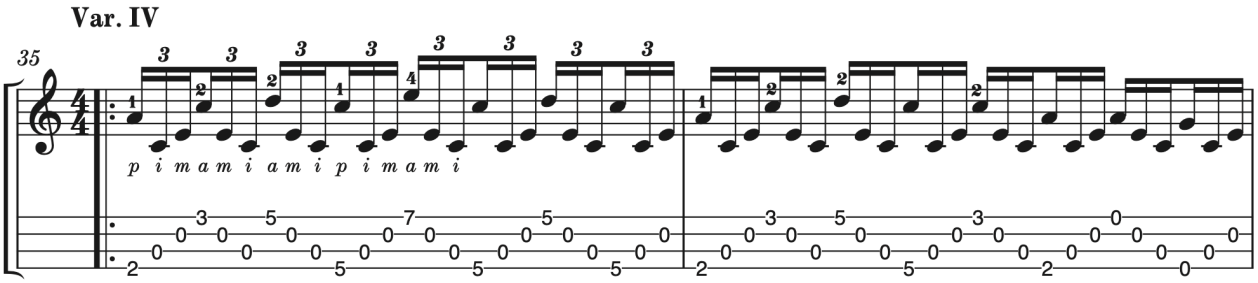
Figure 2.1.28. The theme of Variations on The Dowie Dens of Yarrow



[75] For audio see Folder C.1. For full Score see Folder B.1.

Variation IV merges campanella and arpeggio techniques, as in *The Falling Rain*, but this time the arpeggio starts on the fourth string to create a picking pattern of *p-i-m-a-m-i-a-m-i-p-i-m-a-m-i* (see figure 2.1.29). Each picking hand finger is assigned to a string (*p* = 0,4; *i* = 0,3; *m* = 0,2; *a* = 0,1) and once again the *a* finger is essential to the pattern. The campanella melody falls under the thumb and *a* finger while the *i* and *m* fingers provide the accompaniment. Although the harmonic structure of this piece is very simple with A minor, C major and G major chords predominating, the campanella effects add colour and interest.

Figure 2.1.29. The opening of Variation IV from Variations on The Dowie Dens of Yarrow



A Conspiracy of Ravens

Form

This work began as a campanella arrangement of the English folk ballad *The Three Ravens,* first published in Thomas Ravenscroft's *Melismata* of 1611.[76] The ballad comprises numerous verses but the melody itself lasts less than a minute; making it too short for an effective solo. I thought about creating a set of variations using the theme as I had done with *Variations on The Dowie Dens of Yarrow* but eventually came up with the idea of creating a work that merged old and new techniques through arrangement and composition.[77]

The work is structured as three pieces in two parts:

1. *White Raven I & II*
2. *Blue Raven I & II*
3. *Black Raven I & II*

[76] Manuscript downloaded from http://www.pbm.com/~lindahl/music.html

[77] For audio see Folder C.1. For full Score see Folder B.1.

The melody is first stated in campanella style while the second part is an original composition based on the melody, or theme. Campanella style seemed like an obvious choice for the melody, not only because of its simplicity and beauty, but because both the ballad and technique date from the 1600s.

Instrument

The melody sounded particularly effective played on my five string tenor ukulele made by DJ Morgan in 2014. This instrument was originally commissioned to play the four course guitar repertoire of Adrian Le Roy (1520-98). The term five string is slightly misleading as it implies an instrument with five individual strings but it is still, essentially, a four string instrument; the difference being there are two fourth strings, one high G and the other low G, that are struck and played *as one* to sound the octave.

The idea of double strings stems from the four course Renaissance guitar and, as previously seen, was also a feature of the five course Baroque guitar. According to Christopher Page, in the sixteenth century the four course guitar was usually strung with double fourth, third and second courses plus a single string, or chanterelle, at the top. This made a total of seven strings. (Page, 2015, 6) The five string ukulele with its double fourth string is a useful instrument for playing Renaissance guitar repertoire. It should be noted, however, that Page considers the most common tuning for the four course guitar to be linear, with the fourth course strung as a unison. (Page, 2015) Nevertheless, the five string ukulele is commonly strung in octaves with the high G positioned on the outside so that it is struck first. The two strings can be easily swapped so that the low G is struck first. It should be pointed out that striking the thicker string first means the inner string is easily missed. This stringing can be very effective when playing the works of Adrian Le Roy, particularly in pieces which use a repeated drone note. Despite these considerations *A Conspiracy of Ravens* uses the typical five string ukulele tuning with the high G placed on the outside. This stringing allows the campanellas to ring more clearly.

Keys

The work began as a series of campanella arrangements of the melody using different keys. This enabled a comparison of the different fingerings and sounds. The keys of E minor, D minor and C minor were particularly effective and this evolved into the idea of stating the campanella melody three times using these keys. Thus, *White Raven* is in E minor (see figure 2.1.30), *Blue Raven* in D minor (see figure 2.1.31), and *Black Raven* in C minor (see figure 2.1.32). Each time the melody

appears it is given a different direction. Hence, *White Raven* is marked *Grave*; *Blue Raven* is marked *Misterioso*; and *Black Raven* is marked *Doloroso*. While these changes of mood are subtle the aim is to encourage the performer to explore the expressive possibilities of campanella on the ukulele.

Figure 2.1.30. The opening of White Raven I in the key of E minor



Figure 2.1.31. The opening of Blue Raven I in the key of D minor

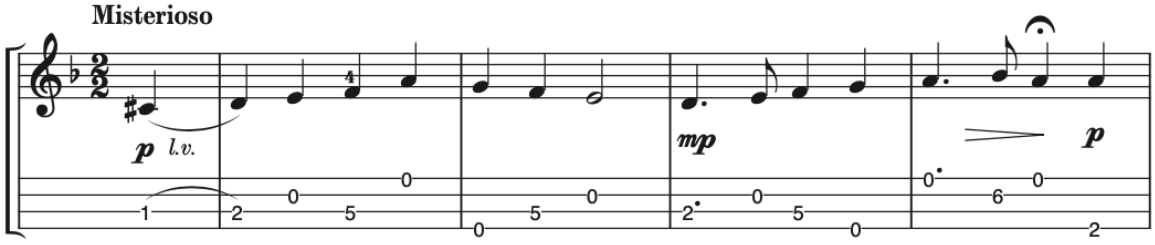
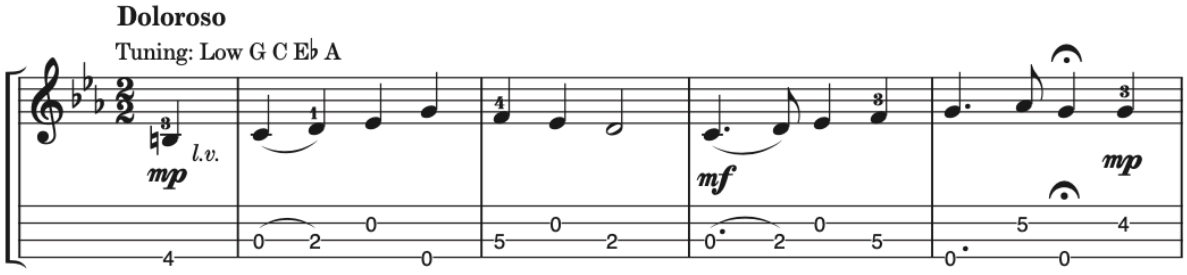


Figure 2.1.32. The opening of Black Raven I in the key of C minor



Lyrics

In Ravenscroft's edition of 1611 the lyrics tell of three ravens 'blacke as they might be' who are discussing what to have for breakfast (see figure 2.1.33). One of them observes that 'downe in yonder greene field' lies a slain knight. The knight, however, is guarded by his hawks and his hounds. As the ravens watch 'a fallow Doe' comes along and 'kist his [the knight's] wounds so red'. The doe, who is 'great with yo[u]ng', is clearly a metaphor for the knight's pregnant mistress. She carries him on her back and buries him near a lake. Bereft of her lover she then dies, presumably of grief, before 'euen-song time' *(even-song time)*. Despite the tragedy the song ends with the hope that God sends every gentleman 'such haukes, such hounds, and such a Le-man' *(such hawks, such hounds and such a mistress)*.

Figure 2.1.33. The first page The Three Ravens from Melismata, 1611



COVNTRY PASTIMES.

Treble. 2. 4. Voc.

Here were three Rauens ſat on a tree, Downe a downe, hay down, hay

downe. There were three Rauens ſat on a tree, with a downe, There were three

Rauens ſat on a tree, they were as blacke as they might be, with a downe derrie,

derrie, derrie, downe, downe.

The one of them ſaid to his mate,
down adowne hey downe,
2 *The one of them ſaid to his mate,*
with adowne:
The one of them ſaid to his mate
Where ſhall we our breakefaſt take?
with adowne dery downe.
Downe in yonder greene field,
downe adowne hey downe,
Downe in yonder greene field,
with adowne.
Downe in yonder greene field
There lies a Knight ſlain vnder his ſhield,
with a downe.
His hounds they lie downe at his feete,
downe adowne hey downe.
4 *His hounds they lie downe at his feete,*
with adowne.
His hounds they lie downe at his feete
So well they can their Maſter keepe,
with adowne.
His Haukes they flie ſo eagerly
downe adowne.
5 *His Haukes they flie ſo eagerly*
with adowne.
His Haukes they flie ſo eagerly,
There's no fowle dare him come nie.
with a downe.
Downe there comes a fallow Doe,
downe adowne.
6 *Downe there comes a fallow Doe*
with a downe.
Downe there comes a fallow Doe,
As great with yong as ſhe might goe.
with adowne.
She lift vp his bloudy hed,
downe adowne.
7 *She lift vp his bloudy hed,*
with

The use of slow tempos and minor keys reflects the solemn mood of the lyrics. *Black Raven* calls for the second string to be tuned down a step to E ♭ in order for the lower three strings to sound the open C minor triad in the order G (0,4), C (0,3), E ♭ (0,2). In normal tuning the first three notes of bar 1 (see figure 2.1.32) would have been placed on the third string thus losing the campanella effect. Similarly in bars 2 and 3 the open E ♭ allows an edgy semitone campanella between the E ♭ and D which would not be possible with normal tuning. Where notes cannot be placed on alternate strings, such as the first notes of *Blue Raven I* (see figure 2.1.31) and again at the start of bar 1 in *Black Raven I* (see figure 2.1.32), the notes are slurred to create a softer, more subdued tone.

Open Strings

As can be seen in figures 2.1.30, 2.1.31 and 2.1.32 open strings are a prominent feature of campanella style.[78] This is because open strings can ring on while the fretting hand is moving into position. In the examples above the only bar where open strings are not employed is bar 4 of figure 2.1.30. Here the placement of the notes (7,2 - 3,1 - 7,2 - 4,4) involves a stretch for the fretting hand. As campanella pieces often require large stretches, the shorter scaled soprano is a popular choice of instrument. As previously discussed, this work was specifically for the five string tenor but could be played on any sized ukulele. If played on a four string ukulele the tuning should be re-entrant to accommodate the campanella effects.

Titles

Although the pieces are based on the ballad *The Three Ravens* the individual titles (*White Raven*, *Blue Raven*, *Black Raven*) reflect the bringing together of old and new ideas. The colours tell the story of raven's transition from white to black and added a new, individual twist to the music. According to myth, when Apollo sent a white raven to spy on his lover Coronis he was so incensed to learn that she had indeed been unfaithful, he scorched the raven black in his fury. The middle movement *Blue Raven* represents the heat of the flames and is suggestive of the Australian raven (a bird from my childhood in Australia), which although black, shimmers deep blue against the sky. As the collective noun for a group of ravens is a conspiracy I felt this was the perfect title for the work, particularly as it was written over lockdown amidst all the Partygate scandals.

[78] As previously mentioned Pierre Bensusan considered a balance of fretted notes and open strings essential to the harp effect.

Original Ideas

Each arrangement of the theme is followed by an original composition. The idea came after studying the manuscripts of Adrian Le Roy. Pieces, such as a *Bransle*, are often followed by a *diminué*, or diminution, in which shorter notes are added as an elaboration. Although in Le Roy's manuscripts the *diminué* only involves a few extra runs it gave me the idea of creating a second part for each Raven which is an elaboration and extension of the melody. Once again campanella was a prominent feature.

In *White Raven II* (see figure 2.1.34) the notes move constantly across the strings to create campanella effects. This highlights the interplay of consonance and dissonance while creating campanella patterns that fall naturally under both hands. The white raven was envisaged as a pure bird, unfettered by the formal constraints of time signature and bar lines. The idea was to give each individual player the freedom to explore rubato and to give the piece a spontaneous, improvisatory feel.

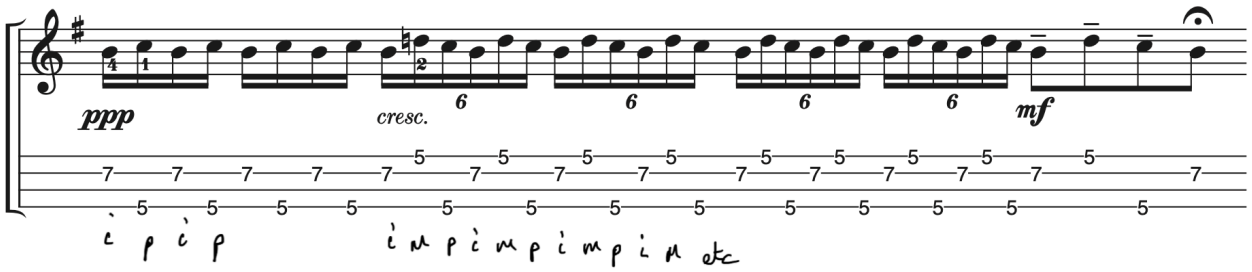
Figure 2.1.34. Campanella effects in the opening of White Raven II



When composing original works for the ukulele repeated campanella motives can be highly effective. For example, as shown in figure 2.1.35, the campanella pattern is arpeggio-like in its repetition. What distinguishes it as a campanella is that the notes are melodic, rather than harmonic. Beginning with just two notes alternating between the second (7,2) and fourth (5,4) strings the idea

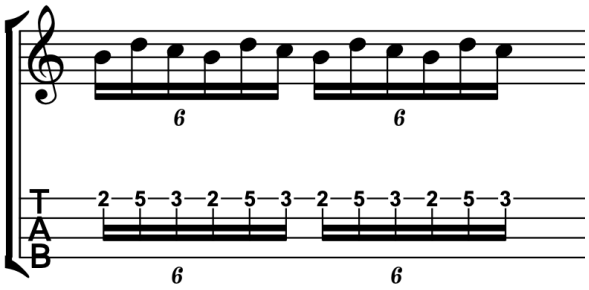
expands into a three note pattern (7,2 - 5,1 - 5,4) which is played as three groups of sextuplets. The picking hand fingering moves from alternating *i* and *p* to an arpeggio pattern of *imp-imp.*

Figure 2.1.35. The arpeggio-like campanella notes in White Raven II



If the notes of the sextuplet were played on one string, as in figure 2.1.36, the campanella effect would be lost. This demonstrates how the re-entrant tuning can be used to create unique, idiomatic effects on the ukulele.

Figure 2.1.36. How the sextuplet figure could be played on one string (see tablature)



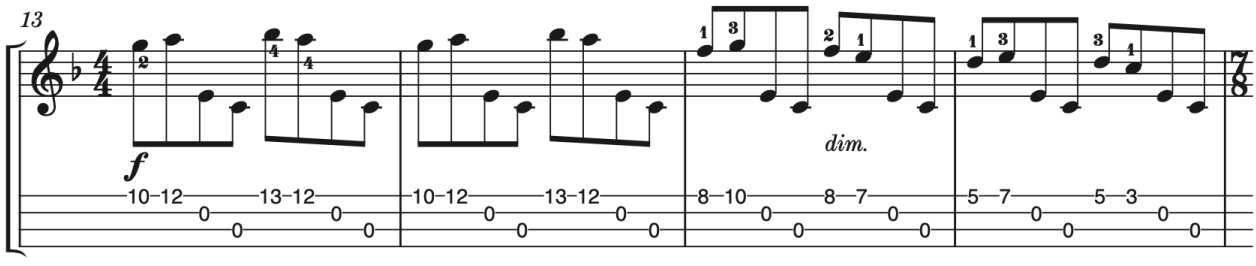
In *Blue Raven II* campanella is used more sparingly, despite the majority of notes moving as a single melodic line. While there are some double stopped notes, such as in the opening three bars (see figure 2.1.37), the piece plays with the contrasting ideas of campanella and linear motion. In bar 4, for example, (see figure 2.1.37) the notes are played as a campanella while from bars 5 to 8 the descending scale-like pattern is played only on the first string.

Figure 2.1.37. Bars 1 to 8 of Blue Raven II



Switching between campanella and linear techniques creates a playful but slightly agitated mood. In bars 13 and 14 (see figure 2.1.38) the first two notes could be played as a campanella between strings 4 and 1 with the notes positioned at 12,4 - 12,1. However, I deliberately placed these notes on the A string so that the first note is always stopped when the next note is played. The short attack gives a sense of disruption to the melody. This is immediately countered by the open E and C strings which are allowed to ring freely.

Figure 2.1.38. Bars 13 to 16 of Blue Raven II



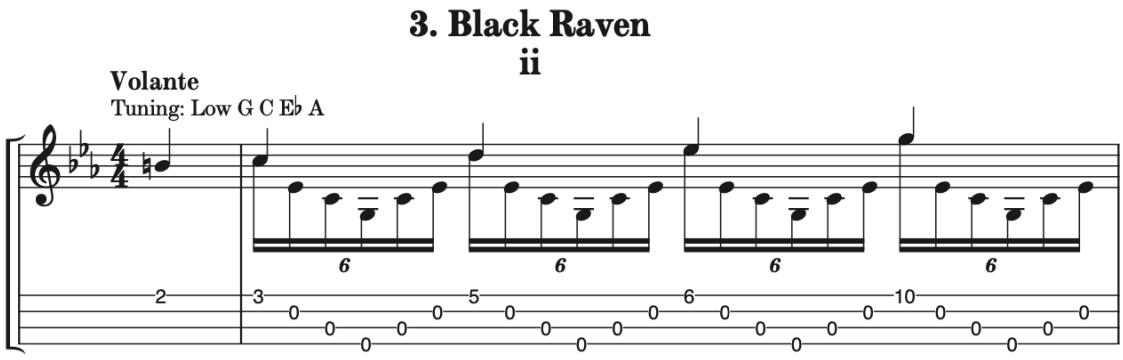
Blue Raven II concludes with a descending cross-string pattern (see figure 2.1.39) which operates as a dialogue between campanellas (melodic notes) and arpeggios (harmonic notes). In bar 64, for example, the first three notes (D-C#-A) form a campanella (circled in red), while the next three notes (A-E-C#) form an A major triad (circled in blue) which is played as an arpeggio. The idea is repeated, once again in 6/8 time, and then reimagined in 4/4 time in bar 66. The whole idea is then repeated, initially as a descending pattern moving down the fingerboard from fifth, to third, to first positions. The piece then concludes with the pattern moving up to second position and a final D minor chord in fifth position.

Figure 2.1.39. Bars 64 to 66 of Blue Raven II



In *Black Raven II* the melody is incorporated into an arpeggio pattern. Although campanella is not used in this piece an example is included here (figure 2.1.40) to clarify the difference between an arpeggio and a campanella. The melody is always played on the top string and the over-ringing is created by the accompanying arpeggio pattern which is played *a-m-i-p-i-m*.

Figure 2.1.40. The opening of Black Raven II



The Importance of Using Tablature

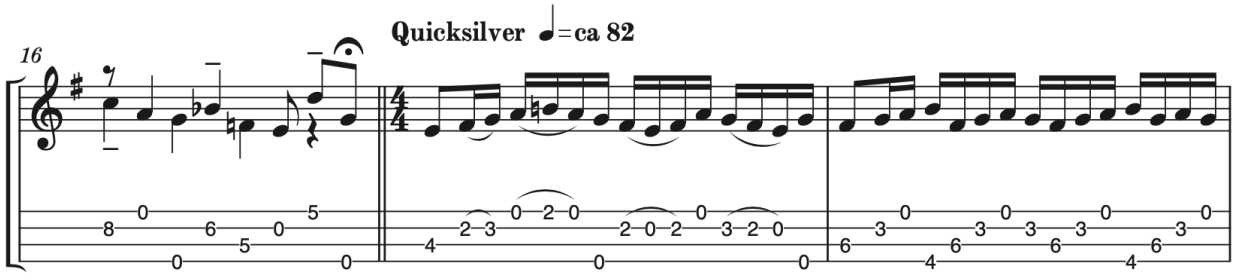
The use of tablature in *A Conspiracy of Ravens* clearly indicates which sections are to be played as campanellas and which are not. Furthermore, using tablature allows this piece to be played on either a five string or a four string ukulele. The full the work lasts approximately eight minutes but the pieces can also be played as individual pairs.

Mercurius Suite in 3 Movements

This piece was started in 2017 and evolved over a period of three years into a three movement work. The overriding aim was to create a multi-movement concert solo that explored the idiomatic features of the ukulele. Once again campanella played a central role. The title *Mercurius* (Latin for the Roman God, Mercury) is intended as a pun on 'me curious' as it expresses my own curiosity about the ukulele. *Mercurius* also embodies the various attributes of Mercury, both as an important God in the Roman pantheon and as one of the seven metals of alchemy. Composing for the ukulele is like bringing together various elements, or techniques, to reveal the instrument's 'gold'. Campanella features prominently in movements one and two, but the work was another opportunity to experiment with, and combine, different techniques in order to express the ukulele's potential.

In the *Quicksilver* section of the first movement *Mercury*,[79] campanellas and slurs are combined to evoke the honey-like movement of liquid silver (or quicksilver). The semiquavers are mostly moving in stepwise motion (see figure 2.1.41) and the combination of these two techniques allows the notes to be played smoothly and fluidly.

Figure 2.1.41. The start of the Quicksilver section from bar 17

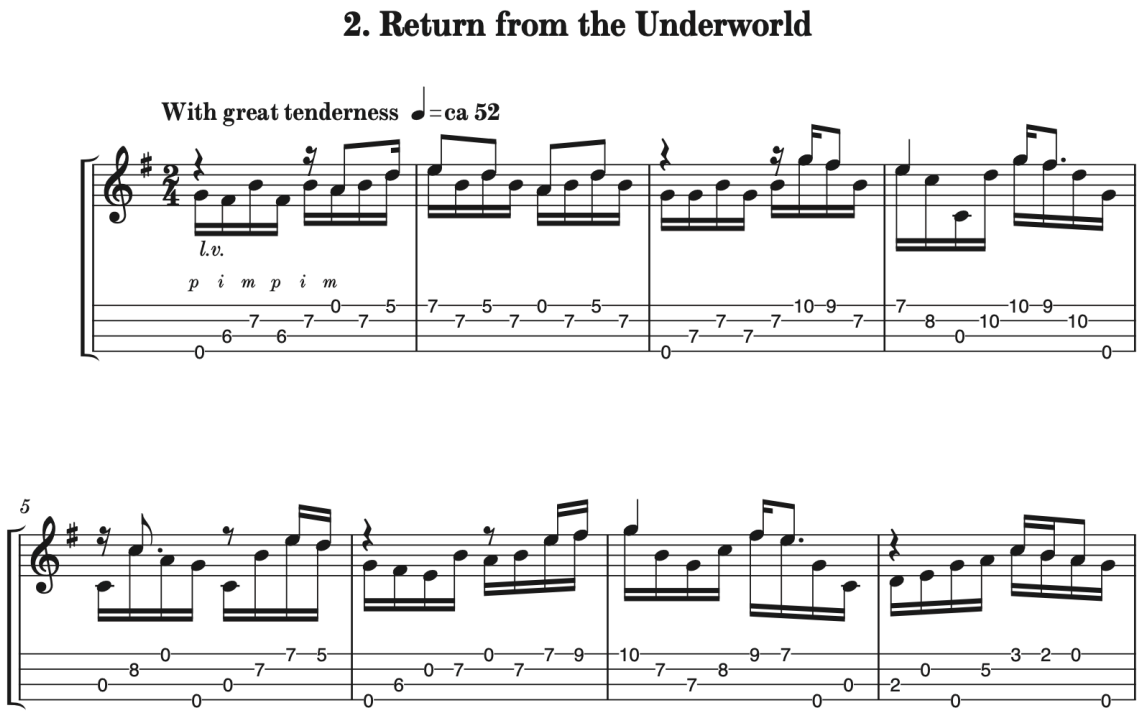


The second movement, *Return from the Underworld*, uses campanella throughout. The constant movement of the notes across the strings means the melodic and accompanying notes are so tightly

[79] For audio see Folder C.1. For full Score see Folder B.1.

interwoven it is often difficult to determine one from the other (see figure 2.1.42). Therefore, notes which need more emphasis and are often placed on the same string and notated with their tails going up.

Figure 2.1.42. The opening of Return from the Underworld

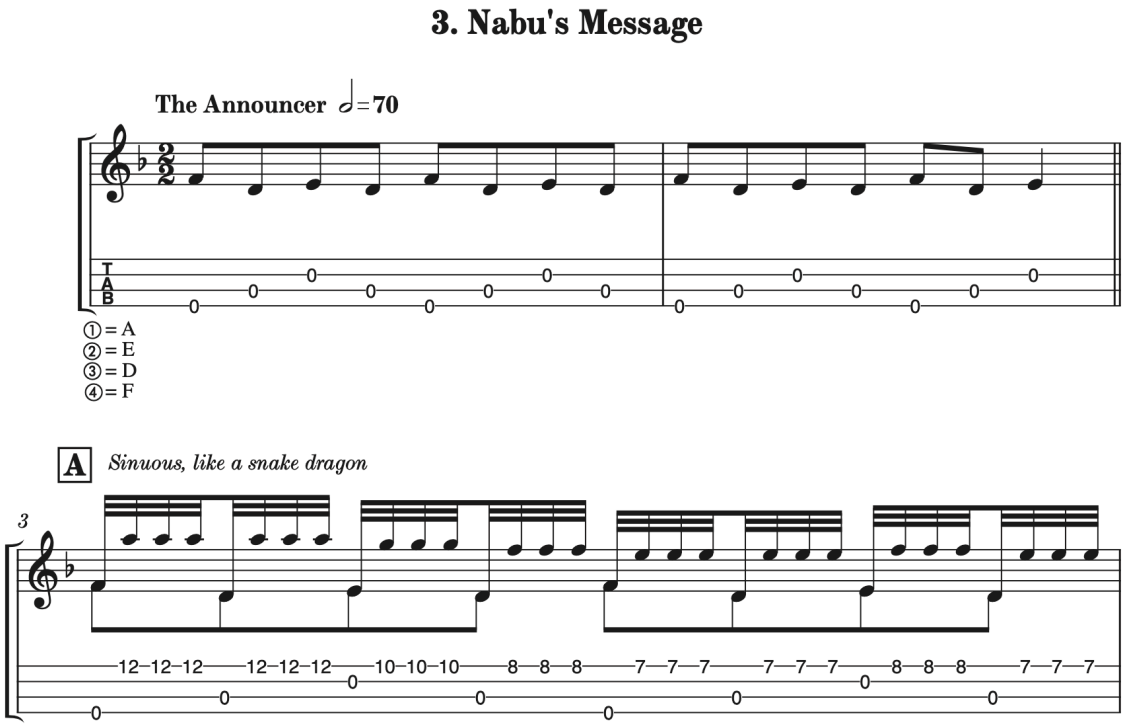


The god Mercury has many duties, one of which is to escort souls to the underworld. In *Return from the Underworld* he returns to the brightness of the living world and is overwhelmed by the fragility of life. The subtle interplay of consonance and dissonance sparked by the interplay of campanella notes is intended to express the delicate balance between light and dark, life and death.

The third and final movement of the suite, *Nabu's Message*, uses a four note tremolo. This technique is discussed in full in chapter 2.6. A brief mention should, however, be given to the repeated open string accompaniment which creates a campanella effect. In order to achieve this the third string is tuned to D and the fourth string to F so that the accompaniment, played by the thumb, forms a repeated four note pattern F-D-E-D across strings 4-3-2-3 (see figure 2.1.43). The close proximity of the notes creates the intervals of a falling minor 3rd, a rising major 2nd and a falling major 2nd. Thus, this second voice forms a campanella drone. The drone is constantly repeated to

spell the word F-D-E, or D-E-F, which is intended as a pun on the word DEATH. The idea of the piece is that if humanity remains deaf to the suffering of the planet it will ultimately lead to our collective death.

Figure 2.1.43. The opening of Nabu's Message



Campanella in Works by Other Composers

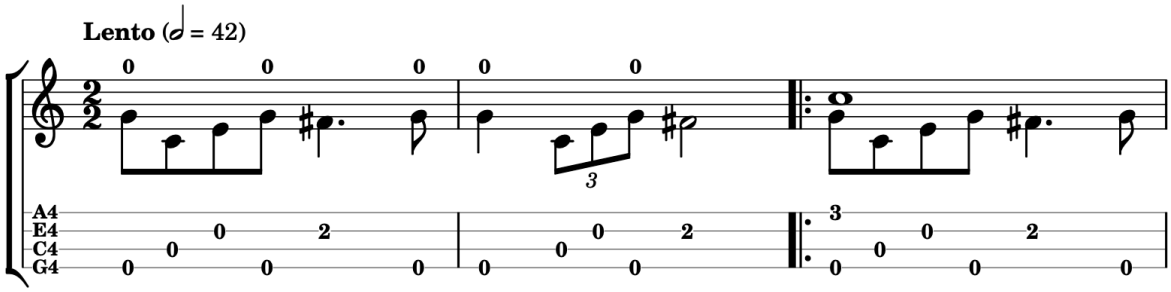
Twenty-six Days by Choan Gálvez

The Spanish composer Choan Gálvez sent me *Twenty-six Days* at the end of 2021 and I premiered it in a YouTube video on 25th January 2022.[80] This is a slow moving, sultry piece which uses over-ringing effects throughout. In the first section (see figure 2.1.44) open strings are used to create an arpeggio based on the C major triad. The sudden introduction of the raised 4th (F♯) on the third beat

[80] For audio see Folder C.2. For full Score see B.4.

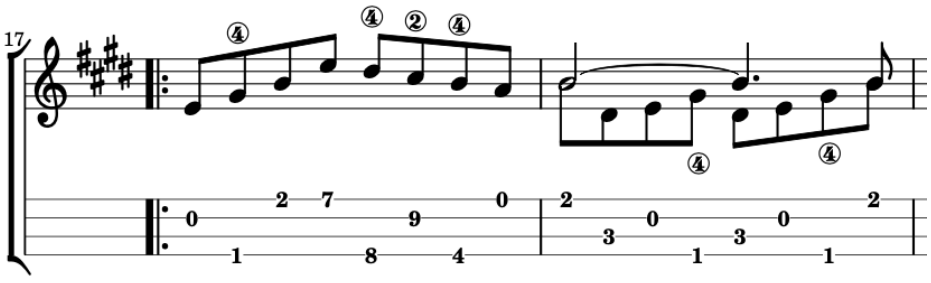
adds an element of dissonance and uncertainty which, as the composer explained, was part of the inspiration behind the piece as it reflected the twists and turns of life.

Figure 2.1.44. The opening of Twenty-six Days by Choan Gálvez



The second section, which modulates to E major, uses campanella techniques in both the melody and the accompaniment. The ascending then descending campanella run in bar 17 (see figure 2.1.45) is answered by the accompaniment in bar 18. Meanwhile, the melodic note B rings in the top voice. In bar 18 it is interesting to note how the accompaniment notes (D♯, E and G♯) are deliberately placed on different strings (see figure 2.1.45). While there is no indication that these notes should be held, their positioning on the fingerboard (which is clearly shown in the tablature) indicates that it is the composer's intention that they be treated as a campanella. The open E might easily have been placed on the third string to avoid the clash with the D♯, and the G♯ could easily have been played at 4,2. When allowed to over-lap these notes create interesting harmonic dissonances and give the music extra fullness and resonance.

Figure 2.1.45. Bars 17 and 18 of Twenty-six Days by Choan Gálvez

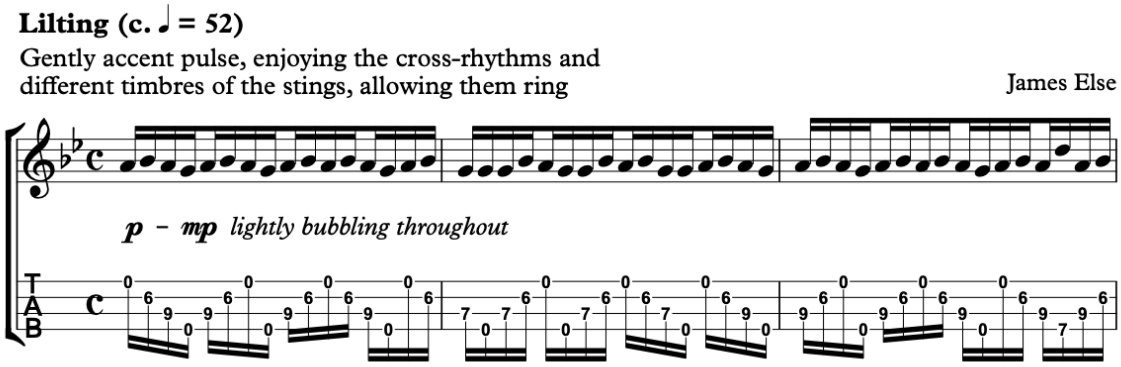


Parting Voices by James Else

This piece was originally commissioned by Donald Bousted. Sadly, Donald passed away in 2021 and I was honoured that the composer asked me to premiere it. The video was posted on YouTube on 29th May 2022.[81]

Although the composer does not use the word campanella in his instructions, the tablature clearly shows that each note is intended to be played on a different string (see figure 2.1.46). Else simply specifies that the performer should enjoy the 'cross-rhythms and different timbres of the strings, [while] allowing them to ring on.' The constant semiquaver movement coupled with the movement across the strings creates an effect of the notes 'lightly bubbling throughout' (see figure 2.46).

Figure 2.1.46. The opening of Parting Voices by James Else



Parting Voices demonstrates how timbral nuances are created by playing the same note in a different position on the fingerboard. For example, in bar 1 of figure 2.1.46 the open A (0,1) sounds naturally brighter than the same A played at 9,3. The A at 9,3 sounds more mellow, or subdued and can also be played with vibrato, thus adding a further harmonic *wobble* to the over-lapping soundscape.

Initially I found the constant changes in the picking pattern quite challenging as it drew attention to the natural imbalances between the fingers. In bar 2, for example, the repeated Gs are positioned 7,3 - 0,4 - 7,3 which results in the picking pattern of *i-p-i.* The downward motion of the thumb *(p)* being naturally stronger, meant the second note had more emphasis despite falling on the

[81] For audio see Folder C.2. For full score see Folder B.4.

weaker beat. Nevertheless, as the composer's direction indicated, the resulting cross-rhythms and timbral nuances enhanced the overall musical effect and I came to appreciate these subtle distinctions were a central element of the piece. The endless stream of semiquavers might easily become monotonous but the piece has a feeling of constantly shifting sand, an effect which is emphasised by the natural imbalance in the picking hand. The intention of the piece is not to achieve evenness but to highlight the delicate ambiguities between strings, positions and fingers. In so doing Else demonstrates how working with a limited palette of notes can, through the use of campanella technique, create music that is filled with subtle shades and colours.

Conclusion

Campanella is undoubtedly one of the most distinctive and idiomatic techniques that can be used on the ukulele. Although campanella can be applied to instruments with linear tunings, re-entrant tuned instruments such as the Baroque guitar and the ukulele offer more scope for moving the melodic notes across a great number of strings. It is for this reason that my preferred tuning for the ukulele is re-entrant. By maximising the over-ringing potential of the ukulele through campanella techniques the instrument offers a truly unique and distinctive voice. However, as John King recognised, this new voice is deeply rooted in historical practice and, thus, represents a bringing together of the old and the new. As one of the first composers to apply campanella techniques in original compositions, my new works are helping to establish a sophisticated, idiomatic and contemporary repertoire for the ukulele.

2.2.a. Tintinnabuli

Introduction

Having made extensive use of campanella in my arrangements and compositions I decided, in 2020, to see how tintinnabuli, another technique inspired by the sound of ringing bells, might also be applied to the ukulele. In 1976 the Estonian composer Arvo Pärt (b.1935) emerged from a period of self-imposed reflection[82] with a new style of composition which rejected the complexity of modernism and embraced a more reductive and tonal approach. (Hillier, 1997) During the 1980s Pärt had experimented with serial and collage techniques but, following an intensive study of Medieval and Renaissance music, (Shenton, 2012, 2) he developed a new and distinctive style which he called tintinnabuli.[83] Pärt describes this technique in the following way:

> I work with very few elements - with one voice, with two voices. I build with the most primitive materials - with the triad, with one specific tonality. The three notes of a triad are like bells. And that is why I call it tintinnabulation.[84]

The word tintinnabuli stems from the Latin *tintinnabulum* which means ‘a small tinkling bell’ (OED). As demonstrated in the previous chapter the re-entrant tuning of the ukulele allows the bell-like effects of campanella to occur naturally on the instrument. In figure 2.2.1 the opening of *Haere* (1916) is a reminder of how Ernest Ka’ai placed the notes on alternating strings to instinctively create campanella, bell-like, effects.[85]

Figure 2.2.1. The opening of Haere by Ernest Kaai



[82] According to Andrew Shenton 1968-76 was a ‘period of artistic reorientation’ for Arvo Pärt. (Shenton xviii)

[83] See also Hillier’s (1997) introduction, p. xi

[84] Pärt quoted in the sleeve-notes of *Tabula Rasa* (CD), ECM, (2010)

[85] For more on Ka’ai and campanella see Chapter 1.2 ‘Classical Approaches to the Ukulele’

The arrangements in John King's *The Classical Ukulele* (2004) demonstrated how, by using the campanella technique, the ukulele could accommodate complex and beautiful works by composers such as JS Bach. As discussed in the previous chapter this portfolio builds on the pioneering work of King by demonstrating how campanella techniques can be used to create new and challenging solo repertoire for the ukulele. Having created numerous new solo works[86] using campanella technique it occurred to me that tintinnabuli, with its bell-like objectives, might be a further opportunity to extend the boundaries of the ukulele. I was particularly drawn to the reductive nature of tintinnabuli and the delicate beauty of works such as *Für Alina* and *Spiegel Im Spiegel*. As Paul Hillier recognised, in Pärt's hands a single triad could be used 'to illuminate an entire work'. (Hillier, 1997, 20) My initial objective was to arrange the relatively simple piano piece *Für Alina* for the ukulele. How to capture the spaciousness and delicate beauty of tintinnabuli on the ukulele would, however, prove more problematic than anticipated but ultimately lead to new insights and innovations.

An Overview of Tintinnabuli

In the introduction to *The Cambridge Companion to Arvo Pärt*, Andrew Shenton describes the tintinnabuli technique as 'seemingly simple, comprising just two musical lines, one of which moves in largely stepwise motion and the other which moves through the notes of a principal triad.' (Shenton, 2012, 1) One line is the melody, which Hillier calls the M-voice, and the second line is the triad or T-voice. (Hillier, 1997) The basic principles of tintinnabuli are found in *Für Alina*, the small piano piece which 'announced' (Hillier, 1997, 87) Pärt's tintinnabuli style in 1976. According to Hillier:

> *Für Alina* highlights the basic manner of tintinnabuli composition: there are two parts in a fixed relationship, one moving mostly by step, and the other filling in notes from the tonic triad; it is homorhythmic; there is no chromaticism and no change of key or tempo. (Hillier, 1997, 87)

Between 1976 and 1977[87] Pärt composed fifteen tintinnabuli works including *Tabula Rasa* (1977), *Cantus In Memory of Benjamin Britten*, *Fratres* (1977) and *Für Alina* (1976). (Shenton, 2012, xviii) The tintinnabuli approach is simple but as Shenton acknowledges, 'What is remarkable is that the

[86] See *The Falling Rain*, *Variations on The Dowie Dens of Yarrow*, *A Conspiracy of Ravens*, & *Mercurius Suite*.

[87] The sleeve-notes of *Tabula Rasa* actually give the dates 1974 to 1976 as the period in which Pärt 'came out' of his 'creative silence'. (ECM, 2010)

method can offer such rich possibilities for compositional variety, all within a music that has broad aesthetic appeal.' (Shenton, 2012, 1) Pärt belongs to a unique group of twentieth century composers, including John Tavener (1944 - 2013) and Henryk Górecki (1933 - 2010), who have combined the simplicity of minimalism[88] with the inspiration of religious subjects. Hillier describes the essential features of this music as 'directness of feeling, transparency of form, austerity of mood, [and] economy of gesture'. (Hillier, 1997, ix) While Pärt's music reflects his own deeply held religious beliefs, derived from the Russian Orthodox Church, 'the music does not', according to Hillier, 'insist upon ontological discussion'. (Hillier, 1997, 1) Nevertheless, the diaphanous nature of the music, combined with the frequent use of slow tempos, inevitably inspires both the listener and the performer to a state of inner quietude and contemplation.

While choral works such as *Miserere* (1989/1992) are set to liturgical hymns, with explicit religious texts, many of the titles of Pärt's instrumental works are non-specific or at least ambiguous enough to allow the listener, and for that matter the performer, to engage with the music in a deeply personal way. For example *Tabula Rasa*, or clean slate, may be perceived as a philosophical reference to the idea that at birth the human mind is a blank slate waiting to receive and respond to ideas provoked by the senses or learnt from experience;[89] or it may simply indicate a fresh start, a new beginning. Given that *Tabula Rasa* was composed in 1977 at the beginning of Pärt's tintinnabuli journey it might also be seen as representing a new beginning for the composer.

Cantus In Memory of Benjamin Britten is a tribute to the British composer Benjamin Britten whose death in 1976 deeply affected Pärt at the very time he was emerging with his new tintinnabuli style. (ECM, 2010) Despite the dedication to Britten the poignancy of the music creates a feeling of loss and suffering that is at once deeply personal yet universally experienced. As Laura Dolp points out in her essay 'Arvo Pärt in the marketplace' (Shenton, 2012) film makers, such as Michael Moore, have often used Pärt's music to add potency to their narrative. In the shocking opening of Moore's documentary 'Fahrenheit 9/11' (2004), which shows live footage of the terrorist attacks on the Twin Towers, the visual horror is sharply contrasted with an ethereal soundtrack that plays the opening of *Cantus In Memory of Benjamin Britten.* (Shenton/Dolp, 2012, 177) In these

[88] Although the terms holy minimalism and sacred minimalism are often applied to Pärt's music Hillier concludes 'that while the *word* 'minimalist' is highly appropriate to describe some important aspects of Pärt's music, the *label* 'minimalist' is misleading, too culturally determined, to stand uncontested or at least unexplained.' (Hillier, 1997, 16)

[89] In Western philosophy the concept of tabula rasa is first found in Aristotle's (384 BC - 322 BC) *De Anima, and* later in the philosophy of John Locke (1632 - 1704).

few moments time seems to be suspended on the mournful chime of the bell which sounds like William Wordsworth's (1770-1850) 'still, sad music of humanity'.[90]

The Challenge

The principal challenge in applying tintinnabuli to the ukulele was determining how to capture its spacious and expressive essence on an instrument renowned for its short, percussive attack. It was, however, also an opportunity to demand new and unexpected things from the ukulele. The aim was to arrange Pärt's *Für Alina* for the ukulele. Although the piano score is relatively simple, comprising just two single lines of music (the M-voice and T-voice) plus a pedal, the slow tempo and long, sustained notes are essential to the contemplative mood. Without the benefit of a sustain pedal the acoustic sound of the ukulele was not naturally suited to this music. The problem was *how to overcome the natural limitations of the instrument while maintaining the ethereal nature of tintinnabuli?*

Music As Meditation

Für Alina is not a typical piano piece full of chords and virtuosic passages. As Hillier acknowledged the piece is technically easy. (Hillier, 1997, 87) Nevertheless, its tranquility and tenderness 'also exemplifies that quality, which distinguishes so many tintinnabuli pieces, of sounding both ancient and fresh at the same time'. (Hillier, 1997, 87) Despite its simplicity, *Für Alina* is a significant work for, as Pärt himself recognised, 'That was the first piece that was on a new plateau. It was here that I discovered the triad series, which I made my simple, little guiding rule.'[91]

I was particularly drawn to the minimalist or reductive nature of *Für Alina*. The score offers few notes (see figure 2.2.2) but Pärt's direction 'Ruhig, erhaben, in sich hineinhorchen' *(quiet, sublime, listen to your inner self)* suggests something distinctively personal in the music. By using the possessive pronoun *your* I felt Pärt was speaking directly to me. It was, therefore, not just a process of arrangement (which, as we will see, led to composition) but a process of contemplation and introspection.

In coming to terms with Pärt's unique approach I was able to draw on my experience of having studied John Tavener's *Chant*[92] (1984) for solo guitar, another work which challenges the

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[90] From the poem *Lines written a few miles above Tintern Abbey*.

[91] Pärt quoted in the sleeve-notes of *Tabula Rasa* (CD), ECM, (2010)

[92] Muir (2012)

performer in unconventional ways.[93] *Chant*, like *Für Alina*, also presents the performer with a score that is, by contemporary guitar standards, sparse (see figure 2.2.3). The piece both looks and sounds quite different from other major guitar works. Consisting of few chords and one largely unadorned melodic line the score looks more like an idea or sketch. Stripped of many of the usual cyphers, such as time signature and bar lines, even the layout defies convention. Perhaps it is for this reason that the publisher (Chester Music) chose not to typeset the music but simply to print it in Tavener's spindly, sometimes difficult to decipher hand - further adding to the personal nature of the music.

Figure 2.2.2. The opening of Für Alina by Arvo Pärt

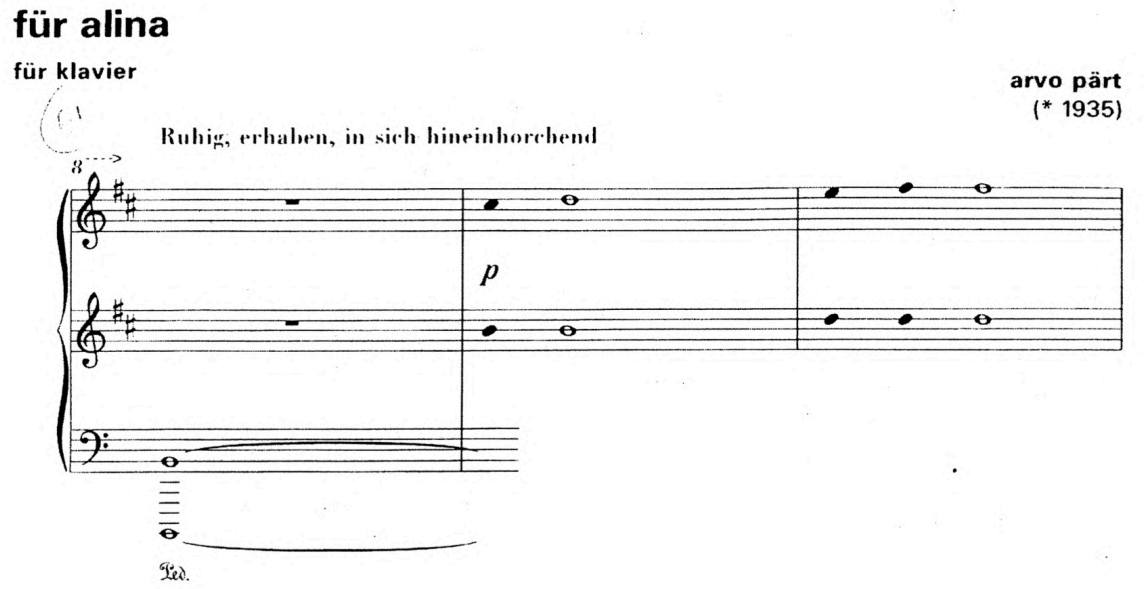


Figure 2.2.3. The opening of Chant for solo guitar by Sir John Tavener



[93] Muir (2014)

Tavener described *Chant* as a meditation on the ancient Byzantine hymn, *Fos Ilaron*. Once again the composer is seen to assume the role of spiritual guide; the instruction at the beginning is to play 'freely, but with great inner stillness.' While the piece is not technically difficult it is long by guitar standards - lasting approximately 10 minutes. I found it surprisingly challenging to engage in this state of 'inner stillness' in a live performance where the scarcity of notes left me feeling exposed with time to worry as to whether the audience was engaged, or just bored.

Rather than overt technical fluency, *Chant* like *Für Alina,* demands a careful control of pacing, texture and space. The aim is to transport the audience into a state of inner reflection rather than rouse them to outward acclamation. Harry Christophers, the conductor of The Sixteen, when asked about performing Taverner's *Shunya* for choir and Tibetan bowl, explained the challenge:

> Every time we perform John's music we have to enter into his soul. Into his way of thinking. His music is very still. It's actually very difficult to sing. He stretches the limits of each voice....Everything is mirroring the world and heaven and hell.[94]

Chant also contains very low notes and very high notes (see figure 2.2.4), as well as harmonics (see figure 2.2.5), thus utilising the full expanse of the guitar's range. As G below the stave features prominently in a figure of four Gs rising in octaves (see figure 2.2.6) I decided to tune the fifth string down a tone in order to play the lowest G as an open string.

Figure 2.2.4. Showing the range of Chant

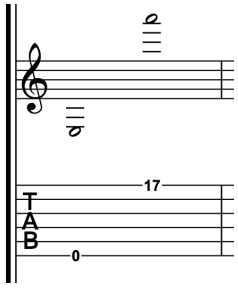
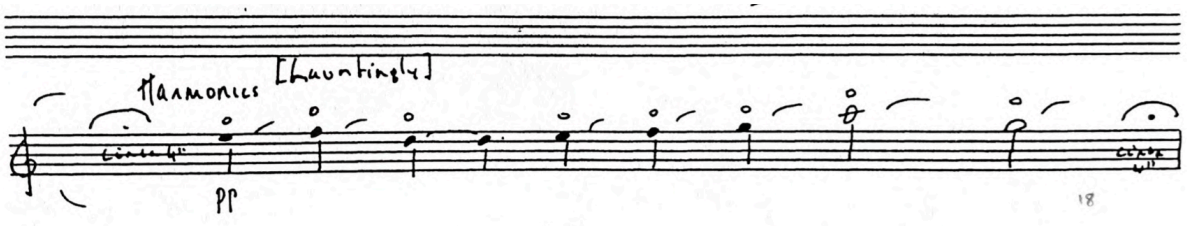


Figure 2.2.5. The harmonics section of Chant



[94] *Towards The Musical Perennis*, BBC programme, *The Music of John Tavener*

This allowed the note to ring on for longer than if it were fretted and mirrored the opening figure of four Es rising in octave leaps and beginning with the open sixth string (see figure 2.2.7). It also adhered to Tavener's instruction, shown in figure 2.2.7, to let the strings vibrate *(laissez vibrer)*.

Figure 2.2.6. The rising G figure in Chant



Figure 2.2.7. The rising E figure in Chant



Both of the above figures are repeated throughout the piece and the extra resonance of the open strings proved particularly effective in allowing the low notes to ring on for their entire duration. While there are some flurries of fast notes in *Chant* there are also moments of intense stillness where long, held notes fade to the point of silence. No other guitar piece (that I have played) has relied on this delicate balance between music and silence so poignantly. I often imagined the piece as a musical rosary where the notes, like beads, were moved very carefully along a thin thread of silence - each note as sacred as a prayer. How to convey this effect to an audience without the music becoming self-indulgent is, however, a difficult task. The fragile relationship between music and silence might be profound and moving or it might be awkward and uncomfortable.

For this reason *Chant* has always been, for me, a more risky piece to perform than, for example, a more technically demanding piece, such as the *Allegro Solemne* from *La Catedral* by Barrios. With rapid semiquavers running up and down the length of the fingerboard for approximately three minutes the technical demands of this piece (velocity, accuracy and fluency) are very specific and can be achieved with careful and considered practice. While it is a nerve

wracking piece to perform the virtuosic excitement of cascading notes means that the odd fluffed note is barely noticeable as both the performer and the listener are swept along by its relentless tension and excitement. In stark contrast the ethereal, other-worldly nature of *Chant,* and the fact that it can last anywhere between 9'11 minutes (Simon James' recording of 1985) and 14'15 (Gerard Cousin's recording of 2008), makes it a long and challenging experience for both performer and listener.[95] I have always felt that the slightest mis-hit note or unwanted squeak would shatter the glass-like fragility of the music and ruin the whole effect. I have often wondered if other guitarists have had similar reservations and whether this might explain why *Chant*, despite being composed by one of the most important British composers of the 20th century, has not achieved the same status as works by Lennox Berkeley, William Walton and Benjamin Britten.[96]

Where Is the Music?

It was interesting, and reassuring, to read in the sleeve-notes of *Tabula Rasa* (ECM, 2010) that other musicians had initially found the minimalist approach confusing. In discussing the first performance of *Tabula Rasa*, which is scored for two violins, prepared piano and strings, Pärt commented that 'When the performers first saw the score, they cried out: "Where is the music?" But then they went on to play it very well. It was beautiful, it was silent and beautiful.'[97] It is interesting to dwell, momentarily, on the reaction of the performers who were clearly disconcerted by the austerity of the score. I had initially reacted in the same way to *Chant*. The single line of music looked more like a part from an ensemble piece than a fully realised concert work. On reflection it does seem odd that formally trained musicians can be so confused, perhaps disorientated, by a score which contains so few notes and has such an intimate relationship with silence. This leads me to the conclusion that, according to convention, more notes represent greater accomplishment while fewer notes imply a low level of ability on the part of the performer. Composers such as Pärt and Tavener challenge our perceived notions of concert music by asking both the performer and the listener to re-evaluate their relationship, not only with each other, but with the music.

[95] My own YouTube performance from 2012 lasts just under 10 minutes.

[96] The main recordings of *Chant* are Simon James *Daybreak* (Premier) 1985; Eleftheria Kotzia *Mediterraneo* 1994; Andrew Keeping *British Guitar Music* 1996; Jonathan Richards *Elegy* 1998; Gerard Cousins *A Gift* 2008.

[97] Pärt quoted in the sleeve-notes of *Tabula Rasa* (CD), ECM, (2010)

Musical Archetypes

Leopold Brauneiss' essay 'Musical Archetypes: the basic elements of the tintinnabuli style' gives a unique insight into Pärt's way of thinking. Brauneiss' analysis of music (such as tintinnabuli) which is constructed from 'very simple basic elements or patterns such as scales', is that these elements act as archetypes. He uses the word archetype in the Greek sense of an 'original or primal image (*arche* = beginning, source).' By going on to draw an analogy with the analytical psychology of Carl Jung he suggests that these archetypes are part of a 'collective unconscious' which is not 'individual but universal.' (Shenton/Brauneiss, 2012, 49) Although the psychology of music is beyond the realms of this commentary, Brauneiss makes some interesting points which enable a fuller appreciation of the minimalist approach.

Quoting from Pärt's acceptance speech for the International Bridge Prize of the European City of Görlitz, Brauneiss links the concept of the archetype, or scale, to Pärt's 'idea of the human soul possessing a net-like basic pattern'. (Shenton/Brauneiss/Pärt, 2012, 50) The scale, in all its myriad forms, represents the basic building block, or pattern, of music. Scales are 'globally present' and 'pre-existent' (Shenton/Brauneiss/Pärt, 2012, 54) and, as such, contain all the attributes of the musical archetype. Through the scale, or the archetype, we are transported to Pärt's 'passionately longed-for "magic island," where all people (and for me [Pärt], all sounds) can live together in love.' (Shenton/Brauneiss/Pärt, 2012, 50)

It is perhaps surprising that such spiritually inspired music is so popular today, but as Laura Dolp points out, this is a reflection that despite our complex and busy lives what we all yearn for is the 'simpler life' (Shenton/Dolp, 2012, 189). Pärt presents the 'simpler life' to us in the form of the scale, or musical archetype, which is part of our 'human geometry'. (Shenton/Brauneiss/Pärt, 2012, 50) On a fundamental level the scale unites us. It is also Pärt's musical pathway to the realisation that 'we are all so similar that we could recognise ourselves in any other person'. (Shenton/ Brauneiss/Pärt, 2012, 49) Such ideas lead to the conclusion that while pieces such as *Für Alina* and *Chant* require a state of inner calm from the performer the aim is to transcend the self, and the ego, and transport the audience to this shared place of beauty; the 'magic island' imagined for us by the composer. To return to the question w*here is the music?* the answer might then be *within us all.* In this scenario the purpose of the performer is to tap into the musical archetype and, like a fisherman, cast a net of beauty over the audience.

Beauty and Unity

Pärt often uses the word beauty in relationship to his music. One of his most famous quotes is 'I have discovered that it is enough when a single note is beautifully played. This one note, or a silent beat, or a moment of silence, comfort me.' (Pärt, 2010, ECM) It seems almost contradictory that such a highly regulated, yet simple system as tintinnabuli, can sound so ethereal and beautiful. But, as Pärt is frequently also quoted as saying, 'The complex and many-faceted only confuses me, and I must search for unity. What is it, this one thing, and how do I find my way to it?' (Pärt, 2010, ECM) Again we return to the concept of the musical archetype; the scale as a simple path that has the potential to lead us to this 'one thing'. Although the 'one thing' is never made explicit the mind seeks to understand it through prayer, contemplation, art, poetry, or through music. The purpose is not to discover the 'one thing' but to continue to search for it.

In Pärt's acknowledgement that 'tintinnabulation is an area I sometimes wander into when I am searching for answers - in my life, in my music, in my work' (Pärt, 2010, ECM) it occurs to me that the musician does not so much as *perform* the music of Pärt as *enter* into it. Each note becomes a search for this one beautiful thing. Applause becomes superfluous, perhaps even a distraction, as the shared experience of the music, passing from the composer to the performer to the listener, is the ultimate gift. In the words of Pärt 'It [the music] is there to transport us towards something that has never been heard before.' (Pärt, 2010, ECM)

The popularity of Pärt's music is testimony to his idea that at the core of all human existence there lies a fundamental desire to live in peace and harmony. Through Pärt's music we are able to experience a few moments on the 'magic island'. Although the performance may end, perhaps leaving us with a sense of loss or sadness, there is also the comfort of knowing that other performances continue the journey long after we are gone. Pärt's music often reminds me of the poem *Ithaka* by Cavafy which begins 'As you set out for Ithaka, hope your road is a long one.' Being able to remain on the long road of pieces such as *Chant* and *Für Alina*, whether as listener or as performer, can be a beautiful, calming and enriching experience. The final words of Cavafy also seem to resonate with Pärt, as despite the transcendental intentions of his music it is also deeply humble.

> And if you find her poor, Ithaka won't have fooled you.
> Wise as you have become, so full of experience,
> you'll have understood by then what these Ithaka's mean.[98]

[98] Sourced from the Poetry Foundation website.

Für Alina on the Ukulele

In order to determine if, and how, Pärt's *Für Alina* could be played on the ukulele the first step was to establish how the range might work. Assuming Pärt's original key of B minor was used the highest note of the M-voice (D6) could be played on a ukulele with 17 frets[99] (see figure 2.2.8). The lowest note of *Für Alina* (B3) could only be played on a ukulele using low G tuning. Even with low G tuning, however, the T-voice could not accommodate the lowest notes - F#1 (see figure 2.2.9).

Figure 2.2.8.
The range of the M-voice in Für Alina

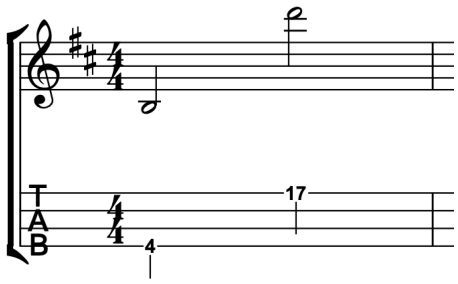
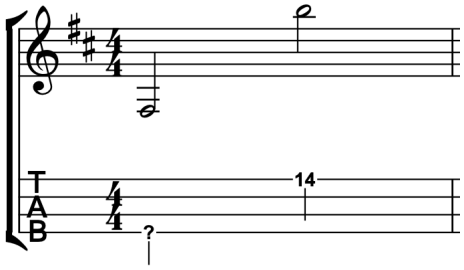


Figure 2.2.9
The range of the T-voice in Für Alina



Assuming a ukulele with low G tuning was used, just one note in the T-voice would need to be raised an octave. Although low G tuning initially seemed like a solution, it created a new problem in that the higher T-voice notes could no longer be played along the fourth string. The tablature in figure 2.2.10 shows how the notes could be positioned using standard re-entrant tuning while figure 2.2.11 shows how, with low G tuning, these notes would have to be moved to another string. As can be seen in figure 2.2.11 the high Bs in bars 6 and 8 would need to be played at the nineteenth fret of the second string. This, however, was not practical as, although my low G ukulele has nineteen frets, the increasingly close proximity of the frets beyond the fourteenth fret meant the sound was not clear. Furthermore, the instant decay of notes played this high up the fingerboard created a sound best described as an inelegant *plink plink plink.*

[99] Nowadays most luthiers are building instruments, even the smaller bodied soprano, with more than twelve frets. The number of frets is, however, far from uniform. For example my DJ Morgan sopranos have 19 frets while my Romero Creations tenor has only fifteen.

Figure 2.2.10. Using standard tuning the T-voice could be played on the 4th & 3rd strings

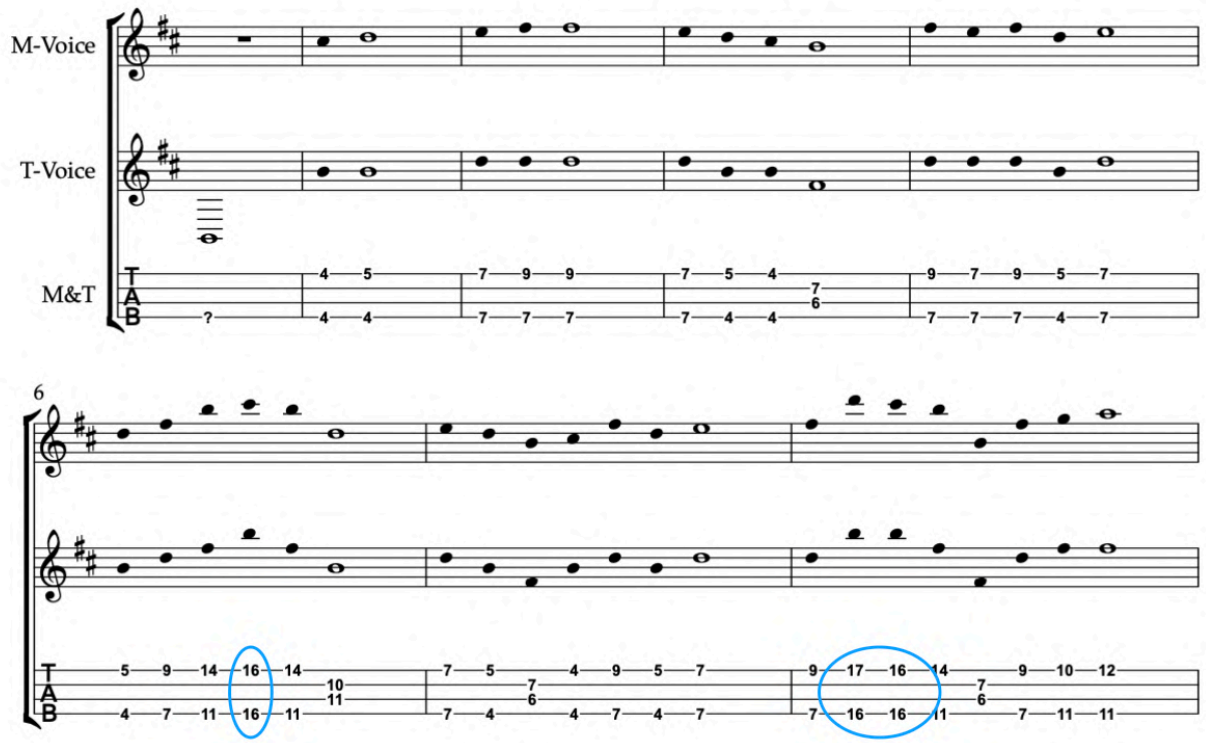
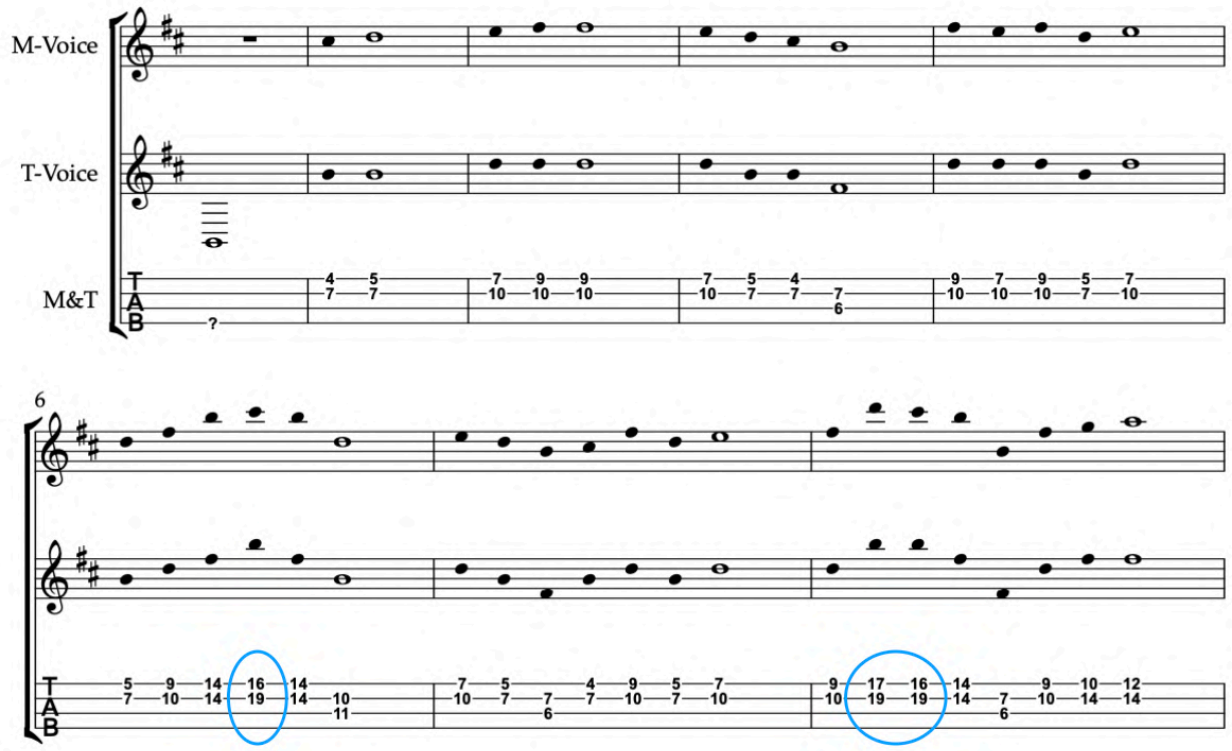


Figure 2.2.11. Using low G tuning the T-voice could be played on the 2nd & 3rd strings



A further problem was what to do about the low B pedal (see figure 2.2.12) which was impossible to play on any ukulele. At this stage I decided to pause working on the arrangement in order to reflect on my findings.

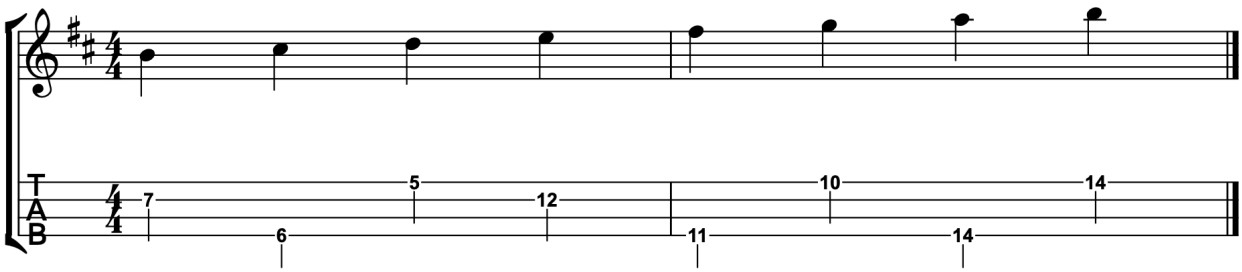
Figure 2.2.12. The low B pedal at the beginning of Für Alina



Comparing Campanella and Tintinnabuli

My research had revealed that although campanella and tintinnabuli are united by the common aim of creating bell-like effects, the techniques are very different. As the scale example in figure 2.2.13 shows campanella requires just one melodic line of music with the notes placed on alternating strings to create the over-ringing effects.

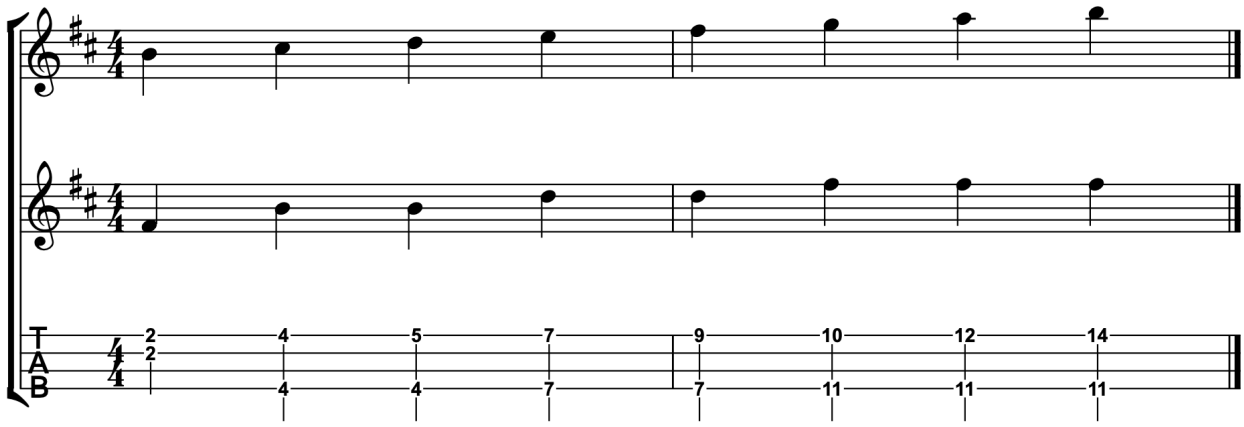
Figure 2.2.13. The B minor scale in campanella style



Meanwhile the bell-like effect of tintinnabuli is created by an accompanying line, or T-voice, which is usually based on the tonic triad. In figure 2.2.14 the top line (once again the B minor scale is used to demonstrate) is the melody, or M-voice, while the second line is the triad, or T-voice, which is

generated from the B minor triad. The tablature line shows how both voices are played on the ukulele.

Figure 2.2.14. The B minor scale in tintinnabuli



According to Brauneiss the tintinnabuli voices are linked to sin (M-voice) and forgiveness (T-voice). (Shenton/Brauneiss/Pärt, 2012, 57) Brauneiss goes on to quote part of a conversation between Pärt and Enzo Restagno (2003) in which the composer said, 'Maybe I could say that the melody represents my sins and my imperfect being, whereas the second voice is the forgiveness that is granted to me.' In another interview from 2010 Pärt expressed it this way: 'One line [M-voice] is who was are, and the other line [T-voice] is who is holding and taking care of us.' (Shenton/ Brauneiss/Pärt, 2012, 57) The melody, which has greater potential to wander, represents humanity. Meanwhile, the triad, which remains fixed to just three notes, represents the steadfastness of God. In this way Pärt can be seen to achieve his goal of connecting time (the brief and wandering melody of our lives) and timelessness (the constant presence of God).[100]

As demonstrated in the above figures it is possible to play the M-voice on one string of the ukulele and the T-voice on another. As previously stated, however, when applying this principal to *Für Alina* the over-ringing of the T-voice, due to the short attack of the ukulele, was not a feature. Every time the T-voice note moved position the over-ringing was interrupted by the release and repositioning of the fingers. An additional problem was that without the constant sound of the pedal note the piece lacked its unifying, omniscient presence. Although the pedal is barely audible in the piano version, it sounds like a constant mantra or repeated "om". Without the pedal, each bar sounded detached and isolated, and the resulting silences awkward rather than ethereal. My

[100] Pärt quoted from the sleeve-notes of *Tabula Rasa*, 2010, ECM

conclusion was that while *Für Alina* could be played on the ukulele, the lack of sustain meant the notes did not contain the bell-like quality essential to tintinnabuli.

Searching for Solutions

Given that the basic principals of tintinnabuli are just two voices I felt there should be a way to make it work on an instrument with four strings. Using Brauneiss' model of the tintinnabuli positions of the A minor triad (Shenton/Brauneiss, 2012, 56), I returned to basics by applying his model to the ukulele. In examples 2.2.15 to 2.2.19 the M-voice and T-voice are given separate staves in the staff notation while the tablature line shows how both voices can be played on the ukulele. In the inferior positions the T-voice is generated by locating either the first triad note beneath M-voice (first position inferior) or the second triad note beneath the M-voice (second position inferior).

Figure 2.2.15. A minor in 1st position inferior

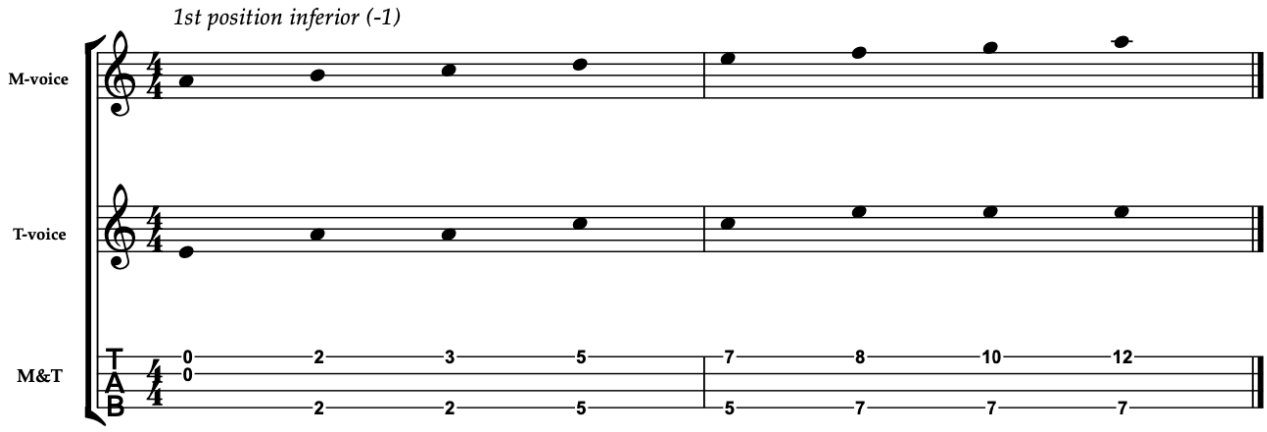
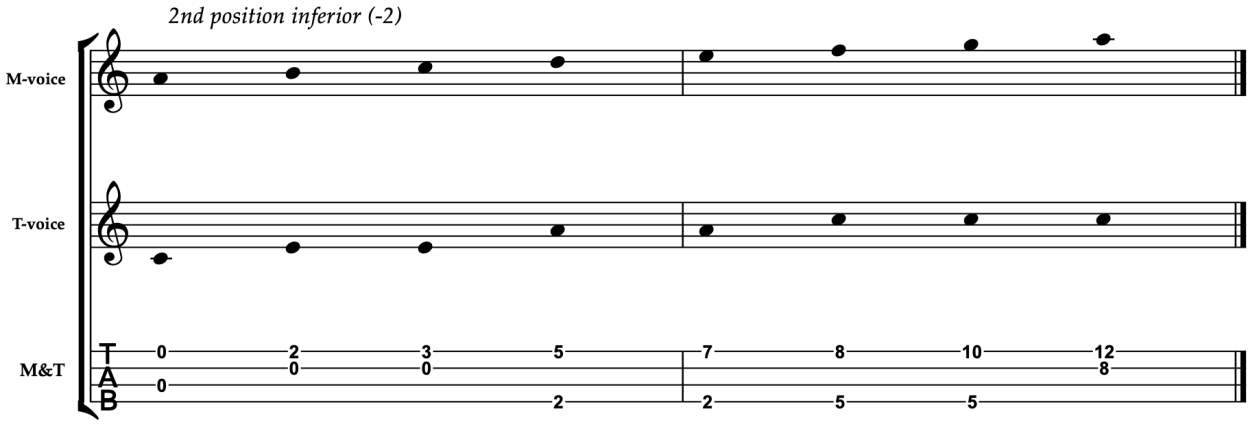


Figure 2.2.16. A minor in 2nd position inferior



In the superior positions the T-voice is located by finding either the first triad note above the M-voice (+1) or second triad note (+2). See figure 2.2.17 for first position superior and figure 2.2.18 for second position superior. For the superior positions I assumed low G tuning would have to be used in order for the voices to sound in the correct order.

Figure 2.2.17. A minor in 1st position superior

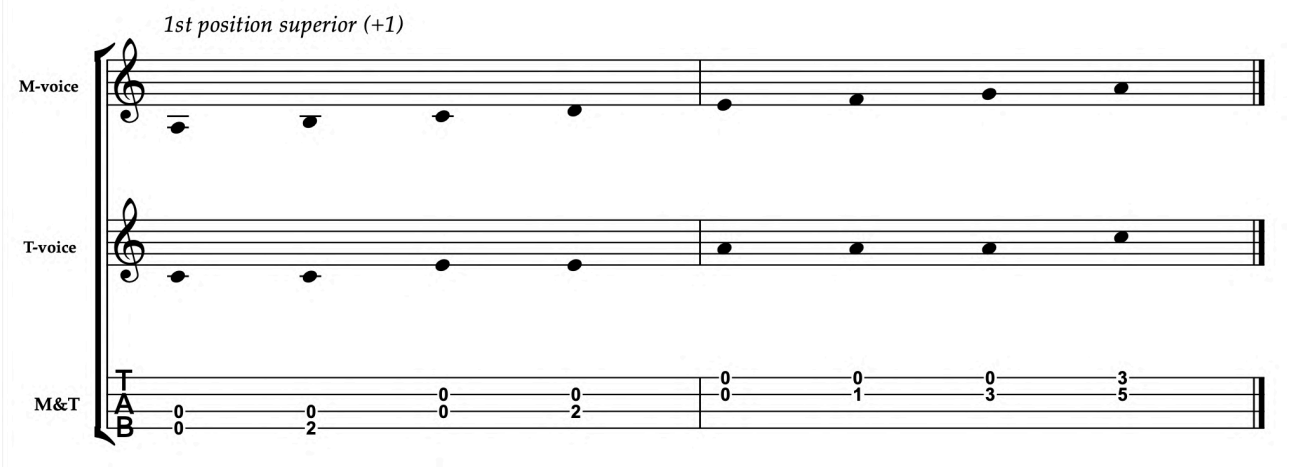
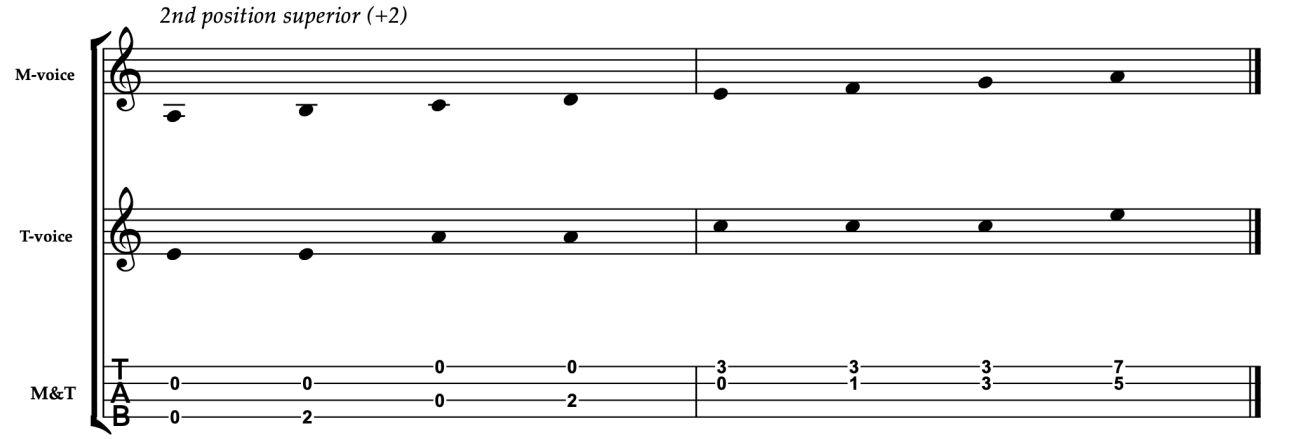
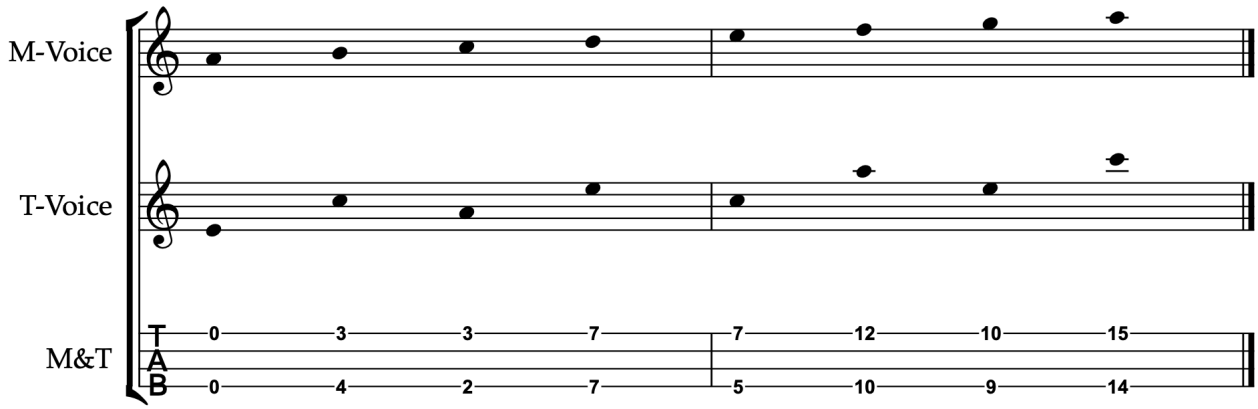


Figure 2.2.18. A minor in 2nd position superior



Brauneiss gives a third example (see figure 2.2.19) of the first position alternating between inferior and superior positions. In other words, if A is our melodic note (M-voice) the first triad note (T-Voice) below this is E. If our next M-voice note is B then first triad note above is C etc.

Figure 2.2.19. A minor, 1st position alternating inferior (-1) & superior (+1)

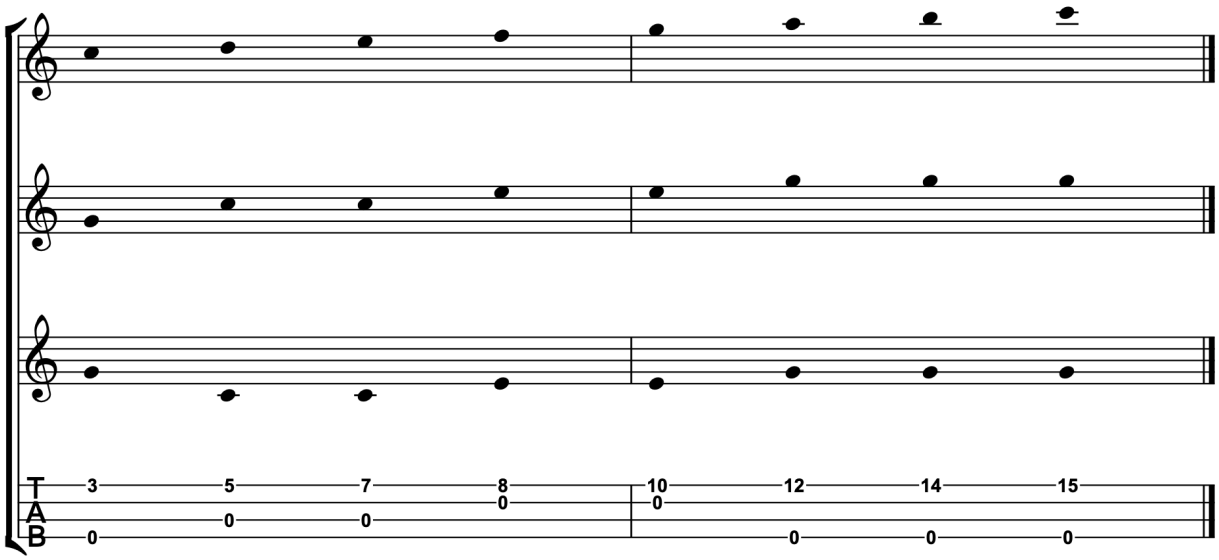


Over-ringing: A Solution

My conclusion was that the lack of over-ringing in the T-voice was largely due to the constant use of closed notes. In campanella style, as in the harp-effect of Pierre Bensusan, the use of open notes is both musically and technically desirable as these notes are able to ring-on naturally, thus allowing the fretting hand the freedom to shift position without disrupting the resonance. It seemed only logical that if more open notes could be used in tintinnabuli the over-ringing potential would be greatly enhanced.

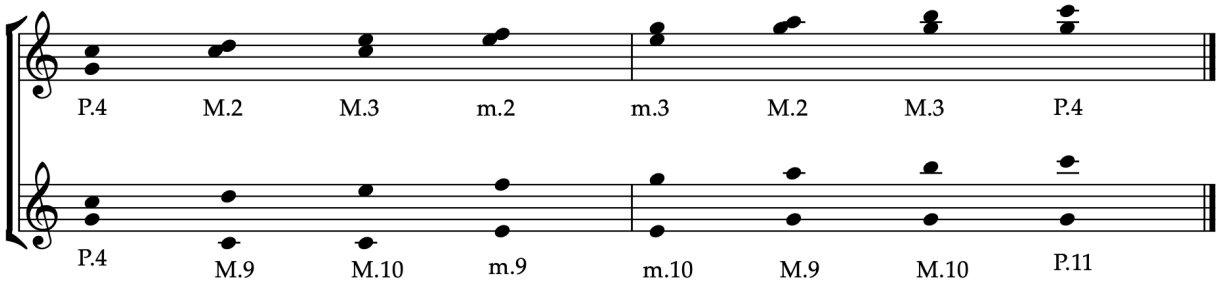
As the open G (0,4), C (0,3) and E (0,2) strings of the ukulele are most commonly tuned to form a C major triad an obvious, albeit simplistic, solution would be to use the open strings as the T-voice while placing the M-voice on the top string. The example in figure 2.2.20 shows the first line of notation as the M-voice. The second line is the T-voice in first position inferior (-1). The third line shows the T-voice re-imagined as the open strings of the ukulele - the notes are the same but seven out of the eight notes have been lowered an octave. The tablature line shows the M-voice (C major scale) placed on the first string with the re-imagined T-voice using the open fourth, third and second strings. Only the first note of the T-voice, however, remains at the original distance from the M-voice.

Figure 2.2.20. C major in 1st position inferior with the open strings assigned to the T-voice



While this idea did generate more resonance in the T-voice it changed the interval relationships between seven of the eight notes. As shown in figure 2.2.21 seconds became ninths, thirds became tenths and so on.

Figure 2.2.21. The tintinnabuli intervals in 1st position inferior & how they are altered when reimagined as open strings



The use of open strings was successful in creating the over-ringing that had previously been missing. The resulting sound was pleasing and, despite the changes to the intervals, this seemed like a good, practical solution. The over-ranging T-voice achieved what Hillier described as the ‘sound-image’.

> When a bell is struck, it continues to sound indefinitely: the ear cannot detect the point at which it ceases to vibrate. This sound-image may be compared to Pärt’s manner of

> articulating the triad from within a musical process, so that the sonority which accumulates is intrinsically clear yet contains overtones and undertones far more dense than the notes on paper would suggest. The combination of melodic lines and triadic notes provided two sorts of continuity: one which leads step by step from one thing to the next, and one which rotates like a single object being viewed from different perspectives. (Hiller, 1997, 86)

The use of open strings meant each T-voice note could ring on, or live, for its natural duration. Free to fade into the other notes it would thus ‘sound indefinitely’. In contrast, when the T-voice notes were fretted (at their original pitch) the sound was interrupted by the constant movement of the fingers. The free ringing of the open strings, however, provided ‘continuity’ while moving within the ‘single object’ of the triad. The price, however, was the change to Pärt’s original intervals and how this impacted the overtones and the undertones mentioned by Hillier.

While Hillier does not specify what he means by overtones and undertones, he is presumably talking about the upper and lower partials, or harmonic tones, which make the music sound ‘far more dense than the notes on paper would suggest.’ (Hiller, 1997, 86) Further insight into what this means is provided by Margueritte Bostonia in her essay *Bells as inspiration for tintinnabuli* (Shenton, 2012, 128) in which she compares the harmonic series of a vibrating string with tuned bells and untuned bells. Bostonia’s conclusion was that ‘dissonant chords or non-functional harmony are often descriptive of the vertical tintinnabuli sonorities which incidentally result from the expanding melodic M-voice interacting in union with the triadic T-voice.’ (Bostonia/ Shenton, 2012, 137) Although my re-imagined T-voice had the same pitch class, the intervals sounded different harmonically. Thus, the ‘vertical tintinnabuli sonorities’ were altered, this impacting what Hillier calls the ‘harmonic framework’ of the piece:

> Similarly, in tintinnabuli music, where the harmony does not ‘move’, the harmonic framework has been tilted sideways to form a musical line, and the relationship between the two different kinds of melodic movement creates a harmonic resonance which is essentially the triad and the fluctuating attendance of diatonic dissonances. (Hillier, 1997, 90)

In my re-imagined T-voice (see figure 2.2.210 seven out of the eight intervals are altered by the distance of an octave; thus becoming compound intervals. The seconds which would otherwise occur on the 2nd, 4th and 6th degrees of the scale become ninths. Meanwhile, the thirds (3rd, 5th

and 7th degrees) become tenths. Furthermore, the final note of the scale becomes an eleventh rather than a 4th. While the quality of major, minor and perfect is maintained the aural difference between these intervals is quite distinct. Although, for example, both the minor 2nd and minor 9th are classed as dissonant intervals the minor 2nd sounds more dissonant than the minor 9th. In Pärt's tintinnabuli vocabulary the tension of the minor 2nd is a distinctive feature. In my reimagined T-voice I had made a compromise: prominence was given to the over-ringing while the intervals were diluted. Changing the intervals may, in the mind of the purist, be seen as a corruption of Pärt's approach but in so doing it had allowed the re-imagined T-voice to occur idiomatically on the ukulele. Although the minor 9th does not contain the same harmonic tension in the T-voice as the minor 2nd the use of open strings added to the resonance and created additional over-ringing effects.[101]

Expanding Narratives

Part of my research into the tintinnabuli technique involved a series of compositional sketches which used open strings as the T-voice. While based on tintinnabuli style they are, however, an adaption and modification of this technique. The compositional sketches eventually formed a set of four short pieces called *Lines of Flight.*[102]

1.Holloway[103]

The first sketch *Holloway*, which was named after the book by nature writer Robert Macfarlane (2014), refers to the ancient sunken paths of South Dorset. There is a network of these ancient paths near to where I live in the neighbouring county of Somerset. I spent many hours over lockdown wandering these paths while listening to the music of Arvo Pärt on my headphones. It was these rambles that sparked the idea of trying tintinnabuli technique on the ukulele. The holloways, like Pärt's music, are timeless - acting as links between the ancient and the modern. Pärt's tintinnabuli style was inspired by Gregorian chant while the holloways have been carved into the landscape by the passage of people and animals over many hundreds, perhaps thousands, of years. In the 2002 documentary *24 Preludes for a Fugue Pärt* explains how *Für Alina* is a journey beyond the black and white keys and how, by focusing on each sound, we are transported to a place where 'every

[101] For audio see Folder C.5. For score see Folder B.2.

[102] The title comes from Deleuze's and Guatarri's *line of flight* in A *Thousand Plateaus*. See p.10 for further discussion about how I used this idea in my research model.

[103] For audio see Folder C.1. For full score see Folder B.2.

blade of grass would be as important as a flower.' (Pärt, 2002) In Pärt's terms the simple melody of *Für Alina* is a musical pathway. The title *Holloway* seemed appropriate as I was both seeking to follow in the footsteps of Pärt while searching for my own, new path.

Figure 2.2.22. The opening of Holloway & the different positions of the T-voice -1/-2/-3 marked in red



Using *Für Alina* as the model for *Holloway* I located the T-voice based on what sounded good rather than using Pärt's fixed system (see figure 2.2.22). This allowed the freedom to experiment with the open strings and the natural resonance of the ukulele. I used a minor key, like Pärt, and, as in 'Black Raven' lowered the E string down a semitone to E♭ to sound an open C minor T-voice. The idea of using alternate tunings was further developed in the other sketches.

The score of *Holloway* mirrors the layout of *Für Alina* with the M-voice written on the top stave and the T-voice on the second stave (see figure 2.2.22). The extra tablature line combines both

voices. Like *Für Alina* the parts are homorhythmic but in *Holloway* the T-voice is constantly shifting positions between -1, -2 and -3 (marked in red in figure 2.2.22). The asterisk next to the number indicates that the note has been lowered an octave.

Another similarity with *Für Alina* is that *Holloway* also uses the idea of expansion and contraction with each bar increasing by one note until a maximum of eight notes is reached and then reducing by one note per bar. Interestingly, Pärt breaks from this pattern in very last bar of *Für Alina* by suddenly expanding to three notes rather than reducing to one note. It is a simple but brilliant way of suggesting that the music has no beginning or ending but is constantly moving in and out like the breath, or the tide. Just as the ancient pathways encounter many passing feet so the notes are played by many passing hands; it is not just one journey but many journeys constantly moving and converging.

Pärt's original hand-written score of *Für Alina*, and also the published version, use stemless note heads. No time signature is given and bar lines are used more as dividers to separate or contain each group of notes rather than define them rhythmically. Thus, the first bar contains one note, the second two and so on. Although I used this format in *Holloway* I found the lack of a pedal note impacted the coherence of the piece. In *Für Alina* the pedal is barely audible but it is a constant and unifying force. On the ukulele even an open string will only sound for a limited time due to the instrument's short attack. As soon as a note is struck it begins to decay. My solution to the problem of the pedal was to introduce each bar of *Holloway* with an open third string C, the lowest note possible on a standard tuned ukulele. Although the C has to be re-played every bar it creates the illusion of continuity. The repeated Cs are more like a drone than a pedal but the intention, to create a unifying presence and reassert the tonic, is effectively the same.

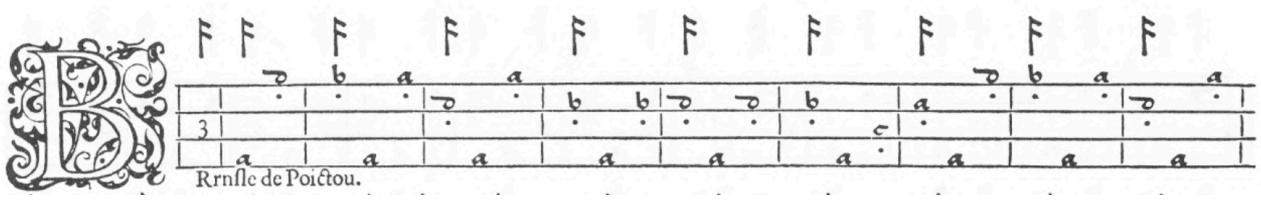
The C drone is notated as a semibreve but is meant to ring on for as long as possible before the other voices emerge from the very edge of silence. The following direction is given at the beginning of the piece to clarify the role of the drone:

> All bars begin with the C drone. The M & T voices enter just before the C decays. The duration of the drone may vary according to attack, acoustic, resonance of the ukulele & mood of the player.

In *Holloway* the constant drone was reminiscent of the Renaissance guitar pieces of Adrian Le Roy. As shown in figure 2.2.23 Le Roy often used the open fourth course as a repeated drone. As Pärt

had drawn on an ancient style of composition, so my own re-imagining of tintinnabuli borrowed from an ancient guitar technique.[104]

Figure 2.2.23. The opening of Bransle de Poitou by Adrian Le Roy from Tiers Livre de Guiterre, 1552



The C drone of *Holloway* represents stability and continuity but inevitably, like the ancient sunken pathways of Dorset and Somerset, even the drone note cannot escape the passage of time. Although it represents timelessness its passage is inevitably transient. The small, open circles of the semibreves contrast with the closed circles of the stemless crotchets; on the one hand time can be seen as open and infinite, like the semibreves, and on the other it is closed and finite like the crotchet note heads. Thus, the stemless note heads of *Für Alina* are a powerful symbol of the composer's intention: appearing to float across the stave they are at once detached from time yet inevitably rooted to its passage. Applied to the score of *Holloway* the stemless note heads are like footprints on the stave, or echoes from another age. Thus, while the score of *Für Alina* is very precise and mathematical in its structure, it is transcendental in its appearance and dénouement. When listening to this piece played on repeat there is no concept of beginning or end, the music simply ebbs and flows.

Although *Holloway* is not truly in the tintinnabuli style it demonstrates how the ukulele can be used to create music that is slow and spacious; a stark contrast from its more normal role as an instrument that is strummed. In the final analysis tintinnabuli was the inspiration behind a style of composition that was transformed and adapted to suit the ukulele. Further experimentation gave rise to three other pieces.

Inspired by the Rhizome

The idea of a rhizomatic approach to theory and research, or one which is essentially non-hierarchical and non-linear, was put forward by Gilles Deleuze and Félix Guattari in *A Thousand*

[104] For audio see Folder C.4.

Plateaus (1980).[105] They used the term *ligne de fuite* (line of flight) to illustrate their idea of the rhizome that can send out roots at any point and is also known for its ability to connect with other species to create mutually beneficial partnerships. I first came across this concept in Robin Nelson's *Practice As Research in the Arts*. According to Nelson

> The shift from modernism into postmodernism marked also a change of conceptual metaphors from a surface/depth model (with a single deep taproot) to Deleuze and Guattari's rhizome (with a tangle of interconnected roots). The rhizome taken as an 'image of thought' lends itself to multiple strands without hierarchy in contrast to deep mining within a linear tradition. (Nelson, 2012, 54)

The tintinnabuli project mirrored my search for a methodology that would connect and justify the different components of my portfolio. Nelson's model involves an 'opening out of disciplines' rather than 'deep mining' a specific line of thought. My approach might, however, best be described as deep mining the style of classical ukulele through the multiple perspectives of the arranger/ composer, performer, teacher and researcher.

My research into tintinnabuli was a further reflection of the rhizomatic idea of Nelson as what began as an arrangement of *Für Alina* soon developed into four original compositions. I called the collection *Lines of Flight* in acknowledgement of how practice imbricated with theory can lead to new creative outcomes. (Nelson, 2012)

2.Patientia

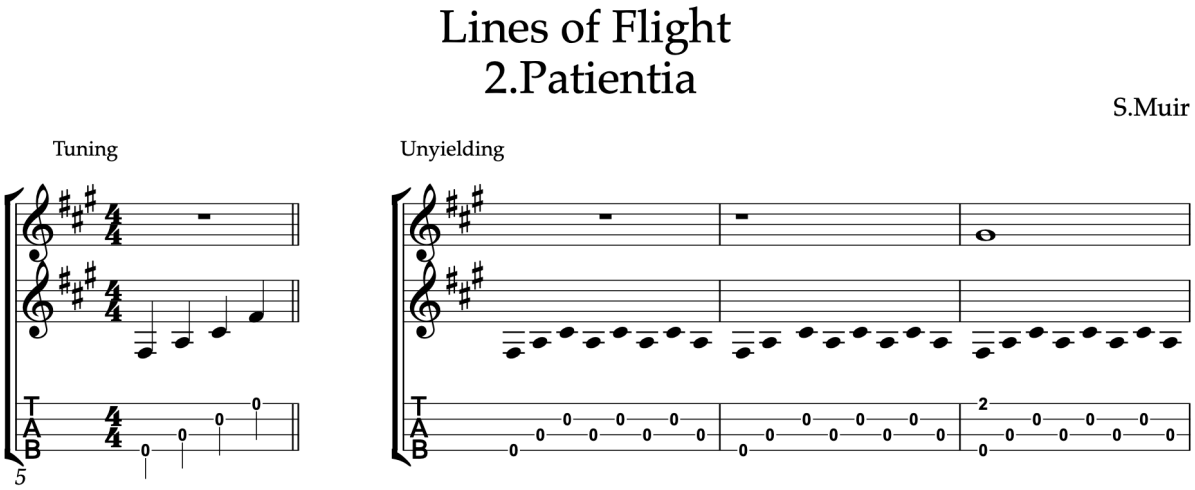
Patientia, the second piece of *Lines of Flight*, used the same score layout as *Holloway* with a stave of notation for each voice plus a third tablature line. Once again stemless notes, the idea of expansion and contraction in the top line, and a minor key were used. In order to use the open string T-voice in a different key, a radical alternate tuning was necessary. This time the key of F♯ minor was used. This meant the open strings were tuned F♯ A C♯ F♯ (see figure 2.2.24).

Further variety was achieved by using my five string tenor ukulele (as in *A Conspiracy of Ravens*) with the fourth string sounding the octave. The notation, however, only shows the low F♯ as I felt the octave stringing was optional and the piece would also work with low G tuning. This time the T-voice moved through the F♯ minor triad like an Alberti bass (see figure 2.2.24) meaning it was, once again, alternating between the inferior positions. The constant repetition of the T-voice

[105] See also p.10 for further discussion.

created a different mood to *Holloway*. Rather than the M-voice emerging from the edge of silence it interrupts the constant, almost monotonous movement of the T-voice. The notes are constantly moving but the repetition of the T-voice also implies fixity, or stasis. The title *Patientia,* which is Latin for patience/suffering/endurance, was a response to the trials of the COVID-19 pandemic.

Figure 2.2.24. The alternate tuning and score layout of Patientia



3.The Red Angel

Once again the T-voice was assigned to the open fourth, third and second strings. The key of G minor meant the open tuning was G B ♭ D B ♭ (see figure 2.2.25). The score is notated for low G tuning but in my recorded version (*New Works for Classical Ukulele Volume 1*, 2022) I decided to use the five string tenor as the octave tuning provided extra resonance and interest. As in *Patientia* the T-voice of *The Red Angel* formed a repetitive pattern of open string arpeggios, this time moving between 6/8 and 3/8 time.

The use of tremolo in the M-voice meant the score was much busier than the previous pieces and I opted for a more conventional layout with just one line of notation, plus the tablature line. As in the other pieces the M-voice moves along the top string while the T-voice alternates across the inferior positions. The tremolo was an opportunity to combine techniques (tintinnabuli and tremolo) in order to achieve new and innovative results. Initially the tremolo was three notes (as in Giovanni

Albini's video) but I later experimented with a four note tremolo.[106] The tremolo sounded more effective with the M-voice notes doubled - beginning with two cycles of tremolo notes and increasing by two cycles per bar until a maximum of sixteen is reached; at which point the M-voice then reduces by two tremolo cycles per bar.

Figure 2.2.25. The tuning and the 4 note tremolo version of The Red Angel



The inspiration for the title came from Season 2 of *Star Trek: Discovery*. (Paramount, 2019) In this series a mysterious time travelling entity known as The Red Angel makes fleeting appearances as she seeks to warn the present about a threat from the future. The idea of simultaneous timelines is expressed in the music by the T-voice acting as the present - with its continuous repetitive movement - and the M-voice representing the fleeting appearances of The Red Angel. The M-voice sometimes moves in 6/8 in accord with the T voice while at other times the feel is 3/4 in order to

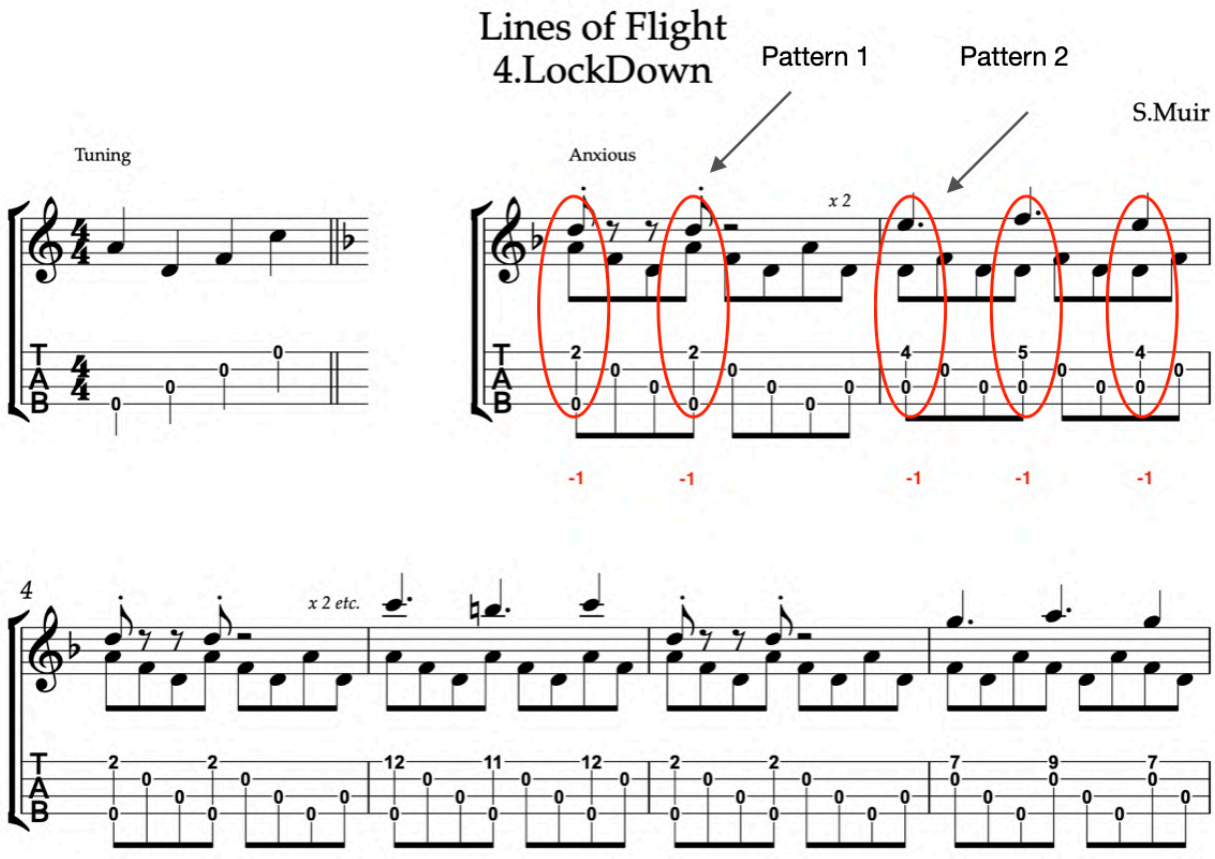
[106] I used the 4 note tremolo in the recording for the album *New Works for Classical Ukulele Volume 1*.

suggest the conflicting timelines. The piece was premiered by Giovanni Albini on YouTube on 20th February 2021 using low G tuning and the original three note tremolo.

4.LockDown

The fourth and final piece of *Lines of Flight* reflects the angst and frustration of being caught in a seemingly endless cycle of lockdowns during the COVID-19 pandemic. My aim was to create a piece that broke away from the repeated bass pattern idea of *Patientia* and *The Red Angel* and adhered more strictly to the rules of tintinnabuli by exclusively using first position inferior for the T-voice (-1). This time the ukulele was tuned A D F C using high-G tuning (see figure 2.2.26) with the open lower strings operating as the T-voice using the D minor triad.

Figure 2.2.26. The tuning and score layout of LockDown with the 2 rhythmical patterns circled in red



Although the notes of the T-voice sound less predictable, repetition remains a central feature. This can be seen in the constant quaver movement of the T-voice and the two rhythmical patterns of the M-voice which are circled in red in figure 2.2.26. Despite the 4/4 time signature the T-voice quavers are grouped 123,123,12 in a rhythmic pattern used by Jorge Cardoso in his well known *Milonga* for

the guitar. The syncopated rhythm is emphasised in the M-voice, marked as pattern 2 in figure 2.2.26.

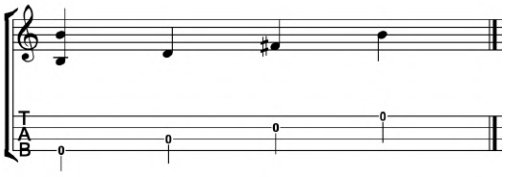
The repeated staccato Ds in the M-voice (see pattern 1) are a musical representation of the restriction of lockdown. Thus, the two patterns are in a constant state of struggle as the music is always striving to break free but cannot. In bars 18 to 34 the Ds are played an octave higher as the mood becomes increasingly hysterical.

The anxious and urgent mood of *LockDown* provides a stark contrast to the other, more reflective and gentle pieces. It is interesting to note, however, that the only piece in *Lines of Flight* which truly represented the tintinnabuli technique was so edgy and perturbed. Perhaps, it also reflected some of my own anxiety about composing in such a locked system. My natural inclination, as evidenced by the other pieces, is to wander. Adhering to the formula of first position inferior felt tyrannical rather than redeeming or comforting. An interesting addition to the end of the YouTube video was the sound of a helicopter flying overhead. Rather than re-record I decided to keep this version as it added to the frenzied and mechanical nature of the music and was oddly reminiscent of *Helikopter-Streichquartett* (1995) by Karlheinz Stockhausen (1928 - 2007).

Für Alina[107]

The *Lines of Flight* sketches were an opportunity to explore how tintinnabuli could be adapted to suit the ukulele. What emerged was how alternate tunings could be used to create open string T-voices in order to enhance the over-ringing potential of the instrument. Applying this idea to *Für Alina* resulted in the open strings being tuned B, D, F♯, B (see figure 2.2.27) in order to maintain Pärt's original key of B minor. The problem of the low B pedal was partly resolved by using the five string tenor as every time the fourth course was struck it sounded the octave (see figure 2.2.28) and added a deeper resonance. A further solution was offered by Milton Mermikides who added live electronics[108] to my audio to further enhance the sonorous and spacious atmosphere.

Figure 2.2.27. The B minor open tuning used in Für Alina with the fourth course sounding the octave Bs



[107] For full score see Folder B.2. For audio see Folder C.5.

[108] Pärt/Muir/Mermikides (2021)

Figure 2.2.28. The opening of my arrangement of Für Alina



A Collaboration

After uploading my arrangement of *Für Alina* to YouTube I discovered that Italian ukulele player Giovanni Albini[109] had also arranged this piece for the ukulele. We then collaborated on a joint presentation and paper that discussed our different approaches to arranging and adapting *Für Alina* for the ukulele. The paper[110] was presented at the conference Vlado Milošević: Tradition As Inspiration, that took place at the Music Academy in Banja Luka on 16-18 April 2021.

Conclusion

Arranging *Für Alina* for the ukulele presented new and distinct challenges which were ultimately solved through a series of compositional sketches. The sketches provided a creative space in which I was able to experiment with alternate tunings and different ways the T-voice may be imagined, or re-imagined, on the ukulele. This in turn led to the T-voice being adapted and modified to maximise

[109] Professor of Music Theory, Solfège and Music Perception and head of research at the Conservatory "Antonio Vivaldi" of Alessandria, Italy

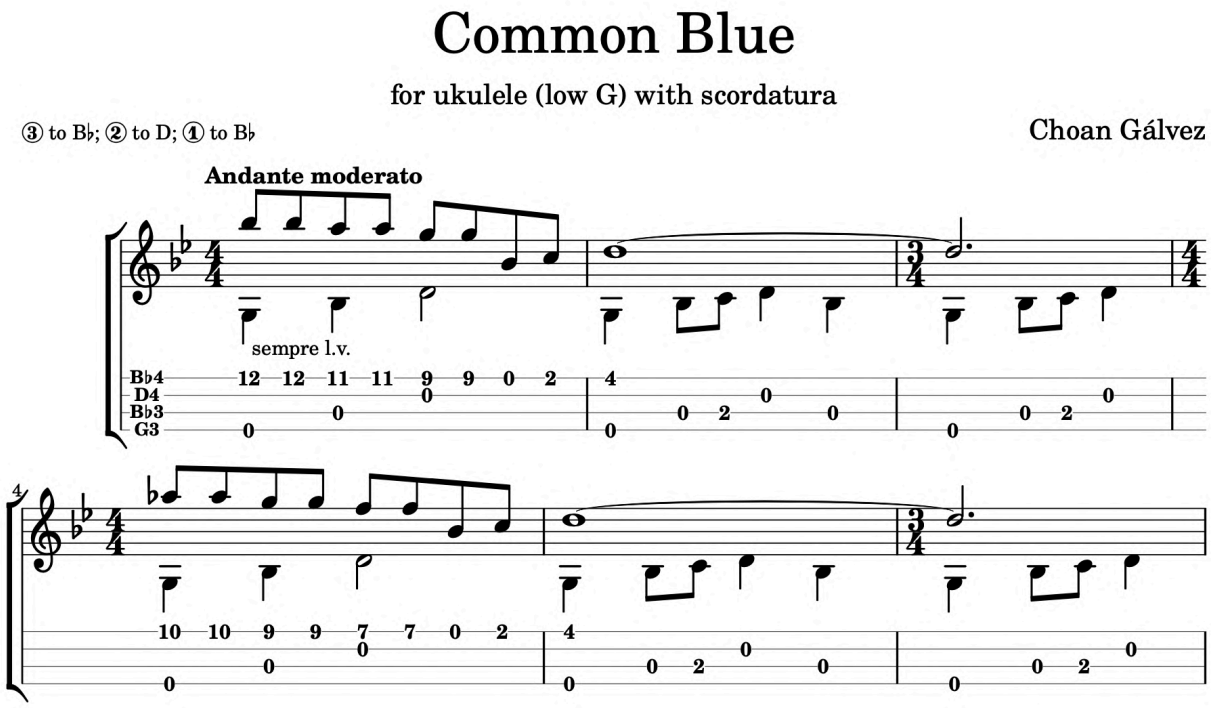
[110] For full text see Folder A.2.

the over-ringing effects of the ukulele. A consequence of this was that the intervalic relationship between the M-voice and the T-voice was often changed with simple intervals becoming compound intervals. As such my sketches were not always true to Pärt's original model of tintinnabuli. My re-imagining of tintinnabuli, however, was guided by the strengths and limitations of the ukulele - the strengths being the ukulele's capacity to accommodate alternate tunings in order to play the T-voice and the weaknesses being its short attack and limited range. Ultimately the use of open string tunings led to the creation of some interesting short pieces which successfully demonstrated the expressive potential of the ukulele.

Over time the *Lines of Flight sk*etches evolved into a set of four original pieces. The various tunings would, however, make the set difficult to perform in a live situation. But, as the pieces were composed during the COVID-19 lockdown, their purpose was to be presented as a set of YouTube videos. To this end the project can be seen as a direct result of, and response to, the restrictions imposed by the COVID-19 pandemic. Unable to travel or perform at live events I focused on creating repertoire uniquely tailored for YouTube. This enabled greater experimentation with alternate tunings and encouraged me to share my work, even though experimental, on my YouTube channel. The videos prompted support and interest from other players. In October 2021 the Spanish composer and ukulele player Choan Gálvez sent me a new piece using the same alternate tuning as *The Red Angel.* This beautiful and haunting piece, called *Common Blue*,[111] (see figure 2.2.29) further developed the alternate tuning idea by using fretted notes on the lower strings to create a simple but effective second voice. It was interesting to observe how, over time, the *Lines of Flight* project developed - beginning with the simple goal of arranging *Für Alina* for the ukulele and resulting in new works, by myself and other composers.

[111] For full score see Folder B.4. For Audio see Folder C.4.

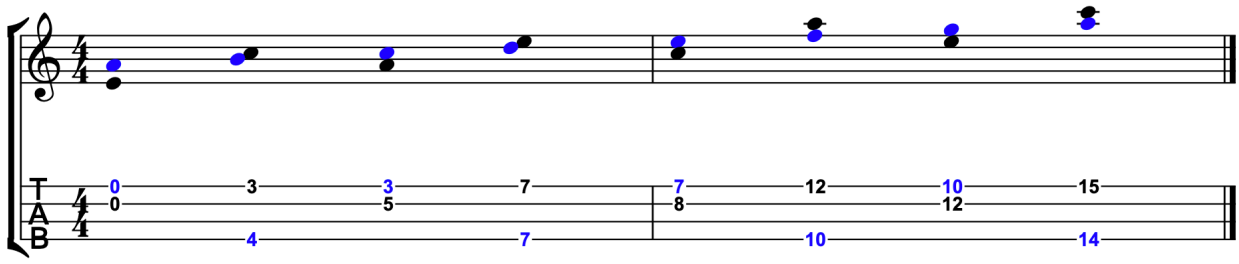
Figure 2.2.29. The opening of Common Blue by Choan Gálvez



2.2.b. Combining Tintinnabuli & Campanella

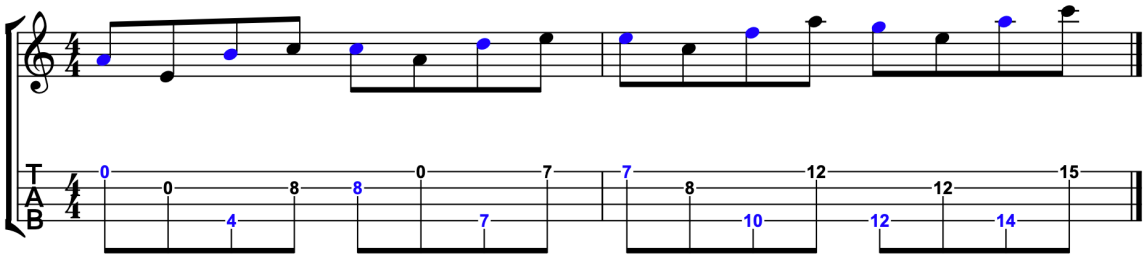
The idea of composing a new piece that combined both campanella and tintinnabuli occurred while reflecting on my individual use of these techniques. The aim was to write a piece that used standard re-entrant tuning and adhered more strictly to Pärt's tintinnabuli formula. In other words the ukulele interface would not be altered as in the other *Lines of Flight* pieces and the intervals would not be changed to facilitate the use of open strings. I adopted the alternating first position (-1/+1) formula offered by Leopold Brauneiss in his essay on the basic elements of tintinnabuli style. (Brauneiss/ Shenton, 2012, 56) This formula is demonstrated in figure 2.2.b.1 with the A natural minor scale (representing the M-voice) shown in blue and the T-voice (triad notes) in black. Rather then use separate staves the voices were combined on one stave - as Pärt had done in his piano piece *Variations for the Healing of Arinuschka* (1977).

Figure 2.2.b.1. The A minor scale as the M-voice (blue) & the T-voice (black) as -1/+1



The M-voice and T-voice were either played simultaneously, as in figure 2.2.b.1, or separately as in figure 2.2.b.2. Separating the voices created additional over-ringing effects as they could be played on alternating strings, or campanella style.

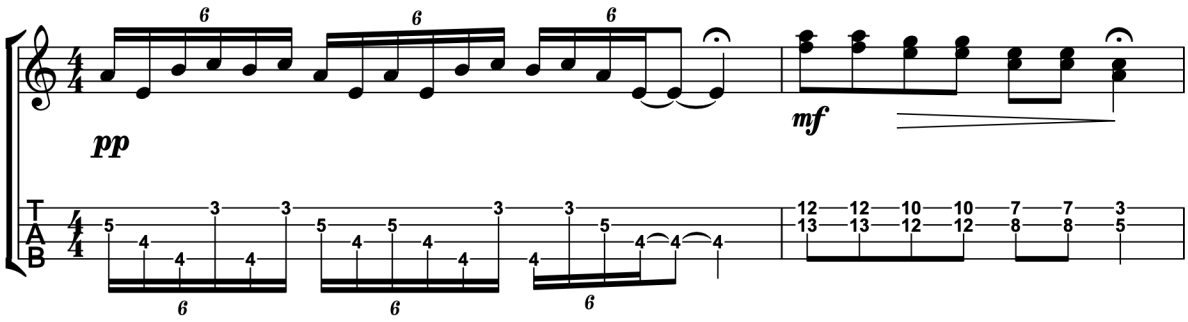
Figure 2.2.b.2. The A minor scale with the M-voice (blue) & the T-voice (black) separated. By placing the notes on alternate strings the campanella effect is also created



This new piece was called *Moths in the Moonlight*[112] as it was inspired by the fluttering wings of moths on a warm summer evening. The piece constituted a new approach to composition for me. Rather than composing with the ukulele in hand, as per usual, I used the above models to first generate the musical ideas in Sibelius. This was a stricter, more formulaic approach to composition than I was accustomed to, but it made me think more about the structure, shape and how devices such as repetition and reflection could be used.

As shown in figure 2.2.b.3 the opening section incorporates two different ideas. The first idea is based on figure 2.2.b.2 and comprises one bar of fast semiquaver sextuplets played as a campanella. Although this figure basically alternates between the M-voice and the T-voice using the -1/+1 formula, repetition is also a feature. For example, the notes B and C are stated and then repeated, as are the next notes of A and E. The pattern for the entire bar is thus -1, +1, Repeat +1, -1, Repeat -1, +1, Repeat +1, -1. The second idea is a simple figure of falling thirds based on figure 3.2.b.1. While the first idea represents the fluttering wings of the moths, the second represents the cool tranquility of the moonlight.

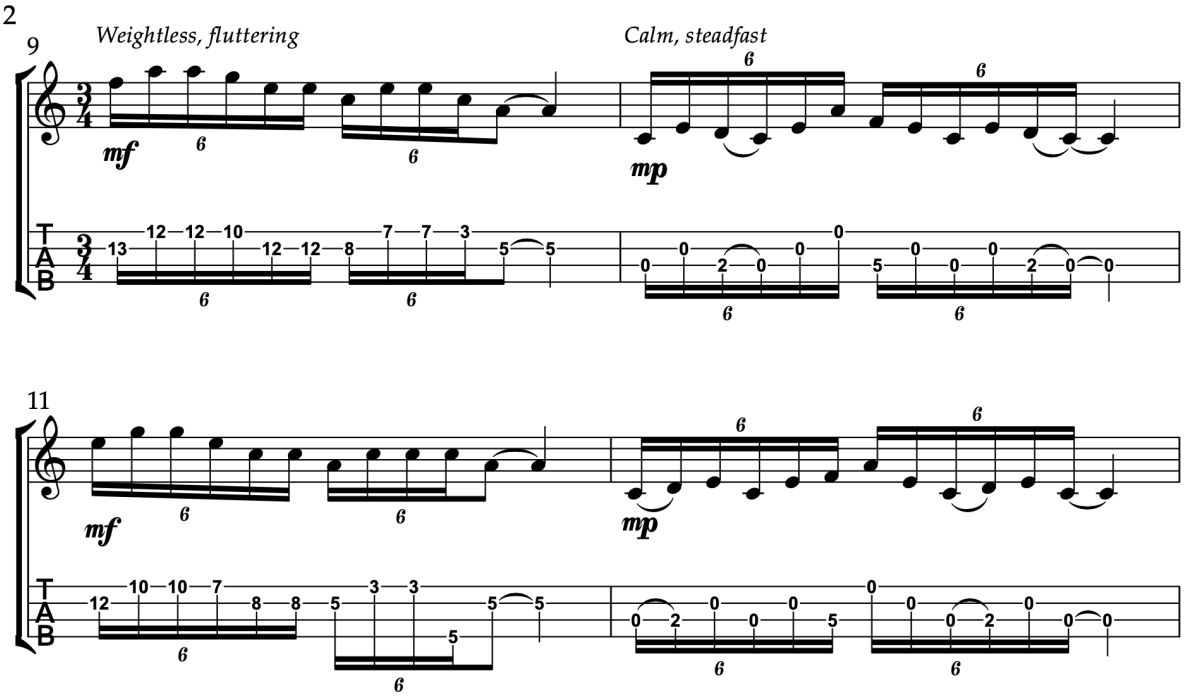
Figure 2.2.b.3. Bars 1 & 2 of Moths in the Moonlight showing the two different ideas



The second section, beginning at bar 9 (see figure 2.2.b.4), expands on the above ideas and uses both sextuplets and repetition. In bars 9 and 11, for example, the T-voice is doubled to create a fluttering two note tremolo effect. Bars 10 and 12 answer with a calmer, more restrained voice once again indicative of the cool tranquility of the moonlight. Bar 12 mirrors bar 10 with the notes written in retrograde to suggest the reflection of the moonlight.

[112] For audio see Folder C.5. For full score see Folder B.2.

Figure 2.2.b.4. Bars 9 to 12 of Moths in the Moonlight



In section three the M-voice and T-voice are played alternately and this time adhere to the +1/-1 pattern. The M-voice is simply a descending A natural minor scale which, when answered by the T-voice, forms a meandering, but inevitably descending line (see figure 2.2.b.5). Harmonics were used to extend the range in bar 17. In bar 20, however, the notes following the open C (0,3) had to be raised an octave as the open C is the lowest note in standard re-entrant tuning.

Figure 2.2.b.5. Bars 17 to 20 of Moths in the Moonlight



From bar 21 to 30 (section 4) the notes alternate between two note intervals and single campanellas. Both ideas maintain the tintinnabuli pattern of -1/+1 (see figure 2.2.b.6). The faster tempo and upbeat rhythm (crotchet followed by two quavers) created a more joyful, dance-like feel as the moths and the moonlight are engaged in a kind of rustic dance.

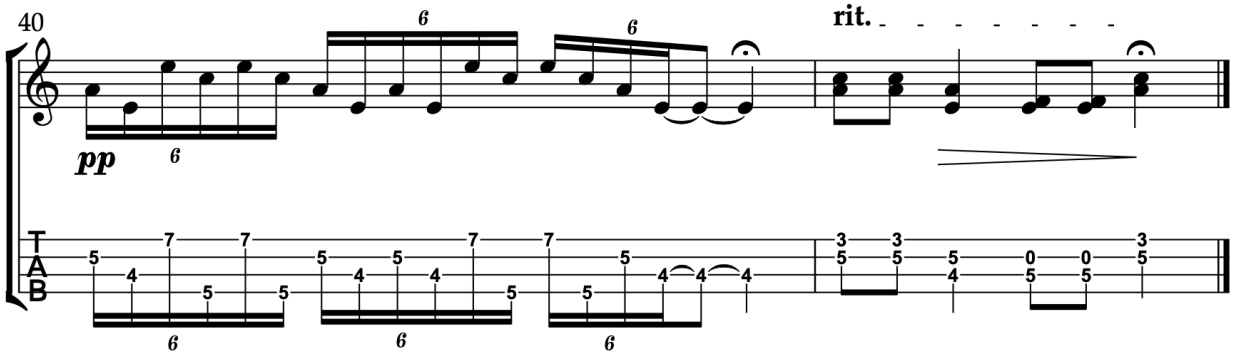
Figure 2.2.b.6. Sections 4 & 5 of Moths in the Moonlight



From bars 30 to 34 (section 5) the M-voice forms another descending scale (see figure 2.2.b.6). This time the scale is in the Phrygian mode starting on E. Once again the tintinnabuli formula means the notes are weaving up and down but the overall movement of the scale is descending. Once the E at 0,2 (bar 33) is reached the final harmonic lifts us back up to A in preparation for the final section.

The last section is a repeat of the first section with the exception of the final bar. On beat three of bar 41 (see figure 2.2.b.7) minor seconds (E-F) replace the major thirds (C-E) of section 1. The resulting semitone dissonance is then answered with minor thirds (A-C). The final interval (A-C) implies the tonic key of A minor but if the previous F at 5,3 is held, there is a subtle suggestion of F major. Whether or not the F is held is left to the discretion of the performer.

Figure 2.2.b.7. The final bars of Moths in the Moonlight



Conclusion

To the best of my knowledge I am the first composer to have united tintinnabuli and campanella techniques. This piece, therefore, represents the development of a compositional style that is truly unique to the ukulele and is the culmination of my research into both techniques. On a personal level it was also a step towards a more controlled and thoughtful process of composition in which theoretical and practical knowledge were united.

2.3. Arpeggios

Introduction

Although arpeggios are one of the staple techniques of guitar players they are, for many ukulele players, a new approach. This became increasingly apparent when I first started publishing ukulele arrangements in 2013. At this time there was only a limited amount of published material available for classical and fingerstyle ukulele. Furthermore, the main classical ukulele exponents at this time, Tony Mizen and Rob Mackillop, both came from classical guitar backgrounds. What this indicated was that classical ukulele players were migrating from the guitar rather than developing within the ukulele scene. I was yet another classical guitarist turning to the ukulele and, as such, came to the ukulele with a developed finger picking technique.

My first publication, *The Little Book of Carulli* (2013), contained six arrangements, including the example of the *Allegretto* shown in figure 2.3.1. Realising that ukulele players without a guitar background were not used to using the *pima* system to differentiate the picking hand from the fretting hand, I included an explanation of *pima* and created a picking hand fingering map for each piece. Figure 2.3.1 shows the opening bars of the *Allegretto* and figure 2.3.2 shows the corresponding picking map. The map was simply the picking pattern of the piece realised as open strings. My intention was that it would encourage ukulele players to focus on the picking technique and was an approach I often used with guitar students.

Figure 2.3.1. The opening of an Allegretto by Carulli arranged for ukulele from my 1st publication

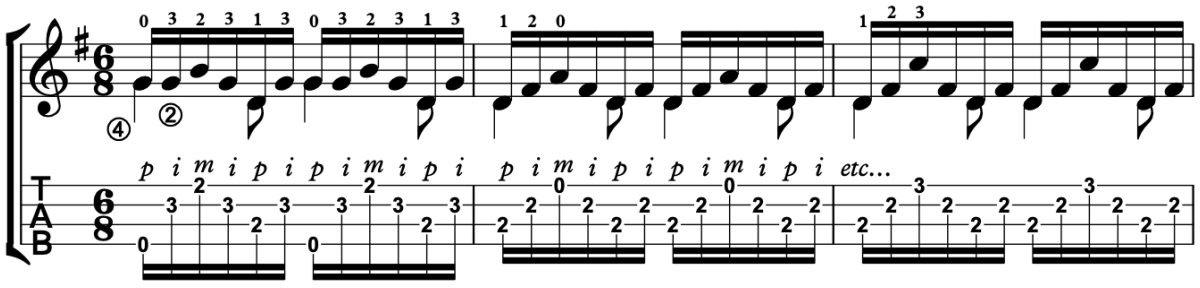
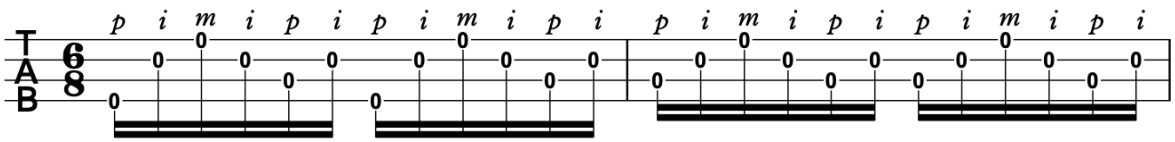


Figure 2.3.2. The picking hand fingering map of the same Allegretto by Carulli



The Little Book of Carulli was self-published as a PDF download (available on Gumroad and PayHip) and promoted by creating YouTube videos of all the pieces. I was surprised by the interest it created, particularly after it was featured on the popular website Ukulele Hunt.[113] Following the success of the Carulli booklet I went on to create a series of '*Little Books…*' comprising arrangements by Sor, Giuliani, Carcassi, Carulli Volume 2 and, after multiple requests from students and social media followers, a booklet of arpeggio exercises.

Soon after starting my PhD studies in 2017 the booklet of arpeggio exercises was revised and published by Les Productions d'Oz as *100 Arpeggio Exercises for Ukulele* (2019).[114] The model was Giuliani's well known *120 Right Hand Studies for Guitar Opus 1a*. In the 2019 version I expanded my original booklet to include exercises in 3/4 and 4/4 time, and introduced simple chord shapes for the fretting hand. As shown in figure 2.3.3 each arpeggio pattern is presented as a pair; firstly as an open string exercise and then with simple chords. The patterns become increasingly complex as the book progresses.

Figure 2.3.3. Exercise 1 & 1a from 100 Arpeggio Exercises for Ukulele



In the same year Productions d'Oz also published my *The Classical Ukulele Method*[115] which focuses on developing the techniques needed to play arpeggios and campanella style. Figure 2.3.4 shows a simple arpeggio exercise from the method based on the chords of G, C and D. Video tutorials on the method are featured on my Patreon site. The videos were a way of providing

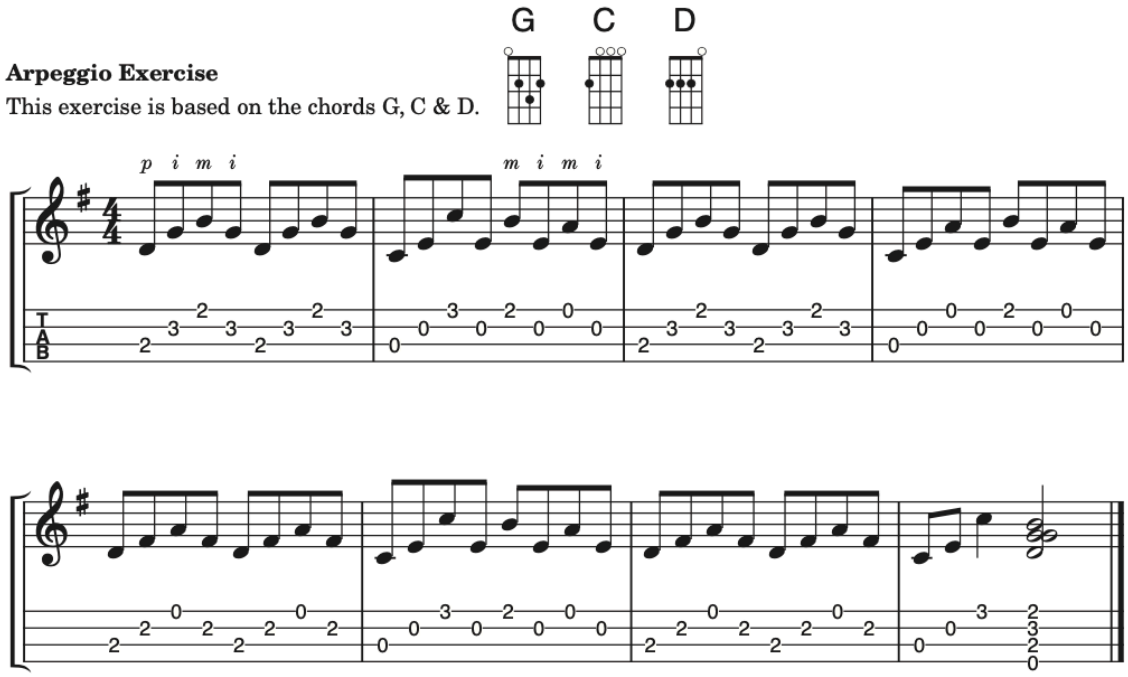
[113] http://ukulelehunt.com/

[114] For full score see Folder B.1.

[115] For full score see Folder B.1.

practical support as they enabled me to elaborate on, and demonstrate, the importance of tone control, rhythm and dynamics. *The Classical Ukulele Method* was the first ukulele method to specifically focus on classical and campanella techniques using the *pima* system.

Figure 2.3.4. An arpeggio exercise from The Classical Ukulele Method



A List of Classical and Fingerstyle Publications

In 2017 I began to index all the classical and fingerstyle publications[116] I could find as the list would enable me to better understand the development of classical and fingerstyle approaches to the ukulele since *The Classical Ukulele* by John King in 2004.[117] The information exposed areas which needed further development, in particular the need for more didactic and pedagogical material aimed at developing techniques such as how to play arpeggios. Since 2017 there has been a significant increase in learning material with books such as Daniel Ward's *Arpeggio Meditations* (Self-published, 2017) and Colin Tribe's *Discovering Fingerstyle Ukulele* (Schott, 2018). My own contribution includes the aforementioned method and book of arpeggio exercises, as well as *12 Progressive Studies* (Les Productions d'Oz, 2018) and *21 Studies for Ukulele* (Les Productions

[116] The list is comprehensive (and regularly updated on my website) but does not pretend to be definitive.

[117] As previously mentioned King's book appears to be the first publication to use the term classical ukulele in 2004.

d'Oz, 2018). Other notable publications include Loretta Notareschi's *Five Etudes* and four books of fingerstyle etudes by Elisabeth Pfeiffer.

12 Progressive Studies

12 Progressive Studies[118] (2018) is a collection of short pieces written for students. The studies explore a number of techniques, including how to play arpeggios. *Flow River Flow* focuses on playing an arpeggio pattern that moves from strings 4-3-2 to 3-2-1 while maintaining the *p-i-m* pattern (see figure 2.3.5). A popular myth amongst many ukulele players is that when finger picking each finger is assigned to, and remains fixed on, one string. In other words the thumb always plays string 4, the index finger string 3, the middle finger string 2 and the ring finger string 1. While this pattern can be very useful, it can also be hugely restrictive if a player does not learn how to reposition the picking fingers. The aim of *Flow River Flow*, therefore, was to help players develop confidence in moving a picking pattern, in this case *p-i-m*, across the strings.

Figure 2.3.5. The moving p-i-m arpeggio pattern in Flow River Flow



The inclusion of didactic material in this portfolio is testimony to the fact that classical ukulele is only just emerging as a recognised approach to the ukulele and, consequently, the extant material remains limited. Furthermore, my engagement with the ukulele community has provided first hand experience of the technical and musical needs of classical ukulele players and this has influenced my contribution. Thus, the new works not only present the ukulele as a virtuosic classical instrument but seek to encourage other ukulele players to learn the classical ukulele style. Only through the development of a rich and varied repertoire will classical ukulele become self-sufficient rather than reliant on players and teachers with guitar backgrounds. The didactic material in the portfolio is intended to offer a foundation for others to build upon.

[118] For full Score see Folder B.1.

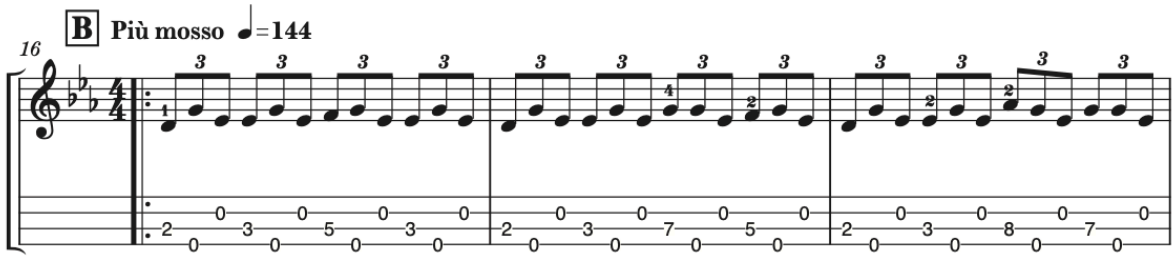
Advanced Arpeggio Techniques in Original Works

This section looks at how I used advanced arpeggio techniques in my own compositions. Chapter 2.1 has already discussed how campanellas and arpeggios have often been combined to create melodic and harmonic interest so this section will focus on other approaches to using arpeggios.

The Falling Rain

In the B section of *The Falling Rain* the melody, which is played on the third string, is incorporated into an arpeggio pattern. The idea was to explore how a melody could be played on the lowest string while using the second and fourth strings to create an accompaniment. Given that the tuning for this piece is G C E ♭ A ♭ the open string accompaniment was based on the minor 3rd created by the open G (0,4) and E ♭ (0,2). The combined effect is a repeated triplet arpeggio pattern (see figure 2.3.6). The melodic note is the first note of each pattern, thus resulting in a movement across the strings in the order of 3-4-2. This constituted the slightly odd arpeggio pattern of ***i***-*p*-*m*. Odd because the index finger plays the melody while *p* - the naturally strongest digit - is assigned to the weaker beat. In order to bring out the melody the index finger needs to be given more weight while the thumb and middle fingers play in a lighter, more relaxed way.

Figure 2.3.6. The triplet arpeggio pattern in section B of The Falling Rain



The melodic notes could be made stronger through the use of rest stroke, but this technique would result in the open G (0,4) being damped and, therefore, spoil the over-ringing effect of the arpeggio. In this respect over-ringing can be seen as a central feature of both campanella and arpeggio techniques. Despite the ***i***-*p*-*m* pattern being repeated, and thus bringing uniformity to the picking hand, the balance becomes increasingly difficult to maintain as the section progresses and tension increases in the hand. If the open notes are over-played the melody is difficult to distinguish.

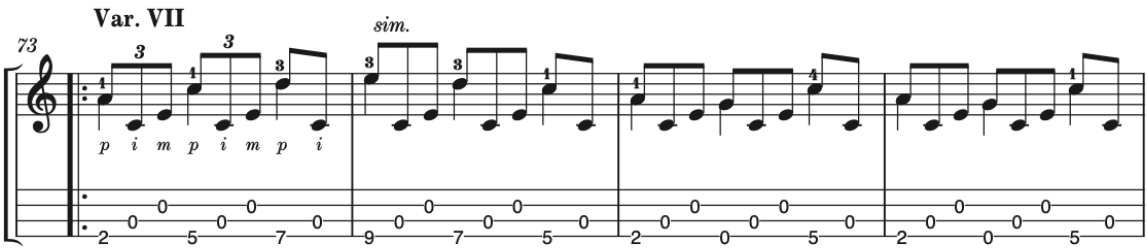
An important point to make here is that the linear tuning of the guitar, together with the availability of rest stroke and free stroke techniques, enables a balance that is idiomatic to both the

instrument and the player. On the other hand the re-entrant tuning of the ukulele disrupts the order of low to high strings and means the picking fingers are either constantly changing roles, as occurs in campanella, or engaged in patterns which conflict with the natural strengths of the hand. For these reasons the re-entrant tuning often sounds and feels counter-intuitive. Such concepts defy the common notion that the ukulele is an easy instrument as even a relatively simple looking arpeggio pattern, such as in figure 2.3.6, demands careful control of the strings in order to achieve the correct balance and tone. In classical and fingerstyle playing the technical and musical challenges of the ukulele are subtle yet specific.

Variations on The Dowie Dens of Yarrow

The *p-i-m-p-i-m-p-i* arpeggio pattern in *Variation VII* of the *Variations on The Dowie Dens of Yarrow* feels more natural for the picking hand. With the melody placed on the fourth string it is played by the stronger thumb (see figure 2.3.7) while the accompanying notes are assigned to the fingers. The entire pattern moves across the strings in the pattern **4**-3-2-**4**-3-2-**4**-3 per bar with the melodic notes highlighted in bold. The arpeggio pattern for each bar is ***p***-*i*-*m*-***p***-*i*-*m*-***p***-*i* which, despite the break at the end of the pattern to ***p***-*i*, feels quite natural for the hand.

Figure 2.3.7. The opening of Variation VII in Variations on The Dowie Dens of Yarrow



Blinter[119]

Blinter was composed for, and premiered by, Giovanni Albini as part of the HighScore Festival 2021. Giovanni included it on his CD *A Contemporary Ukulele* (Da Vinci Classics (2022). The entire piece is built on repeated arpeggio patterns and is a good example of how the boundary between arpeggios and campanellas is often blurred. The first 16 bars use the pattern ***p***-*m*-*i*-***p***-*m*-*i*

[119] Muir (2021). For full score see Folder B.2.

moving across the strings in the order of **4**-1-2-**3**-1-2 (see figure 2.3.8). The 6/8 time signature means the thumb plays on beats one and four to create a pulse of two. Although this pattern is guitar-like in nature the re-entrant tuning of the ukulele means the first note of each pattern, which is placed on the fourth string, is ambiguous in terms of whether it is melodic or harmonic. On the linear tuned guitar it would be regarded as a bass note as it would sound an octave lower. On the ukulele, although still played by the thumb, the re-entrant tuning means this note sounds an octave higher. From bars 1 to 14 the notes on the fourth string are, in fact, the second highest notes in each bar while in bars 15 and 16 these notes are the highest.

Figure 2.3.8. The first 16 bars of Blinter

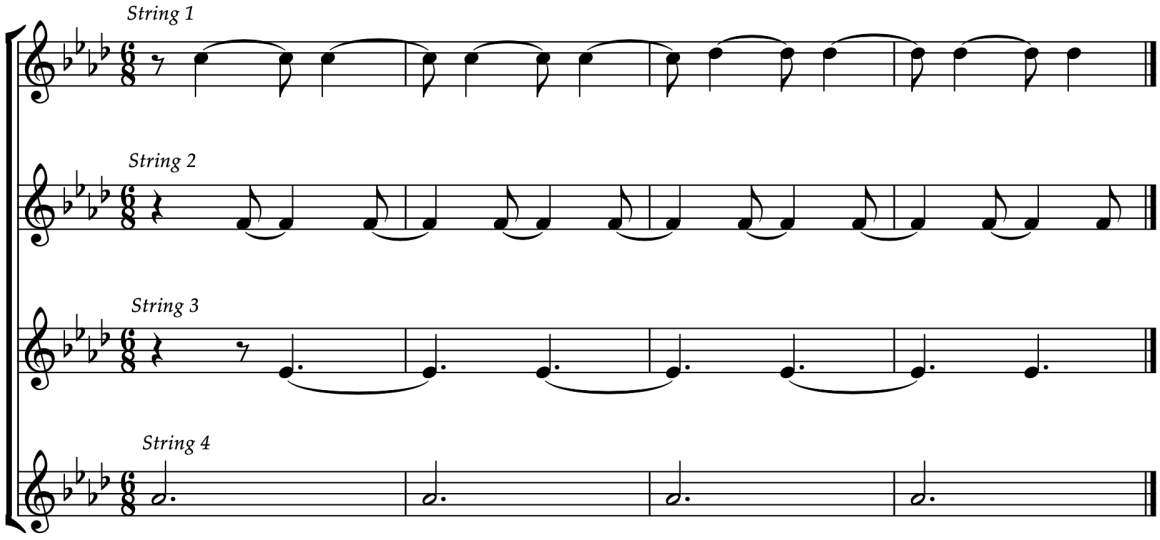


The above example illustrates how in re-entrant tuning the roles of melody and harmony are often difficult to determine both musically and technically. As a guitarist I am used to *feeling* the bass

under my thumb and, on the guitar, the sound corresponds.[120] On the ukulele my thumb might still *feel* like it is playing the bass but the fourth string rarely *sounds* like the bass. On the other hand, the third string both *sounds* and *feels* like a bass note as it is sounds a 5th lower than the fourth string. In this topsy-turvy world my mind is constantly reassessing the role of the thumb and making the appropriate technical adjustments: more weight for the melody, less for the accompaniment.

In the case of *Blinter* the main purpose of the arpeggios is to create moving blocks of colour. The use of chord shapes means very few open notes are used. Whilst holding down chord shapes that utilise all four strings is technically challenging, the objective is still to create a harp-like effect. The notes are intended to be held for as long as possible in order to maximise the over-ringing effects. Figure 2.3.9 shows how *Blinter* would look if each string was notated on a separate stave. As can be seen the over-ringing objective is the same as campanella, but while campanella uses a melodic line, the arpeggio pattern of *Blinter* uses chords.

Figure 2.3.9. How the opening of Blinter would look if each string were notated on a separate stave



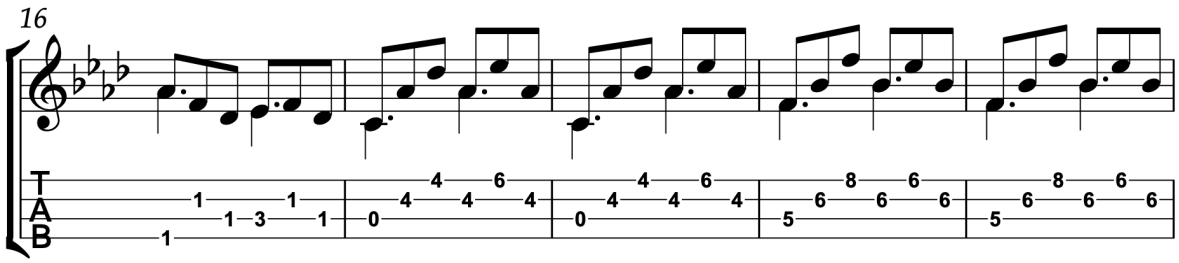
Thus, despite having some campanella characteristics, the repeated picking patterns and harmonic nature of *Blinter* make it essentially an arpeggio piece. It seems, however, that the re-entrant fourth string inevitably suggests campanella effects between the fourth and first strings - owing to them being tuned just a tone apart. Is, for example, the A ♭ in figure 2.3.8 intended to be melody or bass?

[120] In pieces, such as *Preludes I* & *IV* by Villa-Lobos, the thumb is assigned the melody and the fingers play the accompaniment. Even so, their roles remain clearly defined.

The question is left hanging and the resulting ambiguity underlines the often playful and capricious nature of the ukulele.

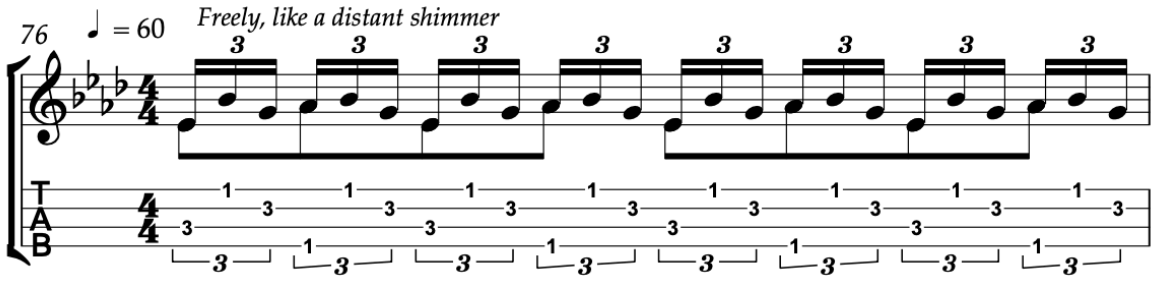
When the troublesome fourth string is not used, such as in figure 2.3.10, order is re-established as the top three strings adhere to the linear hierarchy. The thumb is clearly playing the bass note on the third string and, as this is undeniably the lowest note in the pattern, there is no uncertainty about its function. The melody, meanwhile, sits clearly on the top string.

Figure 2.3.10. The arpeggio pattern from bar 17 with the voices clearly defined



From bar 76 to the end of the piece the arpeggio pattern is the same as the beginning but the triplets are now played as semiquavers, thus doubling the tempo (see figure 2.3.11). The idea of doubling the pattern was suggested by Giovanni Albini. It created a shimmering effect evocative of distant stars while adding a virtuosic element to the arpeggio pattern.

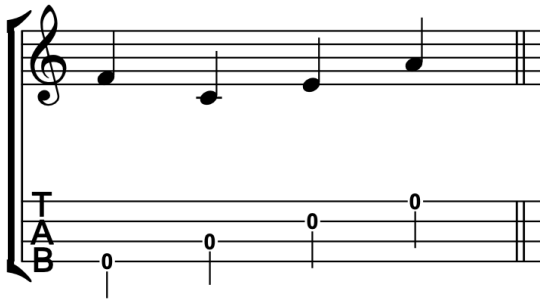
Figure 2.3.11. The same arpeggio pattern as the opening but with the triplets played as semiquavers



Whistlepipe Gully[121]

Composed in 2020 this piece uses the simple idea of an open string arpeggio accompaniment on strings 4, 3 and 2 while the melody is played on string 1. The fourth string was tuned down a tone to F (see figure 2.3.12) to create the open tuning of F major 7.

Figure 2.3.12. The tuning of Whistlepipe Gully



Whistlepipe Gully is a minimalist piece which features constant repetition in both the melody and the harmony. The arpeggio pattern is similar to *Blinter* with the thumb moving between strings 4 and 3 and the middle and index fingers strings 1 and 2. However, as figure 2.3.13 shows, the arpeggio pattern is disrupted by subtle changes in the rhythm. For example, the crotchet F on beat four of bar 1 is played by the thumb at 0,4. The following C (0,3) is also played by the thumb in order to maintain the idea of an alternating bass pattern - an idea which recurs throughout the piece. The open strings create a subtle interplay of consonance and dissonance as the open F (0,4) clashes with the open E (0,2) while the open C (0,3) sounds consonant with both notes. Despite the close proximity of the F and the E the overall effect is harmonic, rather than melodic, as the pattern also includes the open C in order to create an arpeggiated F major 7 chord.

Meanwhile, the idea of falling and rising perfect 4ths (F-C-F) gives a sense that the music is constantly searching for resolution. The piece was inspired by a bush walk I did in 2019 while on tour in Western Australia in an area called *Whistlepipe Gully*. The track meandered through the bush, sometimes ending for no apparent reason and at other times dividing into two or more possible tracks. There were no sign posts and we sometimes ended up in the same place via a different track, or maybe it was a different place that just looked the same. The purpose of the path seemed to be to enjoy the scenery from various points rather than heading to a specific destination. When composing this piece my intention was to capture this idea of meandering and enjoying the moment.

[121] Muir (2020). For full score see Folder B.2.

Figure 2.3.13. The opening of Whistlepipe Gully



Conclusion

Arpeggio techniques can be applied to the ukulele in novel and distinctive ways. While there are many parallels with the guitar, notably in the picking technique, the re-entrant tuning of the ukulele presents opportunities which are unique to the ukulele. The role of the fourth string is often ambiguous as notes that *feel* like the bass may *sound* like the melody. This misalignment of voicing is, however, a natural feature of the re-entrant tuning and pieces such as *Blinter* and *Whistlepipe Gully* seek to allow the ukulele to speak with its own distinctive voice.

2.4. Chord Melody Approach

Introduction

The term chord-melody is used by ukulele players to indicate an instrumental, usually solo, style of playing that combines both the melody and the chords. Chord-melody is not a new approach and can be found in many early ukulele methods by players such as Ernest Ka'ai (1906/10) and Vadah Olcott-Bickford (1920). More recently, however, Jake Shimabukuro (b.1976) has taken the chord-melody approach to a whole new level with instrumental covers of well-known pop and rock songs such as *Bohemian Rhapsody* and *While My Guitar Gently Weeps*. Another well-known advocate of the chord-melody style is Canadian ukulele artist James Hill (b.1980) who, like Shimabukuro, has produced numerous instrumental arrangements of popular songs. The application of chord-melody in a classical sense, however, requires a different technical approach as the notes are, generally speaking, plucked with the fingers rather than strummed or played with the thumb. My own work has sought to apply and adapt ideas from classical guitar technique to the ukulele. Before discussing these works, however, a brief overview of two different approaches to chord-melody will provide some historical and popular context for this technique.

The Bickford Method

One of the earliest examples of a classical approach to the ukulele is found in *The Bickford Method* (1920). Figure 2.4.1 shows an example from *Cupid's Pranks*, a mazurka, on page 21 of the method. Unlike most other methods from this time the Bickfords detail the plucking hand fingering. A small cross indicates the thumb, one dot the index finger and two dots the middle finger.

Figure 2.4.1. The opening of Cupid's Pranks from The Bickford Method (1920)



The fingering indicators clearly show that the three note chords in figure 2.4.1 are intended to be plucked and not strummed. But towards the end of the piece the four note chords, shown in the second bar of figure 2.4.2, have down and up symbols indicating these chords should be strummed. By clearly marking their scores the Bickfords can be seen to have utilised picking and strumming

techniques. Unfortunately, not all authors were as specific as the Bickfords. Ernest Ka'ai, for example, did not indicate whether chords in his methods were intended to be strummed or plucked and appears to have left this to the discretion of the player.

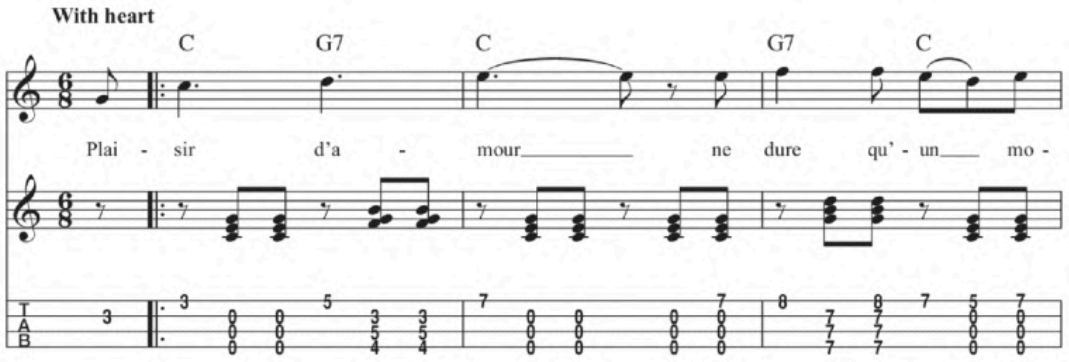
Figure 2.4.2. The strummed chords in Cupid's Pranks from The Bickford Method (1920)



A Modern Approach to Chord Melody

James Hill[122] offers a different approach to playing chord-melody in his *Duets for One* (Flea Market Music, 2017). In this popular book he explores the potential of playing counterpoint on the ukulele. As can be seen in figure 2.4.3 the melody and the accompaniment are written on separate staves, to clarify their roles, while the tablature combines both parts. Quite remarkably Hill uses only his thumb to play both the melody and the accompaniment.

Figure 2.4.3. The opening of Plaisir D'Amour arranged by James Hill (2017)



A feature of Hill's approach is that the accompaniment is given its own harmonic and rhythmic patterns and operates as a separate, clearly distinguished part. In comparison *The Bickford Method* offered a more blended approach with chords used to fill-out the melody while adding harmonic

[122] Hill is a dedicated teacher well known for his teaching resources (*Ukulele in the Classroom* and *The Ukulele Way*) and pedagogy (the JHUI Teacher Certification Program).

interest. As previously mentioned they also incorporated both strumming and plucking techniques. To summarise, Hill's approach is more uniform and predictable but allows each part its own distinctive voice. Meanwhile, the Bickfords offered greater variety by incorporating a range of techniques; an approach which is stylistically more typical of the guitar

My Own Use of the Chord-melody Approach

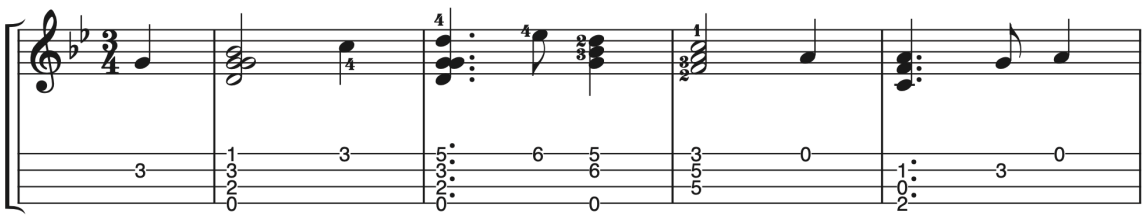
Due to my classical guitar background the *pima* picking technique is the most natural way for me to play the ukulele and this has undoubtedly influenced the way I arrange and compose. As such my approach is more like that used by the Bickfords than the approach used by James Hill. As will be demonstrated, however, my main objective has always been to find new ways of exploring and understanding the unique features of the ukulele and, consequently, I have adapted my technique accordingly.

Chord-melody in My Arrangements

Greensleeves[123]

The first part of my arrangement of the well-known Renaissance tune *Greensleeves* uses a simple chord-melody approach. The first beat of each bar is harmonised by a chord of usually three or four notes. As shown in bar 4 of figure 2.4.4, when the melodic note F moves to the second string it is no longer the highest note in the chord. Once again the re-entrant tuning can be seen to defy convention as the A on the fourth string (2,4) is now the highest note. As the tablature shows, however, its position on the fingerboard clarifies its role as an accompanying note. In linear tuning this note would sound an octave lower and its role would be obvious.

Figure 2.4.4. The opening of my arrangement of Greensleeves

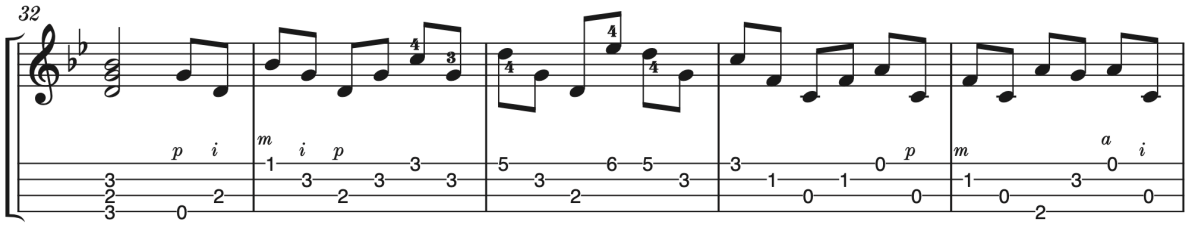


[123] For full score see Folder B.1. Or *12 Traditional Tunes for Fingerstyle Ukulele,* Les Productions d'Oz, 2019

The idea was to introduce the melody in a simple and uncluttered way. No picking fingering was included as it seemed each individual player should decide how to play the chords. My preference is to either roll each chord using the picking hand fingers, or play the chord straight (unbroken) also with the fingers. In other words all the chords were played with either *p-i-m-a* for the four note chords, or *p-i-m* for the three note chords.

When the tune is then repeated at bar 32 the chords are broken to form an arpeggio pattern that weaves around the melody (see figure 2.4.5). Thus, the arrangement offered contrast between the elegant simplicity of the rolled chords and the constant movement of the arpeggio.

Figure 2.4.5. The arpeggiated repeat of Greensleeves

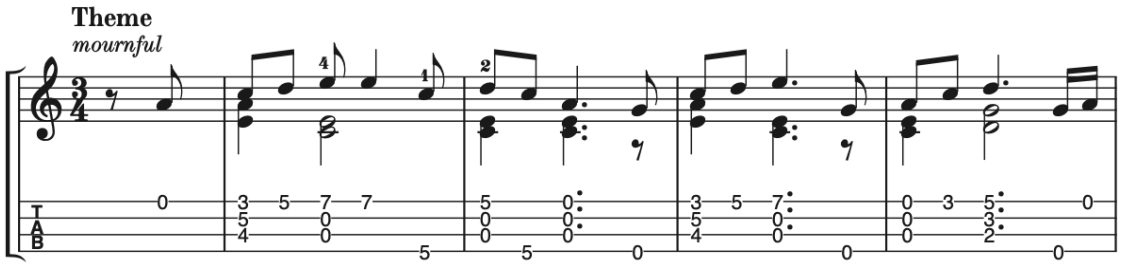


Chord-melody in My Compositions

Variations on The Dowie Dens of Yarrow

One of the advantages of using the *pima* picking technique is that it is very easy to move between different compositional (and arranging) techniques. This can be seen in the opening of *Variations on The Dowie Dens of Yarrow* (see figure 2.4.6) where the theme is arranged in a simple chord-melody style with the melodic notes placed mostly at the top of the chord. With the chords played on the top three strings the voicing hierarchy is maintained as the awkward re-entrant string is not involved. Nevertheless, single melodic notes are occasionally placed on the fourth string (see figure 2.4.6) to

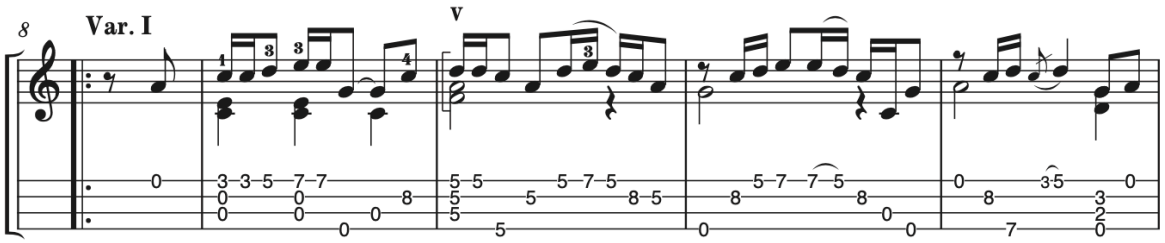
Figure 2.4.6. The opening of the theme of Variations on The Dowie Dens of Yarrow



create subtle campanella effects. Thus, over-ringing is created by both the chords and the campanellas.

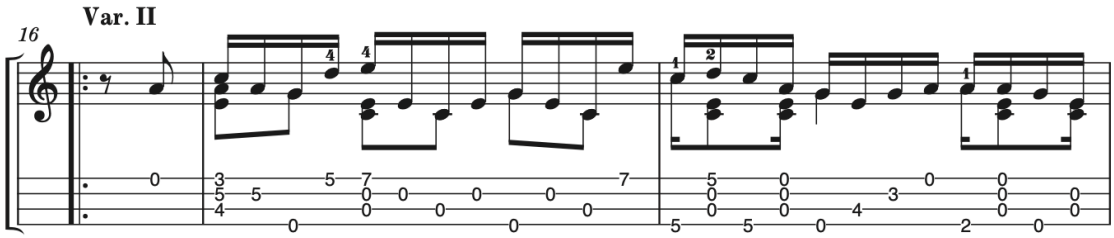
In *Variation I* (see figure 2.4.7) three approaches are combined - chord-melody, arpeggios and campanellas. The tablature clarifies the different positions and roles of the notes. For example, a five note campanella starts on the final quaver in bar 9 with the notes C, D, D, C, A moving 8,2 - 5,1 - 5,1 - 5,4 - 5,2. The campanella idea, made explicit in the tablature, is incorporated into the chord-melody idea as the D (5,1) on beat one of bar 10 is also the tonic of the D minor chord which is played with *p-i-m*.

Figure 2.4.7. The opening of Variation I of Variations on The Dowie Dens of Yarrow



In *Variation II* the chord melody-approach is, once again, merged with other techniques to create a busy, fast moving section. In figure 2.4.8 a simple progression of chords (Am, C, C9, Am, Am) provides the framework for the melody while the single notes form an arpeggio accompaniment. In bar 17 the final note E (7,1) leads into a brief campanella exchange in bar 18. The C at the beginning of bar 18 is played at 5,4 while the next melodic note (D) is played at 5,1. This same melodic note (D) is also part of the C9 chord. The following C maintains its campanella position at 5,4 and the next A at 0,0 is played as part of an A minor chord. In this bar the thumb plays the Cs at 5,4 while the *ima* fingers pluck the chords to create a punchy exchange between fingers and thumb.

Figure 2.4.8. The opening of Variation II of Variations on The Dowie Dens of Yarrow



The following notes (from beat two) form a more fluid arpeggio pattern across the strings (played *p-i-m-a)* before another exchange between the thumb and fingers.

The relentless semiquaver movement of this variation with its constant changes from chords to single notes is made possible by the *pima* system. The finger picking approach also allows chords to be played in a number of ways. For example straight chords give a punchy, rhythmical effect while rolled chords are more harp-like and lyrical. Thus, musical ideas on the ukulele can be seen to be linked to technical approaches that find their roots in the guitar. The most important point, however, is how the use of the *pima* system on the ukulele facilitates more variety and expressive potential in the compositional approach.

Conclusion

Although the chord-melody approach is not new to the ukulele, I have applied it in innovative and extended ways in order to make full use of the idiomatic features of the instrument. As demonstrated in this chapter, the flexibility of the *pima* system, when applied to the ukulele, allows the merging and blending of different techniques such as chord-melody, campanella and arpeggio which in turn has enabled me to explore and develop musical ideas that are both original and idiomatic to the instrument.

2.5. Approaches to Playing Chords

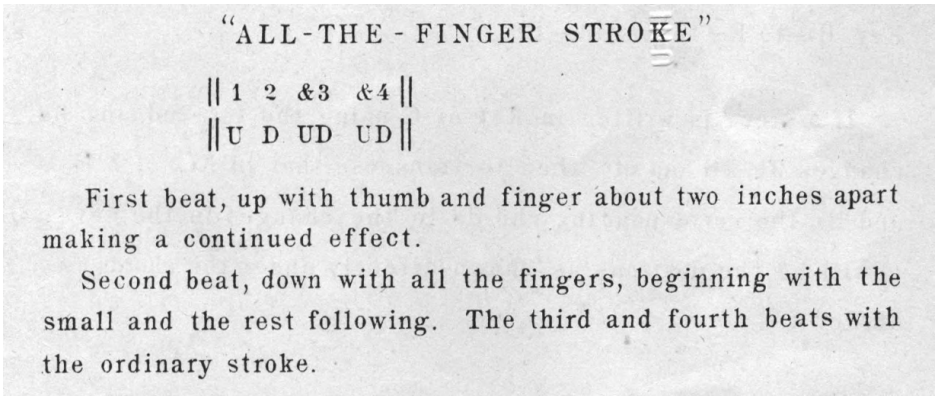
Introduction

The ukulele is most popularly regarded as a strummed instrument but as seen in the previous chapter there are many other ways a chord might be played on the ukulele. To date no research has been done on the different strumming techniques used by ukulele players, past or present. The typical down-up strumming pattern so often used to accompany songs belies the many diverse and often intricate patterns that have been in use since the ukulele first gained prominence in the early twentieth century. While the purpose of this discussion is to explore the different approaches to playing chords used in my own arrangements and compositions, this chapter will begin with a brief background on chordal approaches on the ukulele.

Background

The tradition of strumming chords on the ukulele stretches back to the earliest methods.[124] Ernest Ka'ai dedicated the majority of *The Ukulele: A Hawaiian Guitar & How to Play It* (1906/10) to playing what he calls the 'strokes'. According to Ka'ai the common or ordinary stroke 'is made with the forefinger of the right hand running it rapidly across all the strings with a down and up movement of the wrist…' (Kaai, 1910, 4). Other strokes, such as the 'all-the-finger-stroke' shown in figure 2.5.1, were more intricate.

Figure 2.5.1. Ka'ai's instructions for playing one of the more intricate strokes



"ALL-THE-FINGER STROKE"

‖1 2 &3 &4‖
‖U D UD UD‖

First beat, up with thumb and finger about two inches apart making a continued effect.

Second beat, down with all the fingers, beginning with the small and the rest following. The third and fourth beats with the ordinary stroke.

[124] Holstein's *Chords of the Taro-Patch Guitar* of 1894 is the earliest known method for either the ukulele or the taro-patch (which at this time was the Hawaiian term for the Madeiran rajão). This 16 page booklet comprises chords diagrams and an explanation of how to play them: 'Use the index finger of the right hand and run it rapidly across and back against the strings after the manner of thrumming [strumming]. (Holstein, 1894, 4)

Although Ka'ai's updated and expanded method of 1916[125] also dedicated much time to the strokes, a number of instrumental arrangements and compositions were also included. These pieces are historically important as they represent a move towards the creation of a solo instrumental repertoire similar to that of the classical guitar. Figure 2.5.2 shows the opening of *El Recuerdo* which begins with a scale passage leading to a full C9 chord. In an earlier piece Ka'ai specifies that 'single characters must be picked, group[l]ets tremoloed, and those with the (strummed.' (Ka'ai, 1916, 28) Thus, the open bracket over the chord in bar 2 of figure 2.5.2 indicates it should be strummed.

Figure 2.5.2. The opening of Ka'ai's fingerstyle piece El Recuerdo from his 1916 method



It is not clear, however, how the chord should be strummed. Should it be played as a fast downward strum with the back of the nail of the index finger? Or, should it be played as a fast sweep across the strings with the side of the thumb? Alternatively a slower, downward roll using the thumb would create yet another effect. As any player will know all three approaches yield quite different sounds. Ka'ai may, of course, have left the finer details to the discretion of the performer. My own interpretation would be to play the chord as a slow, arpeggiated roll with the side of the thumb in order to produce a softer, more harp-like affect. The next full chord, in bar 4, does not contain a bracket but as the idea is similar to bar 2 the most obvious conclusion is that Ka'ai intended it to be played in a similar fashion and the missing bracket was either an editing error, or deemed unnecessary.

Other solo pieces, such as the *Banjo Schottische* (see figure 2.5.3), give no indication at all as to how the chords should be played. But, in order to maintain the tempo and feel of this upbeat

[125] *The Ukulele and How It[']s Played* by Ernest Ka'ai, 1916, Hawaiian News Co., Honolulu, Hawaii

and lively piece, a fast downward strum with either the index finger or the thumb would seem to be the most obvious solution. This may seem like a small detail but the fast down-strum with the back of the nail produces a very different attack from that of the slow roll played with the side of the thumb used in *El Recuerdo.*

Figure 2.5.3. The opening of Ka'ai's Banjo Schottische from his 1916 method



The main point here is to demonstrate the diversity of techniques most likely employed by Ka'ai. As the first recognised ukulele virtuoso what is apparent is that his approach incorporated picking techniques comparable to the guitar as well as a variety of 'strokes'. Furthermore, by utilising single notes and chords in instrumental solos he can be seen to have pre-empted what today is called the classical ukulele approach.

Introduction to Using Chords in My Own Work

My approach to playing chords on the ukulele is based on my classical guitar training, but with some refinement and adaptation. The table in figure 2.5.4 provides an overview of my different approaches to playing chords on the ukulele. The list shows fifteen different ways of plucking, or strumming chords that have been used in order to achieve a variety of effects - from percussive to harp-like. The list is offered as an over-view and specific uses are then discussed in the context of how they have been applied in different pieces.

Figure 2.5.4. An overview of my different approaches to playing chords

Ways of Playing a 4 Note Chord* on the Ukulele	**Feel**
Fast down-strum with index finger	Straight, rhythmic, bright, accented, percussive
Fast up-strum (flick) with index finger	As above but less intense, not all strings may be hit
Rasgueado (fan stroke) down-strum with fingers	Broken, explosive, rhythmic, percussive
Fast down-strum with felt pick	Straight, rhythmic, accented, rounded tone, percussive
Fast up-strum with felt pick	As above but less intense, not all strings may be hit
Slow down-strum with felt pick	Broken, harp-like, full but more rounded tone than nail
Slow up-strum with felt pick	As above but may be less intense
Slow arpeggiated or rolled chord with *pima***	Broken, harp-like, elegant
Slow arpeggiated or rolled chord with side of thumb**	Broken, harp-like, elegant but muted
Slow arpeggiated or rolled chord with nail of thumb**	Broken, harp-like, bright
Fast arpeggiated or rolled chord with *pima***	Broken, harp-like, declamatory
Fast arpeggiated or rolled chord with side of thumb**	Broken, muted, brushed, percussive
Fast arpeggiated or rolled chord with nail of thumb**	Broken, bright, brittle, percussive
Straight chord plucked with *pima*	Precise, rhythmic, even
Reverse roll with index finger**	Broken, harp-like
Punch chord with the back of the fingernails	Percussive, slightly abrasive sound
* many of these approaches could be applied to 2 or 3 note chords.	
** arpeggiated or rolled chords may either start on the beat or end on the beat depending on the musical aim.	

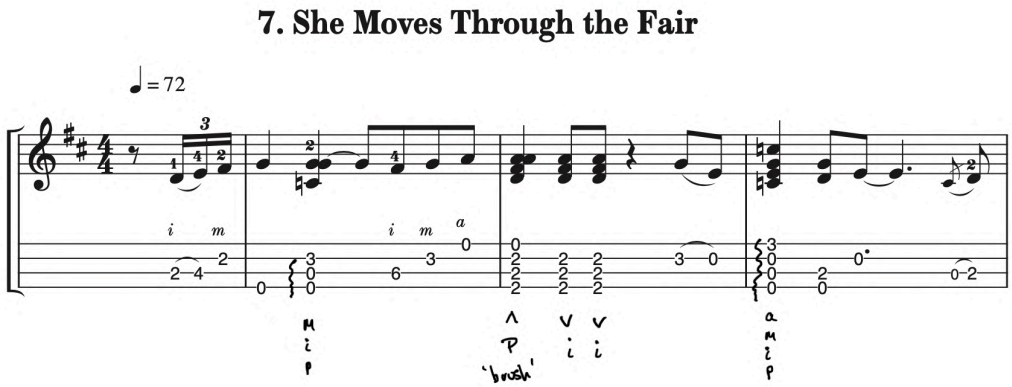
Chord Approaches in My Arrangements

She Moves/Moved Through the Fair[126]

Adapting and arranging this traditional Irish song as an instrumental solo[127] provided another opportunity to explore how chords could be integrated with campanellas and arpeggios. The musical ideas were influenced by the sinister undertones of the lyrics; what begins as a love song soon turns into something more like a Gothic tale.[128] I wanted to express the sinister undertones of the lyrics by interrupting the legato campanella passages with sudden, but subdued, outbursts of chords.

As shown in figure 2.5.5 the chord on beat two of bar 1 sounds gentle and harp-like when rolled with *p-i-m*. The following three melodic notes form a campanella, thus maintaining the harp-like effect but with the uneasy dissonance of a minor 2nd between F♯ and G. This is partly resolved by the D major chords in bar 2. The naturally happy sound of the major chords can, however, be manipulated by the attack. For example, by playing the first D chord with a fast downward brush of the thumb the effect is shadow-like rather than triumphant. The next chords can be played with two quick up-strokes of the index finger and then immediately damped to stop the sound. The attack of the nail gives a harsh, rasping sound and is intended as an agitated response to the more lyrical melodic notes while the sudden silence created by damping the strings adds an air of suspense.

Figure 2.5.5. My approach to playing the chords at the beginning of She Moves Through the Fair



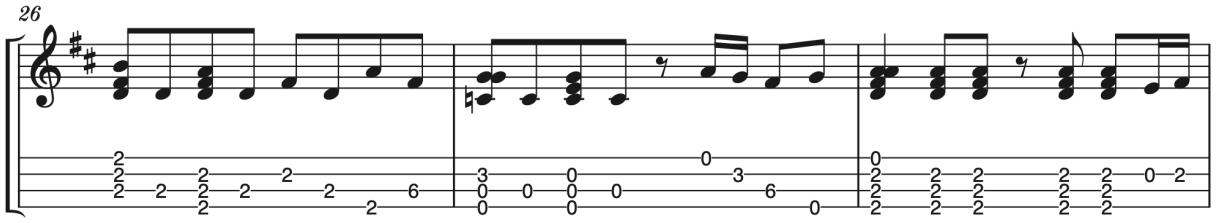
[126] Hughes, H. (1909) *Irish Country Songs: collected & arranged by Herbert Hughes*, Boosey & Hawkes, London

[127] For full score see Folder B.1. or *12 Traditional Tunes for Fingerstyle Ukulele*, Les Productions d'Oz, 2019

[128] The opening lyrics suggest a happy love story as the male voice declares that his young love is looking forward to 'our wedding day'. There is something slightly disturbing, however, in the way she then steps away from him. He watches 'fondly' but the way she 'move[s] here and move[s] there' suggests agitation and gives the impression she is trying to avoid his gaze. At evening when there is 'one star awake' she goes 'homeward'. But rather than suggesting comfort there is a sinister undertone of death. He likens her movement to that of a swan gliding over a lake which on the one hand evokes an image of beauty, grace and chastity but, on the other, implies the paleness and drifting of a ghost. The ghostly image intensifies when she comes to him later that night on feet that 'made no din'. If his lover is a ghost how did she die? Was her earlier agitation because he was her murderer? The final twist comes in the last line when she tells him: 'It will not be long love, till our wedding day.' If she is dead they can only be wedded in death.

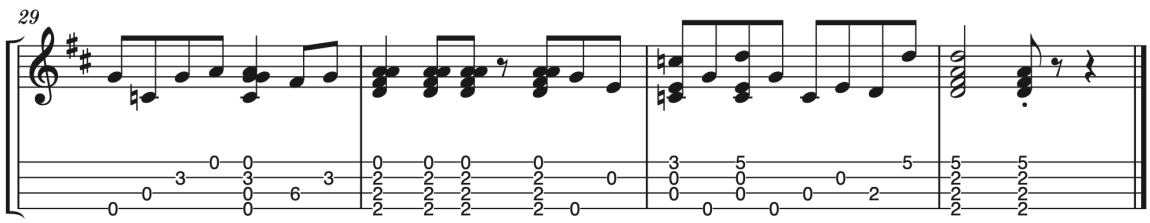
In bar 28 (see figure 2.5.6) the chordal idea is extended as the piece approaches its climax. The two quaver chords, played with a flick of the index finger, are now repeated as an echo effect to imply the ghostly appearance of the woman.

Figure 2.5.6. The chords in bar 28 of She Moves Through the Fair are played with a quick upward flick of 'i'



If the final words of the song ('our wedding day') are a metaphor for the narrator's death, then the joy of unity is shrouded in much darker undertones. Like the lyrics, the D major chords (see figure 2.5.7) imply the happy resolution of marriage, but the upbeat mood can be disrupted by how they are played. Playing the full chords in the final three bars with increasingly aggressive downstrokes counters any suggestion that this is a happy ending. The flurry of quavers in the penultimate bar launches us into the climactic final bar. By playing the first D chord (bar 32) as an aggressive rasgueado an air of arrogant triumph is created. But this mood is immediately cut short by the second D chord which is played as a fast up-stroke using the side of the thumb in order to create a whispering effect. This final 'ghost chord' is then abruptly stopped, or damped, to leave the listener literally 'hanging in the air'.[129]

Figure 2.5.7. The final chords of She Moves Through the Fair



[129] This arrangement, along with a tutorial video, was featured in *Ukulele Magazine* Spring Issue 2023

On the Nature of Daylight by Max Richter[130]

Max Richter's album *The Blue Notebooks* was originally recorded in 2003 but re-released by Deutsche Grammophon in 2018. Richter's idea behind the album was to invite the listener into a peaceful space where they could reflect on the violence and brutality of the world. Originally conceived as a 'quiet protest' against 'loud and vicious rhetoric' the music is as relevant today as it was at the time of the impending war with Iraq (2003-2011) which inspired Richter.[131] *On the Nature of Daylight* is the most well-known piece from the album and has been used extensively in films and TV programmes.[132] The spacious and melancholy atmosphere of the music is perfectly captured in the 2018 video which is featured on Richter's YouTube channel MaxRichterMusic. As of November 2022 the video has amassed an astonishing 10 million views.

The original scoring is for string quintet. The long, held notes at the beginning are ideally suited to this instrumentation but problematic for the short attack of one ukulele. Richter's original key was B♭ minor but I worked from a piano reduction[133] in A minor as this key was more suited to the ukulele. The most challenging section was the opening twenty-four bars which contain one semibreve per bar. The quaver movement from bar 25 was less problematic as the repeated notes and short, scale-like runs were ideally suited to the campanella style.

The chords in the opening section only sustained naturally on the ukulele for approximately half the bar (assuming m.m. ♩ = 66) so the problem was how best to compensate for this loss of sound. One possibility was to play the chords as a tremolo. Rapid repetition of the top note might act as a substitute for the sustain of the strings. I soon abandoned this idea as it sounded too fussy and the music lost its sense of openness. After some reflection I decided I had been overcomplicating the issue. The most obvious answer was to just play each chord as an arpeggiated roll. By using *p-i-m* the dynamic and speed of each roll could easily be controlled. Furthermore, extra emphasis could be given to the top note to highlight the melody (see figure 2.5.8).

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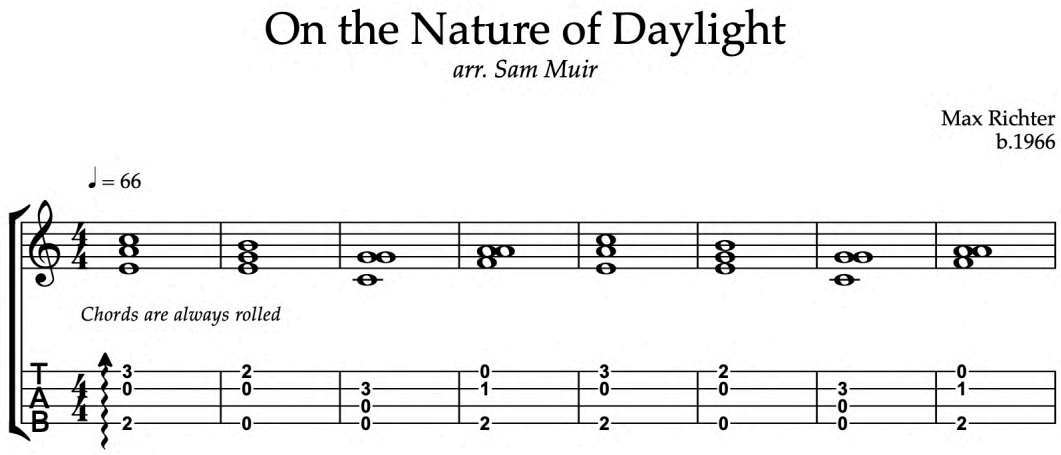
[130] For full score see Folder B.3.

[131] Richter (2022)

[132] For example: Martin Scorsese's *Shutter Island* (2010), Anne Fontaine's *The Innocents* (2016), Michael Winterbottom's *The Face of an Angel* (2014), Henry Alex Rubin's *Disconnect* (2012), Marc Forster's *Stranger Than Fiction* (2006) & Denis Villeneuve's *Arrival* (2016).

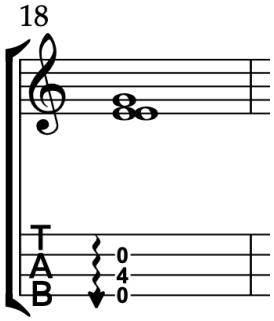
[133] *On the Nature of Daylight* by Max Richter arranged for piano by Jacob's Piano, published under license from Music Sales Corp. 2020, and accessed from Musicnotes website.

Figure 2.5.8. My solution to the opening chords in my arrangement On the Nature of Daylight



Although four note chords were a possibility the triplet effect of three note rolled chords (see figure 2.5.8) sounded the most effective. In bar 18 (see figure 2.5.9) the melodic note was placed on the fourth string and a reverse roll was indicated in order for the G (0,4) to sound as the final note in the chord. This meant the tonic note (E) could be doubled in order to maintain the three note chord. Doubling the E was also a better option than doubling the G which would have happened if the G was played at 3,2.[134]

Figure 2.5.9. The reverse roll used in bar 18



When recording this piece for YouTube I used a metronome set on silent in order to maintain the correct effect of the duration and spaciousness of the opening chords. In order for the top note of

[134] Doubling the G would have made the chord shape 0/4, 4/3, 3/2.

each chord to sound on the first beat of each bar the roll had to start just before the beat. By playing just one rolled chord per bar meant the notes were extremely exposed and there was a temptation to push the tempo (hence the metronome) rather than let each chord dissolve into silence. In the final recording the addition of reverb greatly enhanced the effectiveness of these long, held chords.

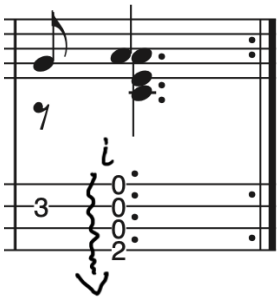
Arranging this atmospheric piece for the ukulele demonstrated how chords can be used to great effect in quiet and reflective music. Due to the short attack of the instrument there is always a tendency to want to fill in all the spaces with notes, but allowing the chords to dissolve into silence led to a more intense and intimate musical experience for both performer and listener.

Using Chords in My Compositions

Variations on The Dowie Dens of Yarrow

As previously seen the reverse roll can be a highly effective way of adding interest to chords. The final Am chord of the theme section of *Variations on The Dowie Dens of Yarrow* (see figure 2.5.10) calls for this technique. In order to play the reverse roll the index finger is dragged across the strings in the order of one to four. Due to the re-entrant tuning the same tonic note of A occurs at both the beginning and the end of the chord. With the As sounding at the same pitch the notes create the intervals of a falling 4th, a falling 3rd and a rising 6th. The sudden change in direction (I like to think of it as an inside out chord) from descending to ascending is what gives this technique a unique twist on the ukulele.

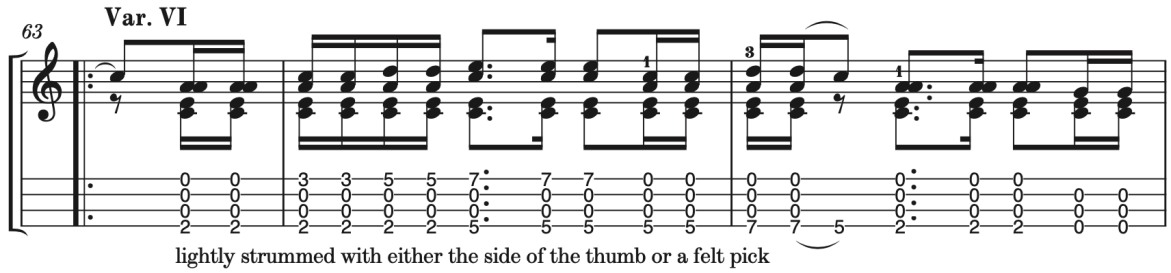
Figure 2.5.10 showing the reverse roll at the end of the theme in Variations on The Dowie Dens of Yarrow



Variation VI is one of the few places in my compositions in which chords are strummed throughout the entire section. In figure 2.5.11 the melodic notes move from the first string to the fourth string. The rising melodic figure A-A-C-C-D-D-E-E-E moves along the first string to the seventh fret (see the tablature in figure 2.5.11) and is then answered by the notes C-C-D-D-C-A played on the fourth string. By placing the melodic notes on the fourth string, the first string sounds the open A in order

to emphasise the A minor tonality. The chords are strummed, either with the side of the thumb or with a felt pick. The melodic notes can be brought out by shifting the weight of the strum from first string to fourth string accordingly.

Figure 2.5.11. Extract of the strummed Variation VI with the melodic notes moving from the first to fourth string



The suggestion to use a felt pick is partly in acknowledgement to ukulele pioneer Jennie Durkee who invented the felt pick around 1918,[135] and partly because I prefer the full but slightly muted sound created by the pick. Durkee's technique, which received great acclaim at the time,[136] was to draw the pick rapidly back and forth across either single or multiple strings in order to create a tremolo effect. While I have not mastered Durkee's tremolo approach, I find the more rounded and even sound of the felt pick an interesting alternative to using the index finger.[137]

The Falling Rain

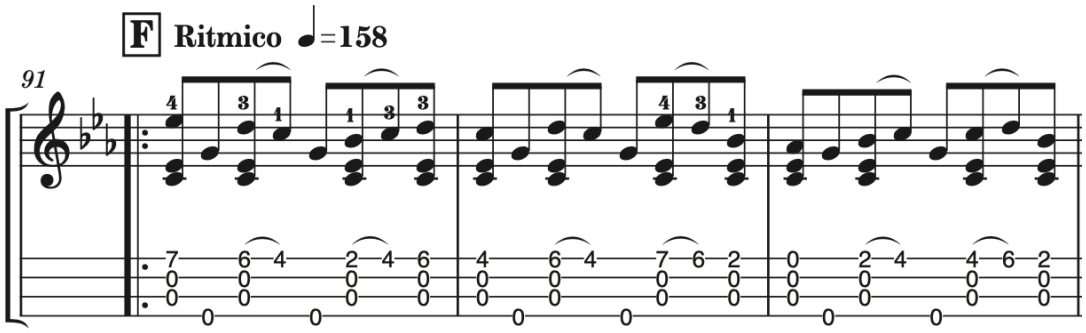
Chords are used in the F and G sections of *The Falling Rain* to represent the increasing intensity of the rain but are played using very different techniques. In the F section three note chords are plucked with the *a-m-i* fingers to create a punchy rhythm. The notes are intended to sound 'as one', rather than broken (as in *On the Nature of Daylight*). In the first part of the bar (see figure 2.5.12) the chords fall on-the-beat while in the second half they fall off-the-beat. The idea was that the shift between strong and weak beats emulated the varying intensity of the rain.

[135] *Crescendo* Volume 10 Number 10, April 1918

[136] See Chapter 1.2 'Classical Approaches to the Ukulele' for more on Durkee.

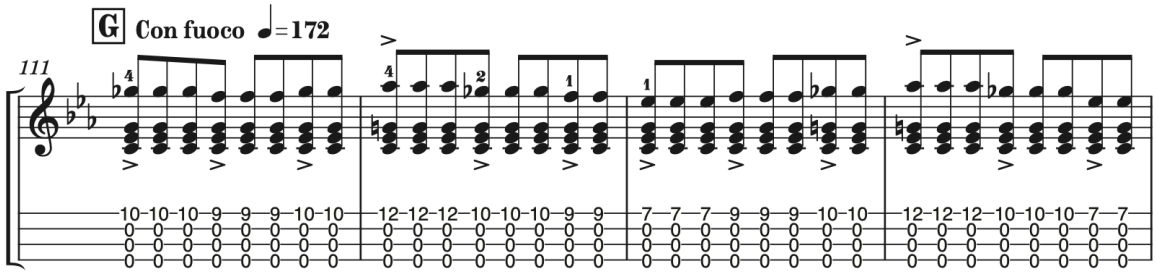
[137] As Durkee realised, using the felt pick also helps to preserve the nail which can be damaged by strumming.

Figure 2.5.12. The chords at opening of the F section of The Falling Rain



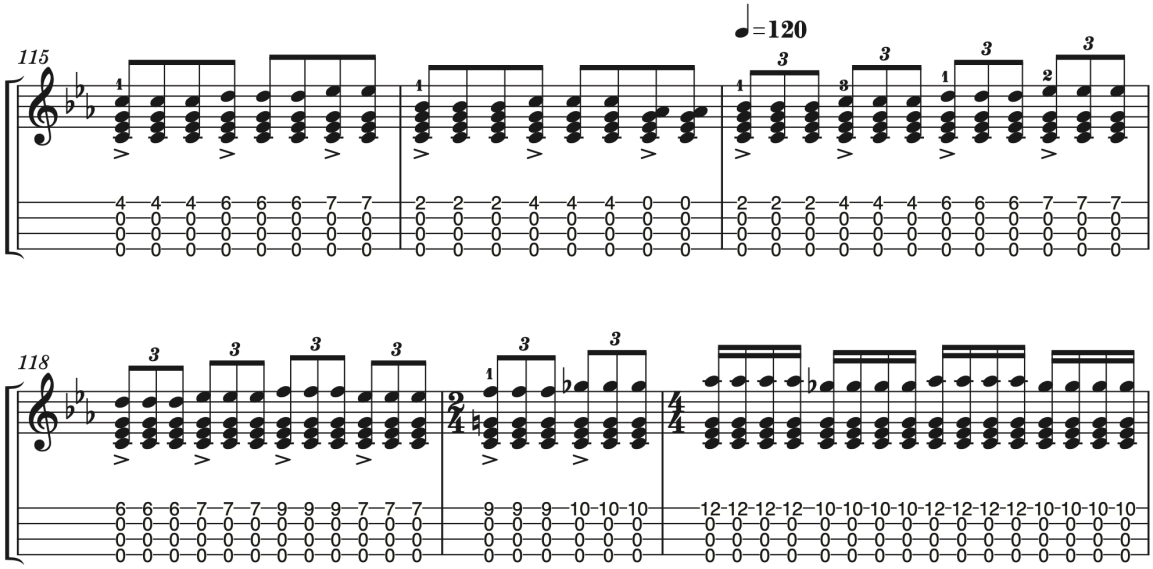
In the climactic G section of *The Falling Rain* the four note chords are intended to be strummed, either with the fingers or a felt pick, to sound wild and aggressive. Although still rhythmical, the notes of the strummed chords are less precise than the clipped, straight chords played by the *a-m-i* fingers in section F. Each bar of section G comprises eight quavers (4/4 time) which are grouped 3-3-2 with the accent falling on the first chord of each group (see figure 2.5.13).

Figure 2.5.13. The full chords in the G section of The Falling Rain



After six bars the rhythm transitions into a triplet pattern of four triplet groups per bar (see bar 117 of figure 2.5.14). Tension builds as the chords move up the fingerboard to the twelfth fret. At this point the rhythm breaks into semiquavers (see bar 120 of figure 2.5.14) with a frantic strumming pattern suggestive of the climax of the cadence in the second movement *Concierto de Aranjuez* for guitar by Joaquin Rodrigo.

Figure 2.5.14. The transition into triplets and then semiquavers in the climactic G section of The Falling Rain

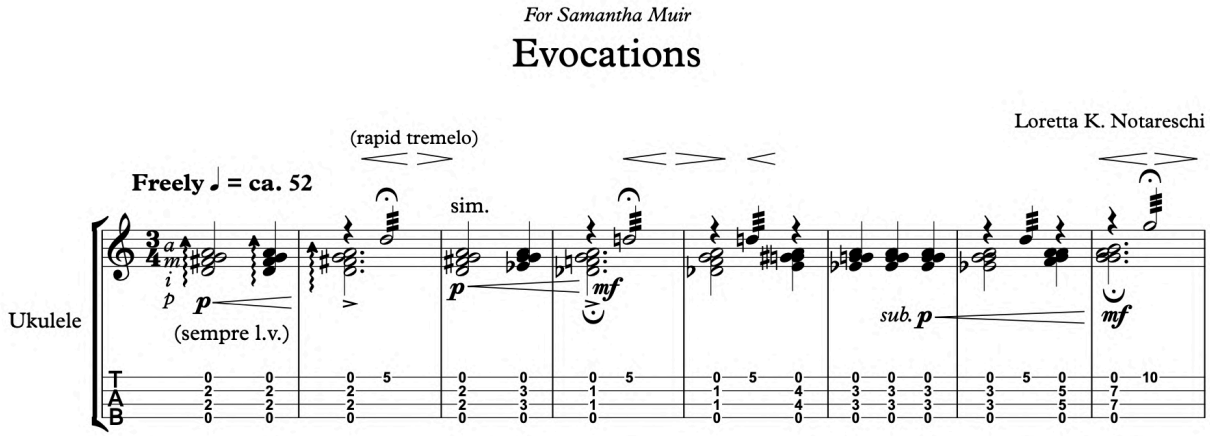


Chords in Compositions by Other Composers

Evocations by Loretta Notareschi[138]

Loretta Notareschi's *Evocations* (2019) uses a number of different approaches to playing chords. The opening bars intersperse four-note rolled chords (using *p-i-m-a)* with rapid tremolo notes (see figure 2.5.15). The music is highly evocative of Spanish guitar music yet sounds modern and idiomatic on the ukulele. Although the ukulele lacks the bass range of the guitar, Notareschi compensates by using the tremolo in the higher register to sound like a distant echo.

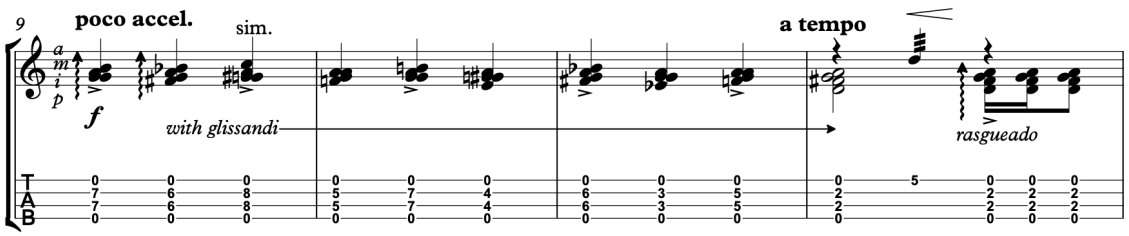
Figure 2.5.15. The opening of Evocations alternating between chords and tremolo effects



[138] For audio see Folder C.2. For full Score see Folder B.4.

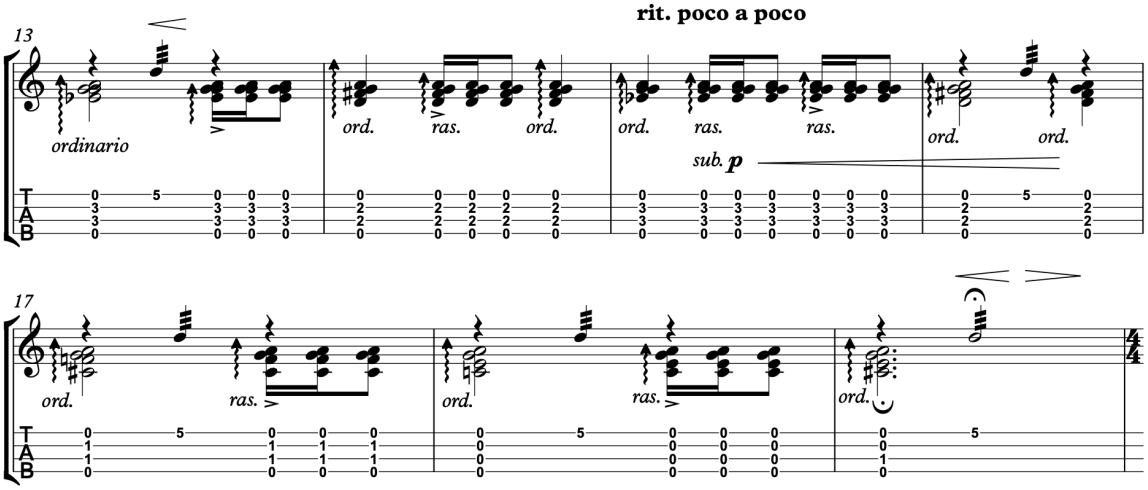
From bar 9 the chords, now played *forte*, become more emphatic as the music presses forward *(poco accel.)*. While the first and fourth strings are always open, the inner voices slide up and down the fingerboard in a *glissandi* effect (see figure 2.5.16). The sound becomes more blurred and the music more intense and unsettled.

Figure 2.5.16. The glissandi chords in Evocations



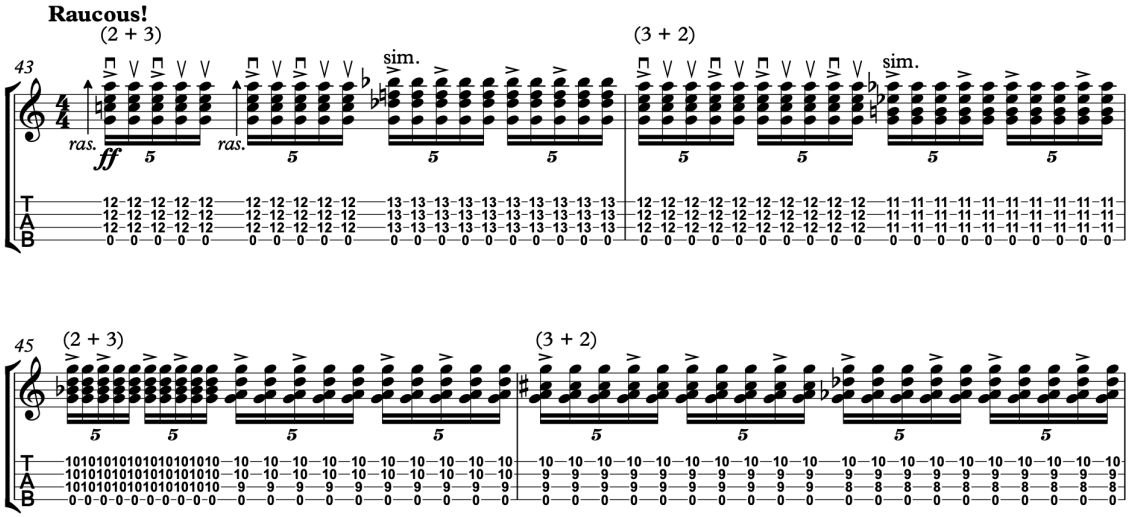
At the end of bar 12 (see figure 2.5.16) the chords become more wild and aggressive. The instruction is to play them rasgueado (with the backs of the fingernails) and this more forceful attack provides a stark contrast to the gentle rolled chords of the opening bars. From bar 13 (see figure 2.5.17) three different approaches (strummed chords, tremolo and rasgueado chords) are combined to form a dramatic end to the first section. Each bar begins with a down strum to create a punchy and rhythmical first chord. The single note tremolo enters like a distant, plaintive voice but is suddenly interrupted by fiery rasgueado chords. Ultimately, the tremolo has the final say as it closes the section in bar 19 with a long diminuendo into silence.

Figure 2.5.17. The climatic end to the first section of Evocations combining chords with tremolo effects



Chords dominate from bar 43 with a ‘Raucous!’ five bars of fortissimo chords grouped in semiquaver quintuplets (see figure 2.5.18). The quintuplets are either played as a group of 2 + 3, as in bar 43, or a group of 3 + 2, as in bar 44. Down strums (D) are used for the accented notes and up-strums (U) for the weaker beats. Hence, the 2 + 3 pattern is D U D U U and the 3 + 2 pattern is D U U D U. I found the changing patterns quite challenging to play but highly effective.

Figure 2.5.18. The quintuplet semiquaver chords in Evocations



Evocations is an excellent example of how different approaches to playing chords can be combined to create different moods and textures. From the harp-like and gentle rolled chords of the opening, to the rhythmical and strident strummed chords of the middle section, *Evocations* uses techniques (tremolo and rasgueado) that suggest the guitar but utilise the idiomatic features of the ukulele. Thus, Notareschi recalls the Spanish origins of the guitar and gives them a new life on the ukulele. A particularly effective technique was interspersing the chords with single note tremolos (as in the opening and closing section) in order to create more resonance and musical interest.

Conclusion

This chapter demonstrates how chords remain an essential feature of the classical ukulele style. When combined with other finger picking techniques such as arpeggios, tremolos and campanellas, chords can be used to add colour and texture to the music. The variety of approaches to playing

chords offered by a fingerstyle approach further extends the expressive potential of the ukulele: from the wild strummed chords of *The Falling Rain* and *Evocations*, to the softly rolled chords of *On the Nature of Daylight* the ukulele can be seen to offer a rich palette of sounds and nuances.

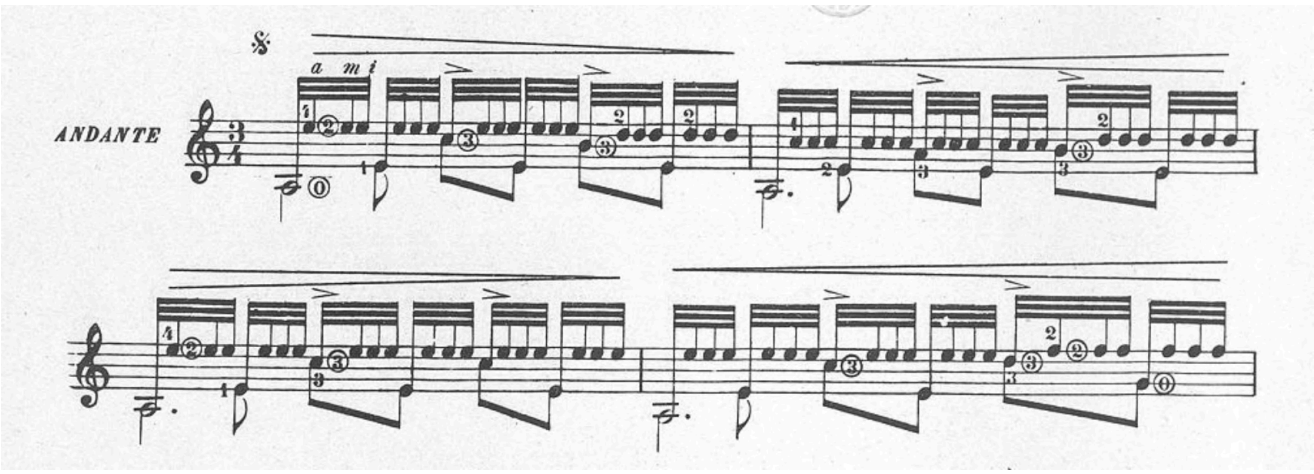
2.6. Tremolo

Introduction

The OED defines tremolo, as applied to music, as ‘a tremulous or vibrating effect produced on certain musical instruments or in the human voice in singing, esp. to express intensity of emotion.’[139] The idea is similar to vibrato but differs in that while vibrato achieves this expression through the manipulation of pitch, tremolo does so through the repetition of notes. There are multiple ways of creating tremolo effects on plucked string instruments, both acoustically and electronically, but my own work has focused on applying what I will refer to as the *Tárrega technique* on the ukulele.

Tremolo on the guitar is a popular technique most commonly associated with Francisco Tárrega’s (1852 - 1909) concert work *Recuerdos de la Alhambra* (*Memories of the Alhambra*).[140] As can be seen in figure 2.6.1 each melodic note is played three times using the *a-m-i* fingers, while the thumb plays the accompanying bass line to create a full sequence of *p-a-m-i*. Expression is achieved through the use of rising and falling dynamics. Bar 1, for example, decrescendos into bar 2 which then crescendos into bar 3, and so on. In Tárrega’s piece the tremolo effect imitates the sound of running water evocative of the many fountains found throughout the Alhambra Palace complex in Granada, Spain. The effect is often described as haunting and beautiful.

Figure 2.6.1. The tremolo technique used in Recuerdos de la Alhambra by Tárrega

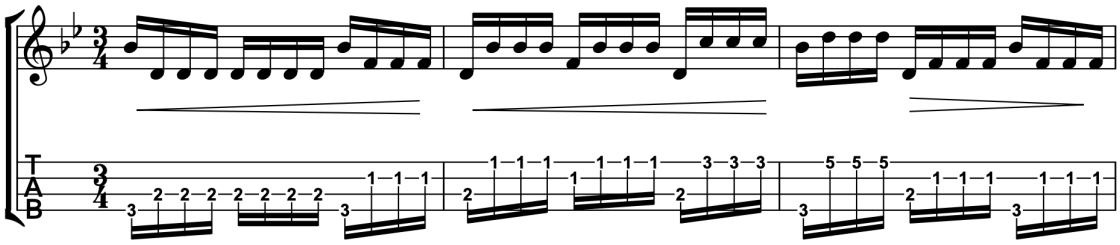


[139] OED online

[140] Accessed from IMSLP website

The first time I used the Tárrega tremolo on the ukulele was in 2015 in a short arrangement of the song *Let Me Call You Sweetheart*[141] (1910) by Leo Friedman and Beth Slater Whitson. The arrangement was a response to a Valentine's Day competition run by the Ukulele Orchestra of Great Britain. The chords and lyrics featured on a UOGB tea towel and the goal was to create the most novel arrangement. Using the Tárrega tremolo (see figure 2.6.2) on the ukulele was a challenging and innovative idea as other players only used a three note tremolo. In my arrangement the melody is in the higher voice and played by the *a-m-i* fingers while the thumb plays an accompaniment. The arrangement, which was presented in a YouTube video, was selected as the winning entry.

Figure 2.6.2. An example of my tremolo arrangement of Let Me Call You Sweetheart



Only after commencing my PhD studies did I realise how novel it was to use the Tárrega tremolo on the ukulele. After studying YouTube videos by leading ukulele players, including Jake Shimabukuro and Tamaine Gardener, I realised that *Let Me Call You Sweetheart* may have been the first time the Tárrega tremolo had been applied to the ukulele. Both Shimabukuro and Gardener, in their YouTube videos, use a three note tremolo. In the video 'Jake Shimabukuro on Three-Finger Tremolo',[142] Shimabukuro starts the tremolo with his middle finger. This is then followed by the index finger on the same string, and finally the thumb plucking another string. The pattern is thus *m-i-p* (see figure 2.6.3). In the video 'Learn How to Play the 'Ukulele: Tremolo Technique with Taimane Gardener,'[143] Gardner also uses a three note tremolo but starts with the thumb on one of the lower three strings. This is then followed by the middle and index fingers plucking the same, usually top, string. Thus, forming a repeated *p-m-i* pattern (see figure 2.6.4).

[141] For full score see Folder B.3.

[142] YouTube: https://www.youtube.com/watch?v=-7Hv0vw4Uts

[143] YouTube: https://www.youtube.com/watch?v=bdcViYO4fNA

Figure 2.6.3. An example of Jake Shimabukuro's 3-note tremolo using m-i-p

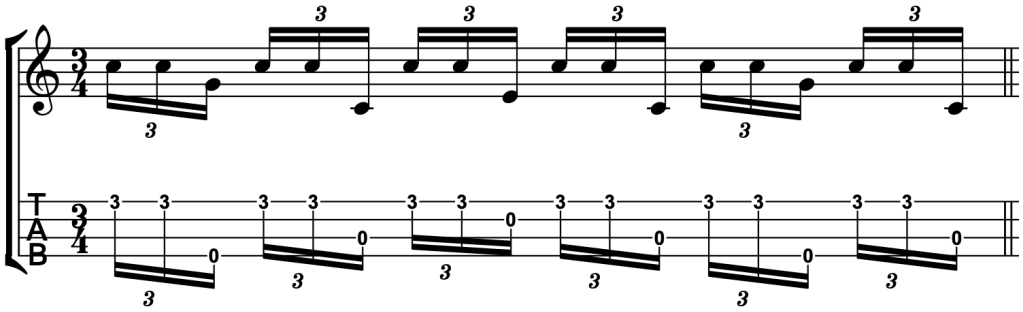
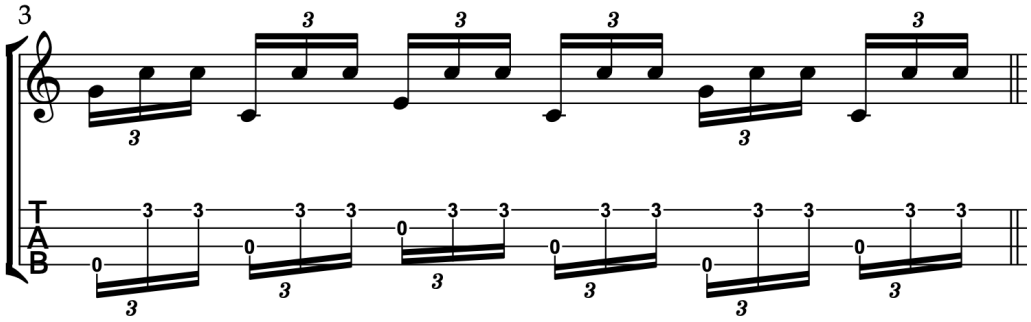


Figure 2.6.4. An example of Taimane Gardener's 3-note tremolo using p-i-m



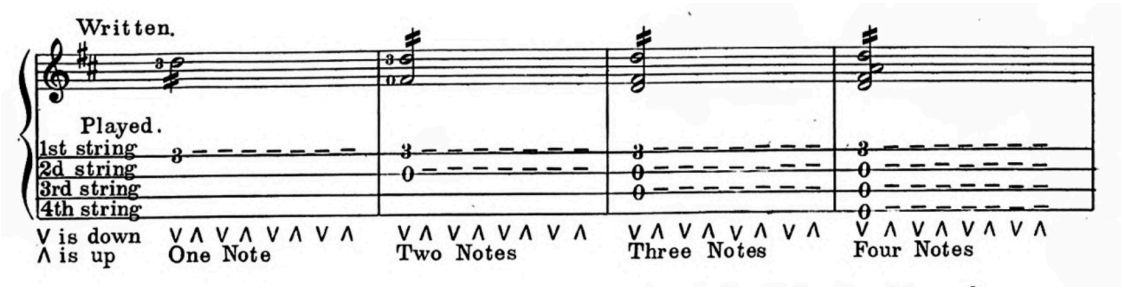
Other approaches to tremolo involve repeated notes on single strings either using a pick or the nail of the index finger. Figure 2.6.5 shows an example from *The Bickford Method* (1920) with the up and down motion of the pick (plectrum) indicated on the score. The authors also suggest that the nail, or tip, of the index finger can be used to emulate the motion of the pick.

Figure 2.6.5. The single note tremolo from The Bickford Method (1920)



A brief mention should also be given to what Jennie Durkee called her ‘American Way’ of playing the ukulele. As outlined in her method of 1917, Durkee applied tremolo to single notes as well as two, three and four note chords (see figure 2.6.6). She achieved this by using a thick felt pick which was swept back and forth across the strings to create a tremolo effect. As previously mentioned, although Durkee achieved rave reviews in the *Crescendo* magazine her style of playing never caught on.

Figure 2.6.6. An example of Jennie Durkee’s ‘American Way’ of playing the ukulele

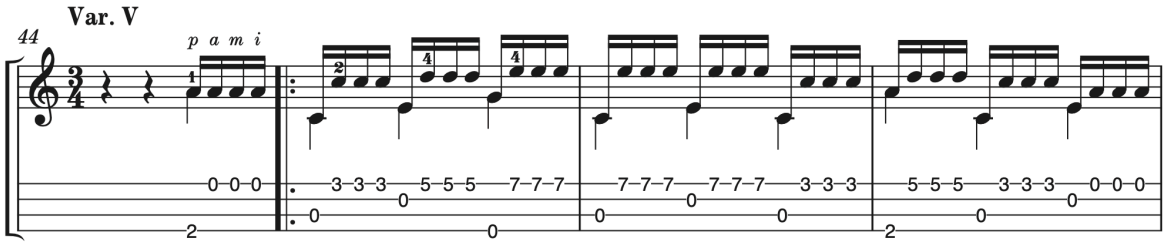


Tremolo in My Compositions

Variations on The Dowie Dens of Yarrow

My first use of the Tárrega tremolo in an original piece for ukulele was in *Variations on The Dowie Dens of Yarrow* (2018). In *Variation V* the melody is mostly assigned to the first string, as shown in figure 2.6.7, while the thumb plays the accompaniment on the lower three strings. The plaintive melody suited the tremolo approach and made an ideal contrast to the busy four string arpeggio of the preceding V*ariation IV* and the punchy strummed chords of *Variation VI*.

Figure 2.6.7. The Tárrega tremolo in Variation V of Variations on The Dowie Dens of Yarrow

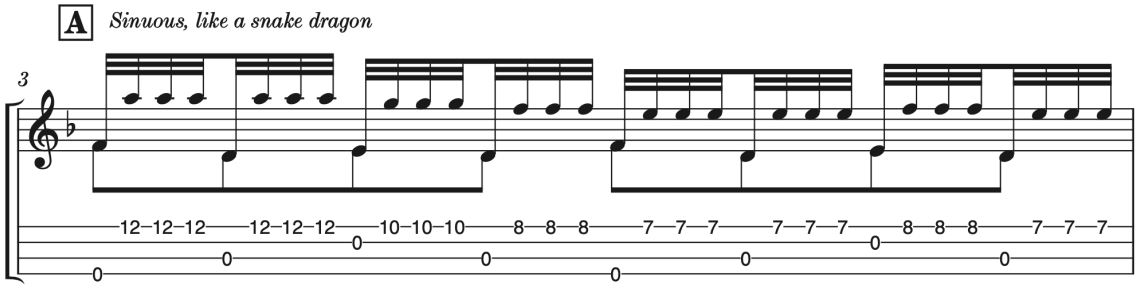


One notable difference in playing tremolo on the ukulele compared to the guitar is that it is more difficult to achieve dynamic variety. This may be because on the ukulele the contact of my *a-m-i* nails with the string is more side-on (due to the hand position) and consequently the sound is naturally softer. If my picking hand is held in a more upright position, like on the guitar, the sound becomes brittle and the increased angle (due to the smaller size of the instrument) puts more strain on the wrist.

Nabu's Message

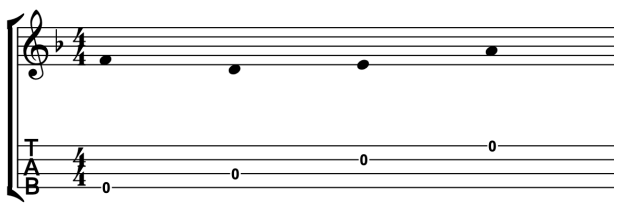
Nabu's Message,[144] the final movement of *Mercurius: A Suite in Three Movements* (2021) uses the Tárrega tremolo to suggest the sinuous movement of the Mushhushshu snake-dragon (see figure 2.6.8). The constant, weaving line of the tremolo seemed an ideal way of portraying Nabu[145] riding through the cosmos on the back of a furious snake dragon.

Figure 2.6.8. The tremolo pattern in Nabu's Message



The alternate tuning has already been discussed in Chapter 2.2.a 'Tintinnabuli' but a brief recap is given here. By lowering string four to F and raising string three to D the open tuning contains Nabu's hidden message F-D-E-A or D E A F. (See figure 2.6.9)

Figure 2.6.9. The DEAF tuning in Nabu's Message



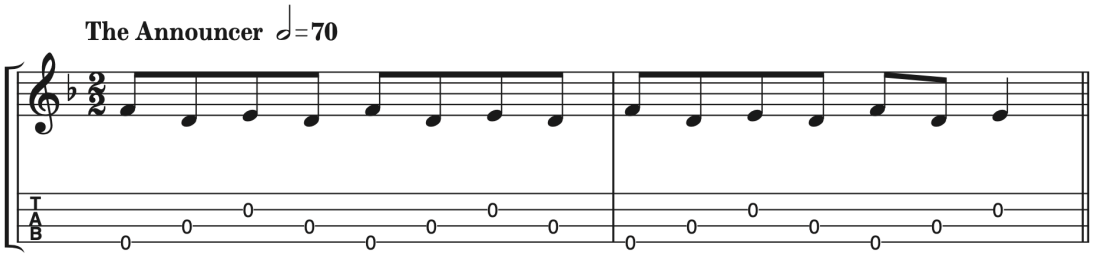
[144] For audio see Folder C.1. For full score see Folder B.1.

[145] Nabu was worshipped by the ancient Babylonians and Assyrians and is associated with the planet Mercury

The DEAF tuning is intended as both a pun and a warning: if humanity remains deaf to the signs of global warning it will ultimately lead to the death of the planet.

As can be seen in figure 2.6.10 the close proximity of the constantly repeated bass notes - F D E D - created the over-ring intervals of a minor 3rd (D-F), a major 2nd (D-E) and a minor 2nd (F-E). Phonetically the notes contain the words DEF (deaf/death) and DED (dead). The effect is eerie and particularly unsettling because in normal tuning these three strings would sound as a more comforting C major triad. My aim was to contrast the constant movement of the tremolo with the jarring sound of the accompaniment in order to create a piece that was hypnotic yet disturbing.

Figure 2.6.10. Nabu's message is hidden in the repeated bass pattern



The Red Angel

As the piece has already been discussed in Chapter 2.2.a 'Tintinnabuli' only a brief mention will be given here. Initially this piece used a slow, three note tremolo played by the *p-m-i* fingers but it was later revised to a four note Tárrega tremolo (see figure 2.6.11). The first performance by Giovanni Albini on YouTube used the triplet tremolo.

A feature of *The Red Angel* is that the tremolo is constantly interrupted or broken rather than played in the more conventional way as a continuous line of notes. The idea reflects the sudden, brief appearances of a time travelling entity known as The Red Angel in *Star Trek Discovery* Season 2. The relentless open string accompaniment played by the thumb represents the continuity of the present timeline. The tremolo enters in short bursts (like the red angel) but each time its duration increases by two beats - from two, to four, to six and so on - until a total of sixteen beats is reached and then the idea is reversed.

Figure 2.6.11. The opening of The Red Angel using a 4-note tremolo & open G minor tuning

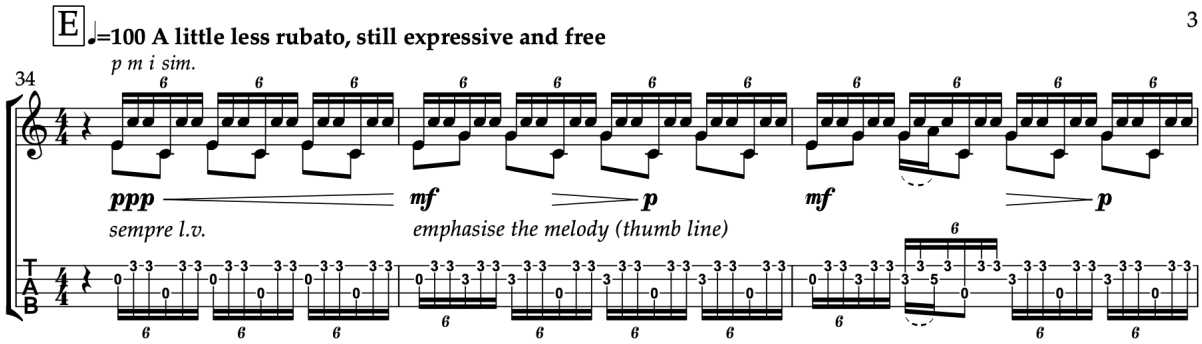


Tremolo in Works by Other Composers

In Every Heart by David John Roche[146]

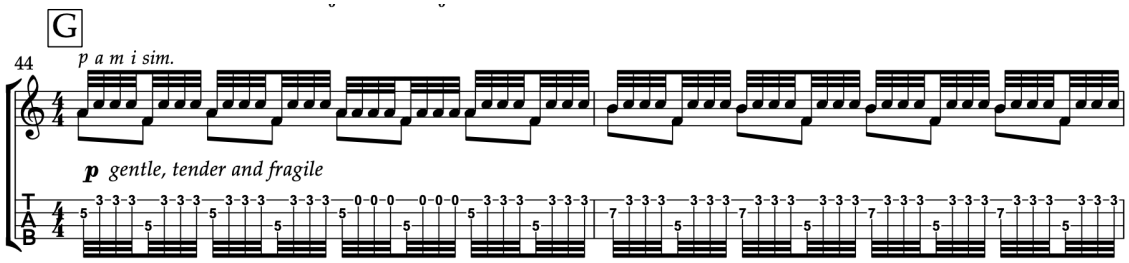
In Every Heart, composed for me in 2021 by David John Roche, uses two approaches to tremolo. Following a slow, chordal introduction the piece moves into a three note tremolo pattern using *p-m-i* (see figure 2.6.12). By assigning the melody to the naturally stronger thumb Roche successfully navigates the problem of dynamic range as the tremolo is always intended to play an accompanying role. At bar 44 the pattern transitions into the full 4-note Tárrega tremolo (see figure 2.6.13).

Figure 2.6.12. An example of the triplet tremolo from In Every Heart by DJ Roche



[146] For audio see Folder C.2. For full Score see Folder B.4.

Figure 2.6.13. An example of the the 4-note tremolo in In Every Heart by DJ Roche

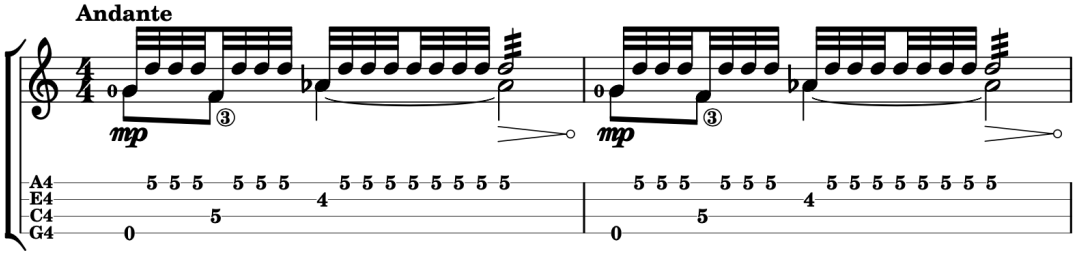


In Every Heart is one of the most effective and expressive pieces in my portfolio of new works. As well as being a successful solo piece, in 2022 Roche used it as the first movement of a new concerto for ukulele.

In Despair by Choan Gálvez

In Despair,[147] composed in 2021, is another piece that uses the Tárrega tremolo. As the title suggests this is a piece full of anguish and the tremolo projects a very different mood from the buoyant, often joyous, *In Every Heart.* In the opening bar of *In Despair* (see figure 2.6.14) the repeated D tremolo contrasts with an edgy campanella style second voice. The second voice moves across the strings from 0,4, to 5,3, to 4,2 (G-F-Ab) until the two voices are finally united on string 1 (see bar 2 of figure 2.6.14). The tremolo then diminuendoes into near silence.

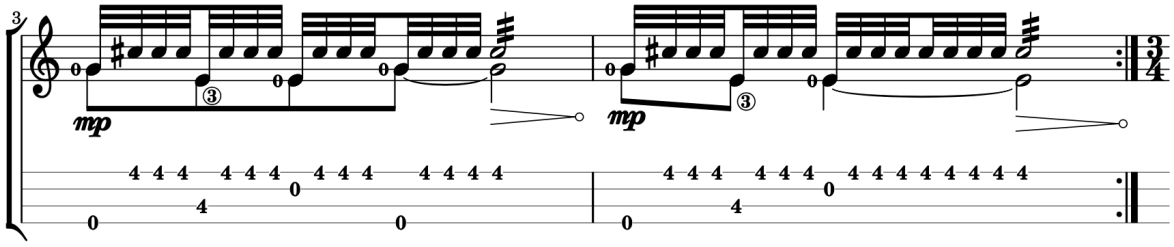
Figure 2.6.14. The opening bars of In Despair



[147] For audio see Folder C.2. For full score see Folder B.4.

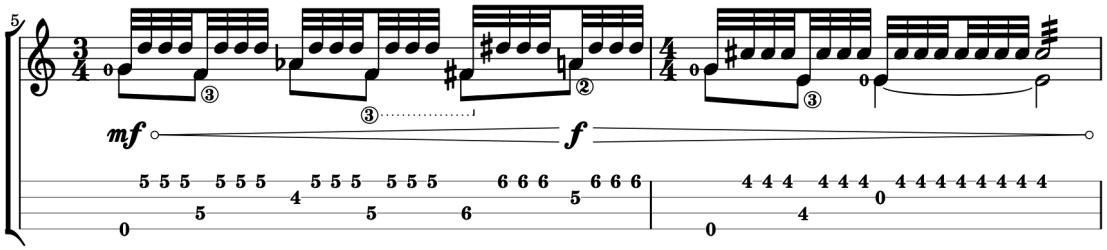
Bars 3 and 4 follow a similar structure but the anguish of the music is intensified by the dissonance of the augmented 4ths (see figure 2.6.15).

Figure 2.6.15. Bars 3 & 4 of In Despair



Bar 5 (see figure 2.6.16) builds on the idea from bar 1 but this time, rather than ending with repeated Ds, the D is raised to D♯ on beat three and the two voices remain separate. Once again the augmented 4th, this time between the A played by the thumb and the repeated D♯, creates a moment of intense agitation. The forte dynamic is a further indication that this is the emotional climax of the first section. Bar 6, which is the same as bar 3, is then repeated three times as the section ends with a plaintive diminuendo.

Figure 2.6.16. Bars 5 & 6 of In Despair



The rest of the piece moves between chromatic campanella ideas and the tremolo idea from section one. Any notion that tremolo pieces are inherently sentimental is quickly extinguished by the often jarring tonal landscape of *In Despair*. In this piece Gálvez effectively combines tremolo and chromaticism to transform the stereotypically happy personality of the ukulele into a darker state of anguish and pain. When playing *In Despair* I was drawn into the music's sense of despondency and consequently, while it was not a piece I could *enjoy* playing, I felt a deep connection with the music and the emotional landscape the composer was trying to achieve. Gálvez poignantly illustrated how

the traditional techniques of tremolo and campanella can be applied to the ukulele in a language that is both modern and idiomatic.

Conclusion

There are multiple ways of playing tremolo on the ukulele. My own work, however, has focused on the four note Tárrega tremolo demonstrated throughout this section. Coming from a guitar background this is the most intuitive approach for me but when applied to the ukulele it presents new technical and musical challenges. Although it is not possible to achieve the same dynamic range as on the guitar, the works presented here (by Muir, Roche and Gálvez) offer a rich diversity of moods and tonalities. From the snaking melody of *Nabu's Message* to the joyful triumph of *In Every Heart* to the emotional outpouring of *In Despair* what emerged from a study of these pieces is how tremolo can be used to create new music for the ukulele that is both technically demanding and emotionally diverse.

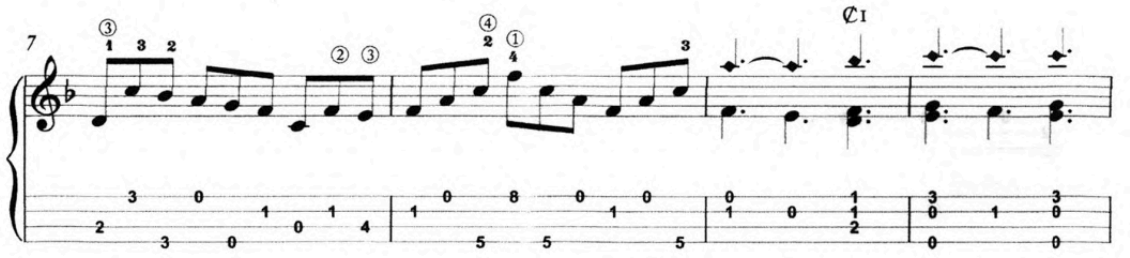
2.7. Harmonics

Introduction

An effective way of extending the dynamic and tonal range of the ukulele is to use harmonics. As on the guitar, harmonics on the ukulele can either be played in the natural positions of the twelfth, seventh and fifth frets or created artificially by lightly touching the string with the index finger of the picking hand at a distance of twelve frets from any stopped note, and then plucking the string with either the thumb or the ring finger.

The first time I encountered harmonics on the ukulele was in John King's arrangement of *Jesu, Joy of Man's Desiring* (2004) in which he used artificial and natural harmonics, often incorporating them with plucked notes (see figure 2.7.1). At the time this was an uncommon technique to use on the ukulele.

Figure 2.7.1. The harmonics in bars 9 & 10 of King's arrangement of Jesu, Joy of Man's Desiring



Using Harmonics in Original Compositions

The Falling Rain

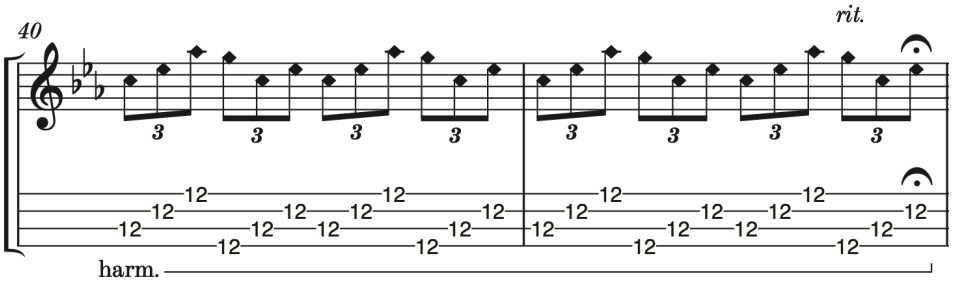
The Falling Rain uses natural harmonics in four places. The first is at the end of the A section where the melody ascends up the fingerboard to the twelfth fret. The run culminates with harmonics at the twelfth and seventeenth frets (see figure 2.7.2).

Figure 2.7.2. The harmonics at the end of section A of The Falling Rain and the final harmonics



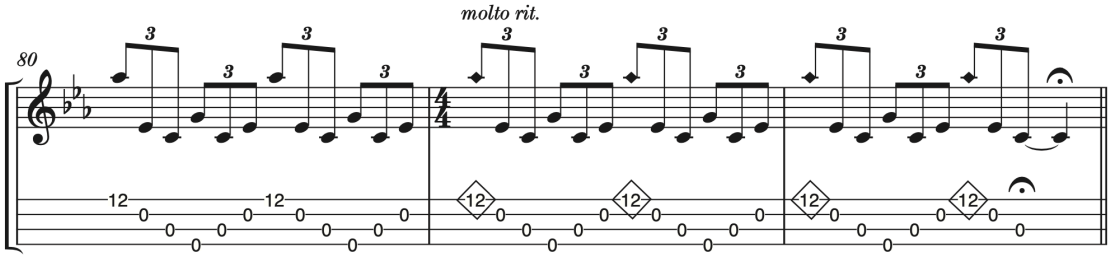
The B section also concludes with harmonics, this time played as a triplet arpeggio figure (see figure 2.7.3). Due to the fast tempo, the fretting finger remains in close contact with the strings causing the harmonic effect to sound more percussive than if the strings were allowed to vibrate naturally. This effect was intentional as it evoked the sound of raindrops thudding on the roof. The *ritardando* at the end of the passage suggests the gradual slowing down of the rain. As the tempo slows the finger is able to release the string and the harmonics can ring freely.

Figure 2.7.3. The harmonics at the end of section B of The Falling Rain



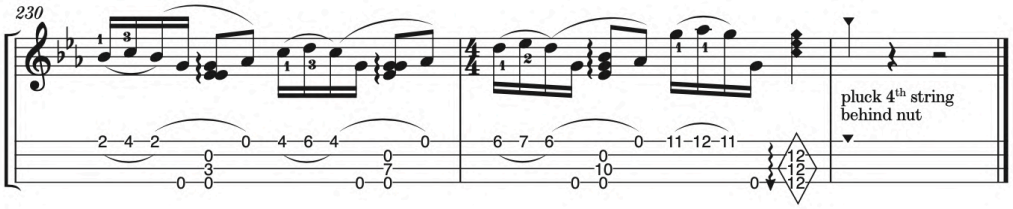
Once again harmonics are used at the end of the D section but this time the other notes in the arpeggio are played as open strings (see figure 2.7.4). The harmonics add a sparkle of interest and colour as the arpeggio, marked *molto rit.* gradually fades away to give the impression of the rain slowly stopping.

Figure 2.7.4. The harmonics at the end of section D of The Falling Rain



The final time appearance of harmonics is at the very end of the piece, or section J. Although this section is essentially a repeat of the A section the ending is different. The final chord is played as natural harmonics at the twelfth fret. The chord is followed by an effect that is created by plucking the fourth string behind the nut (see figure 2.7.5) to create one last percussive 'plink' suggestive of the final raindrop.

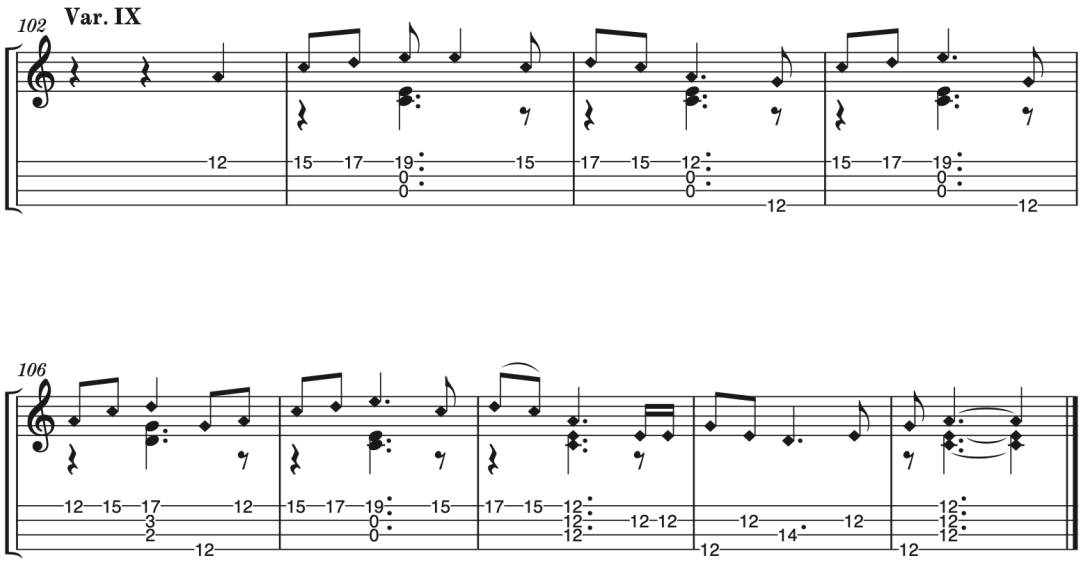
Figure 2.7.5. The harmonic chord at the end of The Falling Rain



Variations on The Dowie Dens of Yarrow

The entire melody of the final *Variation IX* of *Variations on The Dowie Dens of Yarrow* is played using either natural or artificial harmonics (see figure 2.7.6). The delicate nature of harmonics meant they were a perfect way of capturing the melancholy mood of the ending. In the song the lovers both die, hence the use of the word *dowie* (a Scottish word meaning sad or melancholy) in the title. Despite the sad and wistful mood, there is also the hope that the lovers are finally united in death. I particularly liked how the harmonics meant the theme could be presented in a simple and largely unadorned way while adding an ethereal quality to the ending.

Figure 2.7.6. Variation IX with the melody played as harmonics

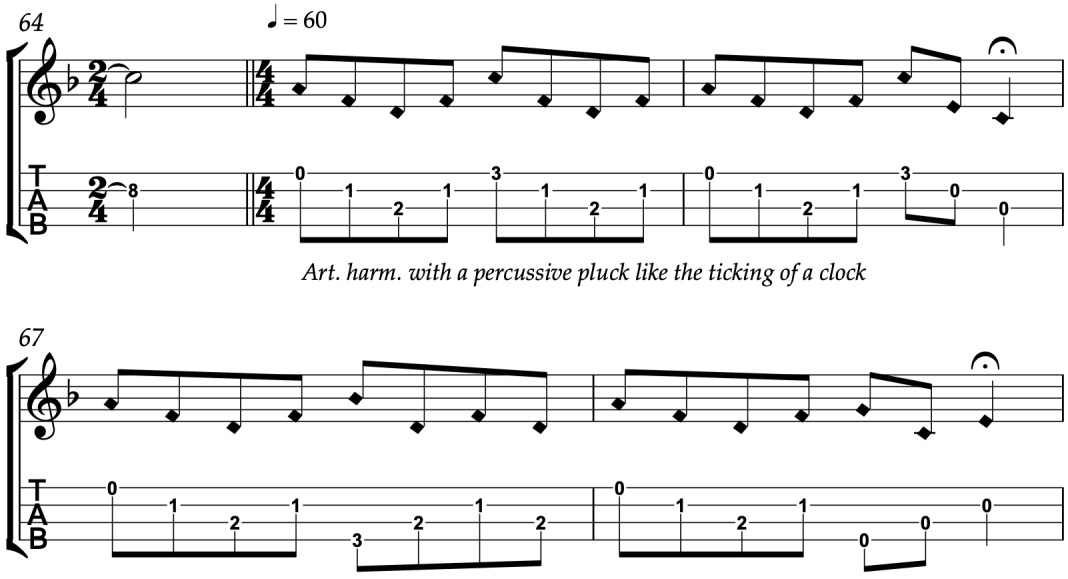


Flower Sleeps at 4[148]

I wrote this piece as a musical tribute to the ornamental plant, *Mirabilis jalapa*, or the *Marvel of Peru.* In English it is often called the four c'clock flower as the flowers usually open at four o'clock. In my friend's garden in Spain, however, the flowers close at four. While staying there I was amused at how the bright little flowers seemed to droop and go to sleep around 4pm, like they were having a little *siesta*.

Overall this is an upbeat, moderately fast piece but I included a slow eight bar harmonic section (see figure 2.7.7) to provide contrast. In this section the harmonics reflect the delicate beauty of the flowers as they are gently nodding off to sleep (closing) in the dazzling heat of a Spanish afternoon. I indicated that all of the harmonics be plucked with a slightly percussive attack to suggest the ticking of a clock.

Figure 2.7.7. An example of the harmonic section of Flower Sleeps at 4



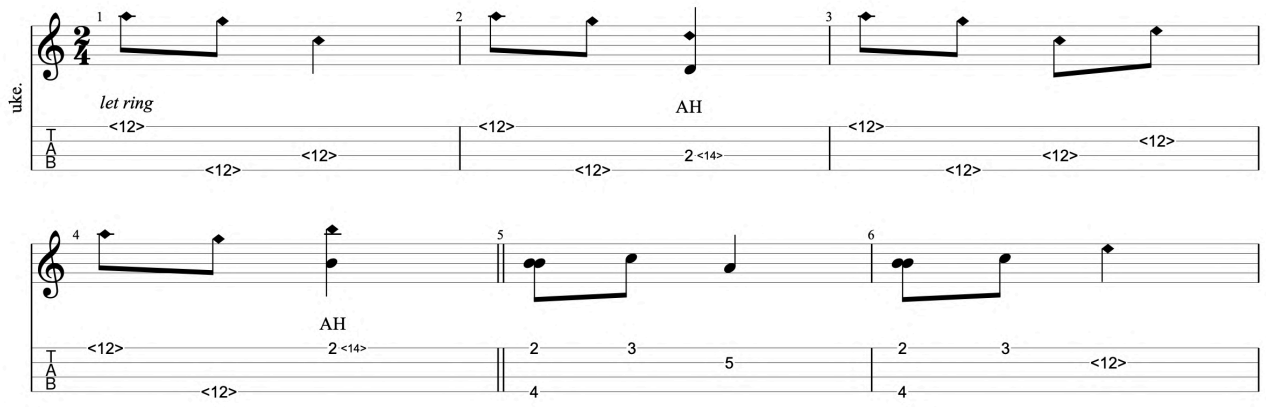
Flower Sleeps at 4 was premiered at the University of Surrey in the final of the Joyce Dixey Composition Competition 2021, and awarded joint second place.

[148] Muir (2021). For full score see Folder B.2.

Tell Me Why[149]

This slow and simple piece is based on the opening notes (see figure 2.7.8) of the song *Why* from the album *Diva* (1992) by Annie Lennox. The song is a soulful ballad about a relationship break-up. I was particularly struck by the lyrics 'Why can't you see this boat is sinking,' and 'I don't think you know what I feel.' The emotional outcry of the song is captured in a three note falling figure A - G - C (see figure 2.7.8) which recurs throughout the piece. These falling notes represent 'the tears we shed'. As the piece was written during the COVID-19 pandemic I thought of it as a breakup between humanity and the God who never answers, rather than a breakup between two people. At this time the entire world was gripped by 'the fear' and 'the dread' of death and isolation. The title *Tell Me Why* was not intended as a question but as a simple, plaintive cry of frustration. I used natural and artificial harmonics combined with campanella affects to give the music a glass-like fragility. My aim was to create a piece of music which demonstrated a delicate and vulnerable side of the ukulele. I was grateful that my friend and fellow ukulele player, Giovanni Albini, agreed to premiere this piece.

Figure 2.7.8. The opening notes of Tell Me Why



Notating Harmonics

What emerged when writing the harmonic sections in these pieces was the different, sometimes confusing, methods used to notate harmonics in music software programmes. While harmonics are

[149] Muir (2023). For full score see Folder B.2.

easily shown in staff notation by the use of a diamond shaped note-head, tablature seems to lack a universal system. In *The Falling Rain* the publisher had two solutions. The fret number was either enclosed in a diamond, or the word *harm.* (harmonic) was written underneath the tablature with a line extending for the duration of the harmonic notes. In *Variations on The Dowie Dens of Yarrow* and *Flower Sleeps at 4* the harmonics are only indicated in the staff notation while the corresponding fret numbers are shown in the tablature. As there is no indication in the tablature that these notes are to be played as harmonics they may be missed by players who only read from the tablature.

Currently, the clearest solution for notating harmonics in tablature seems to be offered by the software program Guitar Pro 7. As can be seen in figure 2.7.8 the natural harmonics are enclosed in a triangular shaped bracket while artificial harmonics are shown with the fret numbers corresponding to both hands. Furthermore, the corresponding picking hand fret is shown in a smaller font as well as being enclosed in triangular brackets. I found this to be the clearest method of notating harmonics in tablature although the general appearance of the Guitar Pro 7 score was less clear.

Conclusion

Harmonics, particularly artificial harmonics, are becoming more popular on the ukulele. In my own compositions and arrangements I have used them for adding subtle nuances and to extend the range of the instrument. The lack of standardisation on how to indicate harmonics in tablature in music software programs is a problem which needs to be addressed.

2.8. Microtonal Ukulele

Introduction

In June/July 2018 I collaborated with microtonal guitar specialist Tolgahan Çogulu[150] to create what is possibly the world's first microtonal ukulele. Under the guidance of Tolga[151] I first learnt how to create microtones on the ukulele and then made an arrangement of the Turkish song *Uzun Ince Bir Yoldayim* (*I Am On A Long and Narrow Road*) by Asik Veysel. A video of the piece was uploaded to YouTube on 26th July 2018.[152]

Adding Fret-lets to the Ukulele

To begin, Tolga sent me five small frets, or fret-lets, by post. He then emailed a link to a YouTube video explaining where and how to place the fret-lets on a guitar. The principal was the same for the ukulele fretboard but the smaller scale of the soprano ukulele meant the placement of the fret-lets was largely a matter of trial and error rather than precise measurement. The idea was to place the fret-lets approximately 1/3 of the distance from the second fret of the second string, and the fourth fret of the third string. A further concern was that as the fret-lets were designed for a guitar they were quite wide and further reduced the distance between the frets. Tolga offered the following advice:

> If you can manage to tape them to the one third of the fret without a buzz, then we can proceed. If there is a buzz, you need to go to a luthier or try to raise or lower the added fret.

The fret-lets were taped to the fingerboard with sticky tape (see figure 2.8.1). Fortunately, the fret-lets worked without any buzzing and I did not need to consult a luthier. The sticky tape, which seemed rather crude, proved surprisingly effective. Not only did it hold the fret-lets in place, it meant that when it came to playing the piece I found the smoothness of the tape assisted in sliding the finger to create the ornamental effects.

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[150] https://www.microtonalguitar.org/

[151] We communicated via email, messenger & by using YouTube videos for sample recordings & learning material

[152] Muir (2018). For full score see Folder B.3. For audio see Folder C.5.

Figure 2.8.1. The position of the fret-lets on the fingerboard of the soprano ukulele



The Arrangement

The second stage of the project was to create a ukulele arrangement of *Uzun Ince Bir Yoldayim*. Tolga emailed me a version of the melody scored in staff notation and guitar tablature (see figure 2.8.2).

Figure 2.8.2. The opening of Uzun Ince Bir Yoldayim arranged for guitar



My initial response was to transfer the melody to the ukulele simply by raising it a 4th. Notes that were too low on the ukulele, such as the first A below the stave, were raised an octave. Before proceeding I made a rough recording in order to get some feedback from Tolga. He thought it sounded ‘beautiful’ and confirmed the microtones were correct but gave the following suggestions:

1) Now that you get the melody, can it be more idiomatic to ukulele? I mean there can be arpeggios, up and down strokes, additional notes with slurs etc. Then, it might be a big hit.

2) Can it be faster? You can bring it to the original tempo of Asik Veysel?

3) Whenever you have a possibility, you have enough time, can you do trill on the microtones (from microtone to the adjacent semitone - not to the semitone apart. For example, from B -35 cents lower to B natural -not to C). It is like softening the harshness of the microtone.

4) Before you repeat the B section, I mean in the last measure of the piece but you are heading to the B section again, can you play eighth notes, D E F microtone and G, starting from the 2 and a half beat.

As soon as I listened to Asik Veysel I realised my interpretation was not only too slow, it also sounded very ‘classical’ and polished. After listening many times to Veysel I came to a better appreciation of the music. I added some chords (see figure 2.8.3) and adopted a more rhythmical, and less precise approach so the piece sounded more rustic.

Figure 2.8.3 shows the melody in the notation and my final version, including chords, in the tablature. I decided to tune the ukulele up a step to A-D-F♯-B as the extra tension in the strings facilitated better control of the trills which were created by sliding the finger rapidly back and forth across the fret-lets. In the YouTube video I added a short introduction of chords struck with the back of the nails to create a harsher, more rhythmic sound. Tolga’s response to the second video was “Just amazing! I’m so happy!” He then shared the video on YouTube and on his Patreon channel.

Figure 2.8.3. The opening of my final arrangement of Uzun Ince Bir Yoldayim arranged for ukulele



Conclusion

The microtonal ukulele project represented very different technical and musical challenges for me. The additional fret-lets meant that by reducing the distance between certain frets the feel and the sound of the fingerboard was altered. The biggest challenge, however, was musical. In arranging *Uzun Ince Bir Yoldayim* I needed to put aside my classical instincts in order to capture the rustic spirit of the music. Ultimately, this was a hugely rewarding and enlightening project which, by uniting folk elements with my classical training and knowledge, added yet another dimension to the ukulele.

2.9. Ukulele and Electronics

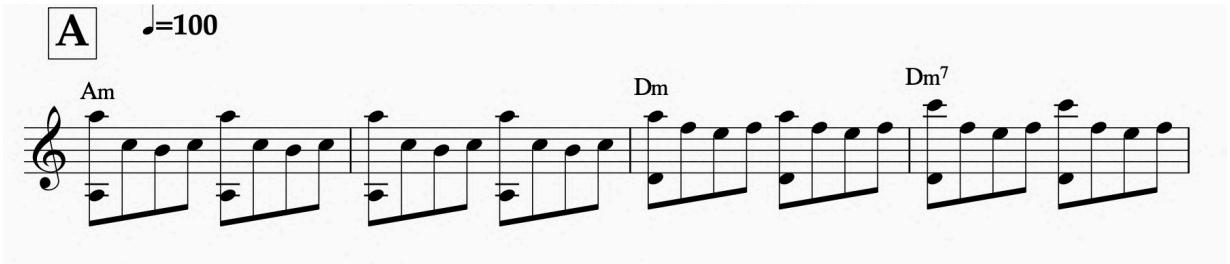
Introduction

In 2019 I collaborated with composer Milton Mermikides to arrange one of his guitar pieces, *Insighted*, for ukulele.[153] My job was to arrange and record the piece. Milton then added the live electronics. This was the first time I had worked with electronics so the project offered some new and unique challenges.

The Arrangement

Milton sent me his guitar version of the score (see figure 2.9.1) which I then adapted for the ukulele (see figure 2.9.2). Initially I treated the upper voice as a scordatura which meant the piece sounded a 4th higher on the ukulele. This worked well for the upper voice but the lower voice was more problematic. For example, in bars 1 and 2 the open fifth string A on the guitar could have been raised a 4th and played as a D at 2,3 on the ukulele. This would have created an octave between voices one and two as in the guitar version. I decided, however, to use A at 5,2 as this is the 5th note in the D minor chord. I felt this extra harmony gave a fullness to the melody that was otherwise missing on the ukulele. While I wanted to maintain the integrity of the original I also wanted to make a version that was unique to the ukulele.

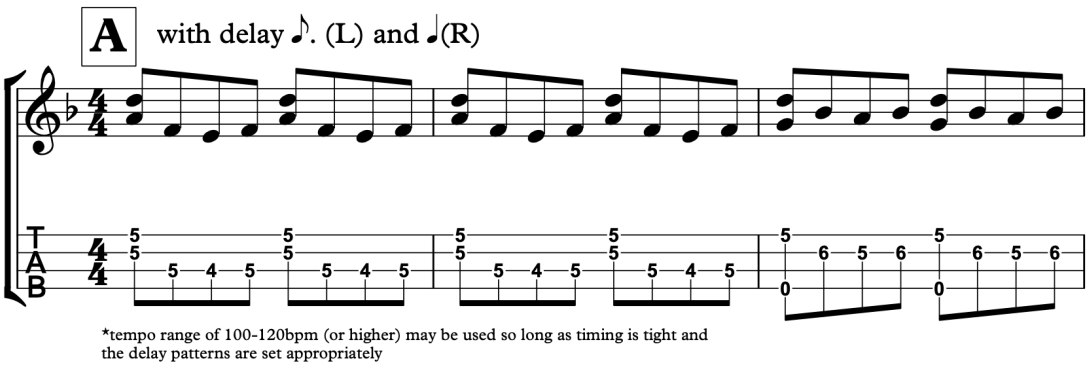
Figure 2.9.1. The opening of Insighted for guitar



Not all the bass notes needed to be altered. In bar 3, for example, I maintained the open fourth string note as in the guitar version (see figure 2.9.2). The note falls on the re-entrant string (0,4 or G) of the ukulele it sounds an octave higher than the corresponding string on the guitar. This made sense musically given that the previous two bars were playing an A in the upper voice, and the G in bar 3 was simply a step down.

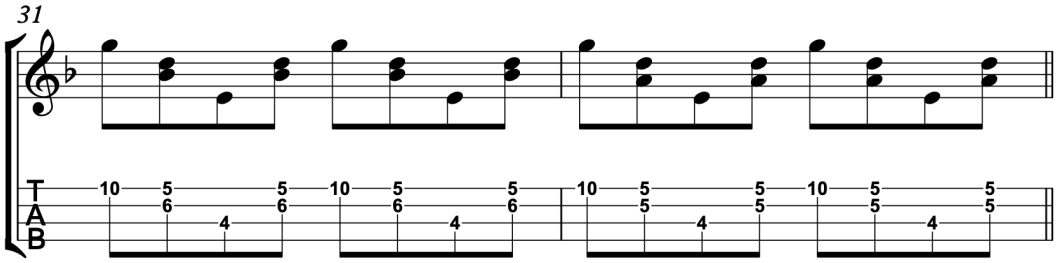
[153] For audio see Folder C.2. For full score see Folder B.4.

Figure 2.9.2. The opening of Insighted arranged for ukulele & electronics



Some bars were quite awkward to play on the ukulele as they required jumps up and down the fingerboard. For example in bar 31 and 32 (see figure 2.9.3) the notes move back and forth between the tenth and fourth frets. This was, however, achievable on the shorter scale of the soprano ukulele, particularly as the tenth fret Gs at 10,1 could be played with a short attack.

Figure 2.9.3. Bars 31 & 32 of Insighted arranged for ukulele & electronics

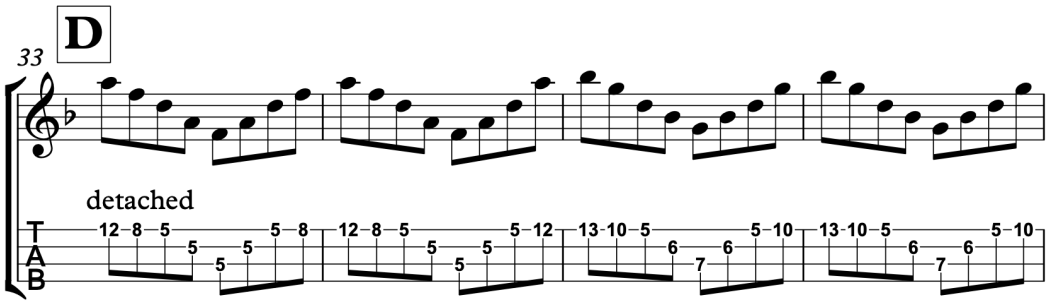


The most challenging section was the D section from bars 33 to 48 (see figure 2.9.4). The direction to play the arpeggios in a detached manner was the polar opposite of playing in campanella style and went against my natural inclination to hold notes for as long as possible to maximise the over-ringing. The muted effect was ultimately achieved, however, by using a combination of muting methods in both hands.

Firstly, I played the arpeggios along one string wherever possible (see figure 2.9.4) as the constant releasing of notes naturally made them sound shorter. Where this was not possible I worked on releasing the fretting hand fingers as soon as the note was plucked. Even so, the sound was still too bright and Milton suggested muting the strings with the side of the picking hand to create a pizzicato effect. I found, however, that the angle of the hand made it particularly difficult to

control the sound on the first and second strings as any slight movement of the hand resulted in the notes either buzzing or being completely muffled. Furthermore, the side-on position of the hand restricted the movement of the fingers - perhaps because this was such an unnatural position for my hand. After some experimentation I found the best sound was achieved by allowing the hand to flatten to a side-on position. However, rather than plucking with the fingers, all the notes were plucked with the side, or flesh, of the thumb. Plucking the strings with the flesh of the thumb produced a naturally more muted sound and this was further controlled by leaning the hand on the bridge to add to the partly muted effect. I found this technique quite demanding as the thumb had to play across all four strings and it was difficult to maintain the tempo and the rhythm.

Figure 2.9.4. The start of the D section of Insighted with the instruction to play the notes 'detached'



In order to maintain the tight groove of the piece Milton suggested I play to a click beat. Although I am used to practising with a metronome it was a very different experience for me to perform an entire piece in this way. At times I felt my musical instincts were being inhibited by the relentless click beat but the discipline was necessary in order for Milton to be able to add the electronics. The ukulele part was recorded while I was on holiday in Spain. I emailed Milton (in London) who then added the electronics.

The following instructions by Milton Mermikides are from the score (Viribus Music, 2019).

Electronics Instructions by Milton Mermikides (2019)

This simplified performance version can be performed with one stereo delay device (hardware or software). This should be set to a dotted quaver on left channel, and a crotchet delay on the right channel. (At 105bpm this would be ≈429ms and ≈ 571ms on left and right channel respectively).

Tape coloration (real or virtual) vs 'clean' digital are at the discretion of the performer. Volume should be set to match the dry signal with a feedback adjusted to hear 1 -3 repeats initially. The feedback however may be adjusted at will by the electronicist (or by performer's controller or pre-programmed) during the piece. Whatever volume this is performed, the dry uke and delayed patterns should be balanced as if in an ensemble. The hard panning of delays with a stereo PA can make for an immersive performance. Some reverb can be added for general blending and flattery, and additional delays, chorus, tremolos etc added as occasional sends so long as the groove is never lost.

Performance Instructions by Milton Mermikides (2019)

This piece lives and dies on a tight groove, the delays acting as a grid for the live performance. This may take some relaxed and enjoyable practice to really sit in the rhythmic feel. That said, there is much timbral and textural variety possible. No dynamics are indicated and are at the discretion of the performer, but a wide dynamic range with some extremely quiet - barely audible - sections can work particularly well.

The title is the word that our then 3-year old daughter used for 'excited'. Originally written for Bridget Mermikides for classical guitar and electronics, this version is adapted for ukulele. Sam Muir's adaptation and adoption of it is much appreciated.

Conclusion

This was a fun but challenging project. Working with electronics was something completely new for me and I would definitely like to have learnt more about how the effects were achieved. Milton and I did discuss further projects and had planned to meet in person but the COVID-19 pandemic, and subsequent lockdowns, put an end to that. It is, however, an area I would like to explore further in the future.

2.10. Damping/Muting Effects

Introduction

Damping or muting effects, also known as pizzicato, occur when the strings are deliberately prevented from vibrating in order to create a muffled, more percussive attack. As discussed in the previous chapter 'Ukulele and Electronics' this effect is the opposite of the campanella approach where over-ringing is an essential feature. To date I have only encountered damping techniques in the works of other composers as the main focus in my own compositions has been on how to apply campanella style. As the previous chapter already discussed the muting method used in *Insighted* by Milton Mermikides for ukulele and electronics, this chapter will look at the damping effects employed in *Meditation on a Small Scale II* by Dimitri van Halderen.

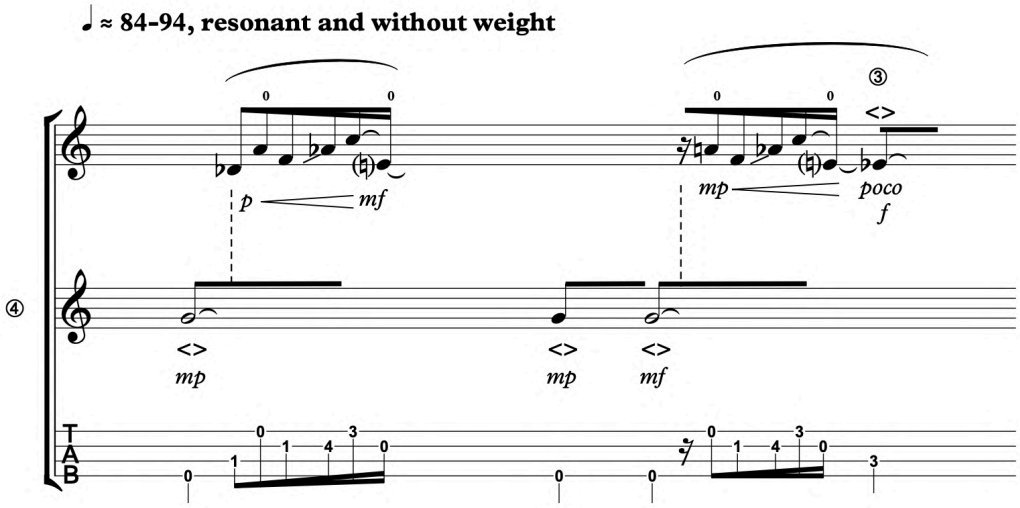
Meditation on a Small Scale II by Dimitri van Halderen

Meditation on a Small Scale II is the second piece of *Meditations on a Small Scale* by Dimitri van Halderen. I originally met Dimitri at the first meeting of the International Guitar Research Centre at the University of Surrey in 2014. Dimitri, who is based in Spain, is a classical guitarist whose musical interests include teaching, performing and composing. We kept in touch through social media and when I put out a call to composers in 2017 for new repertoire for the ukulele, Dimitri was the first person to send me a piece. Dimitri told me this slow and spacious work was inspired by my performance of John Tavener's *Chant* for solo guitar. Initially, *Meditations on a Small Scale* (2018) was one piece, however, a few months later Dimitri sent me a second piece. This highly innovative work was published in 2021 by Les Productions d'Oz. One of Dimitri's greatest innovations was to use a separate stave for the re-entrant fourth string. This really clarified the role of the fourth string when working on the score but, ultimately, the publisher decided not to use this layout due to printing considerations. For the purposes of this discussion I have used Dimitri's original layout.

Meditation on a Small Scale II[154] opens with slow and spacious clusters of notes. As can be seen in figure 2.10.1 the opening clusters are campanella-like in their movement across the strings. Each group of notes begins slowly but accelerates to create short bursts of over-ringing sounds which are then followed by slower single notes. As in Tavener's *Chant* the usual cyphers of time signature and bar lines have been omitted in order to create a score that both looks and sounds free and unrestrained.

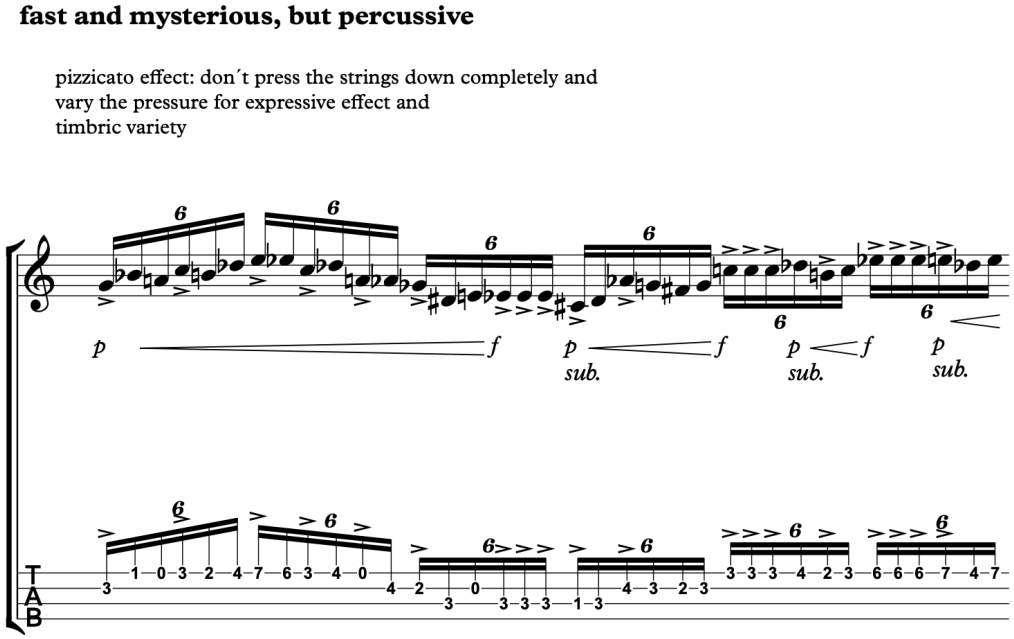
[154] For audio see Folder C.2. For full score see Folder B.4.

Figure 2.10.1. The opening of Meditation on a Small Scale II by Dimitri van Halderen



The slow and resonant opening is followed by a contrasting middle section marked ‘fast and mysterious, but percussive’. As in *Insighted* the percussive sound was challenging to achieve, particularly due to the fast moving semiquaver sextuplets. This section does not use the fourth string so the extra stave is not necessary.

Figure 2.10.2. The opening of the pizzicato section of Meditation on a Small Scale II



I was initially confused by the instruction - ‘don’t press the strings down completely and vary the pressure for expressive effect and timbral variety’. I was not familiar with this technique and initially found it difficult to achieve as my natural tendency was to over-press the strings in order to make the notes sound clearly. I tried using the thumb technique employed in *Insighted* but found the fast tempo was difficult to maintain and the repeated practice resulted in joint pain in my thumb. At this point I decided there had to be a better solution: one that muted the strings but did not cause discomfort in the hand.

Aware that other string players use a device (mute) to dampen the strings, I wondered if something similar existed for the ukulele. Unable to locate anything on the internet I decided to try and make my own mute. Realising that it would require a material firm enough to stay in place but soft enough to muffle the strings, I bought some heavy duty felt furniture pads and cut one into a rectangular strip. The strip was approximately the length of the bridge. I then made slits that aligned with the strings. The idea was that the felt mute could be placed over the strings near the bridge (see figure 2.10.3).

Figure 2.10.3. The felt mute I made for the pizzicato section of Meditation on a Small Scale II



The sound was surprisingly effective but there were two problems. Firstly, the slits needed to be approximately 5mm deep to maximise the muting effect and this meant the overall height of the mute, 12 mm, interfered with my right hand position. With more precise cutting tools I might have

been able to reduce the overall height by a couple of millimetres. As I only had kitchen scissors I settled on cutting an angle on the right side of the mute in to allow more room for the hand. The second problem was that it took some time to position the mute and could not, like a violin mute, be quickly moved into position. Stopping to position the mute in the middle of a performance would have been unacceptable. In a recording session, however, it was easy enough to cut and edit the muted section into the recording and subsequent YouTube video.

By the time of the first live performance at the University of Surrey on 18 May 2022 I had mastered the technique of lightly pressing the strings to create the muted effect. The sound was not as controlled or even as when using the felt mute but was all the more mysterious as it was not immediately obvious how the affect was being achieved.

Conclusion

Using damping or muting effects on the ukulele led to experimentation and innovation. My homemade felt mute, while crude, proved effective if somewhat impractical in a live performance. Nonetheless, it could be a useful tool in a recording session where editing is possible. The more percussive and less resonant sound of the muted passages of *Insighted* and *Meditation on a Small Scale II* were a complete contrast to playing in campanella style. I feel this is an area that I could have experimented more with in my own compositions but am grateful for the demands placed on me by other composers as this technique was definitely outside my comfort zone.

2.11. Alternate Tunings

Introduction

Alternate tuning occurs when one or more open strings are tuned to a different pitch. Many guitar traditions use alternate tunings; from the widely used drop D tuning, to slack key tunings, to the D-A-D-G-A-D tuning widely employed by Pierre Bensusan. On the ukulele alternate tunings have been used by players past[155] and present. Since the early twentieth century many players[156] have used the D6 tuning but this simply results in all the strings being tuned a tone higher meaning the strings retain the same pitch relationships. From fourth to first string these are: a falling 5th, a rising 3rd, and a rising 4th. Alternate tunings are more dramatic as they change the 'pitch-to-place mapping' (De Souza, 2017, 89) of the fingerboard and disrupt 'auditory motor connections'. (De Souza, 2017, 88)

De Souza cites the example of jazz guitarist Kurt Rosenwinkle who used alternate tunings as a way of *unlearning* learned perceptions of the guitar in order to find new and fresh creative outcomes.[157] Rosenwinkle described his approach as an act of 'voluntary self-sabotage' (De Souza, 2017, 88) for although the interface remained the same (De Souza, 2017, 89), well-known patterns such as scales and chord shapes produced new and surprising results.

Due to the limited range and familiar sound of the ukulele's typical C6 tuning I have made extensive use of alternate tunings in my compositions in order to explore and extend the tonal range of the instrument. Nine of the pieces in my portfolio (eight originals and one arrangement) involve altering at least one string. Thus, the expected soundscape of the fingerboard has been both subtly and radically changed. By simply lowering the E string to E ♭ a minor tonality is created (*Holloway*, *Black Raven*). Other tunings, such as in *The Red Angel* and *Nabu's Message*, are more extreme as they require at least two strings to be altered. Figure 2.11.1 provides a full list of the pieces and their various tunings. As all of the pieces are discussed in other chapters there is no need for further detail here. I would, however, make the observation that the inclusion of the tablature line facilitates ease of reading and makes these pieces more accessible and approachable.

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[155] The earliest example would seem to be *Spanish Fandango* by Ernest Kaai from his method *The Ukulele, A Hawaiian Guitar and How to Play It*, 1906/10. In this piece the A string is tuned down to G.

[156] For example: *The Bickford Method* for Ukulele by Vahdah Olcott-Bickford & Zahr Myron Bickford, 1920, Oliver Ditson Co., Boston

[157] De Souza used the example of Rosenwinkle's album *The Next Step* (2001) of which 4 of the 8 tracks use alternate tunings (De Souza, 2017, 88)

Figure 2.11.1. List of alternate tunings used in my compositions & arrangements



2.12. A Ukulele Concerto by DJ Roche

Introduction

The first ukulele concerto was composed in 1999 by Jim Beloff, the owner of Flea Market Music and the author of many ukulele songbooks. Beloff's *Uke Can't Be Serious* has been performed numerous times in the US by both high school and professional orchestras. The music is influenced by the Tin Pan Alley songwriters of the early twentieth century and includes vocalisations. In 2017 Beloff premiered his second concerto *The Dove Tail*. Both concertos were commissioned by Philip Ventre and demonstrate the potential of the ukulele to be both an expressive and an upbeat instrument. The pieces are approximately ten minutes long and represent a significant contribution to the ukulele repertoire.

The most substantial concerto for the ukulele, however, was premiered in June 2015 after the Hawaiian Symphony Orchestra, in collaboration with ukulele virtuoso Jake Shimabukuro, commissioned a ukulele concerto by composer Byron Yasui. In an interview conducted by *Ukulele Magazine* (April 15, 2019)[158] Yasui expressed his belief that the ukulele should be taken seriously, whether in the realms of classical music or jazz. One of the motivating factors in writing his concerto was a desire to prove that the ukulele is not merely a novelty. The concerto features syncopated rhythms and campanella passages. An arrangement of the movement *Tritone* for ukulele, bass and drums is featured on Shimabukuro's album the *Nashville Sessions* (2016).

In the UK the first ukulele concerto was composed by Lindsay Higgs and premiered in 2017 by Ben Rouse (soloist) from the Ukulele Orchestra of Great Britain with conductor Paul Cox.

New Horizons

Following on from the success of David John Roche's solo tremolo piece *In Every Heart* (2021) Roche and I agreed to collaborate on a ukulele concerto project. The project is ongoing and the piece will not be performed until after the completion of this PhD. Nevertheless, the concerto is mentioned here as evidence of my continuing commitment to collaborate with composers in order to further the horizons of the ukulele. Bringing the ukulele into the realm of the symphony orchestra is undoubtedly an ambitious project. The limited range and short attack of the ukulele presents considerable musical and acoustic challenges. We predict that the greatest challenge, however, will be convincing a conductor how a small, unconventional instrument such as the ukulele can successfully navigate a large scale, traditional form such as the concerto. Both Roche and myself

[158] https://ukulelemagazine.com/stories/ukulele-byron-yasui

are excited at the prospect of uniting the small (ukulele) with the large (orchestra) but are deeply aware of the challenges we face in getting this work performed.

As the project is a collaboration between myself (soloist) and David John Roche (composer) I decided to conduct a short, three question interview with Roche in order to get his insights. The questions and answers are presented below. A draft of the score is included in Folder B.4.

SM: When discussing the ukulele concerto you've always expressed a desire to pitch the full force of the orchestra against the ukulele and I've often wondered how this could be successfully achieved given the short attack and limited dynamic range of the ukulele. But I also wonder if there's a clue in your orchestral piece 'The Waves I Ride Have Never Yet Been Crossed' which won First Prize in the 'Dante 700 Orchestral Composition Competition' in 2021. When listening to this piece I find myself swept along on a rising wave of music which, just as it feels ready to break, is suddenly subdued or reigned in. I'm talking about the midway point when the strings and the percussion are steadily building and then suddenly the strings withdraw to leave just the rumbling of the percussion and a sense of something monumental continuing but beyond our reach, or hearing. Your control of the orchestration is really impressive and I wondered if you could tell me about your orchestration for the ukulele concerto and how you envision balancing the ukulele with the orchestra.

DJR: I think the long and short of it is that much of the balance is done in the room (or, if it's a recording, in the studio). The best results (I find) emerge when a conductor takes the lead – they know their orchestra – and asks separately for input from the composer. When input is entirely from one party, I feel that the music is likely to fall short of what it should be. Cumulative effort goes further, and everybody needs the space to be their best.

As for bringing the ukulele out in the texture, I've hedged my bets! There are segments that will require amplification and segments that won't. I feel that contemporary classical music can be naïve with respect to the role of microphones and mixing and, if we were all more upfront, we could use it to lift performances (which has been happening for goodness knows how long anyway, whether people want it or not). I pride myself on creating orchestration that does something interesting or rings out clearly ('The Waves' is a really bad example of clear orchestration – it's dense and muddy on purpose, falling into the interesting category!).

SM: *I was absolutely delighted when I received the concerto and found you have used 'In Every Heart' for the opening movement. It is quite unusual, I think, to use the four note Tárrega style tremolo technique in a guitar concerto. I wondered if you could talk about some of the other techniques you use in the ukulele part and to what extent, if any, you were influenced by guitar techniques.*

DJR: I wanted to write with the uke and away from the uke and, I have to say, it was quite funny seeing you work through the first draft. I thought it was immediately clear which moments were written at the ukulele and which ones weren't. The ones I'd written without the ukulele were unplayable (in parts) and dotted with errors. Without the instrument in your hands, there will be technical problems. Every part needs to be checked by a player… and the ukulele's a nightmare to write for anyway. It's a very restrained, compacted, and intricate musical instrument.

The more nimble movements were derived from my own electric guitar playing. On electric guitars, pickups allow for clearer, longer legato passages – the fretting hand will be picked up loudly even if you're playing at a quiet dynamic. Translate this to the ukulele and you lose some of the electric instrument's capacity to ring out, but you don't lose the space to be nimble. I think my solution was to break up longer legato passages with re-plucked strings. Uke can be fast and it's much easier to perform long, unusual stretches… so it allows for very peculiar opportunities for technical development and adaptation when coming from the guitar. It's exciting.

SM: I think I'm correct in saying that we both feel very positive about this concerto and are confident that once we can get it performed it will be a success. Interestingly, however, most of our discussion has revolved around how we are going to overcome the deeply rooted stereotypical views that exist in the UK that the ukulele is merely a novelty or a comedy prop. While I prefer to avoid the attitude that I am out to prove that the ukulele can be a serious instrument, it seems inevitable that whenever I perform this is, on one level, what I am actually doing. The problem is actually quite complex because when some people see the ukulele they immediately expect George Formby and are disappointed to get Bach; and other people see a ukulele and think 'Eek, George Formby!' and are surprised and delighted to get Bach. How do you think we, as performers and composers,

should address these attitudes? And what can we do to convince an orchestra that they should play a ukulele concerto?

DJR: I don't think it helps to placate negative views – to try to meet people on their downbeat and degrading terms is to negate and denigrate the instrument. I'm being a bit over-the-top here (perhaps), but I feel there is truth in what I'm saying. I think classical music will eventually come around to the ukulele because they won't be able to (or won't want to) ignore its success and financial value. Contemporary Classical Music is an industry in trouble, and it needs to connect with its audiences. The ukulele has an audience. It has stacks of people who love it. If contemporary classical musicians could cultivate these kinds of audiences, they'd have done it aeons ago… but they can't (yet) for whatever reason. Even those who work on larger YouTube channels (see: David Bruce) act more like music communicators than composers (not an insult, I think we need more of it). Throwing all of my comments aside, I've played enough music over the years to know that you'll never please everyone and, after playing countless rock and metal clubs around the UK as a teenager, comments about perceived inadequacies in my harmony and ukulele playing just read as embarrassingly bourgeois. I genuinely find it hard to care. Besides, I'd rather be ridiculed and myself, than praised and disingenuous!

Conclusion

Roche's attitude and approach to the ukulele is refreshing and reflects how the ukulele is, slowly but surely, making an impact on the contemporary classical music scene. While there is still some way to go, it is an exciting prospect to be able to continue my work beyond the confines of this doctorate and I look forward to working with such an inspiring and bold composer.

2.13. Overview of the Didactic Material in the Portfolio

Introduction

As previously discussed in the methodology section, 'the quartet of selves' has represented a holistic approach to classical ukulele by embracing my roles as a performer, composer/arranger, researcher and teacher. While the roles of performing artist and published composer are considered more glamorous, I have always regarded my work in teaching classical ukulele as one of my most valuable contributions to the ukulele.[159] The small selection of didactic work included in this portfolio is testimony to how the 'quartet of selves' has interacted to produce resources specifically aimed at students interested in learning how to play classical and campanella styles (*The Classical Ukulele Method, 100 Arpeggio Exercises for Ukulele*), as well as those who want to further develop their skills (*12 Progressive Studies, 21 Studies for Ukulele*). These works were created in response to the lack of resources for classical ukulele highlighted by my initial review (2017) of the extant publications for classical and fingerstyle ukulele in 2017. Nevertheless, despite the notable increase in resources by myself and others, the majority of well-known classical ukulele players all come from a background in classical guitar.[160]

Background

I started playing the ukulele in 2012 and my arrangements and performances soon became known on YouTube. By 2016 I was being invited to perform and give workshops on classical ukulele at major festivals across the UK (GNUF, UKE East, BUG Day, HUG).[161] Like John King, I adapted my guitar playing skills[162] to suit the ukulele. While King mostly produced advanced repertoire, my own work has sought to create material for players of all levels. King created beautiful campanella arrangements of works by Bach but never wrote about the technical challenges involved in playing this repertoire, or offered advice on how ukulele players might develop the fundamental skills. In 2019 Les Productions d'Oz published my *The Classical Ukulele Method*. The method outlines the basics of a classical fingerstyle approach - beginning with alternating fingers and how to play

[159] A full discussion on ukulele pedagogy has not yet been undertaken and is beyond the scope of this thesis. Anne Ku has done a masters dissertation called 'Paradox of Simplicity: Songsheets for Participative Music Making in Ukulele Groups' but, as the title suggests, this specifically focuses on how ukulele groups use song sheets as a way of learning and participating.

[160] John King, Tony Mizen, Daniel Ward, Jeff Peterson, Paul Mansell, Giovanni Albini, Donald Bousted, Choan Gálvez, Thijs Kevenaar, Rob MacKillop, David John Roche, Michael Waker, Elisabeth Pfeiffer, Samantha Muir, Mustafa Kamalidden

[161] See Folder D for a full list of events & festivals.

[162] I studied classical guitar (performance studies) at the Royal College of Music, London and became an ARCM in 1996

arpeggios, and leading to the more advanced technical demands of campanella playing. To the best of my knowledge this is the first method which fully embraces a classical approach to the ukulele.

The Classical Ukulele Method[163]

My ideas for the method were based on my classical guitar knowledge but adapted to suit the ukulele. As shown in figure 2.13.1 my picking hand is more side-on when playing the ukulele than it would be playing the guitar. In this position the side of the thumb, rather than the nail, is used to pluck strings 3 and 4 and the resulting mellow tone creates what I call the mock-bass effect.[164] As shown in the photos (figure 2.13.1) the right hand wrist is relaxed, rather than arched, and the strings are plucked with a combination of flesh and nail (fingers), or just flesh (thumb). Meanwhile, the fretting hand assumes a position, as in classical guitar, with the fingers curling around the fingerboard to press the frets. As in classical guitar technique there is a clear gap between the hand and the edge of the fingerboard. Due to the small size of the instrument I find it beneficial to use a strap as this supports the instrument while allowing the hand the freedom to move up and down the fingerboard. Using a strap is, however, controversial as many players prefer to allow the neck of the ukulele to rest between the thumb and first joint of the index finger.

Figure 2.13.1. The hand positions as shown in The Classical Ukulele Method (2018)



The first part of the method focuses on playing open strings in order for the student to become familiar with the *pima* system. Since 2012 I have actively promoted this system, both as a

[163] For score see Folder B.1.

[164] While the ukulele lacks a true bass the thumb often assumes the role of playing 'bass' notes. By using the flesh of the thumb, rather than the nail, the sound is fuller and more rounded thus producing a timbral difference that I call the mock-bass.

fingerstyle approach to playing the ukulele, and a way of differentiating the picking hand fingers *(pima)* from the fretting hand fingers (1,2,3,4). The exercises shown in figures 2.13.2 and 2.13.3 demonstrate how simple open string exercises allow the student to focus on the picking technique. The two main techniques are arpeggio technique (see figure 2.13.2) and playing repeated notes with alternating, or walking fingers (see figure 2.13.3).

Figure 2.13.2. An open string arpeggio exercise from The Classical Ukulele Method

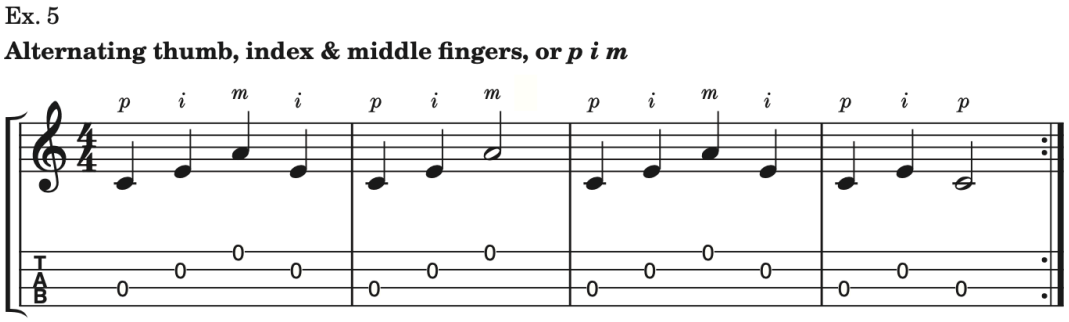
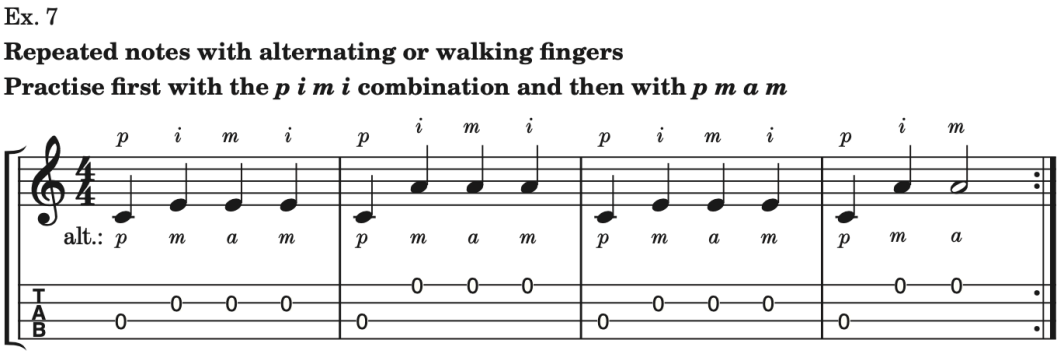


Figure 2.13.3. A walking finger exercise using open strings from The Classical Ukulele Method



Simple melodies in C major are introduced in Part 2 of the method. Right from this early stage students are encouraged to explore how the note G can either be played at 0,4 or 3,2. In Frère Jaques (see figure 2.13.4) two lines of tablature are used to clarify the changes in position of the G. The first tablature line positions all of the Gs at 3,2 while the second line positions them at 0,4. The differences in the picking hand fingering are clearly indicated (see bar 5). In the second tablature line the open G is played by the thumb creating the pattern *p-m-p-i-m-i* across strings 4-1-4-2-2-3. In the first tablature line the pattern is *i-m-i-m-i-p* across strings 2-1-2-2-2-3. By using the open G

the melody moves across the strings to create the over-ringing, or campanella effect, but is more demanding for the picking hand as two fingers and the thumb are required.

Figure 2.13.4. Frère Jaques from The Classical Ukulele Method with 2 options for playing G



In Part 6 of the method students begin playing short pieces, such as *Silent Night* (see figure 2.13.5), in campanella style. Despite the importance of campanella playing the method is not exclusively devoted to this style but incorporates a variety of fingerstyle techniques such as arpeggiation and playing two intervals together. My intention was that ukulele players would be able to develop a range of fingerstyle skills in a progressive and organised manner. By Part 7 students begin to play simple repertoire pieces[165] by Carulli. In the final section (Part 8) the extra material includes a number of scales. *The Classical Ukulele Method* (2019) is the first published method that encourages and enables ukulele players to develop the fundamental techniques needed to play in the classical ukulele style.

[165] These include arrangements of *Waltz* and *Ecossaise* from Op.121 by Carulli

Figure 2.13.5. Silent Night from The Classical Ukulele Method arranged in campanella style



100 Arpeggio Exercises for Ukulele[166]

One of my earliest and most popular self-published ebooks for the ukulele was *The Art Of Arpeggios* (2015). It was written in response to the many requests I received, from workshop participants and through social media, on how to develop finger picking techniques on the ukulele. The booklet, which comprised a number of arpeggio exercises on open strings, focused on learning *pima* technique. As previously mentioned this was a new concept for many ukulele players, particularly the idea of using the ring, or *a*, finger. Due to the popularity of the ebook it was revised in 2019 and subsequently published by Les Productions d'Oz in 2019. The new version expanded on the open string idea by pairing each open string exercise with a second version in which the arpeggio pattern was applied to simple chord shapes (see figure 2.13.6). The initial idea of writing a book of arpeggio exercises was based on Giuliani's *120 Exercises for Right Hand* for classical guitar. I had found the Giuliani exercises particularly beneficial as a young student and wanted to create a similar resource for the ukulele.

[166] For score see Folder B.1.

Figure 2.13.6. Exercises 1 & 1a from 100 Arpeggio Exercises for Ukulele (one using the open strings, the other using simple chord shapes)



12 Progressive Studies[167]

Many of the pieces in *12 Progressive Studies* were originally written as workshop material. Composing pieces, as opposed to arranging pieces, enabled me to control the technical demands of each piece while creating repertoire that used the idiomatic features of the instrument. For example, in the first study, called *Adrian's Song* (see figure 2.13.7), the thumb moves between strings 3 and 4 while the index and middle fingers play repeated notes on the other strings. A common myth about the ukulele is that in fingerpicking technique each finger is assigned to, and remains on, one string. My approach encourages greater flexibility as the fingers and the thumb become accustomed to playing on different strings. By keeping the fretting hand fingering simple by using a lot of open strings the student can fully focus on developing these picking techniques.

Figure 2.13.7. The opening of Adrian's Song from 12 Progressive Studies



[167] For score see Folder B.1.

By the eighth study, *Flow River Flow,* the focus is on the fretting hand moving up and down the fingerboard while the picking hand plays a repeated arpeggio pattern (see figure 2.13.8). Ukulele players often refer to the higher positions as the ‘dusty end’ as they rarely venture beyond the fifth fret. In *Flow River Flow*, fingers 1 and 2 of the fretting hand remain planted on strings 2 and 3 in a fixed pattern while the positions constantly change. The use of repeated patterns allows the student to gain confidence along the full length of the fingerboard.

Figure 2.13.8. The opening of Flow River Flow from 12 Progressive Studies



21 Studies for Ukulele[168]

The *21 Studies for Ukulele* were a further opportunity to explore how classical guitar techniques might be applied and adapted to the ukulele. The aim was to create a set of miniature pieces which were musically stimulating[169] but focused on techniques specific to the ukulele. While I utilised characteristics of the ukulele, such as the re-entrant tuning, I also introduced elements that were less familiar, such as plucking rather than strumming the chords. A selection of examples is given in order to demonstrate the range of techniques used in the studies.

Study No.1 used the open strings to create simple campanella effects but was written in the more unusual time signature of 2/2 and contained elements of syncopation (see figure 2.13.10). The notes use many open strings as the purpose of this study is for the student to focus on controlling the rhythm and pulse with the picking hand.

[168] For score see Folder B.1.

[169] The collection was inspired by the *Études Simples* for guitar by Leo Brouwer.

Figure 2.13.10. The opening of Study No.1 from 21 Studies for Ukulele



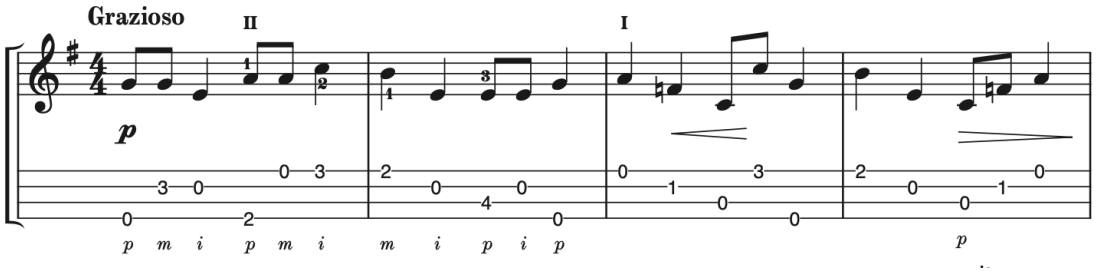
Study No.III on the other hand adopts a linear approach to playing a melody. No open strings are used and the focus is on maintaining a logical fretting hand fingering while shifting between second and fourth position (see figure 2.13.11). The picking hand, meanwhile, uses alternating fingers rather than arpeggio technique.

Figure 2.13.11. The opening of Study No.III from 21 Studies for Ukulele



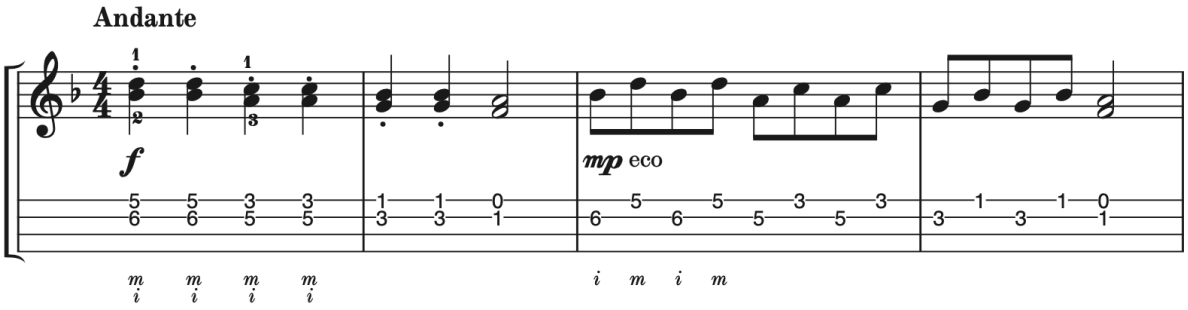
Study No.VI explores how the same note can be played on a different string. This is most apparent in the tablature where, for example, the first G is played at 0,4 and the second at 3,2 (see figure 2.13.12). This promotes knowledge of the fingerboard while allowing the student to recognise, both positionally and aurally, how the same note can be played on different strings.

Figure 2.13.12. The opening of Study No.VI from 21 Studies for Ukulele



Study No.XII involves a number of challenges. In bar 1 two note intervals are plucked with the *i* and *m* fingers while the fretting hand moves from fifth to third to first (b.2) positions (see figure 2.13.13). The same intervals are then played as alternating notes (see bars 3 & 4 of figure 2.13.13). An echo effect is created by the change of dynamic from *forte* to *mezzopiano* while the rhythm changes from crotchets to quavers. An additional challenge is the change from staccato in bars 1 and 2, to legato in bars 3 and 4.

Figure 2.13.13. The opening of Study No.XII from 21 Studies for Ukulele



Study No.XX is written in campanella style but with the added twist of using the unusual time signature of 7/8. The technical and rhythmical challenges of this study make it one of the most difficult pieces in the collection.

The semiquaver figures moving across the strings require precision from both hands in order to create the campanella effects. The fretting hand has the additional challenge of pull-offs. Unlike normal pull-offs when one note is played and the next is sounded by the fretting finger releasing (or pulling) the string, a delayed pull-off occurs when the first note is plucked and held for one or two notes before the string is released. For example, in bar 6 of figure 2.13.14 the delayed pull-off begins with the C on beat four which is plucked and held with finger 2 while the following A is played at 5,2. Finger 2, which has maintained its position on C, then executes the pull-off in order to sound the B (2,1) on beat six.

A similar type of delayed pull-off is used in bar 8 but this time the E at 7,1 is held while the following C and D, which create a little campanella figure, are played at 8,2 and 7,4. The open A on the last beat of the bar is then sounded as the second finger, which has been planted on the E, now releasees the string (see bar 8, figure 2.13.14).

Figure 2.13.14. Bars 5 to 12 of Study No.XX from 21 Studies for Ukulele



The final study (*No.XXI*), which was influenced by *Etude No.XII* for guitar by Villa-Lobos, involves a chord pattern moving rapidly up and down the fingerboard. As shown in figure 2.13.15 the three note diminished chords form a typical shape, but, less typically the chords are plucked with the *ima* fingers, rather than strummed, while the thumb plays a separate open G drone. At bar 16 the chords change to four notes and are strummed with the index finger (see figure 2.13.16). This final study is

very different, both technically and musically, from *Study No.XX*, and was an opportunity to use chords in a more classical context.

Figure 2.13.15. The G drone and 3-note chords at the start of Study No.XXI from 21 Studies for Ukulele

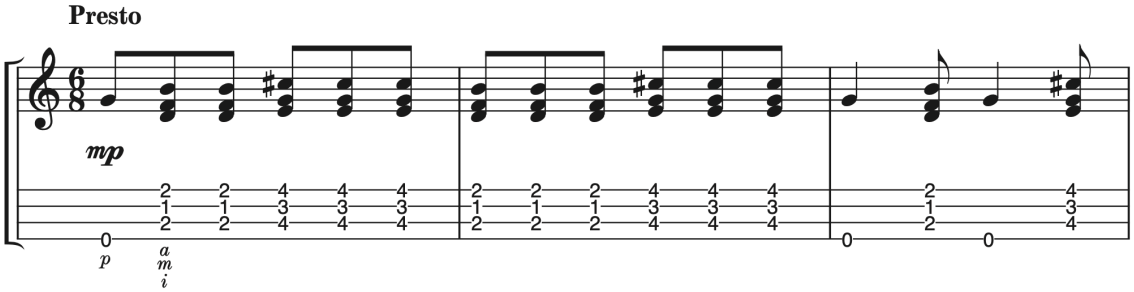
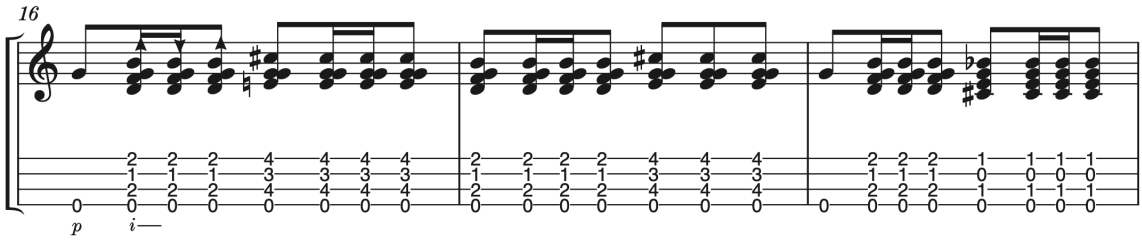


Figure 2.13.16. The 4-note strummed chords in Study No.XXI from 21 Studies for Ukulele



Conclusion

The didactic material presented here represents a combination of experience, knowledge and innovation. I used my experience of teaching both the guitar and the ukulele, and of performing on both instruments, to create material that addresses a range of technical and musical challenges. The over-riding aim was to find new and innovative ways of adapting guitar techniques to the ukulele in order to more fully understand the technical and musical possibilities of the instrument. The process of writing these studies also gave me the opportunity to reflect on aspects of my own playing which had become automatic. Ultimately, rethinking and reassessing technical and musical challenges on a fundamental level led to a deeper understanding and appreciation of the instrument. This, in turn, enhanced my own ability to empathise and communicate with students.

CONCLUSION

A Celebration of the Underdog

One of my favourite primary school memories is sitting under the shade of a big, old eucalyptus tree listening to my teacher recite the bush ballads of A.B. “Banjo” Patterson (1864-1941). Patterson, a much loved Australian poet, is best known as the author of *Waltzing Matilda* (1885) but my favourite poem has always been *The Man From Snowy River* (1890). It’s the classic tale of the underdog, or in this case the ‘stripling’ on the ‘small and weedy beast’, who surprises everyone when he and his weedy pony turn out to have unexpected and extraordinary abilities. Although I never specifically had this poem in mind when I applied to do a PhD on the ukulele, I’ve always been inspired by the idea that appearances can be deceptive. Small guitars, like small horses, should never be underestimated.

In *The Man from Snowy River* the ‘stripling’ is initially told, ‘that horse will never do for a long and tiring gallop…those hills are far too rough for such as you.’ Eventually the lad and his ‘small and weedy beast’ are allowed to join the fray. They then go on to defy all expectations by single-handedly rounding up a herd of run-a-way horses in treacherous mountain terrain. Their moment comes when all the other riders, including the much lauded Clancy of the Overflow (‘no finer horseman ever held the reigns’), give up the chase when the going gets too tough. Just as all seems lost and the old man mutters fiercely ‘we may bid the mob good day, no man can hold them down the other side’, the man from Snowy River comes charging in. By the end of the poem *the man* is declared ‘a house-hold word today’ while the little pony is ‘blood from hip to shoulder from the spur’.

I’ve always felt that the real hero of the story is the battered little horse. After all, it is the ‘small and weedy beast’ who gamely plunges down the mountain ‘like a torrent down its bed’ while the other riders ‘stood and watched in very fear’. It’s lump-in-the-throat stuff as ‘he sent the flint stones flying, but the pony kept his feet, he cleared the fallen timber in his stride.’ Weaving a way through ‘the stringy barks and saplings’ it is the courageous little pony (at the end he can scarcely raise a trot) who carries his rider to glory. I’ve always thought the pony should have had more credit. But, I suppose *The Pony from Snowy River* doesn’t quite have the same ring to it.

To be fair, at the critical moment it is *the man* who realises that his best tactic is to just sit tight and ‘let the pony have his head.’ The simple acknowledgement that the pony knows things that the rider doesn’t, is oddly reminiscent of De Souza’s idea that instruments know things for their

users. The pony *knows* how to safely navigate a mountain descent and, looking back on this research project, I come to the conclusion that the ukulele has always *known* how to be a classical instrument. One only needs to look at the four course guitar works of Adrian Le Roy, or the machete pieces of Candido Drummond, to appreciate the long and distinguished presence of small four string guitar-shaped instruments throughout the history of Western music. Furthermore, most of the original compositions in this portfolio have been written with instrument-in-hand. The ukulele, like the pony, *knows* where I should place my fingers in order to find music that is both pleasing and idiomatic. The pony *knows* what is expected of him. Does the ukulele *know?* I'm not sure, but it often feels like the ukulele is trying to tell me things.

When I play the ukulele is there some echo of the living tree that finds its voice in me? Talking trees may seem like the realm of fantasy but scientists, such as Suzanne Simard, are revealing how trees work and communicate with each other through complex underground systems. Science is giving us the knowledge and the terminology to understand and articulate many things that once seemed fantastic. Of course, the little pony did not have any magical powers, he simply knew where to put his feet because he was born and bred in the mountains. And the rider knew he could trust his steed. Perhaps the most important lesson here is believing in our instrument, be it pony or ukulele. It was belief in the ukulele that set this research project in motion and, to be honest, the ukulele never ceases to surprise me. The exploration into tintinnabuli technique and the resulting arrangements and compositions are a prime example. My early fears that there would be enough to fulfil the academic requirements of a PhD were quickly extinguished by the realisation that there were, and still are, many research avenues left to pursue.

The new works for classical ukulele, many of which have been published, are helping to establish the ukulele as a contemporary classical instrument. The recording *New Works for Classical Ukulele* was deliberately labelled *Volume 1* as other volumes will surely follow. I am mindful, however, that this portfolio is a microcosm of the ukulele's true potential and that the insights presented here have focused exclusively on the classical ukulele style. While this is an exciting and ever growing area of study, it is one part of a much larger field; a field that is only now opening out to include further research into the ukulele. Nevertheless, it is exhilarating to have been the first to have plunged down the mountainside of academia with this 'small and weedy beast'. Odd that it is only after years of study and practice that we finally arrive at a point where we are confident enough to 'let the pony have his head'. Undoubtedly, the real hero of this story is, and will always be, the ukulele.

APPENDICES

Table of Techniques

The table provides an overview of the different techniques used in the new works.

Table of Techniques

	Camp.	Tintin.	Arp.	Chord Melody	Chords	Trem.	Harm.	Alt tuning	Micro-Tonal	Elect.	Damp.	5 String
The Falling Rain	■		■	■	■		■	■				
Dowie Dens	■		■	■	■	■	■					■
Ravens	■		■	■				■				■
Mercurius	■		■			■						
Flower Sleeps at 4	■		■				■					
Binter	■		■									
Whistlepipe Gully			■					■				
Tell Me Why	■						■					
Holloway		■						■				
Patientia		■						■				■
The Red Angel		■				■		■				
LockDown		■						■				
Long & Narrow Rd			■	■	■				■			
Evocations			■		■	■						
In Despair	■					■						
Meditations	■		■	■	■						■	
In Every Heart						■						
Common Blue		■			■						■	■
26 Days	■		■		■							
Fragile	■		■	■	■		■					■
Parting Voices	■		■									
Insighted			■	■						■	■	

Catalogue of Scores

The catalogue of scores includes published and unpublished works, arrangements, original pieces and commissions all of which are contained on Folder B. A summary of the contents is provided in the tables below.

Original Works by S.Muir Published by Les Productions d'Oz		
TITLE	**Catalogue Number**	**Date of Publication**
Variations on The Dowie Dens of Yarrow	DZ3099	2018
The Falling Rain	DZ3288	2019
A Conspiracy of Ravens	DZ 3520	2020
Mercurius Suite in 3 Movements	DZ 3471	2021
Six Pieces for Ukulele	In process	2023

Arrangements by S.Muir Published by Les Productions d'Oz		
TITLE	**Catalogue Number**	**Date of Publication**
12 Progressive Lessons by Sor	DZ3114	2018
12 Traditional Tunes for Fingerstyle Ukulele	DZ3375	2019
From Renaissance to Late Classical Era	DZ3776	2021

Didactic Material by S.Muir Published by Les Productions d'Oz		
TITLE	**Catalogue Number**	**Date of Publication**
12 Progressive Studies	DZ3086	2018
21 Studies for Ukulele	DZ3173	2018
100 Arpeggio Exercises for Ukulele	DZ3264	2019
The Classical Ukulele Method	DZ3420	2019

Unpublished Original Works by S.Muir	
TITLE	**Date**
Holloway	2020
Patientia	2020
The Red Angel	2021
LockDown	2021
Moths in the Moonlight	2022
Blinter	To be published 2023
Flower Sleeps at 4	To be published 2023
Tell Me Why	To be published 2023
Whistlepipe Gully	To be published 2023

Unpublished Arrangements by S.Muir		
TITLE	**Composer**	**Date**
Let Me Call You Sweetheart	Friedman & Whitson	2015
Uzun Ince Bir Yoldayim	Asik Veysel	2018
Für Alina	Arvo Pärt	2020
On the Nature of Daylight	Max Richter	2022
The Blarney Pilgrim*	Trad.	2023
Prelude BWV 1007*	JS Bach	2023

*These pieces will be included in the forthcoming University of West London's *Ukulele Syllabus Grades 6-8.*

Commissions/Collaborations: Solos			
COMPOSER	**Title**	**Date**	**Publisher**
Loretta Notareschi	Evocations	2019	Disegni Music
Choan Gálvez	In Despair	2020	Choan Gálvez
Dmitri van Halderan	Meditations on a Small Scale i & ii	2021	Productions d'Oz
John David Roche	In Every Heart	2021	Ukulele Modern
Choan Gálvez	Common Blue	2021	Choan Gálvez
Choan Gálvez	Twenty-six Days	2022	Choan Gálvez
Marius Fleck	Fragile	2022	Marius Fleck
James Else	Parting Voices	2022	James Else

Commissions/Collaborations: Ensemble			
COMPOSER	**Title**	**Date**	**Publisher**
Tom Armstrong	Shadow Variations	2018	Tom Armstong
Milton Mermikides	Insighted for ukulele & electronics	2020	Viribus Music
John David Roche	Ukulele Concerto (draft)	2022-3	Unpublished

Catalogue of Recordings

The catalogue of recordings includes audio (see Folder C) and video[170] of the new works for classical ukulele. Please note that all of the machete material is organised thematically in Folder A.1. This includes the CD *The Music of Candido Drummond de Vasconcelos* for machete and guitar. Due to copyright restrictions the Portfolio Folders are not available to the public. However, all the audio and/or video is available on Spotify, Bandcamp and/or YouTube. The scores are available from either Les Productions d'Oz or the relevant composer's websites (see below).

AUDIO

• *New Works for Classical Ukulele Volume 1*

Original works composed and performed by Samantha Muir on soprano ukulele, four string tenor ukulele and five string tenor ukulele.

Track List

1. *The Falling Rain*
2. *Variations on The Dowie Dens of Yarrow*
3. *A Conspiracy of Ravens*
    1. *White Raven I & II*
    2. *Blue Raven I & II*
    3. *Black Raven I & II*
4. *Mercurius Suite*
    1. *Mercury*
    2. *Return from the Underworld*
    3. *Nabu's Message*
5. *Lines of Flight*
    1. *Holloway*
    2. *Patientia*
    3. *The Red Angel*
    4. *LockDown*

• Commissioned Pieces

1. *Evocations by Loretta Notareschi*

[170] SamMuir Ukulele YouTube Channel: https://www.youtube.com/channel/UC_6wKay150LzInkitgTkvYg

2. *Fragile by Marius Fleck*
3. *In Despair by Choan Gálvez*
4. *In Every Heart by David John Roche*
5. *Parting Voices by James Else*
6. *Meditation on a Small Scale I by Dimitri van Halderen*
7. *Meditation on a Small Scale II by Dimitri van Halderen*
8. *Insighted by Milton Mermikides*
9. *Twenty-Six Days by Choan Gálvez*
10. *Common Blue by Choan Gálvez*

- A Selection of Pieces from the Methods of Ernest Ka'ai

1. *Banjo Schottische*
2. *El Recuerdo*
3. *Haere*
4. *Hone Aka Wai*
5. *Spanish Fandango*
6. *Ka Wehi*

- A Selection of Pieces for the Four Course Guitar played on a five string ukulele

1. *Passameze from Selectissma Elegantis pub. Pierre Phalèse, 1570*
2. *Bransle de Poitou by Adrian Le Roy, Tiers Livre de Guiterre, 1552*
3. *Troisieme Branle by Adrian Le Roy, Premier Livre de Guiterre, 1551*
4. *Bransle de Poitou by Adrian Le Roy, Tiers Livre de Guiterre, 1552*

- Other Recordings

1. *Prelude BWV 1007 by JS Bach arr. for ukulele by S.Muir*
2. *Brian Boru's March*
3. *The Blarney Pilgrim*
4. *Audio samples of campanella & linear scales*

VIDEO

Please refer to SamMuir Ukulele YouTube channel.

Additional Material

The material in Portfolio Folder D demonstrates how I have promoted the classical ukulele through social media and at various festivals, conferences and other events.

The lists include:

- List of Performances
- List of Ukulele Festivals
- List of YouTube Performance Videos
- List of YouTube Tutorials
- List of Other Events

Contributing Composers

I am hugely grateful to all of the composers who have contributed to this portfolio. For more information on them and how to access their scores please visit their websites.

Loretta Notareschi: www.lorettanotareschi.com

Choan Gálvez: www.choan.es

David John Roche: www.davidjohnroche.com

Milton Mermikides: www.miltonline.com

James Else: www.jameselse.co.uk

Tom Armstrong: www.tomarmstrongcomposer.com

Dimitri van Halderen: see www.productionsdoz.com

Marius Antonin Fleck: see FaceBook and YouTube

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